Supplementary Note

EMToolbox uncovers biased co-option of embryonic EMT programs across cancers

Caleb C. Reagor, CO-AUTHORS, and Igor Adameyko

MesenContext: Run summaries

Analysis summaries for every run in the run ledger (Supplementary Methods), ordered by category, disease, dataset, and run. Each page reports the gate outcomes, manifold geometry, and module signatures produced by the agent for that run. These summaries were harmonized after the full analysis, and their results are consistent with those generated during each run. Two pruned runs are absent: Chen2020 (run 6 of 7) and Ji2020 (run 9 of 10).

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Wang 2019 Glioblastoma — analysis summary

RUN 1 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/10 samples (≥237 cells) intermix on one continuous manifold; no per-patient islands (16 detached cells removed); Harmony drops same-sample neighbour fraction 0.99→0.63, iLISI 1.01→2.00.

Gate 2 — module geography → GO (non-branching)

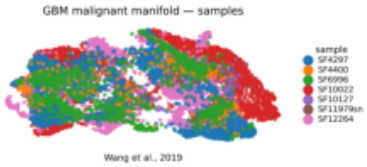
Single developmental axis, not a bifurcation: Gast & pSom co-vary weakly (r=+0.15), Crest is the inverse terminal gradient (vs pSom r=−0.87; Crest-high decile at pseudotime 0.86). No competing arms.

Gate 3 — hub cross-repression → NA

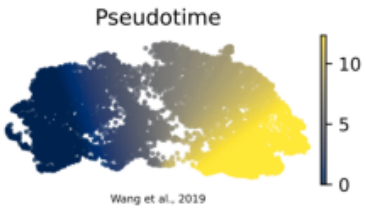
Branching-only test. Non-branching manifold has no Gast/pSom hub bifurcation, so the mutual-exclusivity correlation is not defined here.

STEP FIGURES

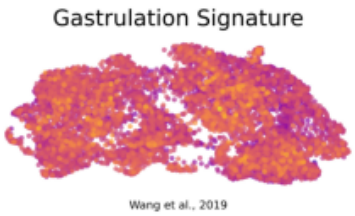
Manifold — samples



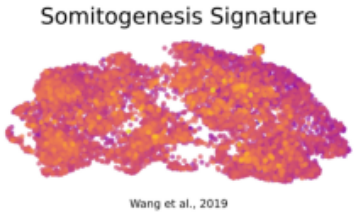
Pseudotime



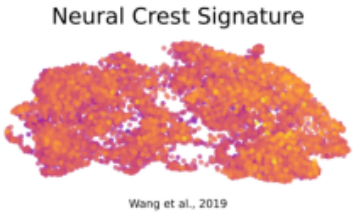
Gastrulation



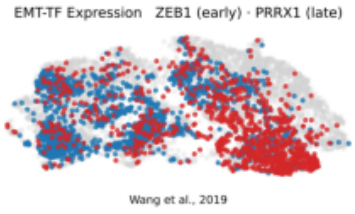
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019), 10x — 10 patients / 16,317 cells (master scope); 9,916 malignant cells, 9,634 after thresholding thin samples and removing detached islands.

Verdict.
Pattern reproduces (non-branching). A single embryonic axis runs from a gastrulation/somitogenesis-high origin to a neural-crest-high terminus; pseudotime is developmental, not proliferative (cycle-score rho +0.02). EMT-TF axis ZEB1 (root) → PRRX1 (terminus).

Parameter choices

Min cells/sample	≥237 (largest multiplicative gap; keeps 7/10)	**PC count**	30 (chord-elbow 8 over-cuts — Chen2020 lesson)
Integration	Harmony on sample, θ=10 (mixing plateau)	**Island removal**	UMAP radius-graph, −16 cells / 1 island
Trajectory	scFates curve N=20 λ=0.3 μ=100 (fold-free, ratio 2.87)	**Root**	curve tip 2 = pSom/Gast-high, Crest-low
Subtypes	3CA meta-programs (Neftel: Astro / MES-Hypoxia / OPC) + Wang 2019	**Geometry**	Linear-pSom/Crest (single axis; pSom strongest globally)

Wang 2019 Glioblastoma — analysis summary

RUN 2 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Judged on the embedding + lobe composition (Kürten fork: never a scalar). Harmony on sample cut the same-sample neighbour fraction 0.99→0.6 and dissolved the θ=2 monochrome patient lobes: one connected component, no lobe a single patient (dominant share 0.35; 6–7/7 samples per lobe). Mixing is partial (iLISI median 1.9/7) — expected GBM CNV identity within a continuous multi-sample manifold, not an archipelago.

Gate 2 — module geography → GO (non-branching)

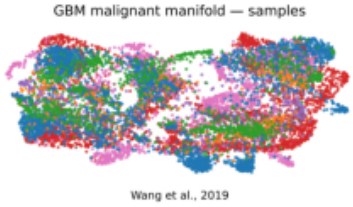
PPT tree over-branches a continuous cloud (forks 5→18 with node count, none stable), and the two arm-candidates Gast & pSom co-vary (spatial r=+0.16) — no bifurcation. Shared Gast/pSom gradient with neural crest as the inverse terminal end (Crest–pSom r=−0.87; Crest-high decile at pseudotime 0.84). Non-branching pattern present.

Gate 3 — hub cross-repression → NA

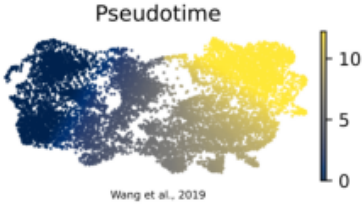
Not applicable — the manifold is non-branching (single principal curve, 0 forks), so there is no hub bifurcation to test Gast↔pSom cross-repression.

STEP FIGURES

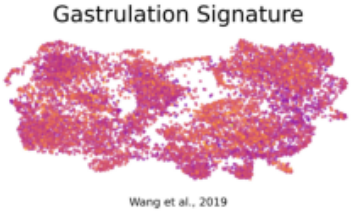
Manifold — samples



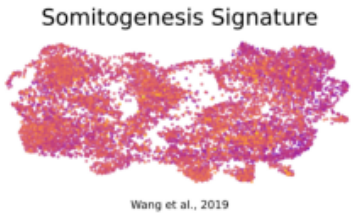
Pseudotime



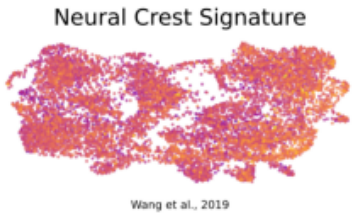
Gastrulation



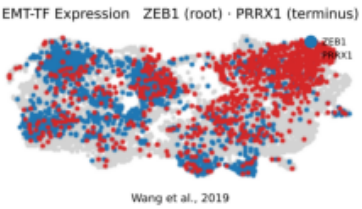
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
IDH-wildtype glioblastoma (Wang et al., 2019), 10x — 9,634 malignant cells (7 samples); non-branching developmental axis.

Verdict.
Pattern reproduces (non-branching): one continuous gastrulation/somitogenesis → neural-crest developmental axis, EMT-TFs ZEB1 → PRRX1. Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Threshold	min 237 cells/sample (largest multiplicative gap 396→143, 2.8×) → 7/10 samples, 9,650 cells	**Integration**	Harmony θ=10 on sample; n_pcs=20; UMAP spread=1.3; overdispersed=2,113
Islands	16 detached cells removed (radius graph) → 9,634 on one component	**Geometry**	Linear-pSom/Crest non-branching curve (ElPiGraph N=20, λ=0.3, μ=100); fold-free (SA 2.8); PPT 0 stable forks
Root	curve tip 2 = Gast/pSom-high, Crest-low origin (neural crest late terminus)	**Scope**	whole manifold; cell cycle orthogonal (|corr|<0.1)
EMT-TFs	ZEB1 (root, 70%) → PRRX1 (terminus, 40%); Snail/Twist/FOXC <5%		

Wang 2019 Glioblastoma — analysis summary

RUN 3 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Samples intermix on one continuous connected manifold. Per-neighbourhood iLISI rises 1.03 (pre-Harmony) → 2.60 (θ=10) and plateaus; n_pcs 20/40 near-identical (not under-powered). One component after island removal.

Gate 2 — module geography → GO (non-branching)

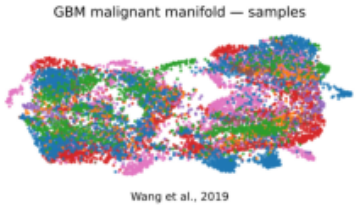
Embryonic module pattern present (non-branching). Gast/pSom co-vary spatially (r=+0.16) along one axis and both oppose Crest (Gast r=−0.49, pSom r=−0.87); Crest is the inverse terminal gradient (high-decile pseudotime 0.84).

Gate 3 — hub cross-repression → NA

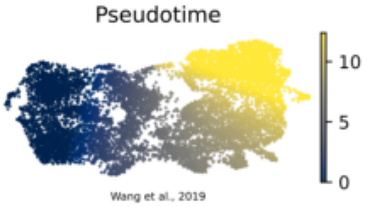
Not applicable — no bifurcation. Gast and pSom co-vary rather than cross-repress, so there is no Gast/pSom hub diverging into two competing arms. Geometry is linear (curve: 2 tips, 0 forks, fold-free ratio 2.82).

STEP FIGURES

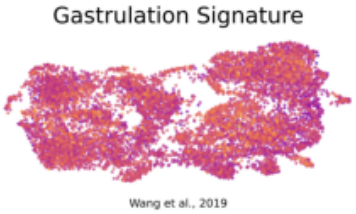
Manifold — samples



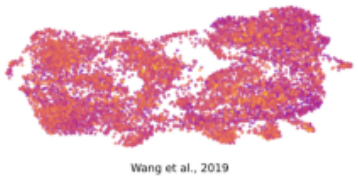
Pseudotime



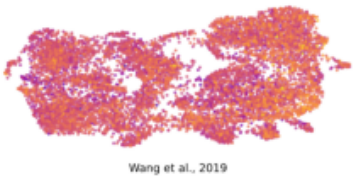
Gastrulation



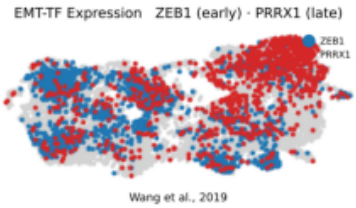
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019), 10x — 9,632 malignant cells, 7 samples.

Verdict.
The embryonic program reproduces as a single collapsed (linear) axis: a Gast/pSom-high developmental origin grading to a Crest-high terminus, EMT-TFs ZEB1 (early) → PRRX1 (late). No Gast/pSom bifurcation → Gate 3 NA. Gate 1 GO / Gate 2 GO (non-branching). Module means Gast −0.002 / pSom +0.008 / Crest −0.007.

Parameter choices

Min-cell threshold	237 (7/10 samples; 9,650 cells); largest ratio gap 396→143 (2.77), cut at geom mean	**Integration**	Harmony θ=10, n_pcs=20; iLISI plateaus 2.4→2.6 (θ 5→10); no sc.pp.scale
Island removal	UMAP radius graph (6× median NN), keep largest component → 18 cells / 2 specks → 9,632	**Subtypes**	AC-like / OPC-like / NPC-like marker scores per Leiden cluster (publication meta-programs)
Geometry / root	Linear non-branching scFates curve; root = Gast/pSom-high, Crest-low tip; continuity +0.98 (diffusion)	**Module means (raw)**	Gast −0.002 / pSom +0.008 / Crest −0.007 (Linear-pSom/Crest)

Wang 2019 Glioblastoma — analysis summary

RUN 4 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/10 samples (≥237 cells, gap-find) on one continuous 10x manifold; per-neighbourhood iLISI 1.03→2.60/7 across θ 2–10, plateaus (θ=20 → 2.59). θ=2 is a per-patient archipelago; θ=10 intermixes into one mass. 16 cells in 1 island removed → 9,634 cells, one component.

Gate 2 — module geography → GO (non-branching)

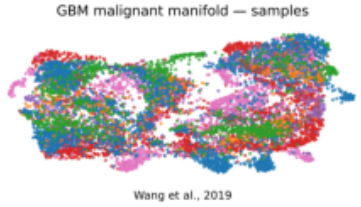
Shared Gast/pSom gradient (co-vary r=+0.16) with Crest an inverse terminal readout (Crest vs Gast r=−0.49, vs pSom r=−0.87). Crest-high decile at pseudotime 0.85 (terminal). No Gast/pSom competing arms; no Crest-on-Gast-arm inversion — no flag.

Gate 3 — hub cross-repression → NA

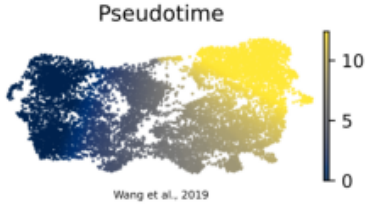
Gast/pSom co-vary (not compete) — single collapsed axis, no bifurcation to test.

STEP FIGURES

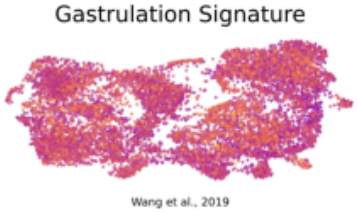
Manifold — samples



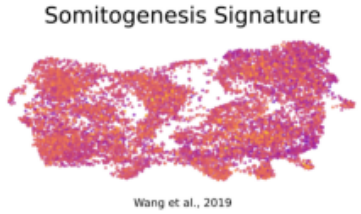
Pseudotime



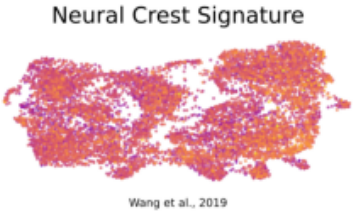
Gastrulation



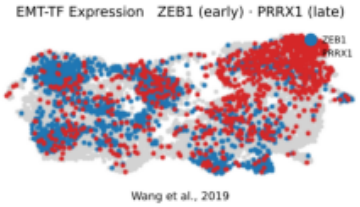
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019); 10x scRNA-seq, single leaf; 9,916 malignant raw → 9,634 final after threshold + island removal. IDH-wildtype, 7 samples.

Verdict.
Pattern reproduces as a non-branching embryonic axis: a Gast/pSom-high origin grading to a Crest-high terminus (pSom ρ=−0.61, Crest ρ=+0.48 along pseudotime), continuity 0.98 checked on an independent diffusion embedding. Linear geometry → Gate 3 NA. EMT-TFs ZEB1 (early) → PRRX1 (late). Module means Gast −0.003 / pSom +0.008 / Crest −0.007. Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Min cells/sample	237 (gap 396→143, ratio 2.77; 7/10)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Overdispersed genes	2,113 (scFates GAM)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Island removal	6×-median radius graph; 16 cells → 9,634	**Root node**	Gast/pSom-high, Crest-low tip
Geometry	Linear (non-branching); fold-free 2.83	**EMT-TFs**	ZEB1 (early) → PRRX1 (late)

Wang 2019 Glioblastoma — analysis summary

RUN 5 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9,916 malignant → 9,650 (7/10 samples, ≥237) → 9,634 on one connected component. Sample iLISI rose 1.0→2.6/7 under Harmony (plateau by θ=10); one 16-cell island removed.

Gate 2 — module geography → GO (non-branching)

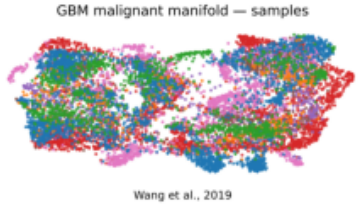
Shared Gast/pSom gradient (r=+0.16, co-vary) with Crest an inverse terminal readout (r=−0.49 vs Gast, −0.87 vs pSom). Curve fold-free (0 forks); Crest-high decile at pseudotime 0.85.

Gate 3 — hub cross-repression → NA

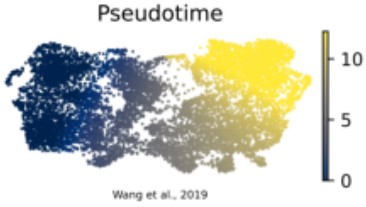
Gast/pSom co-vary (not compete) — single collapsed axis, no bifurcation to test.

STEP FIGURES

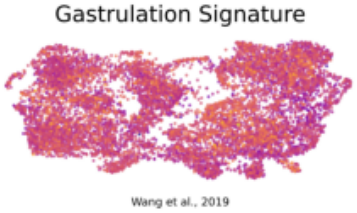
Manifold — samples



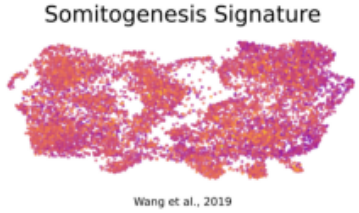
Pseudotime



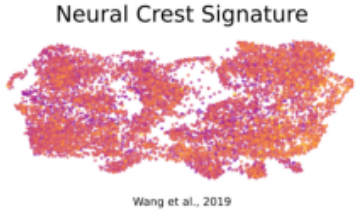
Gastrulation



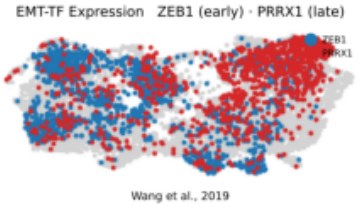
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019); 10x, 10 samples; 9,916 malignant raw → 9,634 final after threshold + island removal. IDH-wildtype primary GBM (single 10x leaf).

Verdict.
Pattern reproduces as a non-branching embryonic axis: pSom-high origin → Crest-high terminus (pSom ρ=−0.62, Crest ρ=+0.49 along pseudotime; Gast flat, ρ=−0.06), continuity confirmed on a diffusion embedding independent of the fitted UMAP (r=0.98). Linear (single collapsed axis; GBM states NPC/OPC → AC/MES span it). Not branching. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB1 (early/root) → PRRX1 (late).

Parameter choices

Min cells/sample	237 (mult-gap 396→143; 7/10)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Geometry	Linear / non-branching (curve)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	pSom-high / Crest-low tip	**EMT-TFs**	ZEB1 (early/root) → PRRX1 (late)

Wang 2019 Glioblastoma — analysis summary

RUN 6 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/10 samples (≥396 cells) intermix on one continuous manifold (radius-graph: ~34 detached cells removed → 1 component; iLISI 1.0→2.35/7). θ=2 archipelago dissolves by θ=10; the pSom↔Crest axis reproduces within all 7 patients (r=−0.50 to −0.64).

Gate 2 — module geography → GO (non-branching)

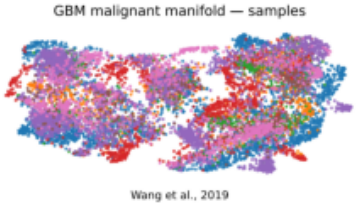
pSom declines along pseudotime (ρ=−0.36) and Crest rises to a terminal readout (ρ=+0.29; Crest-high decile mean pt=0.88); Gast a weak co-gradient at the origin. Non-branching case.

Gate 3 — hub cross-repression → NA

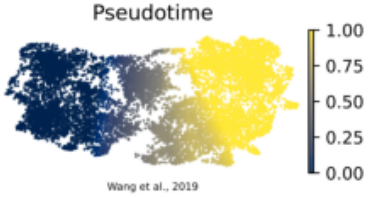
PPT tree fit + tested: forks unstable across N (0–3); node-clusters CO-VARY (Gast↔pSom r=+0.29), hub is co-high not two arms; per-gene r<0 is an XOR-gene artifact (single-module split-half null=+0.5). No cross-repressing branch pair → not tested. Curve, not tree.

STEP FIGURES

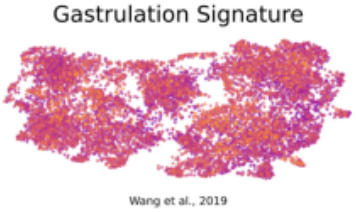
Manifold — samples



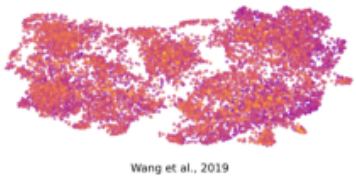
Pseudotime



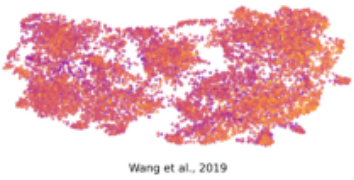
Gastrulation



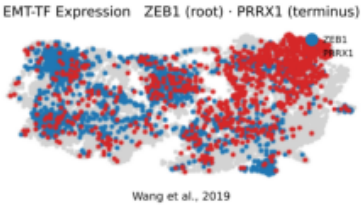
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019), 10x — 10 patients / 16,317 cells (master scope); malignant only: 9,916 loaded, 9,650 across 7 samples (≥396) after the count-gap threshold, ~9,616 after island removal enter the trajectory; whole malignant manifold (no lobe scoping — cycling orthogonal).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The manifold is multipolar (NPC/AC/MES Neftel states), but a fitted PPT tree gives no genuine bifurcation: forks are node-number unstable and its node-clusters CO-VARY in Gast↔pSom (r=+0.29, both opposing Crest), with a single co-high hub rather than a Gast-arm and a pSom-arm. So the embryonic axis is one continuous pSom → terminal-Crest gradient (NPC/OPC origin → MES/hypoxia neural-crest terminus), rooted at the pSom/Gast pole; cell cycle orthogonal (ρ=+0.07), whole manifold kept. EMT-TFs ZEB1 (root) → PRRX1 (terminus); SNAI1/FOXC below the 3% floor (TWIST1 4.3%, weak). Module means Gast −0.003 / pSom +0.008 / Crest −0.007.

Parameter choices

Min cells/sample	≥396 (keeps 7/10; gap 396→143)	**Scope**	whole manifold (no scoping)
Root node	curve tip 2 = pSom/Gast-high, Crest-low	**Integration**	Harmony θ=10; PCA50; n_pcs=20
Trajectory	scFates curve N=6 (tree tested: forks unstable)	**Cell states**	AC/NPC/MES-like (Neftel 2019)

Wang 2019 Glioblastoma — analysis summary

RUN 7 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7 samples share ONE continuous manifold (single kNN component after island removal); per-neighbourhood iLISI rises 1.0→2.3/7 across the θ-sweep; Neftel meta-program lobes intermix, not per-sample islands; cycling a scattered ~19% minority.

Gate 2 — module geography → GO (non-branching)

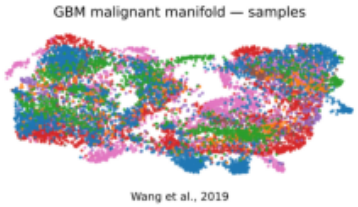
Single collapsed embryonic axis: pSom-high/Crest-low origin → Crest-high terminus (Crest decile mean pt 0.84). Gast/pSom co-vary (+0.16), both oppose Crest (−0.87/−0.49). Non-branching case — GO with Gate 3 = NA.

Gate 3 — hub cross-repression → NA

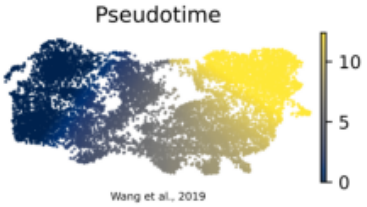
No embryonic-module bifurcation: PPT fork count is resolution-dependent, not stable — harmony-space 1,3,2,1 (Nodes 20/30/50/100), UMAP-space 1,2,2,2,3 (Nodes 20/30/50/80/120); no tip is Gast-positive (tips are pSom-vs-Crest poles). Node-cluster Gast–pSom r=+0.07 sits inside the label-shuffle null [−0.35,+0.28]. Neftel AC/OPC/NPC/MES multipolarity ≠ module bifurcation — no branch to test.

STEP FIGURES

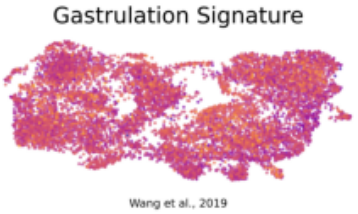
Manifold — samples



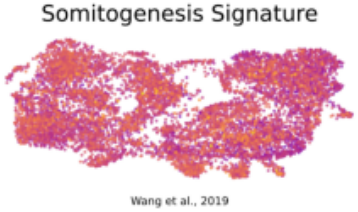
Pseudotime



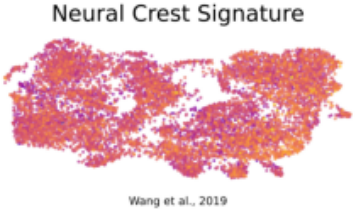
Gastrulation



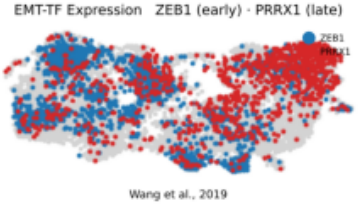
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019), 10x — 9,632 malignant cells, 7 samples (of 10). Malignant cell-state manifold, embryonic-program re-analysis (CCCA meta-atlas).

Verdict.
The embryonic pattern reproduces as a single collapsed axis — a pSom/Gast-high origin grading to a Crest-high terminus — NOT an embryonic bifurcation. Neftel GBM state-multipolarity does not carry a Gast↔pSom branch that survives a fold-stability + independence null. Gate 1 GO / Gate 2 GO / Gate 3 NA. Geometry linear (non-branching). EMT-TFs ZEB1 (early) → PRRX1 (late). Module means Gast −0.003 / pSom +0.008 / Crest −0.007.

Parameter choices

Sample threshold	237 malignant (largest multiplicative count-gap 396→143); 7/10 samples, 9,632 cells	**Embedding**	n_pcs=20; Harmony θ=10 on sample; spread=1.3 (PC1≈PC2≈11%, 2D cloud)
Cell cycle	scored de novo (Tirosh S/G2M absent in source); orthogonal to modules (|r|≤0.12)	**Trajectory**	scFates curve (non-branching), Nodes=20 μ=100; fold-free (SA 2.82, DC1 corr +0.98)
Root	Gast/pSom-high, Crest-low tip (Crest late/terminal)	**EMT-TFs**	ZEB1 (70%) & PRRX1 (40%); VIM excluded (marker); Snail/Twist/FoxC undetectable

Wang 2019 GBM — analysis summary

RUN 8 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/10 samples (≥237 cells; mult-gap 396→143) on one connected component; sample iLISI 2.60/7 across θ=5–20. Cell-cycle & metaprograms scattered — no per-sample archipelago.

Gate 2 — module geography → NO-GO

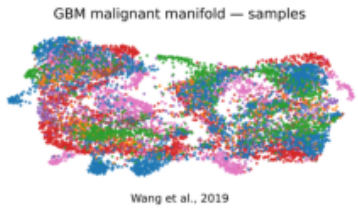
RAW module scores are ~85% noise (spatial-structure ratio 0.12–0.15 vs 0.23–0.39 in the Neftel GBM GO ref) and near-flat across all 3 lobes. Under a NEUTRAL control all three modules co-vary POSITIVELY (Gast-pSom +0.62, Crest-Gast +0.47, Crest-pSom +0.37) — no Crest opposition. The union-pool pSom↔Crest −0.61 was a control-pool artifact amplified by smoothing. No hub/arms.

Gate 3 — hub cross-repression → NA

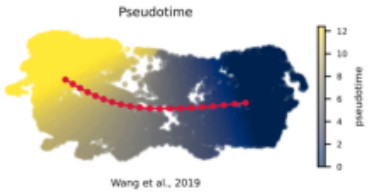
No embryonic geometry present → no trajectory forced, no bifurcation to test.

STEP FIGURES

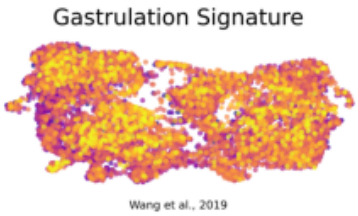
Manifold — samples



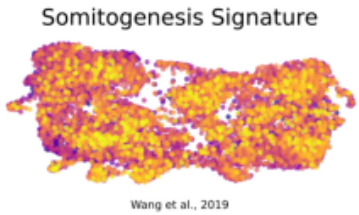
Pseudotime



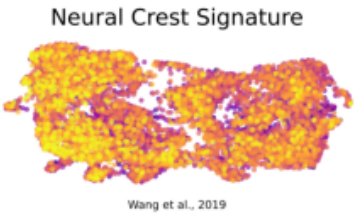
Gastrulation



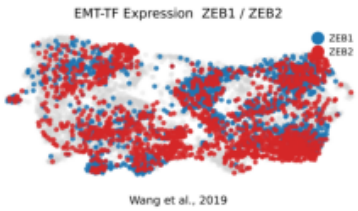
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019), 10x — 10 samples; 9,916 malignant raw → 9,650 after ≥237-cell threshold (7/10 samples) → 9,634 after island removal (16 cells, 1 detached speck).

Verdict.
✗ Pattern does NOT reproduce. The malignant manifold integrates and is continuous (Gate 1 GO), but the embryonic module geography is absent: raw Gast/pSom/Crest scores are noise-dominated and near-uniform, with no hub, no arms, and no Crest opposition once the union-pool scoring artifact is removed. Honest Gate 2 NO-GO — no trajectory forced.

Parameter choices

Min cells/sample	≥237 (mult-gap 396→143; 7/10)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Embedding	native X_umap (no re-embedding)	**Gate 2 test**	raw scores + neutral-control null
Scope	whole manifold (no embryonic lobe)	**EMT-TFs**	ZEB1, ZEB2 (detectable; no order)

Neftel 2019 GBM — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

10x leaf scoped (SmartSeq2 non-integrable: platform iLISI~1.0 across θ=2–20, disjoint samples, UMI vs TPM). 7/10 samples (≥413 cells) on one connected component; sample iLISI 2.9/7.

Gate 2 — module geography → GO (non-branching)

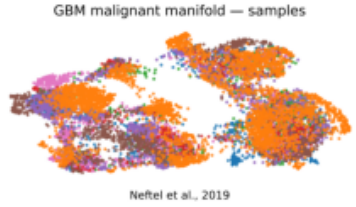
Shared Gast/pSom gradient (r=+0.79, co-vary) with Crest an inverse terminal readout (r=−0.89/−0.90). Curve fold-free (0 forks); Crest-high decile at pseudotime 0.76.

Gate 3 — cross-repression → NA

Gast/pSom co-vary (not compete) → single collapsed axis, no bifurcation to test.

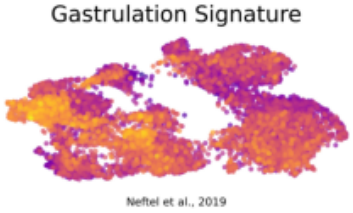
STEP FIGURES

Manifold — samples

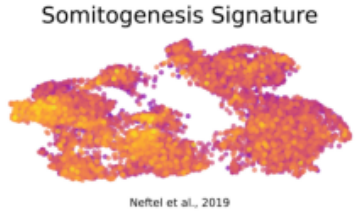


Pseudotime

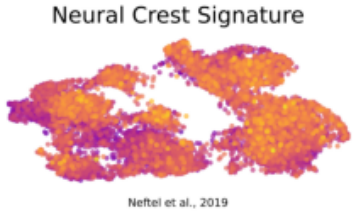
Gastrulation



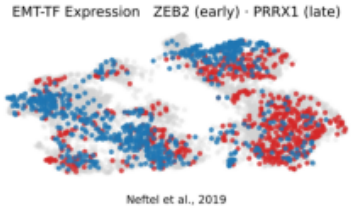
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Neftel et al., 2019); SmartSeq2+10x atlas, 10x leaf analyzed; 11,172 malignant raw → 10,229 final. Platform-exclusive patients make the SmartSeq2 leaf non-integrable with 10x.

Verdict.
Pattern reproduces as a non-branching embryonic axis: Gast/pSom-high origin → Crest-high terminus; developmental order pSom ρ=−0.74, Crest ρ=+0.55 along pseudotime, checked off a diffusion embedding independent of the fitted UMAP. Linear-pSom/Crest (pSom globally strongest).

Parameter choices

Min cells/sample	≥413 (mult-gap 647→264; 7/10)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Scope	10x leaf only (SmartSeq2 non-integrable)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	Gast/pSom-high tip	**Geometry**	Linear (non-branching); 0 forks
EMT-TFs	ZEB2 (early) → PRRX1 (late)		

Couturier 2020 GBM — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample integrates 10 patient tumours: sample iLISI 1.16→4.43 of 10 (plateau; peaks at θ=10). One connected component, 0 detached islands. Judged on the regenerated embedding + per-neighbourhood iLISI: malignant structure is mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

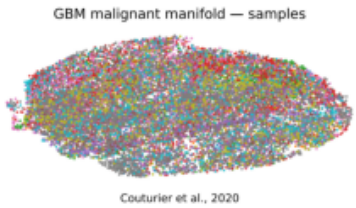
Non-branching: a single Gast/pSom → Crest axis. Gast & pSom co-vary (smoothed r=+0.53), both oppose Crest (Gast↔Crest r=−0.73, pSom↔Crest r=−0.81). Along pseudotime Gast/pSom fall (ρ −0.48 / −0.41), Crest rises to a terminal tip (Crest-high decile mean pseudotime = 0.94).

Gate 3 — cross-repression → NA

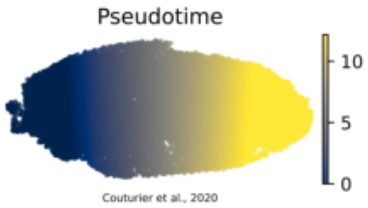
Not applicable: the manifold is non-branching (principal curve, 2 tips / 0 forks), so there is no Gast/pSom-opposed bifurcation to test. Branching-only correlation / modules notebooks are not produced (per spec section 6).

STEP FIGURES

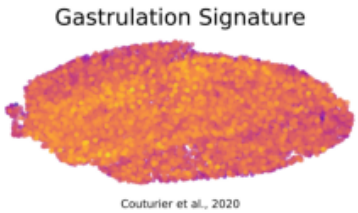
Manifold — samples



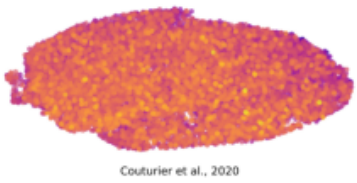
Pseudotime



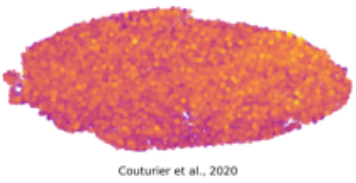
Gastrulation



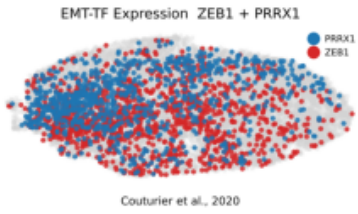
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Couturier et al., 2020), 10x — 16,804 malignant cells (10 patient tumours). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching, Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis rooted at the Gast/pSom-high origin, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs ZEB1 + PRRX1 (both terminus-enriched; no root-end factor).

Parameter choices

Sample threshold	≥418 cells (536→327 gap, ratio 1.64); 10/29 samples, 16,804 cells	**Gene selection / PCA**	scFates find_overdispersed = 2,963 genes; PCA 50 comps
Integration	Harmony on sample θ=10 (iLISI 1.16→4.43 plateau)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	Non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips / 0 forks)	**Root node**	curve tip 0 = Gast/pSom-high, Crest-low (developmental origin)
Module means (raw)	Gast +0.004 · pSom +0.077 · Crest −0.044 (pSom strongest)	**EMT-TFs**	ZEB1 + PRRX1 (both terminus-enriched, distinct families); no root factor

Darmanis 2017 Glioblastoma — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4 samples intermix per-neighbourhood (iLISI 1.09→2.89 of 4) on one continuous manifold (single kNN component, 0 islands); no per-patient islands.

Gate 2 — module geography → GO (non-branching)

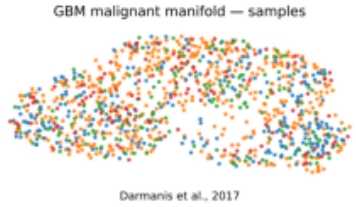
Gast/pSom co-vary (r=+0.66) and both oppose Crest (−0.97 / −0.75): one developmental axis, Crest an inverse terminal gradient. Non-branching case.

Gate 3 — hub cross-repression → NA

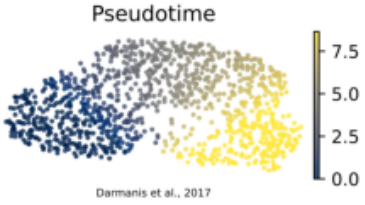
No bifurcation (curve: 2 tips, 0 forks, fold-free across params) → not tested.

STEP FIGURES

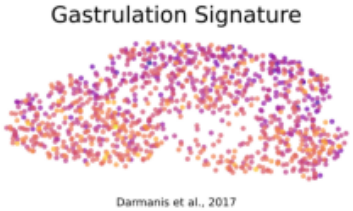
Manifold — samples



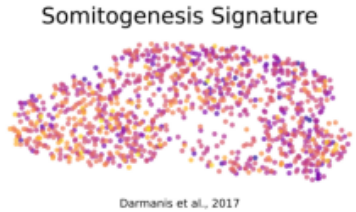
Pseudotime



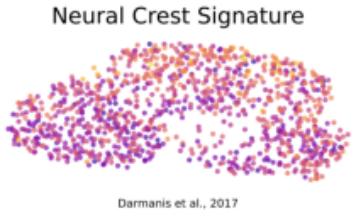
Gastrulation



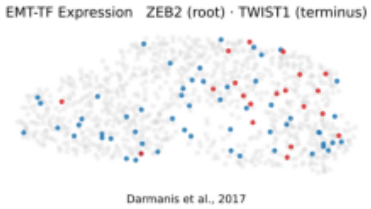
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Darmanis et al., 2017), SmartSeq2 (TPM) — 1,091 malignant cells (4/4 samples).

Verdict.
The embryonic-program pattern REPRODUCES as the non-branching case: a single continuous gastrulation/somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high pole with Crest as an inverse terminal gradient. No Gast–pSom bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. Linear-pSom/Crest geometry (pSom μ=+0.44 strongest). EMT-TFs ZEB2 (root-trending) → TWIST1 (terminus-trending).

Parameter choices

Min-cell threshold	≥100 → 4/4 samples, 1,091 cells (counts 531/268/163/129; ratios <2)	**Expression format**	matrix pre-log-normalized (range 0.7–18.7); undo loader's 2nd log1p (expm1)
Genes / embedding	139 overdispersed (GAM); n_pcs=20; Harmony θ=10	**Islands removed**	0 (single connected kNN component; radius-graph = 1 component)
Geometry	Non-branching — scFates curve N=20 λ=0.3 μ=100; 2 tips, 0 forks, fold-free (2.68)	**Root**	curve tip 0 — Gast/pSom-high, Crest-low origin (Crest late)
EMT-TFs	ZEB2 (51%, root-trending) · TWIST1 (18%, terminus-trending); Snail/FoxC near-absent		

Darmanis 2017 GBM — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4 patients intermix on one continuous manifold (iLISI 2.68 / 4 vs 1.16 pre-Harmony; θ=10, 20 PCs); radius-graph finds 0 detached islands; all 3 sub-lobes reproduce the embryonic relationships.

Gate 2 — module geography → GO (non-branching)

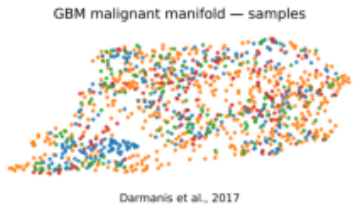
Gast & pSom co-vary spatially (r = +0.72); Crest is the inverse terminal gradient (r = −0.92 vs Gast, −0.59 vs pSom; Crest-high decile at pseudotime 0.98). Single (Gast/pSom)→Crest axis — non-branching case, passes, no bifurcation.

Gate 3 — cross-repression → NA

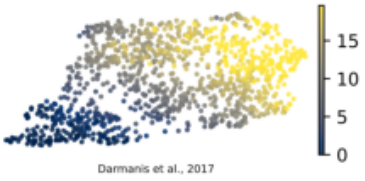
Non-branching manifold: a principal tree shows no Gast-vs-pSom arm divergence (all tips pSom-high, Gast-low), so there is no hub to test. Gate 3 is branching-only — NA.

STEP FIGURES

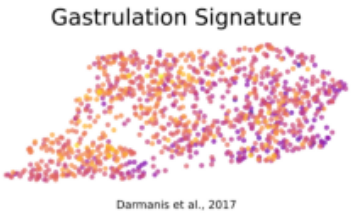
Manifold — samples



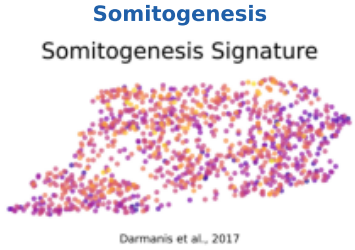
Pseudotime



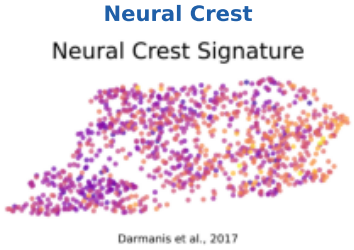
Gastrulation



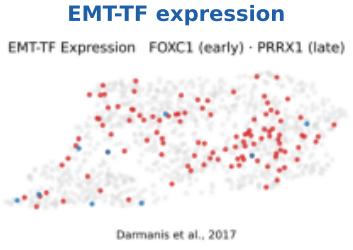
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Darmanis et al., 2017), SmartSeq2 full-length scRNA-seq — 1,091 malignant cells across 4 patients. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces as a single non-branching embryonic axis: a Gast/pSom-high origin resolving into a neural-crest-high terminus. The curve is fit on the integrated PCA representation so the axis reaches the Crest pole, not the pSom edge of the 2-D UMAP. Geometry Linear-pSom/Crest (pSom the globally strongest module). EMT-TFs FOXC1 (early/root) → PRRX1 (late/terminus).

Parameter choices

Sample threshold	all 4 kept (≥100-cell floor; no clean low-end gap)	**HVG selection**	scanpy Seurat top-2000 (pagoda2 overdispersion degenerates on pre-log TPM)
Integration	Harmony on sample θ=10, 20 PCs (iLISI plateaus after θ=5; 1.16→2.68/4)	**Trajectory**	ElPiGraph curve on X_pca_harmony (2 tips, 0 forks, 2.97); non-branching
Root	curve tip max Gast+pSom−Crest (developmental origin; neural crest late)	**Geometry**	Linear-pSom/Crest (Gast μ +0.010 · pSom μ +0.099 · Crest μ +0.041)
EMT-TFs	FOXC1 (early/root) → PRRX1 (late/terminus); 10-factor panel		

Yuan 2018 high-grade glioma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8 samples intermix per-neighbourhood (iLISI 1.0→4.08/8) on one continuous manifold (single kNN component, 0 islands); no per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

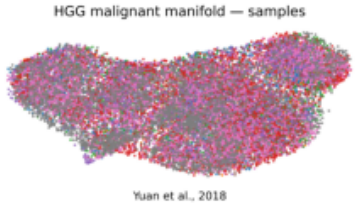
Gast/pSom co-vary (r=+0.23) and both oppose Crest (−0.66 / −0.75): one developmental axis, Crest an inverse terminal gradient. Non-branching case.

Gate 3 — cross-repression → NA

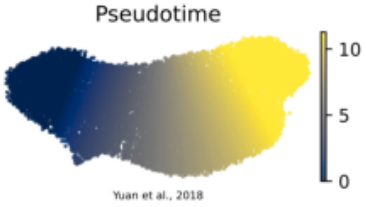
No bifurcation (curve: 2 tips, 0 forks, fold-free across params) → not tested.

STEP FIGURES

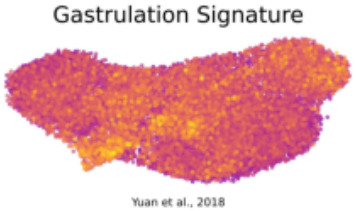
Manifold — samples



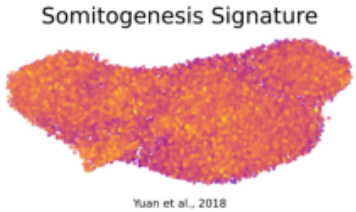
Pseudotime



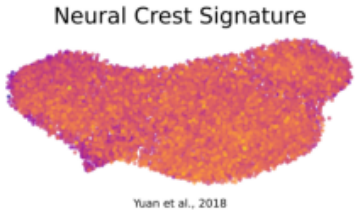
Gastrulation



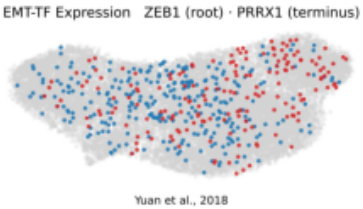
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-grade glioma (Yuan et al., 2018), microwell array-based platform (UMI counts) — 20,080 malignant cells (8/8 samples); embryonic-program cell-state analysis.

Verdict.
The embryonic-program pattern reproduces as the non-branching case: a single continuous gastrulation/somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high pole with Crest as an inverse terminal gradient. No Gast–pSom bifurcation (Gate 3 NA). Linear-pSom/Crest geometry (pSom μ=+0.035 globally strongest).

Parameter choices

Min-cell threshold	≥100 floor (largest gap 2012→598, r=3.36); 8/8, 20,080 cells	**Genes / embedding**	729 overdispersed (GAM); n_pcs=20; Harmony θ=10
Islands removed	0 (single connected kNN component)	**Geometry**	Non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips, 0 forks)
Root	curve tip 0 = Gast/pSom-high, Crest-low (Crest is late)	**Module means (raw)**	pSom μ=+0.035 globally strongest (Linear-pSom/Crest)
EMT-TFs	ZEB1 (14%, root) · PRRX1 (9%, terminus); Snail/Twist/FoxC near-absent		

Venteicher 2017 IDH-mut astrocytoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9 samples intermix per-neighbourhood (iLISI 1.0→4.99/9) on one continuous manifold (single kNN component, 0 islands); no per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

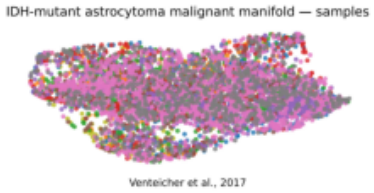
Gast/pSom co-vary (r=+0.60) and both oppose Crest (−0.89 / −0.76): one developmental axis, Crest an inverse terminal gradient. Non-branching case.

Gate 3 — cross-repression → NA

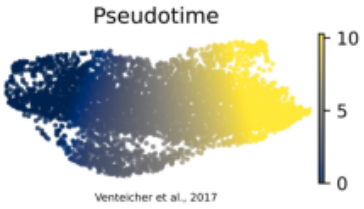
No bifurcation (curve: 2 tips, 0 forks, fold-free across params) → not tested.

STEP FIGURES

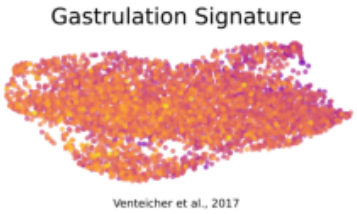
Manifold — samples



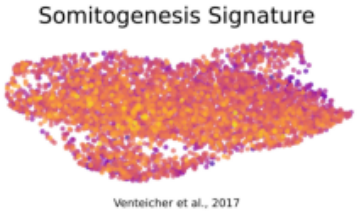
Pseudotime



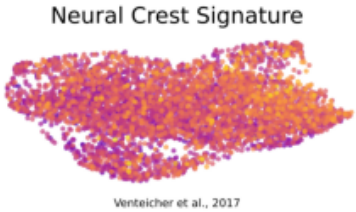
Gastrulation



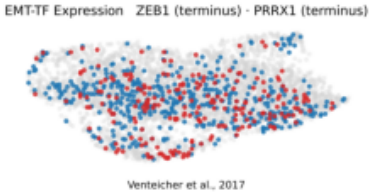
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
IDH-mutant astrocytoma (Venteicher et al., 2017), SmartSeq2 (TPM) — 10 samples; 4,985 malignant cells across 9/10 samples enter the analysis; the whole malignant manifold enters the trajectory (non-branching linear geometry).

Verdict.
The embryonic-program pattern reproduces as the non-branching case: a single continuous gastrulation/somitogenesis→neural-crest developmental axis, rooted at the Gast/pSom-high pole with Crest as an inverse terminal gradient. No Gast–pSom bifurcation (Gate 3 NA). Linear pSom/Crest geometry (pSom μ=+0.21 globally strongest). Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. EMT-TFs ZEB1 (91% pan-tumour) → PRRX1 (terminus-enriched).

Parameter choices

Min-cell threshold	≥146 → 9/10 samples, 4,985 cells (tail gap 192→112, r=1.71; geo-mean cut)	**Genes / embedding**	6,317 overdispersed (GAM); n_pcs=20 (scree elbow~6 rejected); Harmony θ=10
Islands removed	0 (single connected kNN component; radius-graph = 1 component)	**Geometry**	Non-branching — scFates curve N=20, λ=0.3, μ=100 (2 tips, 0 forks; fold-free, self-approach 2.96)
Root	curve tip 0 — Gast/pSom-high, Crest-low (developmental origin; Crest is late)	**EMT-TFs**	ZEB1 (91% pan-tumour) → PRRX1 (terminus-enriched); Snail/Twist/FoxC families near-absent

Tirosh 2016 IDH-mut oligodendroglioma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 6 tumours intermix per-neighbourhood (iLISI 1.02→4.30 of 6 under Harmony θ=10) on ONE continuous manifold (single kNN component, 0 islands removed) — no per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

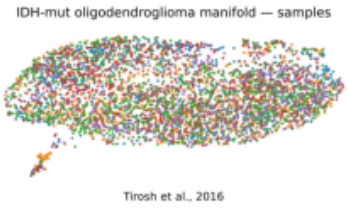
Gast/pSom co-vary (r=+0.38) and BOTH oppose Crest (−0.85 / −0.59): a single developmental axis with Crest an inverse terminal gradient. Non-branching case — GO. Crest enriches the terminus, not a Gast arm.

Gate 3 — cross-repression → NA

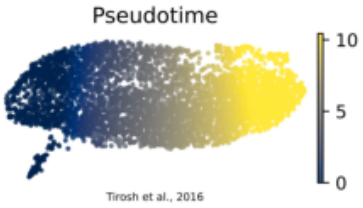
No bifurcation — scFates curve is robustly fold-free (2 tips, 0 forks, self-approach 2.99 across Nodes 15–30).
Hub cross-repression is not defined for a linear axis — Gate 3 not tested.

STEP FIGURES

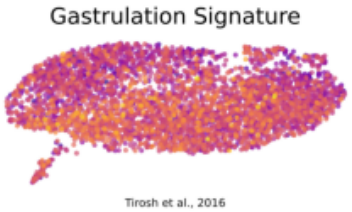
Manifold — samples



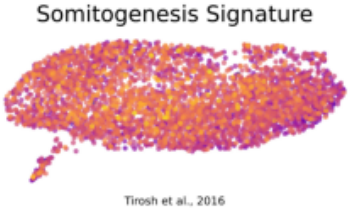
Pseudotime



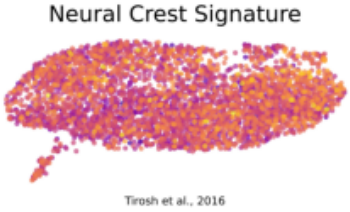
Gastrulation



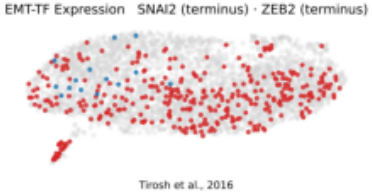
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
IDH-mutant oligodendroglioma (Tirosh et al., 2016), SmartSeq2 (TPM) — 4,089 malignant cells across 6/6 samples. Non-branching Linear-pSom/Crest.

Verdict.
The embryonic-program pattern reproduces as the non-branching case: one continuous gastrulation/somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high pole with Crest as an inverse terminal gradient. No Gast–pSom bifurcation (Gate 3 NA). Geometry Linear-pSom/Crest (pSom μ=+0.16 globally strongest; Gast −0.04, Crest −0.20). EMT-TFs SNAI2 · ZEB2 (both rise to the Crest terminus).

Parameter choices

Min-cell threshold	≥100 → 6/6 samples (4,089 cells); counts descend smoothly 1185→428	**Genes / embedding**	5,800 overdispersed genes; n_pcs=20; Harmony θ=10 on sample
Islands removed	0 (single kNN component; continuous manifold)	**Geometry**	non-branching curve (N=20, λ=0.3, μ=100): 2 tips / 0 forks, fold-free
Root node	curve node 2 = Gast/pSom-high, Crest-low origin	**EMT-TFs**	ZEB1 (82%), ZEB2 (83%), PRRX1 (38%), SNAI2 (3%) rise to Crest; display SNAI2 · ZEB2

Hovestadt 2019 medulloblastoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

25 samples intermix per-neighbourhood on one continuous manifold (iLISI 1.17→7.67; 0 islands removed). Judged on the regenerated embedding across a Harmony {2,5,10} sweep — no per-sample archipelago despite strong medulloblastoma molecular-subgroup batch effect. Cycling compartment present but Gast is not proliferation-confounded (r=−0.13), so the embryonic axis is real and the whole manifold is retained.

Gate 2 — module geography → GO (non-branching)

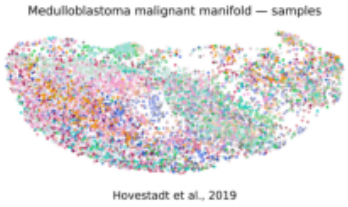
Non-branching case: Gast & pSom co-vary (r=+0.46, overlap — no two competing arms), Crest an inverse terminal gradient (Crest–Gast r=−0.90, Crest–pSom r=−0.67). Single (Gast/pSom)→Crest developmental axis; Crest-high decile at pseudotime 0.82 (terminal). Passes Gate 2 as the non-branching outcome.

Gate 3 — cross-repression → NA

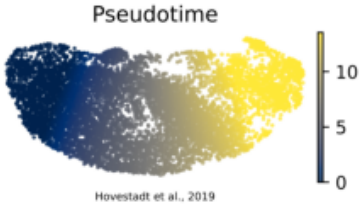
No bifurcation — no hub to test. Gate 3 is branching-only, so it is Not Applicable for this linear manifold.

STEP FIGURES

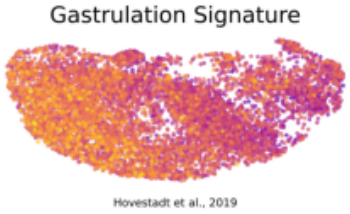
Manifold — samples



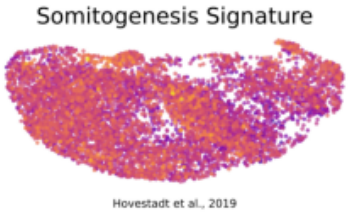
Pseudotime



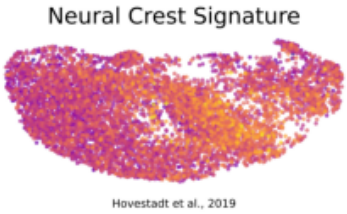
Gastrulation



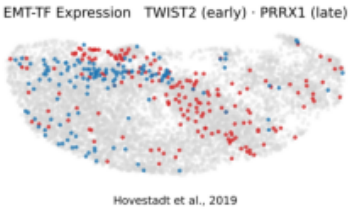
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Medulloblastoma (Hovestadt et al., 2019), SmartSeq2 — 7,921 malignant cells across 25 samples (of 36; ≥100-cell floor). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces as the non-branching case — one embryonic Gast/pSom → Crest axis, Crest inverse-terminal; no bifurcation (Gate 3 NA). Rooted at the Gast/pSom-high pole (Gast early, Crest late); the embryonic axis is not proliferation-confounded so the whole manifold is retained. EMT-TFs TWIST2 (early/root) → PRRX1 (late/terminus).

Parameter choices

Min-cell threshold	≥100 (no cliff) → 25/36 samples, 7,921 cells	**Integration**	Harmony on sample θ=5, n_pcs=20 (mixing saturates 5–10)
Island removal	0 cells	**Geometry**	Linear (non-branching); scFates curve N=20, μ=100, fold-free (2.97)
Root node	curve tip 2 = Gast/pSom-high, Crest-low origin	**Module means**	Gast −0.043 · pSom +0.186 · Crest −0.026
EMT-TFs	TWIST2 (early, pt 0.24) → PRRX1 (late/terminus, pt 0.50)		

Choudhury 2022 meningioma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant cells form a per-patient archipelago. At the best-mixing integration (θ=10) the kNN graph is a single component (iLISI 1.0→3.0) — the scalars a naïve test reads as “integrated” — yet the embedding is organised BY PATIENT: every patient self-clusters locally (self-neighbour enrichment 1.8–7.2×); 3/5 patients cross the 2.5× line (1/3/4 at 7.2/3.4/2.8×) and the two largest (5,6) hold their own territories. Integration strength does not fix it — the enrichment GROWS with θ (patient 1: 7.2→11.9→12.9×), so Harmony over-corrects. Each patient carries the same embryonic modules → inter-patient CNV/clonal heterogeneity (expected for meningioma), not a shared cell-state continuum → do not force a trajectory.

Gate 2 — module geography → N/A

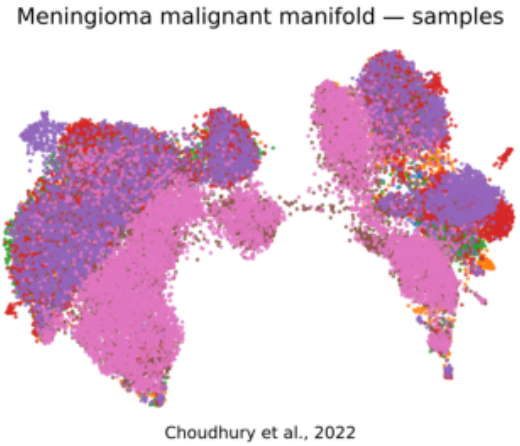
Not reached. Gate 2 tests a Gast/pSom hub and Crest geography on a fitted manifold; with Gate 1 NO-GO no manifold/trajectory is fit, so the module-geography gate does not apply.

Gate 3 — cross-repression → N/A

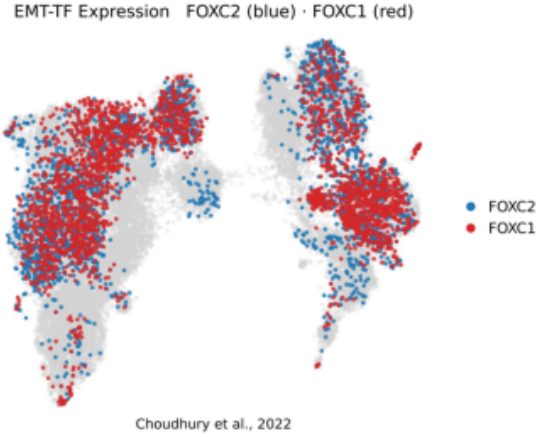
Not reached (branching-only, and downstream of Gate 2). No hub, no bifurcation, no hub r.

STEP FIGURES

Malignant manifold — samples



EMT-TF expression



Dataset.

Meningioma (Choudhury et al., 2022), 10x (UMI counts) — 37,049 malignant cells; 7/10 samples, 5 patients enter preprocessing. First meningioma / first non-glial brain tumour in the atlas.

Verdict.

The embryonic-program pattern does NOT reproduce in Choudhury 2022 meningioma: the malignant compartment is a per-patient archipelago with no sample-integrated, continuous cell-state manifold. Gate 1 NO-GO — analysis stops before the trajectory. A genuine portability NO-GO (integration failure), distinct from a pattern-absent Gate-2 NO-GO. The FOXC mesenchymal TF family is broadly expressed (arachnoid/mesenchymal identity), reported unconditionally without pseudotime.

Parameter choices

Sample threshold	≥780 cells (geo-mean across the 2254→270 gap, 8.35×); keeps 7/10 samples (5 patients)	**Gene / PCA**	2,828 overdispersed (scFates GAM); n_pcs=20 (PC1 alone ≈25% variance, inter-patient/CNV axis)
Integration	Harmony on sample, θ swept 10/20/40; θ=10 best mixing (iLISI 3.0) — diagnostic, not corrective	**Root / geometry**	None — no trajectory fit (Gate 1 NO-GO); geometry not applicable
EMT-TFs shown	FOXC2 (68%) · FOXC1 (67%) — broadest detectable (FOXC/PRRX panel; VIM excluded, not a TF)		

Gojo 2020 pediatric ependymoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ-sweep on sample: sample iLISI 1.5→6.6 (θ=10), one connected component (0 islands); PF/ST/SP subgroups grade continuously across one cloud (subgroup iLISI stays ~1.6 — biology preserved).

Gate 2 — module geography → GO (non-branching)

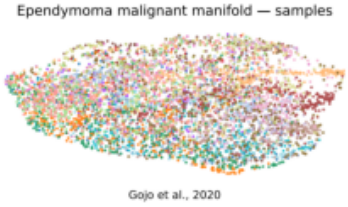
Non-branching: shared Gast/pSom body with an inverse terminal Crest gradient (Crest-high decile mean pseudotime = 0.97). Crest opposes Gast (r=−0.50) and pSom (r=−0.38); Gast~pSom weak (−0.20).

Gate 3 — cross-repression → NA

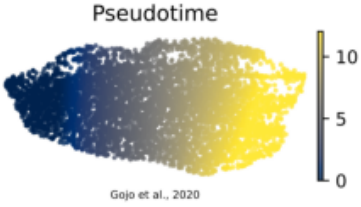
No competing Gast/pSom arms (weak −0.20, no bifurcation) — branching-only test does not apply.

STEP FIGURES

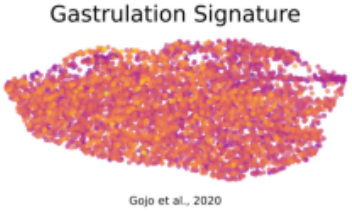
Manifold — samples



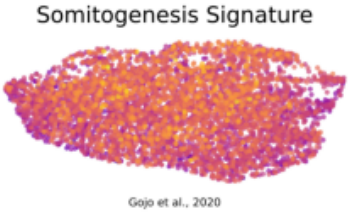
Pseudotime



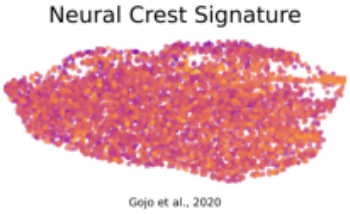
Gastrulation



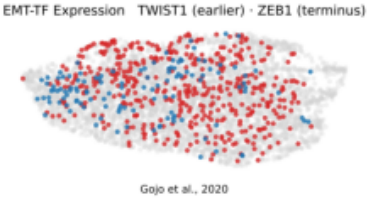
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric ependymoma (Gojo et al., 2020), SmartSeq2, 5,378 malignant cells across 20 samples (threshold ≥110/sample; PF/ST/SP molecular subgroups).

Verdict.
Pattern reproduces as the non-branching case — one continuous gastrulation/somitogenesis → neural-crest axis, no Gast/pSom bifurcation. Geometry: Linear-pSom/Crest. EMT-TFs overlay TWIST1 (earlier) ↔ ZEB1 (terminus).

Parameter choices

Min-cell threshold	≥110 / sample (largest gap 140→88, geometric mean)	**Cells retained**	5,551 → 5,378 (20/22 samples above threshold; 0 islands)
Overdispersed genes	4,997 (scFates find_overdispersed)	**Integration**	Harmony on sample θ=10 (n_pcs=20, spread=1.3); iLISI 6.6
Geometry	Non-branching; scFates curve N=30, λ=0.3, μ=100 (2 tips, 0 forks)	**Root**	Gast/pSom-high, Crest-low tip (Crest is late/terminal)
Module means (raw)	Gast +0.116 · pSom +0.329 · Crest +0.003 (pSom-dominant)	**EMT-TFs**	TWIST1 (earlier) ↔ ZEB1 (terminus); FOXC1/SNAI1/ZEB2/PRRX1 detected

Filbin 2018 pediatric H3-K27M glioma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Malignant structure is sample-mixed AND continuous. iLISI 1.00→3.81 of 6 samples after Harmony(θ=10); one connected component on the independent Harmony-PCA graph (no islands removed). Gast is not proliferation-confounded (prolif–Gast ρ=+0.09).

Gate 2 — module geography → GO (non-branching)

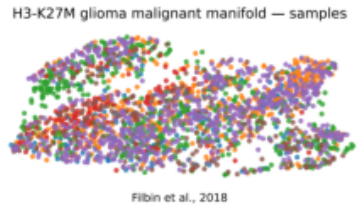
Non-branching embryonic axis. Gast/pSom co-localise (joint enrichment 1.54; spatial r=+0.61) as one developmental origin; Crest is an inverse terminal gradient (pSom&Crest joint 0.22; Crest-high decile pseudotime 0.99). No Gast/pSom hub diverging into two arms — GO (non-branching).

Gate 3 — cross-repression → NA

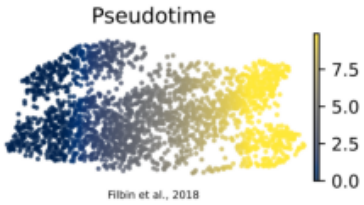
Not applicable — no bifurcation. A single principal curve (2 tips, 0 forks) fits the manifold, so there is no branch point at which to test mutual Gast↔pSom cross-repression.

STEP FIGURES

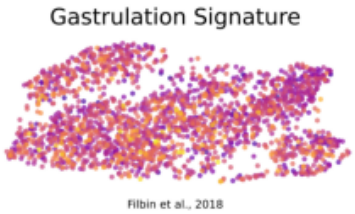
Manifold — samples



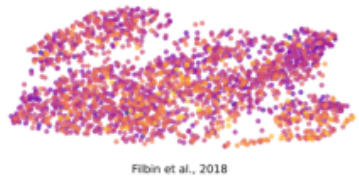
Pseudotime



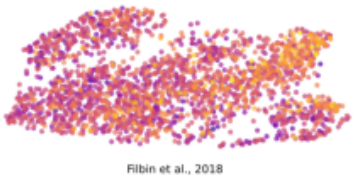
Gastrulation



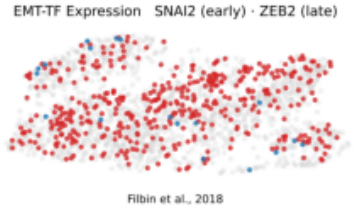
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric H3-K27M mutant glioma (Filbin et al., 2018), SmartSeq2; 2,259 malignant cells across 6 samples (all kept, ≥100 floor). One continuous non-branching manifold.

Verdict.
Pattern reproduces (non-branching). One continuous gastrulation/somitogenesis → neural-crest axis, rooted at the Gast/pSom-high origin; Crest a terminal inverse gradient. G1 GO · G2 GO · G3 NA — Linear-pSom/Crest.

Parameter choices

Min-cell threshold	≥100 floor → all 6 samples kept (2,259 cells); 141–667 smooth, no cliff	**Integration**	Harmony on sample, θ=10, n_pcs=20; patient batch 0.96, plateaus by θ=5
Geometry	scFates curve, Nodes=20 — non-branching (2 tips, 0 forks; self-approach 2.97)	**Root node**	Gast/pSom-high, Crest-low tip 2 — developmental origin; non-cycling
Islands removed	0 — single connected component (UMAP + Harmony-PCA graph)	**EMT-TFs shown**	SNAI2 (early) · ZEB2 (late) — best-demarcating pair; ZEB1/ZEB2/PRRX1 broad

Wu 2021 Breast — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

20/20 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI 1.0→4.96); both lobes (Low-cycling 11,803 / Cycling 12,686) carry near-identical developmental means — no lobe scoping.

Gate 2 — module geography → GO (non-branching)

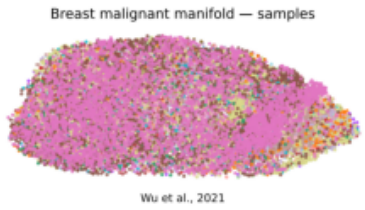
Non-branching signature (spec §1): embryonic programs present with Crest an inverse/terminal gradient — pSom↔Crest r=−0.71, Crest terminal (crest-high decile pt=0.94, rho=+0.55); Gast/pSom body ordered along pseudotime (pt 0.34/0.46), centroids collinear (area/perim²=0.010), cell-cycle-independent (pt↔cyc rho=+0.00). FLAG: Gast/pSom cross-repress (r=−0.47, obs/exp=0.36) — spec-permitted, organized at the body end opposite Crest.

Gate 3 — hub mutual exclusivity → NA

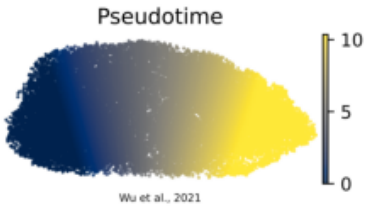
No bifurcation: Gast/pSom repel (obs/exp=0.36) with 0 forks, so the program collapses onto a single axis rather than two arms — there is no branch pair to test for mutual exclusivity.

STEP FIGURES

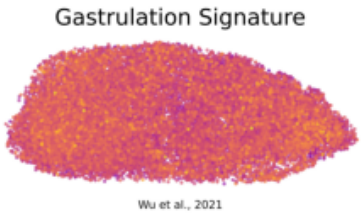
Manifold — samples



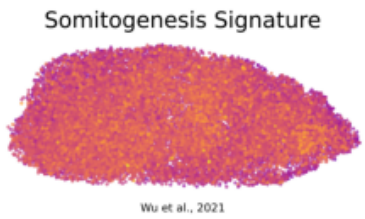
Pseudotime



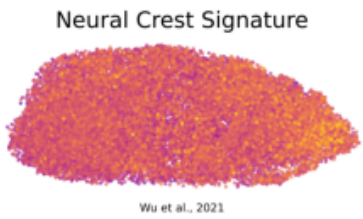
Gastrulation



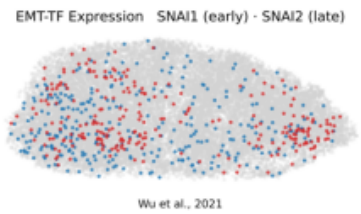
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Carcinoma (Wu et al., 2021), 10x — 20 patients / 100,064 cells (master scope); malignant only: 24,489 cells across all 20 samples (≥100 min-cell floor) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching developmental pattern reproduces — Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. Per spec §1–§2, the non-branching case (embryonic programs present, Crest an inverse/terminal gradient, no bifurcation) passes Gate 2. Wu's breast malignant manifold shows exactly that: a Gast/pSom body opposite a terminal Neural-Crest gradient (crest-high decile pt=0.94, rho=+0.55; dominant inverse gradient pSom↔Crest r=−0.71), the programs organized along one axis rather than resolved into two arms. Module-high territories are collinear (area/perim²=0.010) and ordered Gast(pt 0.34)→pSom(pt 0.46) body→Crest terminus, cell-cycle-independent — developmental signal, not proliferation. Atypical feature, flagged: Gast and pSom cross-repress (r=−0.47, obs/exp=0.36) rather than overlap; the spec explicitly permits some cross-repression in the non-branching case, and here the repulsion is organized — at the body end opposite the terminal Crest — not the disorganized spread of a NO-GO. EMT-TFs SNAI1 (root) ↔ SNAI2 (terminus) are near the detection floor (5.2%/4.1%; Δpt=0.09); the axis is carried by the module scores, not a single canonical EMT-TF.

Parameter choices

Min cells/sample	≥100 floor (gap rule ill-defined)	**Integration**	Harmony θ=10; PCA 50→20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Compartments**	Low-cycling 11,803 / Cycling 12,686

Gao 2021 Breast — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

5/6 breast samples (≥347 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 89 detached removed; kNN = 1 component); both compartments (Embryonic 5,242 / Cycling 1,080) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

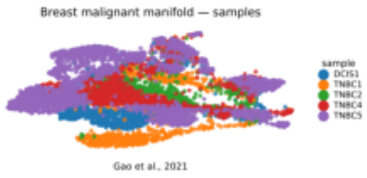
Organised Gast↔pSom cross-repression (r=−0.49) laid out along one axis; Crest anti-correlated with pSom (r=−0.58), rising to a terminal readout (ρ=+0.23; crest-high decile mean pt=0.75). Along pseudotime Gast declines (ρ=−0.31); no Gast/pSom co-expression hub (395 vs 731 exp.).

Gate 3 — hub mutual exclusivity → NA

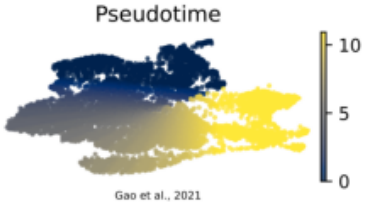
Single cross-repression axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

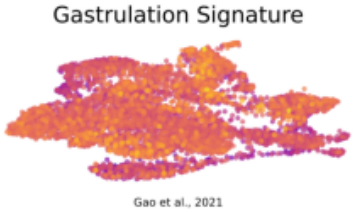
Manifold — samples



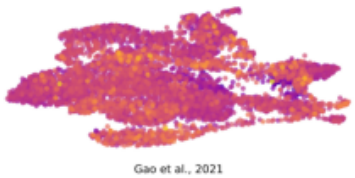
Pseudotime



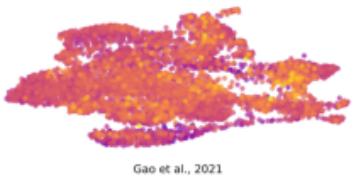
Gastrulation



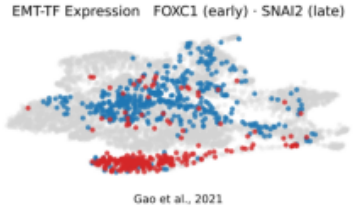
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Gao et al., 2021), 10x — 11 patients / 29,992 cells (master scope, incl. thyroid). Breast-only malignant: 6,606 pooled across both matrix leaves (5 anaplastic-thyroid samples excluded), 6,411 across 5 samples (≥347) after threshold, 6,322 after island removal.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Gao's breast malignant manifold is a single axis of organised Gast↔pSom cross-repression (r=−0.49) with a terminal neural-crest gradient (non-branching, Linear-pSom/Crest); it lacks Puram's hub-and-spoke bifurcation (no co-expression hub), so Gate 3 has no branch pair to test. The proliferation split is orthogonal to the developmental axis (root tercile 4.5% cycling vs terminus 40%), so the whole manifold is kept (Bischoff/Sun/Lee no-scoping precedent). EMT-TFs FOXC1 (early/root) → SNAI2 (late/terminus); ZEB1/TWIST2 near the detection floor (1.1%/0.3%); end-separation weak (Δpt=0.25).

Parameter choices

Min cells/sample	≥347 (gap 618→195)	**Scope**	whole manifold; breast-only
Root node	tip 2 = Gast/pSom-high	**Integration**	Harmony θ=10; PCA50; spread 1.3
Trajectory	scFates curve N=20 λ=0.3 μ=100	**Compartments**	Embryonic 5,242 / Cycling 1,080

Gao 2021 Breast Cancer — analysis summary RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

After Harmony on sample, a single connected kNN graph and iLISI 1.00→1.14 (of max 5) read as 'integrated' BUT the embedding is organised BY PATIENT: per-patient self-neighbour enrichment 2.3–5.6× at θ=10; the worst GROWS with θ (6.0/5.6/6.0/7.6× over θ 5/10/20/40, so Harmony over-corrects rather than merges) and is PC-count invariant (5.6× at n_pcs 20/30/40). Every patient carries the same embryonic modules — identity, not biology. Clusters by sample: do not force a trajectory.

Gate 2 — module geography → NA

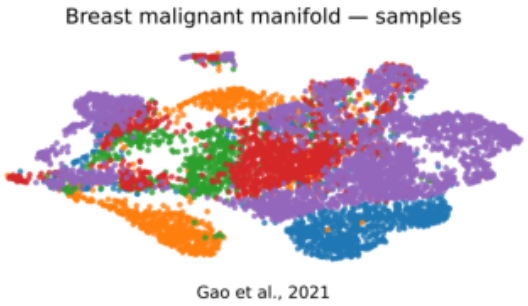
Not evaluated — Gate 1 NO-GO halts the chain before the trajectory / signature steps (a valid sample-integrated cell-state manifold is a precondition).

Gate 3 — hub cross-repression → NA

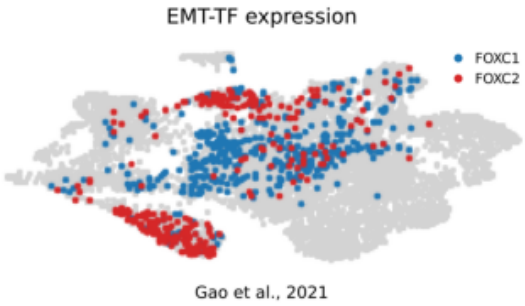
Not evaluated — branching-only, and no trajectory exists.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Triple-negative breast carcinoma + DCIS (Gao et al., 2021), 10x — 6,411 breast malignant cells (5 patients after threshold), scoped out of an 11-sample breast+thyroid bundle.

Verdict.
The pattern does NOT reproduce. The breast malignant compartment is a per-patient archipelago (inter-patient clonal / CNV heterogeneity), not a sample-integrated cell-state continuum. Gate 1 NO-GO — no trajectory fit; downstream notebooks not produced. Global module means Gast −0.024 / pSom +0.016 / Crest −0.042. Descriptive EMT-TF overlay: FOXC1 (28.5%) / FOXC2 (16.5%).

Parameter choices

Lineage scoping	scoped to breast on EPCAM/KRT8/18/19 high vs PAX8/TG~0 (bundle also had ATC thyroid)	**Sample threshold**	347 cells (gap 618→195, ratio 3.17); kept 5/6 breast samples → 6,411 cells (dropped TNBC3)
n_pcs / integration	n_pcs=20 (PC1=31%, inter-patient/CNV axis); Harmony θ swept 5/10/20/40; θ=10 best	**Geometry**	None — Gate 1 NO-GO, no manifold fit
Module means	Gast −0.024 / pSom +0.016 / Crest −0.042	**EMT-TFs**	FOXC1 (28.5%) + FOXC2 (16.5%); widened FOXC/PRRX; VIM excluded; descriptive

Gao 2021 Breast Cancer — analysis summary RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=8) on sample, n_pcs swept 20→40: per-neighbourhood iLISI 1.00→1.59 (of 6 samples), rising then plateauing at θ=8. kNN graph is ONE connected component at every θ (3–10); 1 detached cell removed. Patient-territorial but continuous — not an archipelago — GO.

Gate 2 — module geography → GO (non-branching)

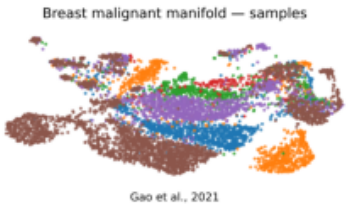
Smoothed geography: Gast↔pSom anti-correlate (r=−0.50, organized cross-repression); Crest opposes pSom (r=−0.55), weakly tracks Gast (+0.19). Along pseudotime Gast falls (ρ=−0.33) and Crest rises (ρ=+0.24; NC-high decile at t=0.77, terminal). A single Gast/pSom → Crest axis with a terminal Crest gradient — not resolved into two competing arms. Crest does not enrich a distinct Gast arm — no inversion flag.

Gate 3 — hub cross-repression → NA

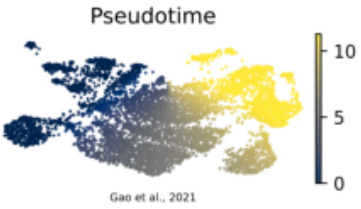
Non-branching topology. Curve: 2 tips, 0 forks (Nodes 15/20/30), fold-free (self-approach 2.7–2.9). A PPT tree returns 2 forks, reconciled as a node-density artifact, NOT a Gast/pSom Y: all arm-segments share one flat profile (Gast−pSom spread 0.009); forks dissolve as ppt_sigma rises (2→2→1→0); the 2 forks sit 2.24 UMAP units apart (median edge 0.84). No bifurcation → NA.

STEP FIGURES

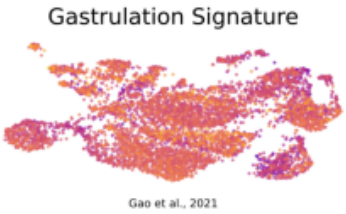
Manifold — samples



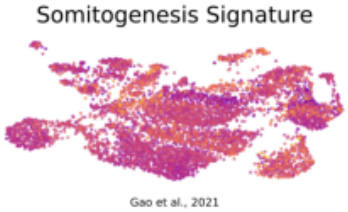
Pseudotime



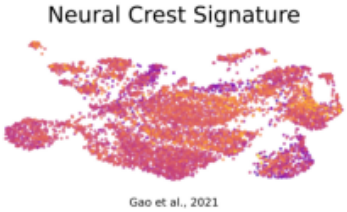
Gastrulation



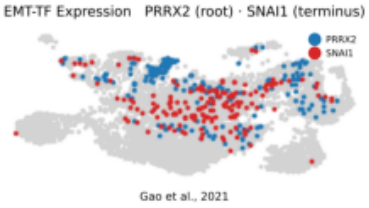
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Gao et al., 2021), 10x — 6,606 malignant cells (6 breast samples; thyroid ATC excluded); 6,605 after island removal.

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis with organized Gast↔pSom cross-repression and a terminal Crest gradient (non-branching); the PPT tree's 2 forks are a density artifact, not a bifurcation, so Gate 3 is NA. Geometry Linear — consistent with the CCCA breast cohort. Gate 1 GO / Gate 2 GO / Gate 3 NA. Raw module means Gast −0.025 / pSom +0.020 / Crest −0.045. EMT-TFs PRRX2 (early/root) → SNAI1 (late/terminus).

Parameter choices

Threshold	100 malignant cells/sample — all 6 breast samples kept (smallest TNBC3=195); thyroid ATC excluded	**Data**	Two leaves inner-joined on 22,164 shared genes; 2,696 overdispersed; no sc.pp.scale
Embedding	Harmony θ=8 on sample; n_pcs=30, n_neighbors=30, UMAP spread=1.3	**Trajectory**	scFates curve (non-branching), Nodes=20 λ=0.3 μ=100; fold-free (SA 2.73)
Root	Gast/pSom-high, Crest-low curve tip; pseudotime vs cycling ρ=+0.40 (secondary)	**Geometry**	Linear; module means Gast −0.025 / pSom +0.020 / Crest −0.045
EMT-TFs	PRRX2 (early/root) → SNAI1 (late/terminus), detection-gated (>5%)		

Gao 2021 breast cancer — analysis summary

RUN 4 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Breast malignant cells form a per-patient archipelago. After best-effort Harmony (θ=10) the per-neighbourhood iLISI reaches only 1.6 of a 3.8 ceiling (42%), same-sample kNN stays 2.9× the mixed baseline, and 4/5 tumours are self-segregated (3.5–5.1×). Not a sample-integrated manifold.

Gate 2 — module geography → N/A

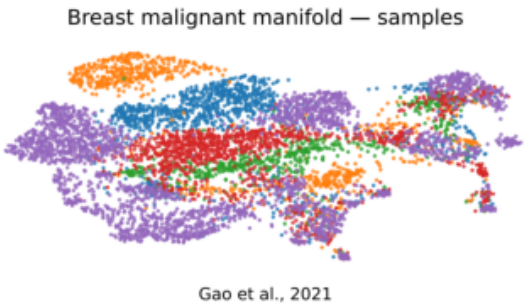
Not evaluated — Gate 1 NO-GO stops the chain. Module scores are flat and low-amplitude everywhere (Gast −0.024 / pSom +0.016 / Crest −0.042); no Gast/pSom hub, no arms.

Gate 3 — cross-repression → N/A

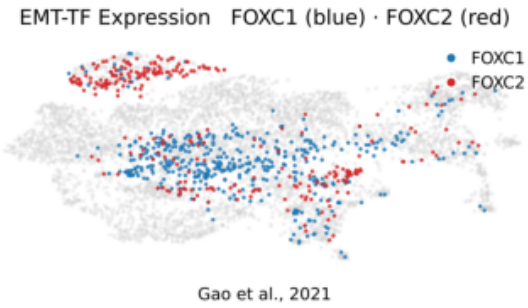
Not evaluated — no trajectory is fit (branching-only gate).

STEP FIGURES

Malignant manifold — samples



EMT-TF expression



Dataset.
Breast cancer (Gao et al., 2021), 10x — 6,411 malignant cells across 5 breast tumours (DCIS1, TNBC1–5) pooled from the /Breast and /Breast_and_thyroid leaves (inner-join 22,164 genes); 5 anaplastic-thyroid (ATC) samples dropped as a different cancer lineage.

Verdict.
The embryonic-program pattern does NOT reproduce. The breast malignant compartment is inter-patient CNV/clonal heterogeneity — a per-patient archipelago no integration strength resolves into one continuous, sample-mixed manifold — so no cell-state trajectory is fit (Gate 1 NO-GO). The only cross-patient axis is proliferation. This corrects prior attempts that forced a curve onto the over-integrated blob.

Parameter choices

Scope	breast-only; 5 ATC samples dropped (different lineage)	**Sample threshold**	≥347 cells (gap 618→195, 3.17×); drops TNBC3 → 5 samples
Embedding	2,663 overdispersed; PCA n_pcs=30 (robust 20/30/40)	**Integration**	Harmony on sample, θ swept 0/2/5/8/10/20; θ=10 plateau
Gate-1 read	per-neighbourhood iLISI / kNN / self-enrichment (not a global scalar)	**EMT-TFs shown**	FOXC1 (28.5%) · FOXC2 (16.5%); descriptive overlay (no pseudotime)

Qian 2020 Breast — analysis summary

RUN 1 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant cells form a per-patient archipelago. A single connected kNN graph appears only at the most generous integration (θ=10), where iLISI is just 1.1 of 7 samples — the scalar a naïve test reads as 'integrated' yet the embedding is organised BY PATIENT: 6/7 patients self-cluster far above chance (self-neighbour enrichment 6.8–23.5×, all past the 2.5× line). θ=5 splits the manifold into two disconnected components, one PURE sample 54 (the dominant 65% tumour). Integration does not fix it — enrichment GROWS with θ and is PC-count-invariant (22× at n_pcs 20/30/40), so Harmony over-corrects. Every patient carries the SAME embryonic modules — inter-patient CNV/clonal heterogeneity, not a shared cell-state continuum — do not force a trajectory.

Gate 2 — module geography → NA

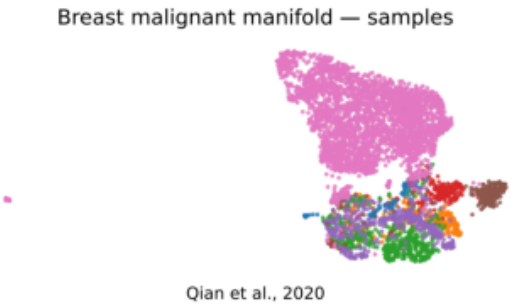
Not reached. Gate 2 tests a Gast/pSom hub and Crest geography on a fitted manifold; with Gate 1 NO-GO no manifold/trajectory is fit, so this gate does not apply.

Gate 3 — hub cross-repression → NA

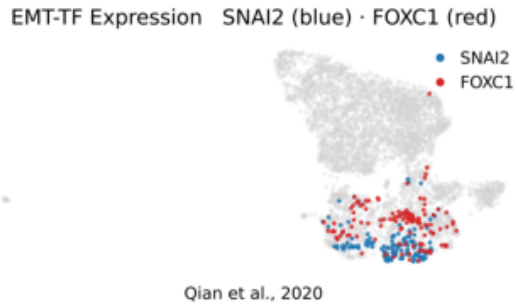
Not reached (branching-only, downstream of Gate 2). No hub, no bifurcation, no hub r.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Breast cancer (Qian et al., 2020), 10x (UMI counts) — 4,818 malignant cells across 7 samples = 7 patients. Gate 1 NO-GO: per-patient archipelago, no integrated continuum.

Verdict.
The embryonic-program pattern does NOT reproduce in Qian 2020 breast cancer: the malignant compartment is a per-patient archipelago — no sample-integrated, continuous cell-state manifold. One patient (sample 54, 65% of cells) even forms a fully disconnected component at moderate integration. Gate 1 NO-GO — analysis stops before the trajectory (Gates 2 & 3 N/A). EMT-TFs SNAI2 and FOXC1 are the broadest detectable factors, reported over the un-integrated manifold.

Parameter choices

Sample threshold	≥100 floor; argmax gap 3111→518 (6.01×) = dominant tumour, not a tail; keeps 7/7	**Gene / PCA**	1,225 overdispersed genes; n_pcs=20 (PC1 21% var, inter-patient/CNV axis)
Integration	Harmony on sample, swept θ 2/5/10/20/40; θ=10 first single component (diagnostic)	**Root / geometry**	none — no trajectory (Gate 1 NO-GO); per-patient archipelago
EMT-TFs shown	SNAI2 (11.3%) · FOXC1 (10.4%), overlaid without pseudotime		

Qian 2020 Breast Cancer — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 7 patients traverse one continuous manifold: the shared developmental axis (signature PC1, 54% var) is spanned by 7/7 patients (minor samples alone cover 78%); module variance from patient identity only R^2=0.05–0.15. iLISI is imbalance-limited by the 65%-dominant tumour (ceiling 2.26 of 7); the call is led by the embedding + biology, not a scalar.

Gate 2 — module geography → GO (non-branching)

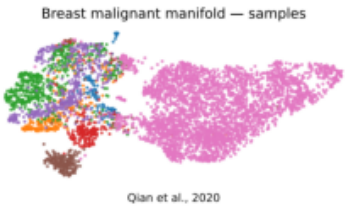
The embryonic programs form one developmental geography: a spatial Gast↔Crest inverse gradient (smoothed r = −0.49, batch-confirmed within patients), Gast & pSom overlapping toward the origin. An honest PPT tree put every tip at negative Gast — no Gast arm, no Gast↔pSom hub. Crest is an inverse gradient, not a Gast-arm enrichment: no inversion flag.

Gate 3 — hub cross-repression → NA

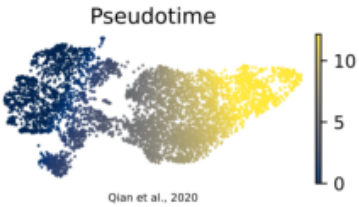
Non-branching manifold — no bifurcation and no Gast/pSom hub, so the cross-repression test does not apply.

STEP FIGURES

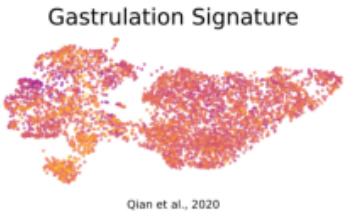
Manifold — samples



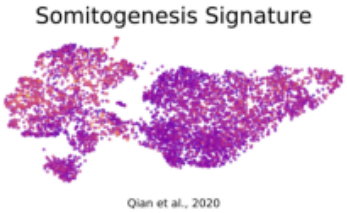
Pseudotime



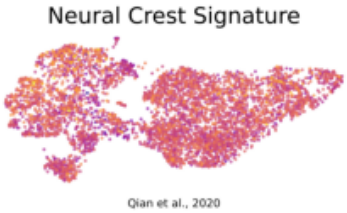
Gastrulation



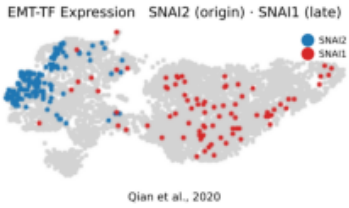
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Qian et al., 2020), 10x — 4,801 malignant cells, 7 patients (treatment-naive).

Verdict.
The embryonic programs reproduce as a NON-BRANCHING manifold: gastrulation and somitogenesis overlap toward a developmental origin, with neural crest forming a spatial inverse gradient to gastrulation (Crest↔Gast r = −0.49) and no Gast/pSom hub or bifurcation. Geometry = Linear-pSom/Crest. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TF axis SNAI2 (early/origin) → SNAI1 (late), detection-gated panel.

Parameter choices

Malignant cells	4,818 raw → 4,801 (thresholded, de-islanded)	**Sample threshold**	≥100 cells (gap rule mis-fires: dominant high-side cliff) → 7/7
Overdispersed genes	1,225 (scFates find_overdispersed)	**Integration**	Harmony θ=10, n_pcs=20 (iLISI-sweep derived)
Embedding	UMAP spread 1.3, n_neighbors 30, seed 1; island removal 17 cells / 1 speck	**Trajectory**	ElPiGraph principal CURVE, 20 nodes (fold-free, self-approach 2.99)
Root	Gast/pSom-high, Crest-low tip (Crest late)	**EMT-TFs**	SNAI2 (origin) · SNAI1 (late), detection-gated

Qian 2020 breast cancer — analysis summary RUN 3 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Harmony on sample cannot integrate the compartment. Across θ 2–20 (confirmatory to 50) × n_pcs 20–40 the regenerated embedding keeps the dominant patient (sample 54, 65% of malignant cells) as a segregated territory: at low θ a literal disconnected component (2,675 vs 2,143 cells), at high θ a lobe the kNN graph merges only as a connectivity artifact. Per-neighbourhood iLISI stays ≤1.5 of 2.26; sample-54 cells keep 92% same-sample 30-NN (chance 65%). Dropping sample 54, the other 6 integrate well (iLISI 3.4/4.86) — the obstruction is the dominant patient, not a generic batch.

Gate 2 — module geography → NA

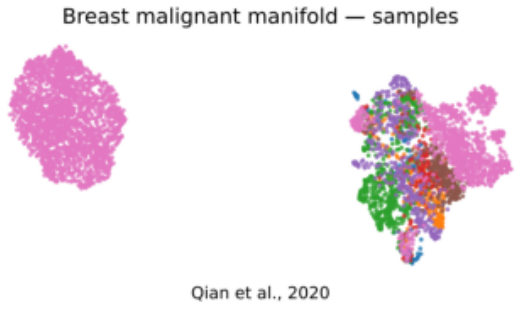
Not reached. Gate 1 NO-GO stops the chain before a manifold/signatures are fit, so module geography is not evaluated (manifold / signatures / correlation / modules notebooks are not produced, per the deliverable spec).

Gate 3 — cross-repression → NA

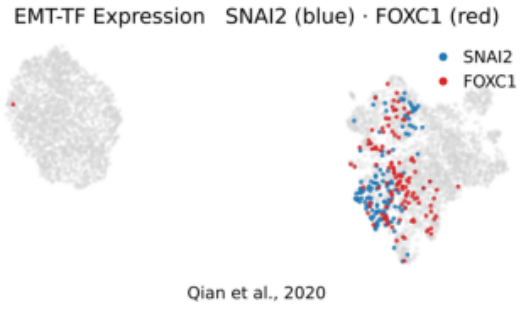
Not reached — no trajectory, hence no bifurcation and no hub cross-repression test.

STEP FIGURES

Malignant manifold — samples



EMT-TF expression



Dataset.
Breast cancer (Qian 2020), 10x, 4,818 malignant cells over 7 patients; sample 54 alone is 65% of the malignant compartment (one dominant patient). No sample dropped (all 7 pass the 10-cell floor).

Verdict.
The embryonic pattern does NOT reproduce here — the malignant compartment cannot be integrated into a continuous manifold: a single dominant patient forms an irreconcilable segregated territory at every integration strength, so no cell-state trajectory is fit. Gate 1 NO-GO / Gate 2 NA / Gate 3 NA.

Parameter choices

Sample threshold	≥10 malignant cells/sample; all 7 pass → 4,818 cells	**Overdispersed genes**	1,225 (scFates find_overdispersed); no sc.pp.scale
Integration	Harmony on sample; θ 2/5/10/20 (+50); best mixing θ=5	**Embedding**	n_pcs swept 20/30/40 (committed 30); n_neighbors=30; spread=1.3
Gate-1 metric	per-neighbourhood iLISI (max 2.26) + UMAP-radius-graph components	**Module means (raw)**	Gast −0.020 · pSom +0.031 · Crest +0.013
Geometry	NA — no trajectory fit (Gate 1 NO-GO)	**EMT-TFs shown**	SNAI2 (11.3%) · FOXC1 (10.4%); FOXC/PRRX in panel

Pal 2021 Breast — analysis summary

RUN 1 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Mixing + continuity judged on the regenerated embedding. iLISI 1.0 → 4.96 of 22 samples (Harmony θ=10); one continuous connected component (11 speck cells removed). A proliferation compartment (76% cycling, 14,165 cells) is resolved before the GO → the trajectory is scoped to the low-cycling developmental Body (45,191 cells).

Gate 2 — module geography → GO (non-branching)

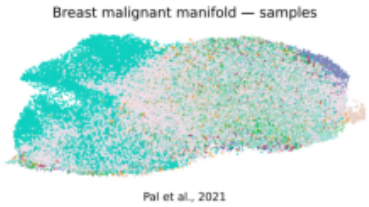
No Gast/pSom two-arm hub; a single embryonic axis runs pSom-high (origin) → Crest-high (terminus). Smoothed pSom vs Crest r=−0.93; joint pSom&Crest enrichment 0.23 (<1, inverse gradient). Crest terminal (high-decile pseudotime 0.92); Gast weak and flat.

Gate 3 — hub cross-repression → NA

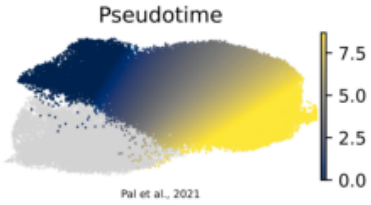
Branching-only gate → not applicable (single linear axis, no bifurcation to test for cross-repression).

STEP FIGURES

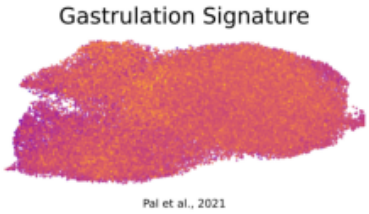
Manifold — samples



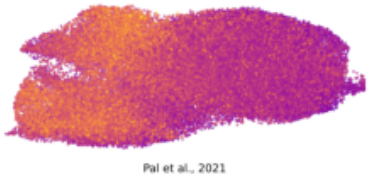
Pseudotime



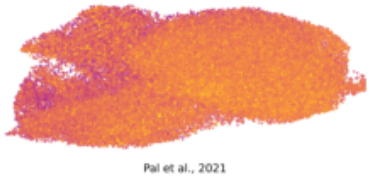
Gastrulation



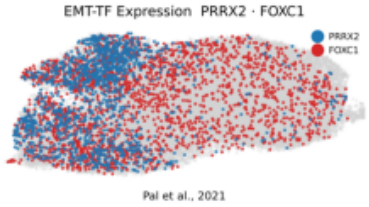
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Pal et al., 2021), 10x — 45 samples / 305,224 cells (master scope); 59,356 malignant cells (22 samples). Trajectory scoped to the low-cycling developmental Body (45,191 cells; 14,165-cell cycling compartment excluded).

Verdict.
The embryonic pattern reproduces as the NON-BRANCHING case — one continuous, sample-integrated malignant manifold carrying a single somitogenesis→neural-crest developmental axis, with no Gast/pSom bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs PRRX2 (strongest root) & FOXC1 (most expressed, 16%) — all detected factors origin-enriched (no terminus factor). Module strength Gast μ=−0.007 · pSom μ=+0.036 · Crest μ=−0.034 → Linear-pSom/Crest.

Parameter choices

Min-cell threshold	≥12 (geom-mean of 56→3 cliff, ratio 18.67); keeps 22 of 26	**Integration**	Harmony on sample, θ=10 (swept 2/5/10); n_pcs=20, spread=1.3, 3733 overdispersed
Sub-region scoping	Leiden res=0.15 → 2 lobes; Cycling lobe (76% cycling) excluded	**Geometry**	Non-branching curve (scFates N=20 λ=0.3); 2 tips, 0 forks, fold-free (SA 2.98)
Root node	curve tip 0 — pSom/Gast-high, Crest-low, 9% cycling; pt vs prolif ρ=+0.01	**EMT-TFs**	PRRX2 (strongest root) & FOXC1 (16%) — all detected origin-enriched; VIM excluded

Pal 2021 Breast Cancer — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Batch mixing + continuity, judged on the regenerated embedding across a θ × n_pcs sweep (per-neighbourhood iLISI). At θ=10/n_pcs=20 mixing plateaus (median iLISI 4.96 of ~22); every UMAP band holds 18–19 samples on one connected manifold (8 island cells removed).

Gate 2 — module geography → GO (non-branching)

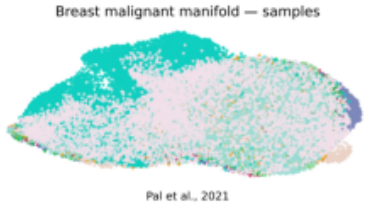
One organized developmental axis: pSom high at the root, Crest high at the terminus (Crest↔pSom spatial r=−0.93; Gast↔pSom r=−0.57), an inverse terminal Crest gradient → GO. FLAG: Crest co-enriches the Gast side (r=+0.40 vs pSom −0.93), an inversion.

Gate 3 — hub cross-repression → NA

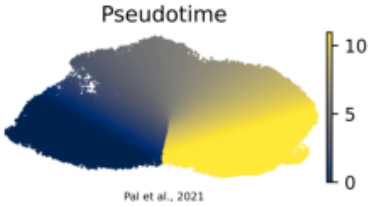
Branching-only. The manifold is non-branching (PPT forks scale with node count 10→21; a curve fits fold-free, 2 tips / 0 forks), so there is no bifurcation to test for Gast↔pSom cross-repression.

STEP FIGURES

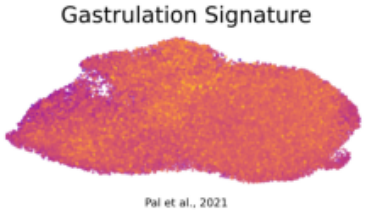
Manifold — samples



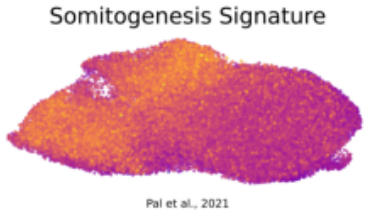
Pseudotime



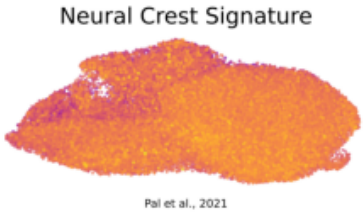
Gastrulation



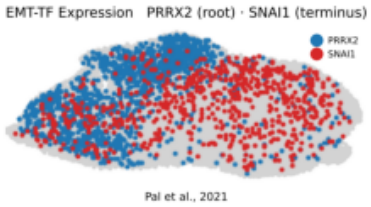
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Pal et al., 2021), 10x Chromium — 59,359 malignant cells · 22 samples (ER+, HER2+, TNBC).

Verdict.
The embryonic pattern reproduces as a non-branching axis (Gate 1 GO / Gate 2 GO / Gate 3 NA): a single gastrulation/somitogenesis origin resolves toward a neural-crest terminus, with no Gast/pSom bifurcation. Crest-arm inversion flagged. Module strengths Gast −0.007 / pSom +0.036 / Crest −0.034 (Linear-pSom/Crest). EMT-TFs PRRX2 (early/root) ↔ SNAI1 (late/terminus).

Parameter choices

Sample threshold	≥12 cells (geom mean of 56→3 gap); 22/26 samples kept	**Integration**	Harmony θ=10 (iLISI plateau 5→10), n_pcs=20
Overdispersed genes	3,733 (scFates find_overdispersed)	**Island removal**	radius graph 6× median-NN, largest component (8 cells dropped)
Geometry	Non-branching (ElPiGraph curve; 2 tips, 0 forks, fold-free SA=2.52)	**Root**	pSom/Gast-high, Crest-low tip (origin; cyc ρ=−0.57)
Module means	Gast −0.007 / pSom +0.036 / Crest −0.034 (Linear-pSom/Crest)	**EMT-TFs**	PRRX2 (early/root) ↔ SNAI1 (late/terminus)

Pal 2021 breast cancer — analysis summary

RUN 3 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

iLISI 1.00→4.95 of 22 samples (per-neighbourhood, θ=10); one connected component on the independent Harmony-PCA and diffusion-map graphs. Mixed and continuous.

Gate 2 — module geography → GO (non-branching)

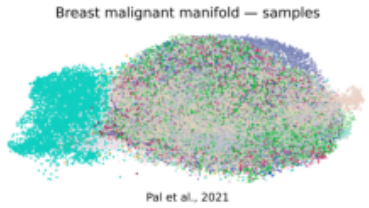
Single organized axis: pSom/Gast-high origin → Crest-high terminus (Crest decile pt 0.93; pSom&Crest enrich 0.23, inverse terminal gradient). Non-branching — no Gast/pSom co-expression hub (enrich 1.03).

Gate 3 — cross-repression → NA

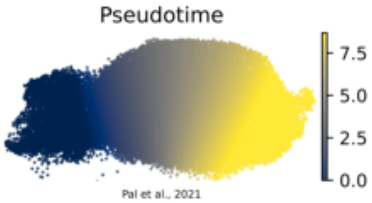
No bifurcation (no hub, no second arm) — not applicable. Gast & pSom show mild spatial cross-repression (smoothed r=−0.32 on the Body), permitted in the non-branching case.

STEP FIGURES

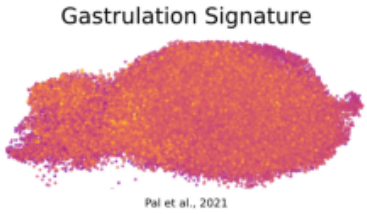
Manifold — samples



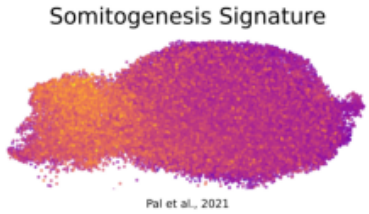
Pseudotime



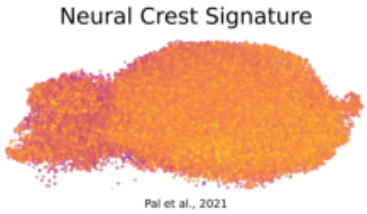
Gastrulation



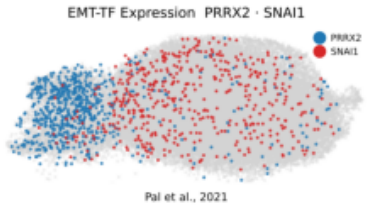
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Pal et al., 2021), 10x — 59,367 malignant cells (22/26 samples); trajectory scoped to the 44,398-cell low-cycling Body. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form — one continuous pSom/Gast → Crest developmental axis, no Gast/pSom bifurcation. G1 GO / G2 GO / G3 NA. EMT-TFs PRRX2 (origin) · SNAI1 (latest) — all 5 detected factors peak at the origin, none Crest-terminal.

Parameter choices

Min-cell threshold	12 (geom-mean of 56→3 cliff); keeps 22/26 samples	**Integration**	Harmony θ=10, n_pcs=20 (iLISI plateau 5–10)
Geometry	Linear (non-branching); scFates curve N=20, 2 tips / 0 forks	**Root**	pSom/Gast-high, Crest-low origin; terminus Crest-high
Scope (cell-cycle)	Body 44,398 (low-cycling); 14,965 proliferative compartment excluded	**Module means (all)**	Gast −0.007 · pSom +0.036 · Crest −0.034; hub enrich 1.03 (none)
Cycle independence	pt vs proliferation ρ=0.10 (cycle-independent axis)	**EMT-TFs**	PRRX2 (origin) · SNAI1 (latest); all 5 detected peak at origin

Bassez 2021 Breast — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) integrates all 41 samples: same-sample neighbour fraction 0.37 → 0.21, iLISI median 5.51 → 9.10 of 41. One connected component (45 cells / 27 specks removed). Read on the regenerated embedding + per-neighbourhood iLISI, not a scalar: mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

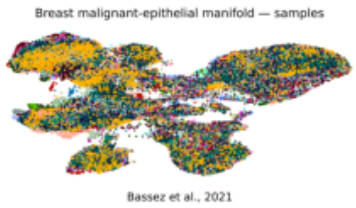
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=−0.21), both oppose Crest (pSom↔Crest r=−0.82, Gast↔Crest r=−0.24). Along pseudotime Gast/pSom fall (ρ −0.11/−0.32), Crest rises (ρ +0.32) to the terminal lobe. Joint pSom&Crest enrichment 0.23 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

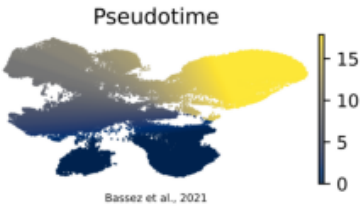
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

Manifold — samples

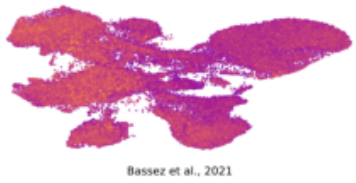


Pseudotime

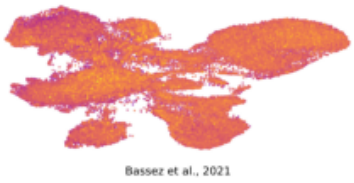


Gastrulation

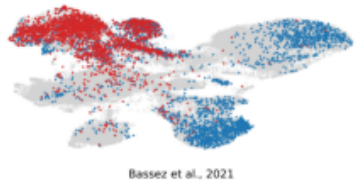
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast tumours (Bassez et al., 2021), 10x — 65,837 malignant cells / 41 samples; single low-cycling body (2% cycling).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (root) → PRRX1 (terminus).

Parameter choices

Sample threshold	geom-mean of 92→3 cliff = 16 (drops 3-cell BIOKEY_34) → 41/42 samples	**Overdispersed genes**	2,498 (scFates find_overdispersed)
Integration	Harmony on sample, θ=10 (mixing flat across sweep)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 45 cells / 27 islands removed → 65,837
Geometry	Non-branching → scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.53	**Root**	tip 2 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast −0.014 · pSom +0.022 · Crest −0.005 (Linear-pSom/Crest)	**EMT-TFs shown**	ZEB2 (root, ρ −0.07) ↔ PRRX1 (terminus, ρ +0.07); detection-gated

Bassez 2021 Breast Cancer — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

41/42 samples intermix on one continuous manifold (per-neighbourhood iLISI median ~9 of 41 after Harmony θ=10 vs 5.6 unintegrated); no per-patient archipelago, 80 island cells removed. Judged on the embedding.

Gate 2 — module geography → GO (non-branching)

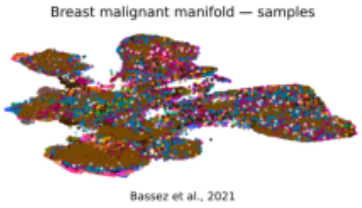
Single pSom→Crest developmental gradient (smoothed pSom↔Crest r = −0.82); Crest terminal, gastrulation weak/flat. A PPT tree fit as an honest branching test put every tip at negative Gast — no Gast arm, no hub.

Gate 3 — hub cross-repression → NA

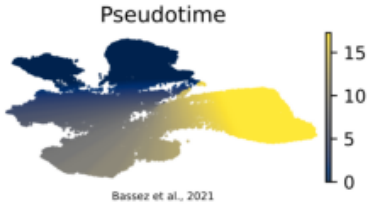
Non-branching manifold — no bifurcation and no Gast/pSom hub, so the cross-repression test does not apply. Crest enriches the trajectory terminus (not a Gast arm): no Crest-arm inversion flag.

STEP FIGURES

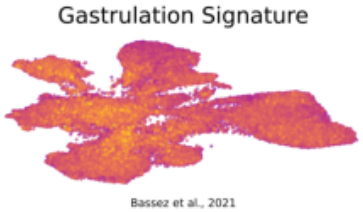
Manifold — samples



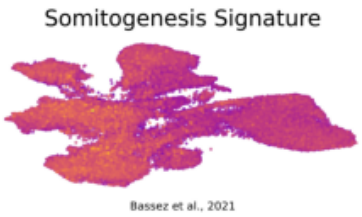
Pseudotime



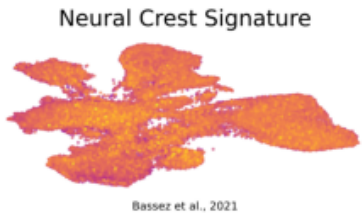
Gastrulation



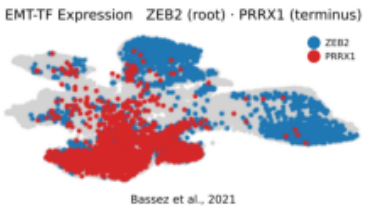
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Bassez et al., 2021), 10x — 65,802 malignant cells, 41 patients (TNBC / ER+ / HER2+, pre/on anti-PD1).

Verdict.
The embryonic programs reproduce as a NON-BRANCHING single axis: a posterior-somitogenesis origin resolving into a neural-crest terminus (pSom↔Crest r = −0.82), with gastrulation weak and no Gast/pSom hub. Geometry = Linear-pSom/Crest. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TF axis ZEB2 (early) → PRRX1 (late).

Parameter choices

Malignant cells	65,885 → 65,802 (thresholded, de-islanded)	**Sample threshold**	16 cells (multiplicative-gap; drops one 3-cell sample) → 41/42
Overdispersed genes	2,498 (scFates find_overdispersed)	**Integration**	Harmony θ=10, n_pcs=20 (iLISI-sweep derived)
Embedding	UMAP spread 1.3, n_neighbors 30, seed 1; island removal 80 cells / 21 specks	**Trajectory**	ElPiGraph principal CURVE, 20 nodes (fold-free, self-approach 2.43)
Root	pSom-high / Crest-low tip (origin; Crest late)	**EMT-TFs**	ZEB2 (root) · PRRX1 (terminus), detection-gated

Bassez 2021 Breast Cancer — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10, n_pcs=30) on sample, swept {2,5,10,20} × n_pcs {20,30,40} on a per-neighbourhood iLISI: median 4.9→8.6 of 41 samples (worst-neighbourhood p10 1.2→3.2), plateaus 2→10 then declines at θ=20. One connected component; 418 speck cells (0.63%) in 38 islands removed. Samples enter as fine intermixed speckle across one continuous, low-cycling body (3.1% cycling) — no per-patient archipelago → GO.

Gate 2 — module geography → GO (non-branching)

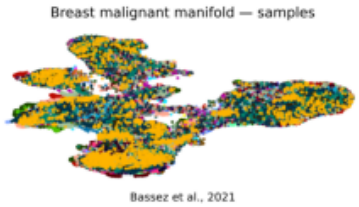
Single (Gast/pSom-origin) → Crest-terminus gradient. Smoothed geography: Gast~pSom −0.20, Crest~pSom −0.82, Crest~Gast −0.24. Along pseudotime pSom falls (ρ−0.27) and Crest rises (ρ+0.27); joint pSom&Crest enrichment 0.23 (<1 = inverse terminal gradient, not a co-expression hub). No Gast-enriched arm at any PPT stiffness. Crest-arm inversion NOT flagged (Gast&Crest enrichment 0.50 <1; Crest tracks the terminus).

Gate 3 — hub cross-repression → NA

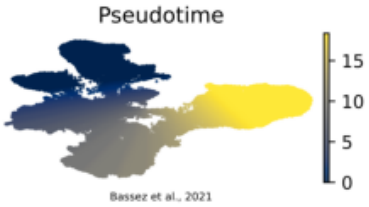
Non-branching topology (curve: 2 tips / 0 forks across N=15/20/30; PPT tree forks lack a Gast arm — max-tip Gast −0.001 vs Crest~pSom −0.87..−0.91, i.e. lobe over-fitting of one axis). No bifurcation to test → NA.

STEP FIGURES

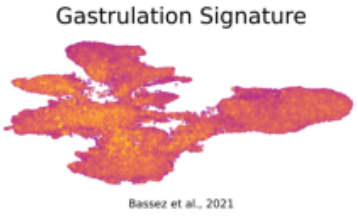
Manifold — samples



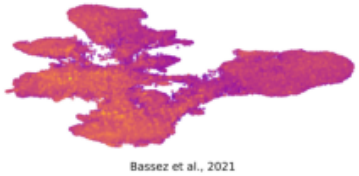
Pseudotime



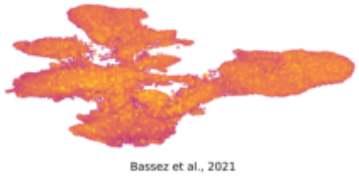
Gastrulation



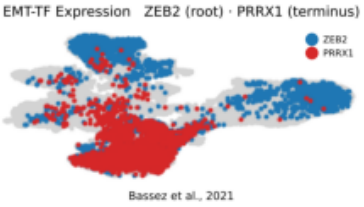
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Bassez et al., 2021), 10x — 65,885 malignant cells (42 samples); 65,464 analysed (41 samples, cell floor 17).

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis (non-branching); no Gast→pSom bifurcation, so Gate 3 is NA. Geometry Linear-pSom/Crest, consistent with the linear outcome across most CCCA datasets. Gate 1 GO / Gate 2 GO / Gate 3 NA. Raw means Gast −0.014 / pSom +0.022 / Crest −0.005. EMT-TFs ZEB2 (early/root) → PRRX1 (late/terminus).

Parameter choices

Threshold	17 cells/sample (geom mean across 92→3 spillover cliff); dropped BIOKEY_34 → 41/42 samples	**Embedding**	PCA 50, no sc.pp.scale; Harmony θ=10; n_pcs=30, n_neighbors=30, UMAP spread=1.3; 2,498 overdispersed
Trajectory	scFates curve (non-branching), Nodes=20 λ=0.3 μ=100; fold-free (SA 2.43); continuity corr=+0.93	**Root**	tip 2 = Gast/pSom-high, Crest-low (origin; Crest LATE); pseudotime vs cycling +0.17
Geometry	Linear-pSom/Crest; raw means Gast −0.014 / pSom +0.022 / Crest −0.005	**EMT-TFs**	ZEB2 (early/root) → PRRX1 (late/terminus), detection-gated

Bassez 2021 Breast Cancer — analysis summary

RUN 4 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

θ=10 Harmony on sample: iLISI 4.9→8.6/41 (p10 3.2). θ-sweep on the embedding: θ=2 fragments (45% main), θ=10 one continuous body (>99.9%). Lobes span all 41 patients (max 0.17). Complete island removal (0.17% specks). GO.

Gate 2 — module geography → GO (non-branching)

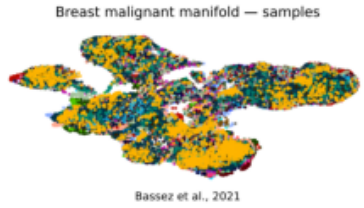
Gast~pSom co-vary (raw −0.10 / sm −0.21), both oppose Crest (sm −0.24 / −0.82). Along pseudotime Gast/pSom fall (ρ−0.12/−0.27), Crest rises (ρ+0.30); joint pSom&Crest 0.23<1 — inverse terminal gradient. One axis, no arms. No inversion.

Gate 3 — hub cross-repression → NA

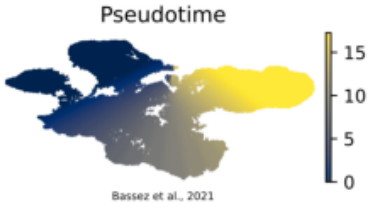
Non-branching (curve 2 tips/0 forks; PPT 0 Gast-enriched tips). No bifurcation → NA.

STEP FIGURES

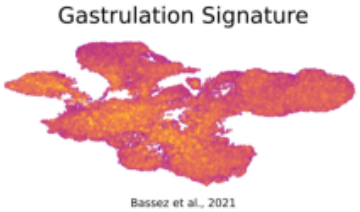
Manifold — samples



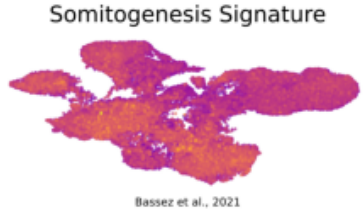
Pseudotime



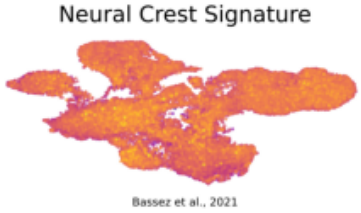
Gastrulation



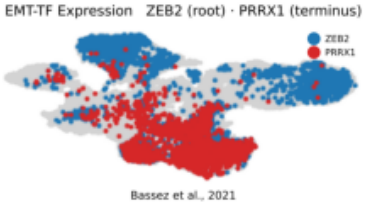
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Bassez et al., 2021), 10x — 65,885 malignant cells; 65,615 analysed (41 samples, ≥20 cells).

Verdict.
The embryonic pattern reproduces as one continuous (Gast/pSom) → Crest axis — a linear (non-branching) manifold. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (early/root) → PRRX1 (late/terminus).

Parameter choices

Threshold	20 cells/sample → 41/42 kept (gap-finding; anchor 100 drops 39 real samples)	**Embedding**	n_pcs=30 (scree, centres 20–40); Harmony θ=10 (iLISI plateau + continuous embedding)
Islands	single-pass radius-graph specks (0.17%); no re-embedding	**Trajectory**	scFates curve, N=20 (2 tips, 0 forks, fold-free self-approach 2.6)
Root	tip 2 = Gast/pSom-high, Crest-low (Crest is late, never root)	**Geometry**	non-branching / linear (Gast/pSom → Crest single axis; corr x_lin~t = +0.94)
EMT-TFs	ZEB2 (root) → PRRX1 (terminus), detection-gated		

Bassez 2021 Breast Cancer — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

41 samples intermixed on one continuous manifold; sample iLISI 6.1→9.1 (of 41) after Harmony θ=10; 363 island cells removed; cycling orthogonal to modules.

Gate 2 — module geography → GO (non-branching)

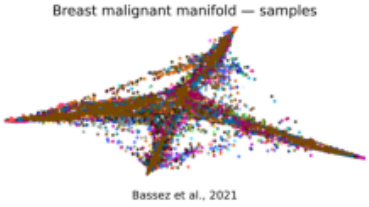
Single collapsed axis: pSom/Gast developmental origin → Crest terminal (pSom↔Crest r=−0.82). No Gast arm at any tip; PPT tree over-branches — not a bifurcation.

Gate 3 — hub cross-repression → NA

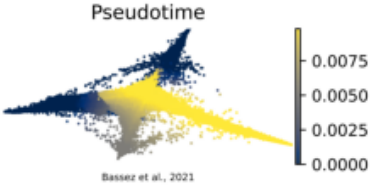
Non-branching geometry — no Gast/pSom competing arms to test.

STEP FIGURES

Manifold — samples



Pseudotime

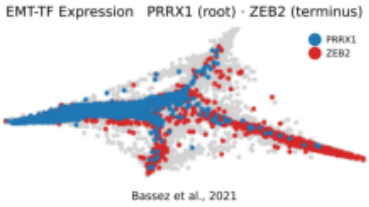


Gastrulation

Somitogenesis

Neural Crest

EMT-TF expression



Dataset.
Breast Cancer (Bassez et al., 2021), 10x — 41 samples, 65,519 malignant cells.

Verdict.
Pattern reproduces as the non-branching case — one embryonic axis, pSom/Gast origin → Crest terminus. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means (raw): Gast −0.014 / pSom +0.022 / Crest −0.005. EMT-TFs PRRX1 (early/root) · ZEB2 (late/terminus).

Parameter choices

Sample threshold	16 (gap-finding) → 41/42 kept	**Embedding**	overdispersed 2498; n_pcs 20; Harmony θ=10 on sample; n_neighbors 30; UMAP spread 1.3; islands 363 removed
Trajectory	scFates curve on diffusion rep, Nodes=15, λ=0.3, μ=100; fold-free (self-approach 2.56)	**Root**	pSom/Gast-high tip (pt 0.28); Crest terminal (0.74)
Annotation	Bassez2021 malignant meta-programs (EMT I dominant) + data-derived compartments	**Module means**	Gast −0.014 / pSom +0.022 / Crest −0.005 (raw)
EMT-TFs	PRRX1 (root) · ZEB2 (terminus)		

Bassez 2021 breast cancer — analysis summary

RUN 6 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample: per-neighbourhood iLISI 1.6→7.8 of 34 samples (θ sweep 1/2/5/10, mixing rises to a plateau). One connected component, 0 detached islands. Samples intermix across one continuous strip — GO.

Gate 2 — module geography → GO (non-branching)

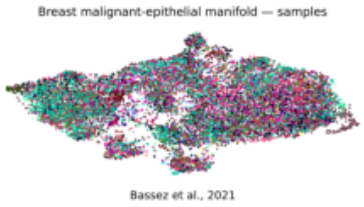
Gast & pSom cross-repress (smoothed r=−0.70; node-cluster r=−0.55) with NO co-expression hub (both-high enrichment 0.11<1). Crest opposes pSom (r=−0.91) as an inverse terminal gradient (NC-high decile pt=0.82; joint pSom&Crest 0.25<1). A single cross-repressive (Gast/pSom)→Crest axis, not two diverging arms — GO.

Gate 3 — cross-repression → NA

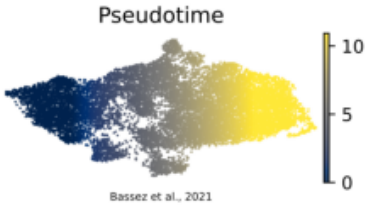
Non-branching topology (curve: 2 tips, 0 forks across N=15/20/30; PPT tree tips all pSom-high/Gast-low — no Gast arm). No bifurcation to test — NA.

STEP FIGURES

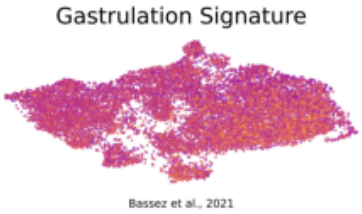
Manifold — samples



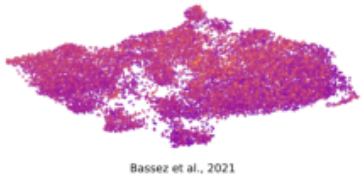
Pseudotime



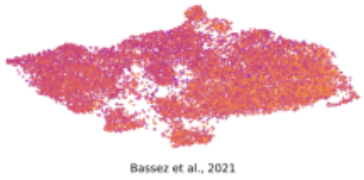
Gastrulation



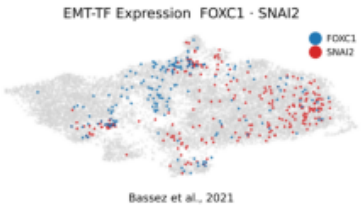
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Bassez et al., 2021), 10x — 13,834 malignant-epithelial cells (scoped from 65,885 CCCA "Malignant", 21% epithelial) across 34 samples (≥20 cells). Non-branching Linear.

Verdict.
The embryonic pattern reproduces on the correctly-scoped malignant epithelium as a single gastrulation/somitogenesis → neural-crest axis with organised Gast/pSom cross-repression (non-branching); no co-expression hub diverging into two arms, so Gate 3 is NA. Geometry Linear. EMT-TFs FOXC1 (early/root) → SNAI2 (late/terminus).

Parameter choices

Scoping	CCCA "Malignant" 21% epithelial → scoped to 13,834 epithelial cells	**Threshold**	≥20 epithelial cells/sample (re-derived); 34/42 samples
Embedding	Harmony θ=10; n_pcs=20, n_neighbors=30, spread=1.3; 2,211 HVG	**Trajectory**	scFates curve (non-branching), N=20, μ=100; fold-free (2.98)
Root	pSom-high, Crest-low tip (pSom early, Crest late); ρ pSom −0.34, Crest +0.44	**Geometry**	Linear; raw means Gast −0.027 · pSom +0.030 · Crest −0.011
EMT-TFs	FOXC1 (early/root) → SNAI2 (late/terminus); VIM excluded		

Griffiths 2021 Breast — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Samples/patients intermix per-neighbourhood on ONE connected manifold. Harmony(sample) same-sample neighbour fraction 0.91 → 0.17 (θ=5); iLISI 1.0 → 13.4 of 67 samples. ~1,100 detached cells (1.0%) removed as islands; largest component retained. Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

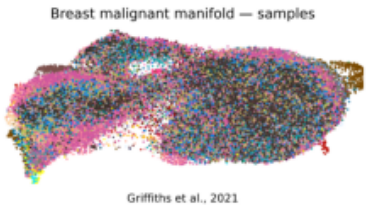
Single Gast/pSom → Crest axis, no hub diverging into two arms. Organized Gast↔pSom cross-repression (smoothed r=−0.42); Crest an inverse terminal gradient (Crest vs pSom r=−0.76; NC-high decile at pseudotime 0.77; joint pSom&Crest enrichment 0.31 < 1). Curve fits clean (2 tips, 0 forks, fold-free); PPT tree only over-fragments (5–13 forks). Crest does NOT enrich a Gast arm (no arms) — no flag.

Gate 3 — hub cross-repression → NA

No bifurcation (non-branching geometry) → branching-only Gate 3 not evaluated; 04_correlation / 05_modules not produced (spec §6).

STEP FIGURES

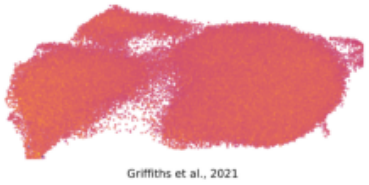
Manifold — samples



Pseudotime

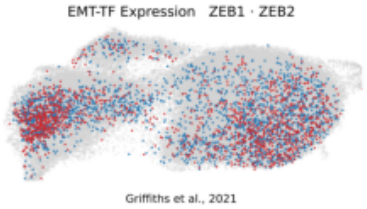
Gastrulation

Somitogenesis



Neural Crest

EMT-TF expression



Dataset.
Breast cancer (Griffiths et al., 2021), 10x + iCell8 — 108,849 malignant cells on the trajectory (67/95 samples, 110,195 cells loaded; two technology leaves 10X 110,568 + iCell8 804, concatenated on 15,691 shared genes; a 246-cell iCell8/cycling lobe scoped out).

Verdict.
The embryonic-program pattern reproduces as the non-branching case: one continuous, sample-integrated gastrulation/somitogenesis → neural-crest axis with organized Gast↔pSom cross-repression and terminal Crest — no Gast/pSom hub bifurcating into two competing arms. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB1, ZEB2 expressed but flat along pseudotime (|ρ|<0.05); overlay the two most-expressed non-directionally.

Parameter choices

Min-cell threshold	≥100 (depth-invariant floor; no dominant cliff) → 67/95 samples, 110,195 cells	**Technology**	10X 110,568 + iCell8 804 malignant, concatenated on 15,691 shared genes
Embedding	n_pcs=30, Harmony(sample) θ=5, spread=1.3, k=30	**Trajectory**	scFates curve N=20, non-branching; fold self-approach 2.95 (fold-free)
Compartment	246-cell lobe (100% iCell8, 70% cycling) scoped out as technology/cycling confound	**Root**	curve tip 0 (Gast/pSom-high, Crest-low); axis cell-cycle-free (pt vs cycling r=−0.04)
Geometry	Linear-pSom/Crest (global means Gast −0.021, pSom −0.052, Crest +0.021)	**EMT-TFs**	ZEB1, ZEB2 expressed but flat (|ρ|<0.05); overlay non-directionally; VIM excluded

Griffiths 2021 Breast Cancer — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony on sample (θ sweep 2/10/20; per-neighbourhood iLISI 1.7→19.2 at θ=10, the peak). The regenerated embedding is one continuous, thoroughly sample- and patient-mixed body; cell cycle is flat (no proliferation lobe). Only a 328-cell iCell8 technology streak + tiny specks detach (island removal 110,195→109,353). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

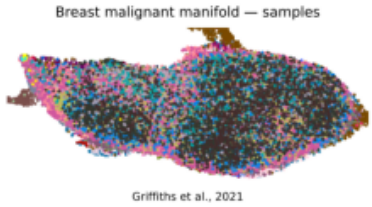
Organised Gast↔pSom cross-repression along one fold-free axis (pseudotime ρ: Gast +0.40, pSom −0.36) with a terminal Crest gradient (ρ +0.28) — the non-branching pattern. FLAG: Crest enriches the Gast-leaning terminus, not pSom (an inversion; passes anyway).

Gate 3 — hub cross-repression → NA

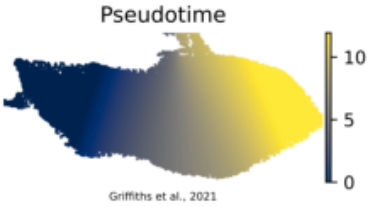
No bifurcation: a PPT tree yields 15 tips (13 branch nodes) all with identical module identity, and Gast/pSom co-express below the independence expectation (7.3% vs 9%) — no convergence hub. Gate 3 is branching-only, so NA.

STEP FIGURES

Manifold — samples

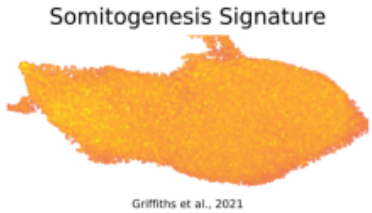


Pseudotime

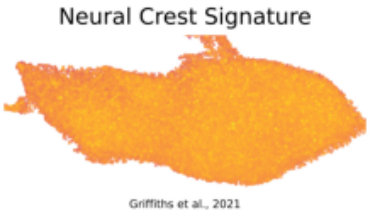


Gastrulation

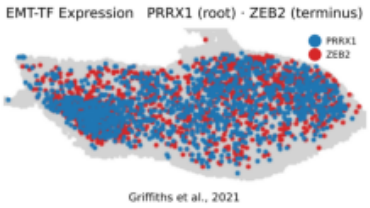
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (ER+ neoadjuvant letrozole ± ribociclib) (Griffiths et al., 2021), 10x + iCell8 — 109,353 malignant cells, 67 samples.

Verdict.
The embryonic programme reproduces as a COLLAPSED bifurcation: a single continuous gastrulation/somitogenesis ↔ neural-crest axis (geometry class Linear-pSom/Crest; module means Gast −0.019 / pSom −0.053 / Crest +0.023). Gast and pSom cross-repress along the axis but never resolve into two competing arms, so there is no hub and no Gate-3 test. Crest is terminal but enriches the Gast side (flag). EMT-TFs near-silent in this treated luminal cohort; the most detectable pseudotime-graded pair is PRRX1 (root) ↔ ZEB2 (terminus). Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Min-cell threshold	≥100 cells/sample (gap-finder degenerate on smooth dist) → 67/95 samples	**HVG / n_pcs**	4,072 over-dispersed genes; n_pcs = 20 (scree elbow ~PC7, ~32% cum. var)
Integration	Harmony θ = 10 (iLISI peak 19.2; θ=20 over-integrates)	**Island removal**	6×-median radius graph, largest component (842 cells / 57 specks removed)
Geometry	Non-branching fold-free curve (fold ratio 2.95); root = Gast/pSom pole	**Root / EMT-TFs**	pSom-high developmental origin; PRRX1 (early) ↔ ZEB2 (late)

Griffiths 2021 Breast Cancer — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample integrates 74 samples across two technologies. Mixing is judged on the per-neighbourhood iLISI (eval-spec metric), median 1.02→11.71 — not on the coarse same-sample-neighbour scalar. The manifold resolves into a large Body (91.8k) and a low-complexity left lobe; the Body reproduces the reference Gast↔pSom cross-repression and is one continuous mass (single kNN-graph component; 96% coherent diffusion-map mass) → GO, scoped to the Body.

Gate 2 — module geography → GO (non-branching)

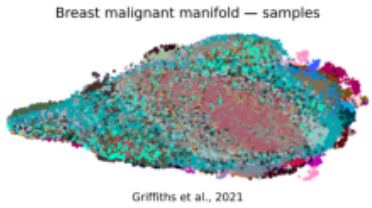
Organized Gast/pSom cross-repression (smoothed r=−0.45). Along pseudotime pSom falls (ρ=−0.65) while Crest rises (ρ=+0.57); NC-high decile at t=0.85 (terminal). Joint pSom&Crest enrichment 0.31<1 — inverse terminal gradient, a single (Gast/pSom)→Crest axis. FLAG: Crest co-enriches the Gast pole marginally more than pSom (0.33 vs 0.31) — arm inversion flagged, pass anyway.

Gate 3 — hub cross-repression → NA

Non-branching topology (principal curve: 2 tips, 0 forks across N=15/20/30; fold-free self-approach 2.86). No bifurcation to test → NA (branching-only correlation/modules notebooks not produced).

STEP FIGURES

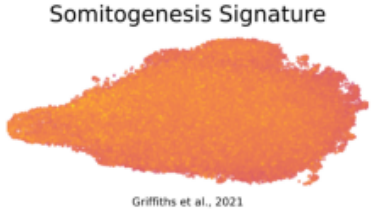
Manifold — samples



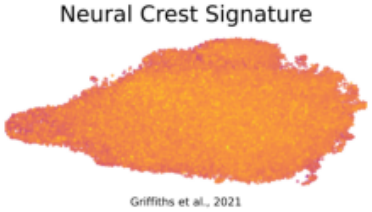
Pseudotime

Gastrulation

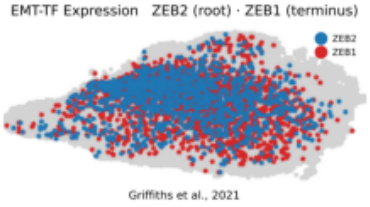
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (ER+ neoadjuvant) (Griffiths et al., 2021), 10x + iCell8 — 91,802 malignant cells analysed (Body compartment; 74 samples ≥50).

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis (non-branching), with organized Gast/pSom cross-repression (r=−0.45) and a terminal Crest gradient; no Gast↔pSom bifurcation, so Gate 3 is NA. Geometry Linear — consistent with the other CCCA breast datasets. Gate 1 GO / Gate 2 GO / Gate 3 NA. Raw module means Gast −0.020 / pSom −0.053 / Crest +0.016. EMT-TFs ZEB2 (early/root) → ZEB1 (late/terminus).

Parameter choices

Threshold	≥50 malignant cells/sample (no dominant count cliff) → 77/95 samples, 110,999 cells; Body = 91,802	**Compartment**	Scoped to the Body lobe; separated left lobe (16,545) is a low-complexity technical partition (NEAT1/MALAT1-led) excluded
Embedding	Harmony θ=10 on sample (removes 10x/iCell8 batch); n_pcs=30, n_neighbors=30, spread=1.3; 4,885 overdispersed; no sc.pp.scale	**Trajectory**	scFates curve (non-branching), Nodes=20 λ=0.3 μ=100; fold-free (SA 2.86); corr=+0.94
Root	Gast/pSom-high, Crest-low tip; axis not proliferation-driven (pt vs cycling ρ=+0.07)	**Geometry**	Linear; raw means Gast −0.020 / pSom −0.053 / Crest +0.016
EMT-TFs	ZEB2 (early/root) → ZEB1 (late/terminus); ZEB1 9.3% / ZEB2 5.3%		

Griffiths 2021 Breast Cancer — analysis summary

RUN 4 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample integrates 67 samples across both technologies (10x+iCell8). Mixing judged on the per-neighbourhood iLISI (median 10.5/67), not the coarse same-sample scalar; continuity on the independent Harmony-PCA kNN graph (1 component). The manifold is one continuous, sample-mixed mass with a proliferative sub-region (not a confounded lineage); only ~1.4% detached specks removed → GO on the full manifold.

Gate 2 — module geography → GO (non-branching)

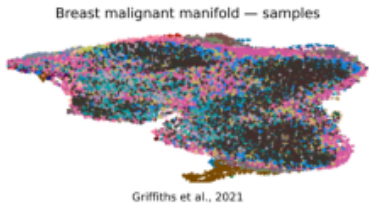
Organized cross-repression: smoothed pSom↔Crest r=−0.72 and Gast↔pSom r=−0.38 (raw pSom↔Crest −0.53). pSom and Crest occupy opposite territories on one continuous manifold — a single (Gast/pSom)→Crest inverse/terminal gradient, present but NOT resolved into two diverging arms → GO. No arm inversion to flag (Crest does not preferentially enrich a Gast arm).

Gate 3 — hub cross-repression → NA

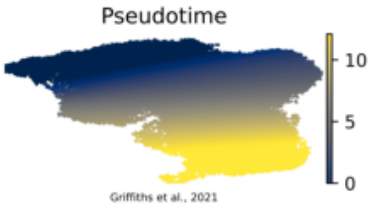
Non-branching topology (principal curve: 2 tips, 0 forks; fold-free self-approach 2.88). PPT 'forks' grow with node count — space-filling, not a real bifurcation. No hub bifurcation to test → NA.

STEP FIGURES

Manifold — samples

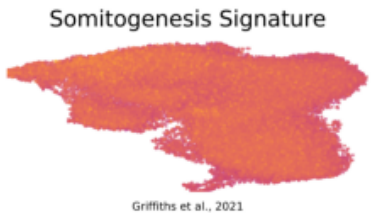


Pseudotime

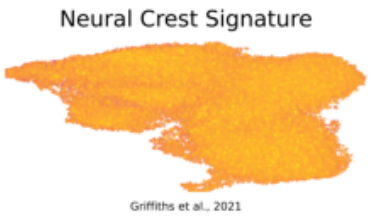


Gastrulation

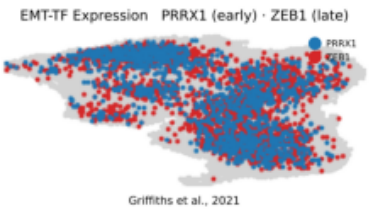
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (ER+ neoadjuvant) (Griffiths et al., 2021), 10x + iCell8 — 108,658 malignant cells analysed (67 samples ≥100).

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis (non-branching), with organized Gast/pSom→Crest cross-repression (pSom↔Crest r=−0.72) and a terminal Crest gradient; no Gast/pSom bifurcation, so Gate 3 is NA. Geometry Linear — consistent with the other CCCA breast datasets, on the full continuous manifold (no lobe removal). Gate 1 GO / Gate 2 GO / Gate 3 NA. Raw module means Gast −0.019 / pSom −0.053 / Crest +0.023. EMT-TFs PRRX1 (early) → ZEB1 (late).

Parameter choices

Threshold	≥100 malignant cells/sample (no dominant count cliff; median 646) → 67/95 samples, 108,658/111,372 cells	**Embedding**	Harmony θ=10 on sample (removes 10x/iCell8 batch); n_pcs=30, n_neighbors=30, spread=1.3; 4,072 overdispersed; no sc.pp.scale
Trajectory	scFates curve (non-branching), Nodes=20 λ=0.3 μ=100; fold-free (SA 2.88)	**Root**	Gast/pSom-high, Crest-low tip (Crest late/terminal); axis not proliferation-driven (cycling vs pt ρ=+0.03)
Geometry	Linear; raw means Gast −0.019 / pSom −0.053 / Crest +0.023	**EMT-TFs**	PRRX1 (early) + ZEB1 (late); all sparse (ZEB1 9%, rest <6%), pseudotime-flat

Griffiths 2021 Breast Cancer — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Two platform leaves (10x 110,568 + iCell8 804 malignant) share zero samples/patients and never mix under Harmony (platform iLISI ~1.0 across the sweep) → trajectory scoped to the 10x leaf. On the 10x leaf, 71 ER+ tumours intermix on one continuous manifold (sample iLISI 1.21→12.63 of 71; independent Harmony-PCA kNN = 1 component; 110,050 cells retained). Cycling is orthogonal to the embryonic axis → whole manifold kept. GO.

Gate 2 — module geography → —

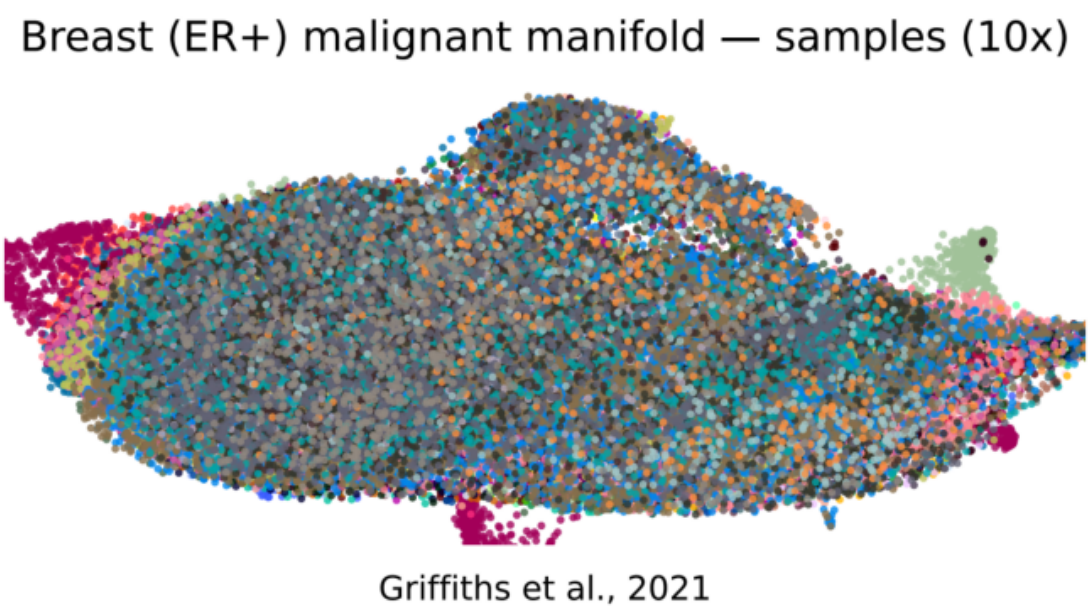
Not evaluated — this attempt stopped after Gate 1 (only 01_preprocessing produced; no trajectory / signature steps committed, no processed.h5ad).

Gate 3 — hub cross-repression → —

Not evaluated — downstream of Gate 2, which was not reached.

STEP FIGURES

Manifold — samples (Gate 1)



Dataset.
Breast Cancer (ER+, FELINE neoadjuvant trial; Griffiths et al., 2021), 10x + iCell8 — 10x leaf scoped: 110,050 malignant cells across 71 samples (≥50 cells).

Verdict.
INCOMPLETE ATTEMPT — reached Gate 1 GO (71 samples form one continuous 10x manifold; cycling orthogonal) but stopped before the trajectory / signature steps; only the samples deliverable was produced. Gate 1 GO / Gates 2 & 3 not evaluated. Superseded by later Griffiths runs.

Parameter choices

Platform scope	10x leaf (iCell8 804 cells never co-embed; platform iLISI ~1.0)	**Sample threshold**	≥50 cells/sample (smooth dist, no clean cliff); retains >99.8%
Integration	Harmony on sample (iLISI-sweep); n_pcs=30 (scree)	**Island removal**	radius graph 6× median-NN, largest component → 110,050 cells
Gate 1	GO — sample iLISI 1.21→12.63 of 71; 1 connected component; cycling orthogonal	**Status**	INCOMPLETE — stopped after Gate 1 (preprocessing only)

Griffiths 2021 breast cancer — analysis summary

RUN 6 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

67/95 samples (≥100 cells) intermix on one continuous manifold; per-neighbourhood iLISI(sample) 1.02→15.3 (θ=5), one kNN component (0 detached). The two density lobes are a proliferation / ribosomal axis (identical module scores), not per-patient islands.

Gate 2 — module geography → NO-GO

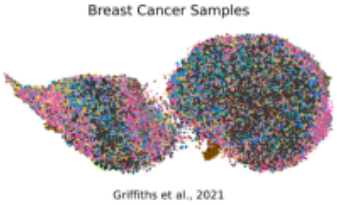
No Gast/pSom hub and no two-arm divergence: Gast-high and pSom-high populations are spatially anti-located (Jaccard 0.09; co-high fraction 0.007, below the 0.01 independence expectation) and stay negatively related after removing the ribosomal axis (partial r=−0.15) — Gast and pSom mark the two housekeeping lobes. The only embryonic relationship, a weak (Gast|pSom)–vs–Crest anti-correlation (r=−0.85), is transverse to the trajectory (|pseudotime ρ|≤0.11) — matches neither GO template. The embryonic pattern does not reproduce.

Gate 3 — cross-repression → NA

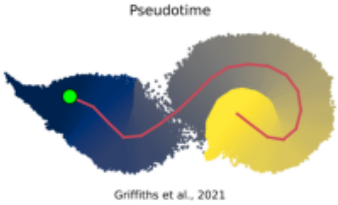
Not reached — Gate 2 NO-GO halts the chain, and the manifold is non-branching, so there is no hub or branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

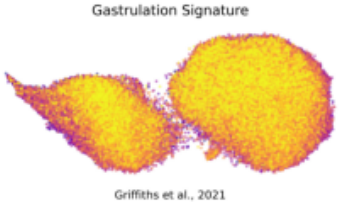
Manifold — samples



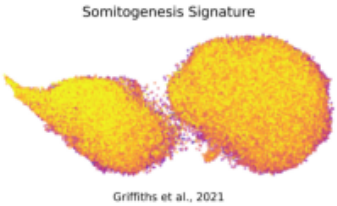
Pseudotime



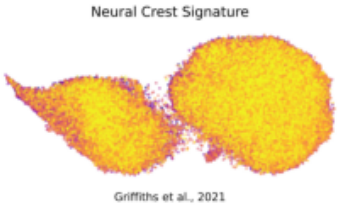
Gastrulation



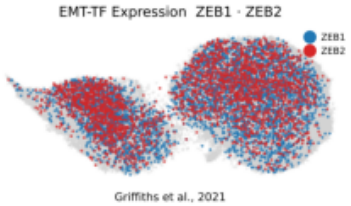
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Griffiths et al., 2021), neoadjuvant letrozole+ribociclib trial, 10x + iCell8 — 41 patients / 111,372 malignant cells. 100-cell floor keeps 67 samples; after stripping 2,324 islands and scoping 5,821 cycling cells, 102,179 cells form one continuous ER+ luminal body carrying the (attempted) trajectory. Non-branching.

Verdict.
Pattern does NOT reproduce — Gate 1 GO, Gate 2 NO-GO, Gate 3 NA. The manifold is one integrated, continuous, non-branching luminal body, but its dominant geometry is a proliferation/ribosomal axis, not the embryonic program. There is no Gast/pSom hub and no two arms — Gast and pSom are anti-located, marking the two housekeeping lobes (co-high 0.007 < chance). The only embryonic relationship is a weak (Gast|pSom)–Crest anti-correlation (r=−0.85) transverse to the trajectory. Corrects five prior attempts that forced a Gate-2 GO on this housekeeping-confounded signal. Module means Gast −0.021 / pSom −0.052 / Crest +0.021.

Parameter choices

Min cells/sample	≥100 floor (67/95; smooth dist, no cliff)	**Integration**	Harmony θ=5 (sample); PCA20; spread=1.3
Scope	cycling + islands out; luminal body kept (102,179 cells)	**Trajectory**	scFates curve (ElPiGraph) N=20; 2 tips / 0 forks
Root node	curve tip, Gast/pSom-high Crest-low	**Geometry**	Linear / non-branching (curve 2 tips)
EMT-TFs	ZEB1 (~8.8%), ZEB2 (~5.0%) det.-gated; flat/non-directional; no VIM	**Overdispersed**	3,444 genes; no scaling

Chung 2017 breast cancer — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample: per-neighbourhood same-sample fraction 0.75→0.25 (chance 0.09); iLISI 3.8→5.9 of 11 samples. One connected component, 0 detached islands. Samples intermix across a single continuous strip — GO.

Gate 2 — module geography → GO (non-branching)

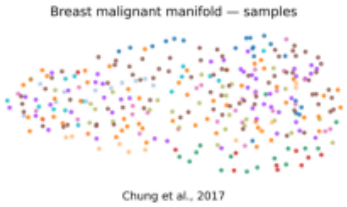
Gast & pSom co-vary (raw r=+0.18) and both oppose Crest (r=−0.65 / −0.54). Along pseudotime Gast/pSom fall (ρ≈−0.95) while Crest rises (ρ=+0.95); NC-high decile at t=0.98 (terminal); joint pSom&Crest enrichment 0.37<1 — inverse terminal gradient, a single (Gast/pSom)→Crest axis. Present but not resolved into two arms — GO.

Gate 3 — cross-repression → NA

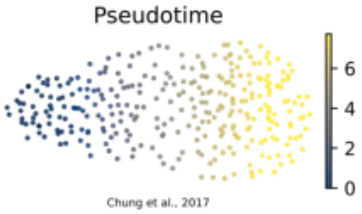
Non-branching topology (curve: 2 tips, 0 forks across N=15/20/30; PPT tree also 0 forks). No bifurcation to test — NA.

STEP FIGURES

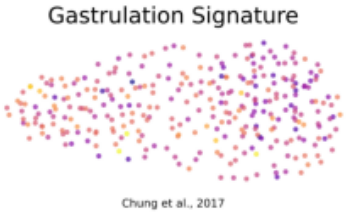
Manifold — samples



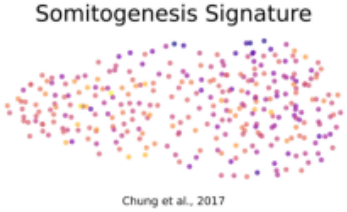
Pseudotime



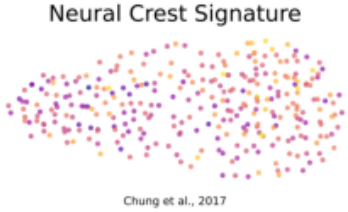
Gastrulation



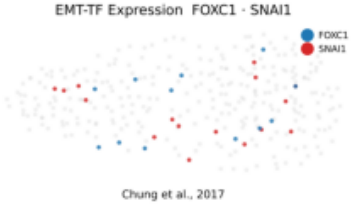
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Chung et al., 2017), SmartSeq2 — 317 malignant cells (13 samples reported) → 309 analysed (11 samples ≥10 cells). Sparse plate-based data. Non-branching Linear-pSom/Crest.

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis (non-branching); no Gast↔pSom bifurcation, so Gate 3 is NA. Geometry Linear-pSom/Crest — consistent with the linear outcome across most CCCA datasets. EMT-TFs FOXC1 (early/root) → SNAI1 (late/terminus).

Parameter choices

Threshold	≥10 cells/sample (SmartSeq2 scale; anchor floor 100 would empty it) → 11/12, 309 cells	**Embedding**	Harmony θ=10; n_pcs=20, n_neighbors=30, spread=1.3; 401 HVG
Trajectory	scFates curve (non-branching), N=20, λ=0.3, μ=100; fold-free (2.98)	**Root**	Gast/pSom-high, Crest-low tip (Gast/pSom early, Crest late); lower-cycling pole
Geometry	Linear-pSom/Crest; pSom globally strongest (Gast −0.276, pSom +0.124, Crest −0.074)	**EMT-TFs**	FOXC1 (early/root) → SNAI1 (late/terminus); 10-TF panel, VIM excluded

Gulati 2020 Breast — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Four samples (≥100 malignant cells) intermix per-neighbourhood on one continuous manifold — iLISI rises 2.77→3.56 of 4 after Harmony (θ=5), one connected component, no islands removed. Read on the regenerated embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

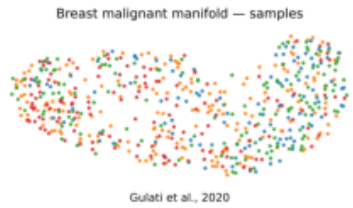
A single (Gast/pSom)→Crest developmental axis. Gast/pSom do NOT diverge into cross-repressing arms (raw r=−0.05), while both oppose Crest (Gast–Crest r=−0.50, pSom–Crest r=−0.60). Crest is the inverse terminal readout (rises along pseudotime ρ=+0.93; NC-high decile at pt 1.00; joint pSom&Crest enrichment 0.31<1). Non-branching case → GO.

Gate 3 — hub cross-repression → NA

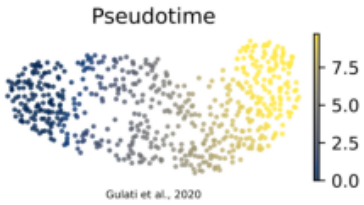
No bifurcation (curve: 2 tips / 0 forks), so there is no Mixed hub to test for Gast↔pSom anti-correlation. Branching-only → NA; 04_correlation / 05_modules not produced.

STEP FIGURES

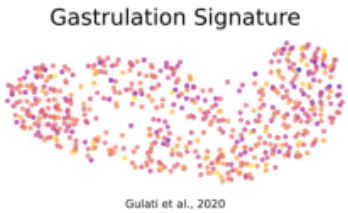
Manifold — samples



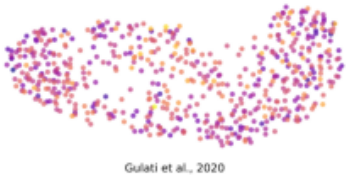
Pseudotime



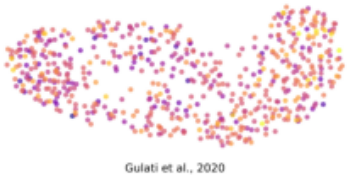
Gastrulation



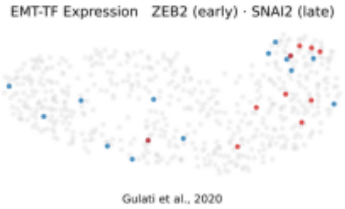
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Gulati et al., 2020), SmartSeq2 — 861 malignant cells (8 samples) → 587 analysed (4 samples ≥100); single low-cycling Body (no proliferative compartment scoped out).

Verdict.
Pattern reproduces (non-branching, Linear-pSom/Crest): a single Gastrulation/Somitogenesis → Neural-Crest developmental axis with Crest as the inverse terminal readout; no Gast↔pSom bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (root-end) → SNAI2 (terminus).

Parameter choices

Malignant / threshold	861 → floor ≥100 (counts smooth, no cliff) → 4 samples / 587 cells; 0 islands
Trajectory / geometry	scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips / 0 forks; SA 2.89
Subtype / annotation	single low-cycling Body (no proliferative compartment scoped out); cell_type = Malignant
Expression / integration	SmartSeq2 TPM (log1p); overdisp=98; n_pcs=20; Harmony(sample) θ=5; k=30; spread=1.3
Root	curve tip 0 (Gast/pSom-high, Crest-low origin); Crest terminal; pt vs prolif ρ=−0.01
EMT-TFs	ZEB2 (root-end) / SNAI2 (terminus) from 10-TF panel; detection-gated; VIM excluded

Gulati 2020 breast cancer — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/8 samples (≥54 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 2.47→4.79 of 7); both lobes (Low-cycling 513 / High-cycling 305) carry the trajectory.

Gate 2 — module geography → NO-GO

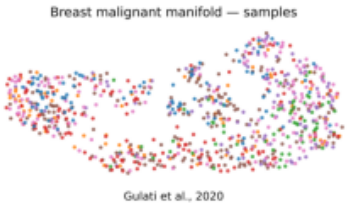
Developmental geography is ABSENT in raw per-cell scores (Gast↔pSom = −0.02) and appears only after graph-smoothing. A permutation null (smoothing random scores on the same graph) reproduces it: smoothed −0.77 within null 95% CI ±0.7; the defining terminal-Crest gradient (ρ=+0.73) is non-significant (p=0.14). The programs are present but not organised into a developmental axis — pattern absent.

Gate 3 — cross-repression → NA

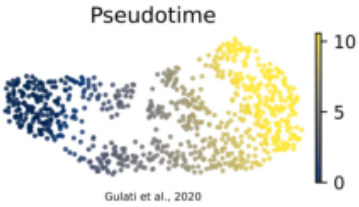
Not reached — Gate 2 NO-GO halts the gate chain; branching-only, and there is no bifurcation or Mixed-type hub to test.

STEP FIGURES

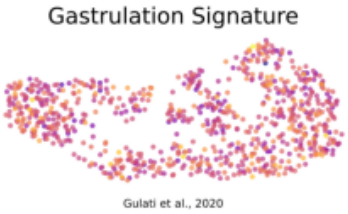
Manifold — samples



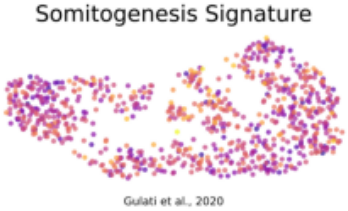
Pseudotime



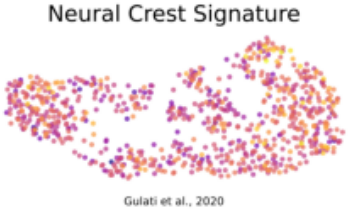
Gastrulation



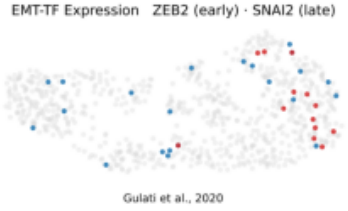
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Gulati et al., 2020), SmartSeq2 — 15 samples / 1,902 cells (master scope); malignant only: 861 load, 818 across 7 samples (≥54) enter preprocessing; whole malignant manifold (no lobe scoping). A trajectory is fit, but the module pattern is absent (Gate 2 NO-GO).

Verdict.
Pattern does NOT reproduce — Gate 1 GO, Gate 2 NO-GO, Gate 3 NA. Gate 1 holds (818 cells intermix on one continuous manifold), but the embryonic-module geography is absent at single-cell resolution: raw Gast↔pSom = −0.02 (no cross-repression), and the smoothed structure (−0.77; terminal Crest ρ=+0.73) is not distinguishable from a graph-smoothing artifact — a permutation null on this 818-cell SmartSeq2 graph gives 95% CI ±0.7, terminal-Crest non-significant (p=0.14). EMT-TFs (unconditional step) ZEB2 (early) → SNAI2 (late).

Parameter choices

Min cells/sample	≥54 (7/8; gap 68→43)	**Scope**	whole manifold (no scoping)
Root node	tip 2 (pSom/Gast-high)	**Integration**	Harmony θ=2; PCA20; spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100 (PPT forks unstable 2→8)	**Compartments**	Low-cyc 513 / High-cyc 305 (cell cycle orthogonal, |r|≤0.03)
EMT-TFs	ZEB2 (early) → SNAI2 (late); mesenchymal factors lean crest-terminal		

Karaayvas 2018 TNBC — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=5) integrates all 5 patients: same-sample neighbour fraction 0.82 → 0.28 (null 0.25), iLISI median 1.97 → 4.12 of 5. One connected component (0 islands). Judged on the regenerated embedding + per-neighbourhood iLISI, not a scalar: samples intermix AND the manifold is continuous.

Gate 2 — module geography → GO (non-branching)

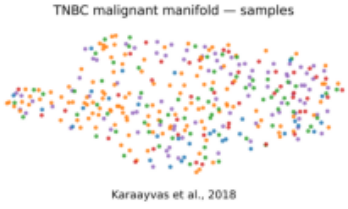
Non-branching case: a single pSom-origin → terminus developmental gradient. pSom falls along pseudotime (ρ −0.41); Gast & Crest both rise to the terminal tip (ρ +0.79 / +0.82; Crest decile pt=0.98). Joint pSom&Crest enrichment 0.37 (<1 = inverse terminal gradient, not a co-expression hub). FLAG (pass anyway): Crest co-enriches the Gast pole (smoothed Gast↔Crest r=+0.59), an inversion of the expected pSom-arm Crest — Gast, not pSom, tracks the neural-crest terminus here.

Gate 3 — hub cross-repression → NA

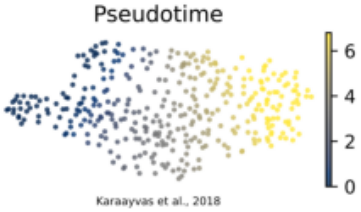
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks, fold-free). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

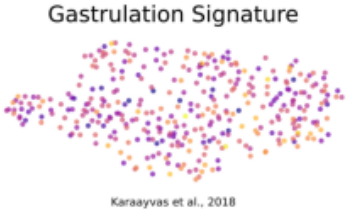
Manifold — samples



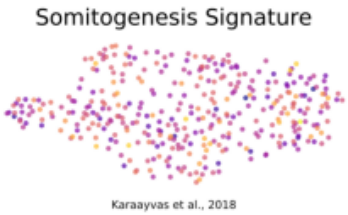
Pseudotime



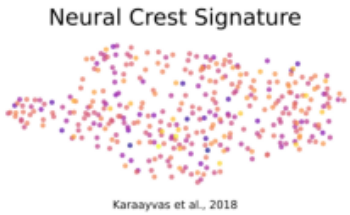
Gastrulation



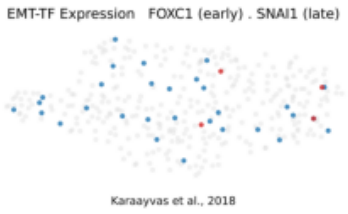
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Triple-negative breast cancer (Karaayvas et al., 2018), SmartSeq2 — 360 malignant cells over 5 patients; single low-cycling body (~14% cycling), cell-cycle-free axis (pseudotime vs cycling ρ −0.19).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous somitogenesis-origin → neural-crest terminus axis. Gate 1 GO / Gate 2 GO / Gate 3 NA, with the Crest-arm inversion flagged (Crest tracks the Gast terminus). EMT-TFs FOXC1 (early/root) → SNAI1 (late).

Parameter choices

Sample threshold	≥12 (geom-mean of 31→5 cliff, ratio 6.2) → 5/6 samples, 360 cells	**Overdispersed genes**	51 (scFates find_overdispersed; sparse SmartSeq2)
Integration	Harmony on sample, θ=5 (mixing near null then flattens)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 1 component, 0 islands removed
Geometry	Non-branching scFates curve N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.91; corr(x,t)=0.97	**Root**	tip 2 = Crest-low, pSom-high origin
Module means (raw)	Gast −0.220 · pSom +0.106 · Crest −0.067 (Linear-pSom/Crest)	**EMT-TFs shown**	FOXC1 (early/root, 33%) ↔ SNAI1 (late/terminus); detection-gated

Karaayvas 2018 TNBC — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample: per-neighbourhood same-sample fraction 0.82→0.27 (chance 0.20); iLISI 1.5→4.1 of 5 samples. One connected component, 0 detached islands. Five samples intermix across a single continuous cloud — GO.

Gate 2 — module geography → NO-GO

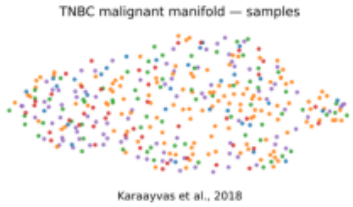
Module geography is ABSENT in the raw per-cell scores (Gast~pt ρ=−0.08, pSom +0.02, Crest −0.03) and appears only after graph-smoothing. A permutation null (360 cells) gives 95% CI ±0.9; every smoothed gradient is non-significant (p≥0.26). Tell: smoothing FLIPS the sign of the one real raw signal (Crest↔Gast −0.43→+0.44, Crest↔pSom −0.55→+0.10) — fabricated, not latent. Pattern absent.

Gate 3 — cross-repression → NA

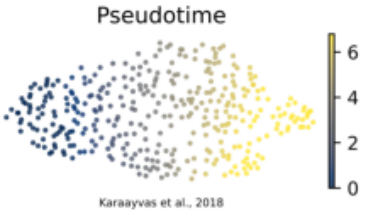
Non-branching topology (curve: 2 tips, 0 forks across N=15/20/30; PPT tree forks unstable 1–4 = spurious in a diffuse cloud). No bifurcation to test — NA.

STEP FIGURES

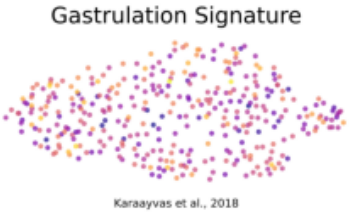
Manifold — samples



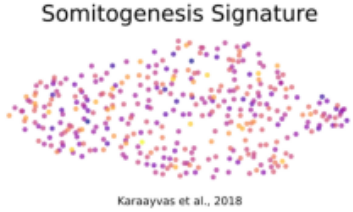
Pseudotime



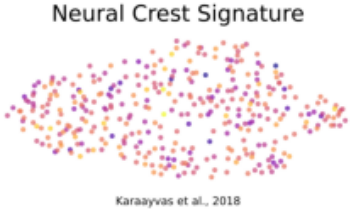
Gastrulation



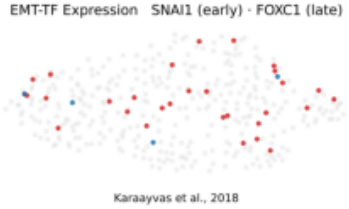
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (TNBC) (Karaayvas et al., 2018), SmartSeq2 — 365 malignant cells (6 samples reported) → 360 analysed (5 samples ≥10 cells). Sparse plate-based data. A trajectory is fit, but the module pattern is absent (Gate 2 NO-GO).

Verdict.
Pattern does NOT reproduce. Gate 1 holds (360 cells intermix on one continuous manifold), but the embryonic-module geography is absent at single-cell resolution: raw pseudotime gradients are ~0 and the smoothed structure is a graph-smoothing artifact (null 95% CI ±0.9, all p≥0.26; smoothing flips the raw Crest signs). Gate 2 NO-GO, Gate 3 NA. EMT-TFs SNAI1 (early/root) → FOXC1 (late/terminus).

Parameter choices

Threshold	≥10 cells/sample (SmartSeq2 scale; floor 100 keeps only 1/6); cliff 31→5 → 5/6, 360 cells	**Embedding**	Harmony θ=10; n_pcs=20, n_neighbors=30, spread=1.3; 51 HVG
Trajectory	scFates curve (non-branching), N=20, λ=0.3, μ=100; fold-free (2.97)	**Root**	Gast/pSom-high, Crest-low tip; not proliferation-rooted (pt–cycling ρ=+0.18)
Geometry	Gate 2 NO-GO — module pattern absent at single-cell resolution	**EMT-TFs**	SNAI1 (early/root) → FOXC1 (late/terminus); 10-TF panel, VIM excluded

Kim 2018 TNBC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample: same-sample neighbour fraction 0.99→0.40; iLISI 3.89 of 9 samples; continuity confirmed off the independent PCA-harmony graph (1 connected component), 0 islands.

Gate 2 — module geography → GO (non-branching)

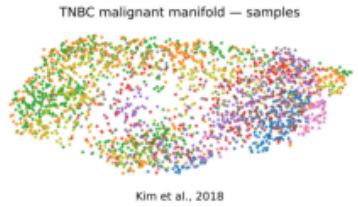
Single axis: pSom-high origin → Gast/Crest-high terminus. Organized Gast↔pSom cross-repression (smoothed r=−0.84); Crest a terminal gradient. FLAG: Crest enriches the Gast pole (inversion) — passes anyway.

Gate 3 — cross-repression → NA

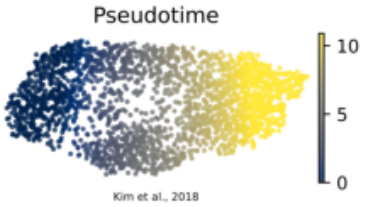
No bifurcation (curve: 2 tips, 0 forks) — branching-only Gate 3 not applicable.

STEP FIGURES

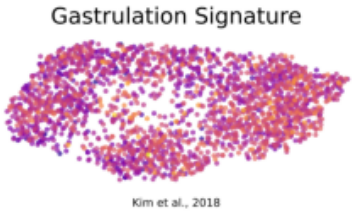
Manifold — samples



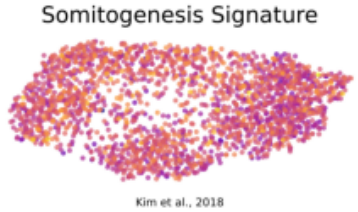
Pseudotime



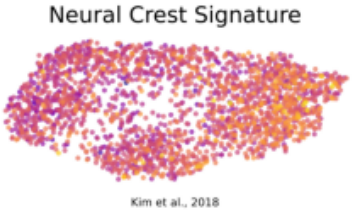
Gastrulation



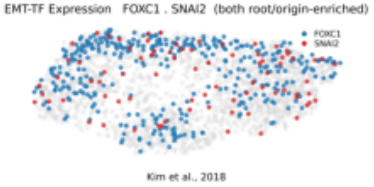
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Triple-negative breast cancer (Kim et al., 2018), Nanogrid — 2,263 malignant cells across 9 samples (9/16 kept at ≥100-cell floor). Non-branching Linear-pSom/Crest.

Verdict.
The embryonic-program pattern reproduces as a non-branching linear axis: a single continuous, sample-integrated manifold from a pSom-high developmental origin to a Gast/Crest-high terminus, with organized Gast↔pSom cross-repression and Crest as a terminal gradient enriched on the Gast pole (inversion flag). No bifurcation → Gate 3 NA. EMT-TF expression (FOXC1, SNAI2) concentrates at the root/origin end; no EMT-TF marks the terminus in this dataset.

Parameter choices

Min-cell threshold	≥100 (floor coincides with real 52→100 inflection) → 9/16, 2,263 cells	**Embedding**	HVG=536; PCA50; Harmony θ=10 (TNBC patient batch); n_pcs=20, spread=1.3
Trajectory	scFates curve (non-branching), N=20, λ=0.3, μ=100; fold-free (2.98)	**Root**	pSom-high/Crest-low developmental origin (Crest is late)
Geometry	Linear-pSom/Crest (raw Gast −0.009, pSom +0.040, Crest −0.085)	**EMT-TFs**	FOXC1 (65%, ρ=−0.41) + SNAI2 (ρ=−0.23); BOTH root/origin-enriched, none terminal

Pelka 2021 colorectal cancer — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

64 tumours integrate to one continuous manifold (same-sample neighbour fraction 0.71→0.10, robust across θ 2–10); a single connected component after island removal.

Gate 2 — module geography → GO (non-branching)

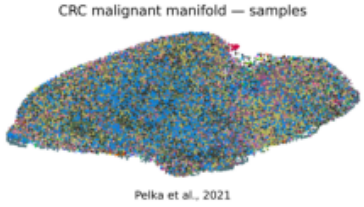
Non-branching (2 tips, 0 forks). Shared developmental gradient with neural crest as the inverse terminal end: pSom↔Crest spatial r=−0.89; along pseudotime pSom ρ=−0.79, Crest ρ=+0.83.

Gate 3 — cross-repression → NA

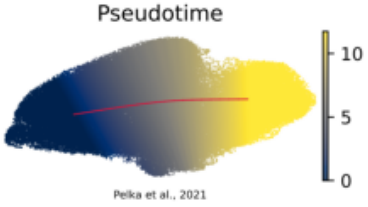
No bifurcation → no hub to test. Cross-repression gate not applicable to the non-branching case.

STEP FIGURES

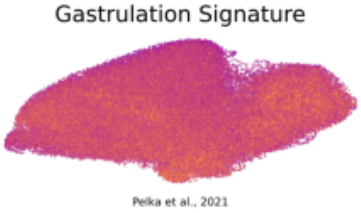
Manifold — samples



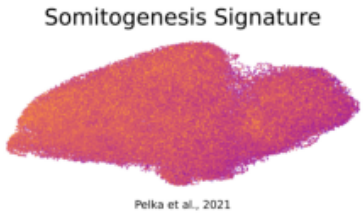
Pseudotime



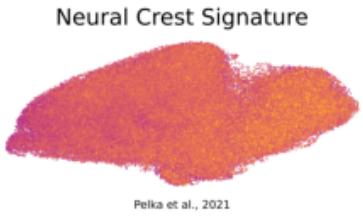
Gastrulation



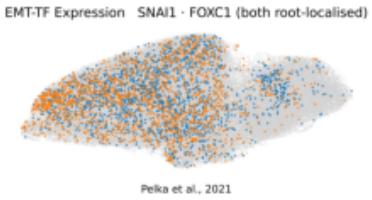
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Colorectal cancer (Pelka et al., 2021), 10x; 108,999 published-malignant cells across 64 tumours, all samples above the atlas 100-cell floor; 108,979 after removing 20 detached-island cells.

Verdict.
Pattern reproduces (non-branching, Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest developmental axis. Gate 1 GO / Gate 2 GO / Gate 3 NA. No competing arms, so no hub r. EMT-TFs SNAI1 and FOXC1 both localise to the root; canonical ZEB1/TWIST1 below the 3% floor.

Parameter choices

Min cells/sample	≥100 floor (gap-rule degenerate 1.58)	**Integration**	Harmony θ=10; n_pcs=20 (4,556 HVGs)
Scope	whole manifold (proliferation orthogonal)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100 (SA 2.99)
Root node	somitogenesis/gastrulation-high tip	**Geometry**	Linear (non-branching); 2 tips / 0 forks
EMT-TFs	SNAI1 · FOXC1 (both root-loaded)		

Lee 2020 CRC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

14/15 samples (≥151 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); both lobes (Embryonic 6,147 / Cycling 4,823) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

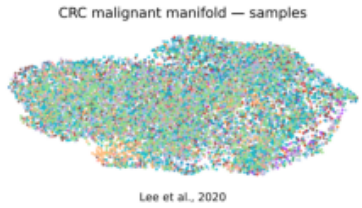
Gast/pSom co-vary (r=+0.49) and decline along pseudotime (ρ=−0.76/−0.56); Crest anti-correlated (r=−0.81/−0.83), rising to a terminal readout (ρ=+0.77; crest-high decile mean pt=0.95).

Gate 3 — hub mutual exclusivity → NA

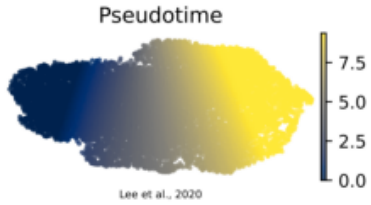
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

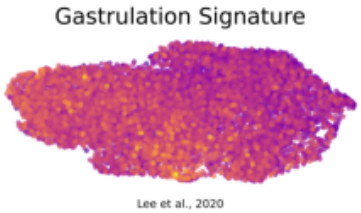
Manifold — samples



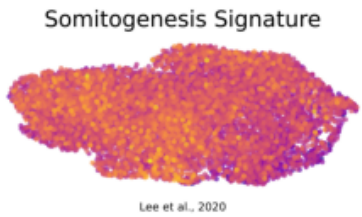
Pseudotime



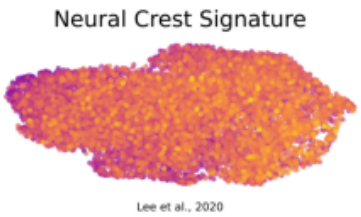
Gastrulation



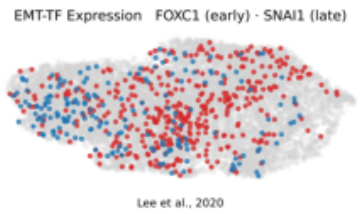
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Colorectal Carcinoma (Lee et al., 2020), 10x — 15 patients / 21,657 cells (master scope); malignant only: 11,075 load, 10,970 across 14 samples (≥151) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Lee's colorectal malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Unlike Kürten, the developmental axis spans both lobes (Crest terminus inside the cycling lobe), so the whole manifold is kept (Bischoff/Sun no-scoping precedent). EMT-TFs FOXC1 (early/root) → SNAI1 (late/terminus); ZEB1/TWIST near the detection floor (0.8%/0.2%); EMT-TF end-separation weak (Δpt=0.22).

Parameter choices

Min cells/sample	≥151 (keeps 14/15; gap 219→105)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Embryonic 6,147 / Cycling 4,823

Li 2017 CRC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9/10 samples (248 malignant cells) intermix on one continuous, elongated manifold (radius-graph: 0 detached islands; kNN: 1 component); whole manifold enters the trajectory.

Gate 2 — module geography → GO (non-branching)

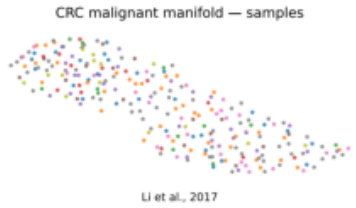
Gast/pSom co-vary (r=+0.89) and decline along pseudotime (ρ −0.82 / −0.37); Crest anti-correlated (r=−0.96 / −0.97), rising to a terminal readout (ρ +0.56; crest-high decile pt=1.00).

Gate 3 — cross-repression → NA

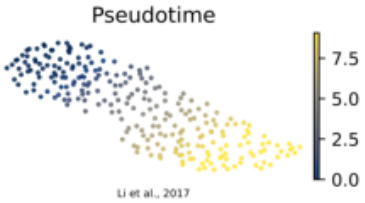
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

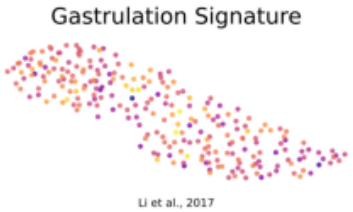
Manifold — samples



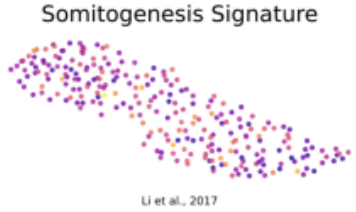
Pseudotime



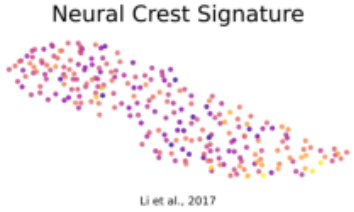
Gastrulation



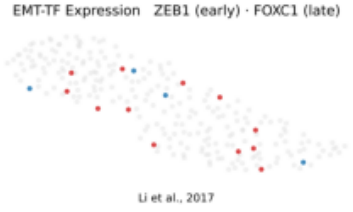
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Colorectal carcinoma (Li et al., 2017), SmartSeq2 — 18 samples / 590 cells (master scope); 254 malignant loaded; 248 across 9 patients (≥8 cells) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory. Non-branching Linear-pSom/Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Li's colorectal malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Cell cycle is orthogonal to the axis (pseudotime↔cycling ρ=−0.01), so the whole manifold is kept. EMT-TFs ZEB1 (early/root) → FOXC1 (late/terminus); end-separation weak (Δpt=0.14) at this SmartSeq2 depth (248 cells).

Parameter choices

Min cells/sample	≥8 (keeps 9/10; smooth counts)	**Integration**	Harmony θ=5; PCA(HVG); spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=15, λ=0.3, μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**HVG**	seurat 2000 (pagoda2 degenerate at n=248)
EMT-TFs	ZEB1 (early/root) → FOXC1 (late/terminus); Δpt=0.14		

Puram 2017 HNSCC — recapitulation summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

10/17 samples (≥100 cells) intermix on one continuous manifold; no per-patient islands (64 detached cells removed).

Gate 2 — module geography → GO

Gast/pSom hub diverges into arms; Crest is a terminal readout at the Basal tip (0.43 > Atypical 0.27 ≈ Classical 0.24).

Gate 3 — hub mutual exclusivity → GO

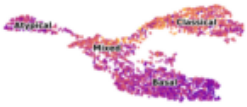
On 466 Mixed-hub cells, per-gene Gast vs pSom correlation vectors anti-correlate: r = −0.47 (ref ≈ −0.49).

STEP FIGURES

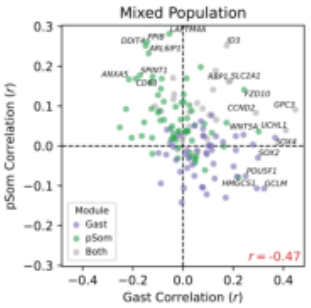
Gate 1: integration

Trajectory / root

Gast

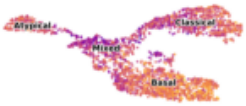


Gate 3: r = −0.47



pSom

Crest



EMT-TFs

Dataset.
Head & Neck SCC (Puram et al., 2017), SmartSeq2 — 18 patients / 5,902 cells (master scope); malignant only: 2,539 load, 2,408 across 4 subtypes enter the trajectory.

Verdict.
✓ Pattern reproduces — all three gates pass. Gast/pSom mutually exclusive within the Mixed hub (r = −0.47 vs ref −0.49), hub at the root of a hub-and-spoke manifold; EMT-TFs ZEB1 (Classical) / PRRX1 (Atypical).

Parameter choices

Min cells/sample	≥ 100 (keeps 10/17)	**Integration**	Harmony on sample, 20 it
Island removal	UMAP radius-graph (−64 cells)	**Trajectory**	scFates PPT N=200 λ=2000 σ=0.5
Root node	85 = Mixed hub (466 cells)	**Subtypes**	markers + Puram 2017
EMT-TFs shown	ZEB1 (Classical) / PRRX1 (Atypical)	**Reference r**	−0.49 (embryonic anchor)

Kürten 2021 HNSCC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9/20 samples (≥182 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); the low-cycling embryonic lobe (5,955 cells) carries the trajectory.

Gate 2 — module geography → GO (non-branching)

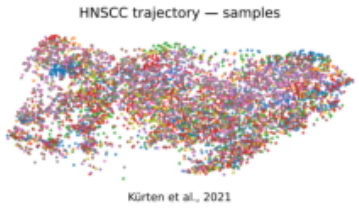
Gast/pSom co-vary (r=+0.33) and decline along pseudotime (ρ=−0.32 / −0.38); Crest anti-correlated (r=−0.77 / −0.64), rising to a terminal readout (ρ=+0.44; crest-high decile mean pt=0.99).

Gate 3 — cross-repression → NA

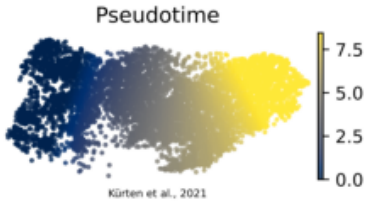
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

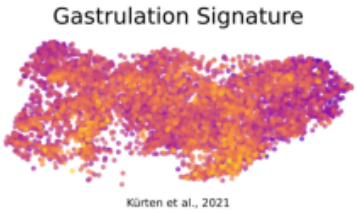
Manifold — samples



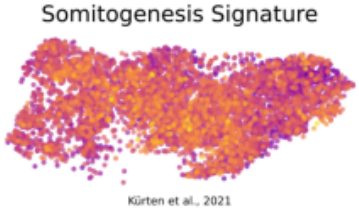
Pseudotime



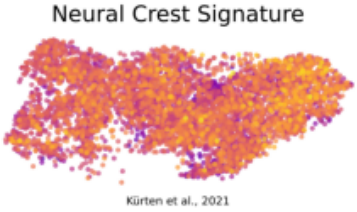
Gastrulation



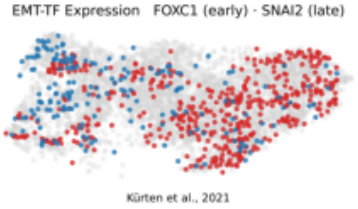
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Head & Neck SCC (Kürten et al., 2021), 10x — 33 patients / 62,700 cells (master scope); malignant only: 11,122 across 9 samples (≥182) enter preprocessing; 5,955 embryonic-lobe cells enter the trajectory. Non-branching Linear-pSom/Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Kürten's malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. EMT-TFs FOXC1 (early/root) → SNAI2 (late/terminus); canonical SNAI1/ZEB1 near the detection floor (2.0%/2.3%).

Parameter choices

Min cells/sample	≥182 (keeps 9/20; gap 89→376)	**Lobe selection**	embryonic (low-cycling 0.22) vs 0.46
Root node	tip 0 = Gast/pSom-high, Crest-low	**Integration**	Harmony(sample) θ=10; PCA 50PC; UMAP spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100	**Compartments**	Embryonic 5,955 / Cycling 5,167
Geometry	Linear (non-branching); no hub	**EMT-TFs**	FOXC1 (early/root) → SNAI2 (late/terminus); SNAI1/ZEB1 near floor

Chen 2020 NPC — analysis summary

RUN 1 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Continuous (1 connected component, 100% of cells) BUT not mixed: across the integration-strength sweep 73%→56%→1% of cells sit in patient-pure Leiden clusters (θ=1/2/3); diff-sample neighbour fraction 0.30 at θ=2 vs 0.69 ceiling. The purest clusters (99% P11) are non-cycling genuine patient-private CNV states, not a shared axis. P11+P01 alone = 72% of cells. θ=3 forces mixing only by over-integrating (a detached speck appears). Do not force a trajectory.

Gate 2 — module geography → NA

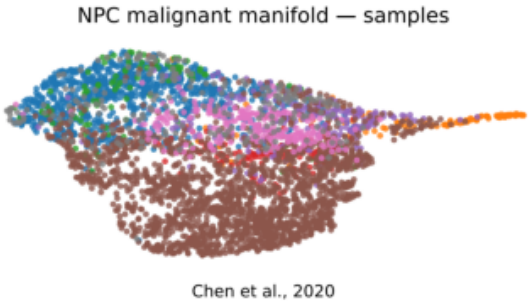
Not evaluated — gate chain stops at Gate 1 NO-GO (no manifold / signatures / correlation / modules notebooks produced).

Gate 3 — cross-repression → NA

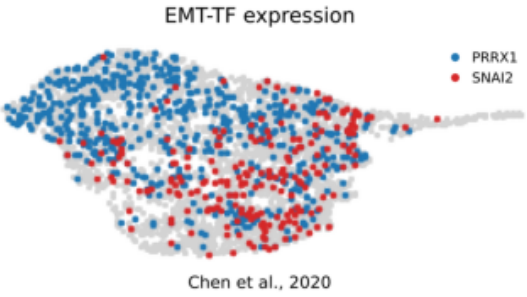
Not evaluated — no manifold, no hub.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
EBV-associated nasopharyngeal carcinoma (Chen et al., 2020), 10x; 3,674 malignant cells across 8 samples.

Verdict.
Pattern does NOT reproduce — Gate 1 NO-GO. The malignant compartment clusters by patient, not into a sample-integrated continuous manifold, so no trajectory is fit.

Parameter choices

Min-cell threshold	62 (median-restricted spillover gap 107→36, ×2.97)	**Overdispersed genes**	1,290 (GAM)
n_pcs	6 (scree elbow)	**Integration**	Harmony θ=2 (balance of the θ=1–3 sweep)
Geometry	none — Gate 1 NO-GO, no trajectory	**EMT-TFs (descriptive, ≥3%)**	PRRX1 62% · SNAI2 24% · FOXC1 13% · ZEB2 13% · ZEB1 7% · TWIST2 6%; SNAI1/TWIST1 below floor

Chen 2020 NPC — analysis summary

RUN 2 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Continuous (1 connected component) but NOT sample-mixed: ~50% of cells in patient-pure clusters at θ=2 (~90% at θ=1), robust across n_pcs {15,20,30} — not the attempt-1 n_pcs=6 artifact. Purest clusters = non-cycling P11 (patient-private CNV). Only θ=3 mixes, by over-integrating.

Gate 2 — module geography → NA

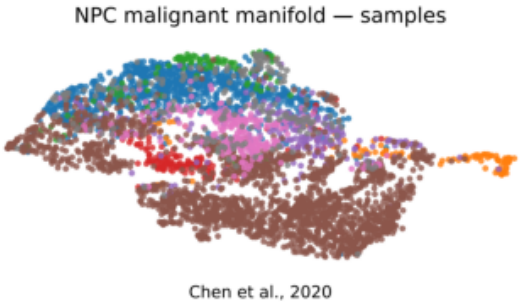
Not evaluated — no sample-integrated continuous manifold on which to fit a trajectory / score module geography.

Gate 3 — hub cross-repression → NA

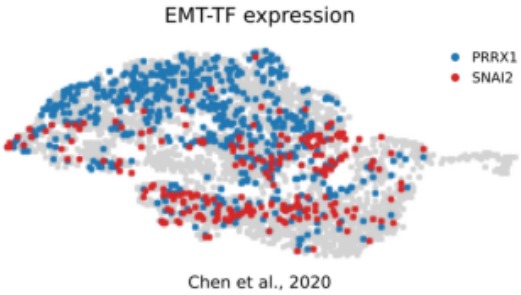
Not evaluated — requires a branching trajectory with an identified hub, which Gate 1 precludes.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 14 patients / 71,896 cells (master scope); malignant only: 3,763 load → 3,674 across 8 patients (≥62 cells) enter the embedding.

Verdict.
Pattern does NOT reproduce — Gate 1 NO-GO. The malignant compartment is patient-segregated (inter-patient CNV heterogeneity), not a shared sample-integrated developmental manifold. No hub, no trajectory, no cross-repression axis to test. EMT-TFs shown for reference: PRRX1 / SNAI2 (FOXC/PRRX families; canonical SNAI1 & TWIST1 below the 3% detection floor).

Parameter choices

Min cells/sample	62 (median-restricted gap; keeps 8/14)	**Genes**	1,290 over-dispersed (GAM)
n_pcs	20 (re-derived; Kürten 20 / Puram 40)	**Integration**	Harmony on sample, θ=2, 20 it
Embedding	UMAP, Leiden res=0.5 → 9 clusters	**Trajectory**	none (Gate 1 NO-GO)
EMT-TF panel	detection-gated ≥3%; PRRX1/SNAI2/FOXC1	**Geometry**	patient-segregated (not a manifold)

Chen 2020 NPC — analysis summary

RUN 3 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/14 samples (≥107 cells) intermix on one continuous manifold: iLISI 2.1 / 3.2 ceiling, metrics plateau at θ≥2 (over-integration null passes); P11-removal control confirms genuine cross-patient mixing; single kNN component.

Gate 2 — module geography → NO-GO

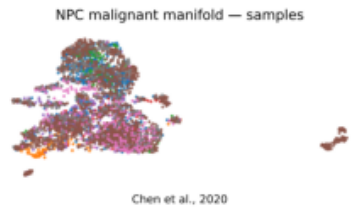
No Gast/pSom hub. Gast and pSom are orthogonal (r=−0.05) — neither co-varying (overlap case) nor cross-repressing (branching); both diffuse across the manifold. Structure is carcinoma-state (cycling, squamous/basal, hypoxia, IFN/MHC-II), not embryonic. Both oppose Crest (−0.54/−0.44); Crest is not a proliferation confound.

Gate 3 — hub cross-repression → NA

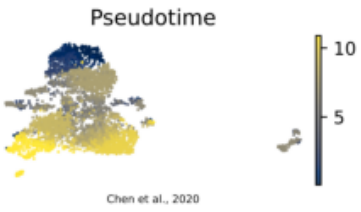
Not tested — no bifurcation and no hub to run 04_correlation / 05_modules on (branching-only gate).

STEP FIGURES

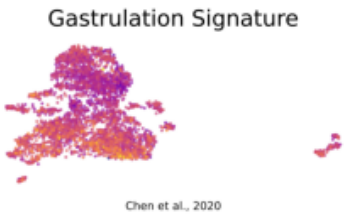
Manifold — samples



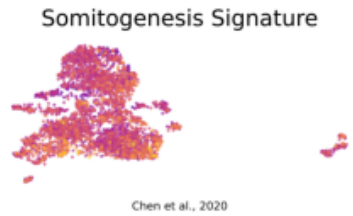
Pseudotime



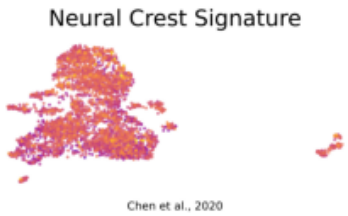
Gastrulation



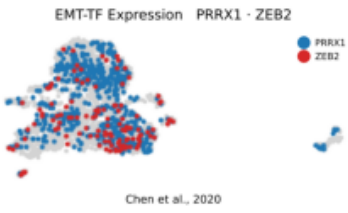
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 14 patients / 3,763 malignant load; 3,674 across 8 samples (≥107 cells) enter the manifold.

Verdict.
Pattern does not reproduce. Gate 1 GO but Gate 2 NO-GO: NPC malignant cells form a continuous carcinoma-state manifold with no embryonic module geography — Gast/pSom independent, no hub, no arms. EMT-TFs PRRX1 (62%) / VIM (53%) broad, no mesenchymal terminus; canonical SNAI1/ZEB1/TWIST1 near-absent (1–7%).

Parameter choices

Min cells/sample	≥107 (keeps 8/14)	**Integration**	Harmony on sample, θ=2, 20 it
Island removal	none (single kNN comp.)	**Trajectory**	scFates PPT N=50 λ=300 σ=0.5
Root node	35 = Progenitor-like	**n_pcs**	19 (>0.5% var each; θ-sweep 0–10)
Scaling	none (log1p, reference-matched)	**Signatures**	score_genes, union control pool

Chen 2020 NPC — analysis summary

RUN 4 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/14 samples pass the min-cell threshold 107; Harmony(sample, θ=2) fully intersperses all 8 patients; 1 connected component after island removal (438 detached cells dropped). Mixed and continuous.

Gate 2 — module geography → NO-GO

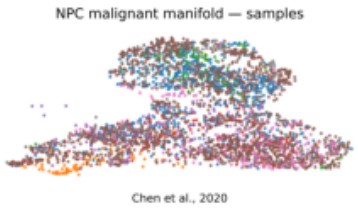
A robust Crest-terminal axis exists (Crest opposes Gast −0.64 / pSom −0.44, SURVIVES cell-cycle correction → not a proliferation artifact). But the defining embryonic BODY is absent: Gast/pSom are decoupled (r=+0.01 vs Kürten GO +0.33 / Puram GO −0.49) and pSom does not participate (r(t,pSom)=−0.17, flat). A two-module Gast/Crest gradient without a coupled Gast/pSom body is not the embryonic pattern.

Gate 3 — hub cross-repression → NA

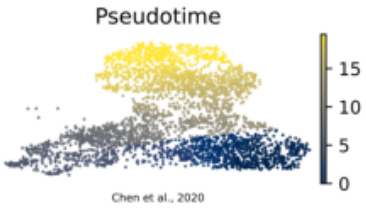
No bifurcation on a linear manifold → no Gast↔pSom arms to test. Correlation + modules notebooks not produced.

STEP FIGURES

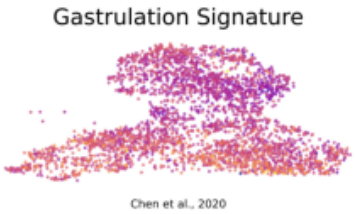
Manifold — samples



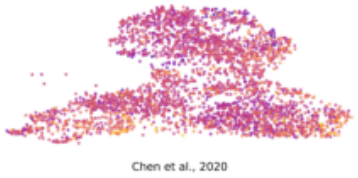
Pseudotime



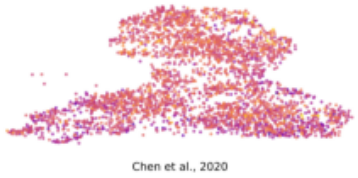
Gastrulation



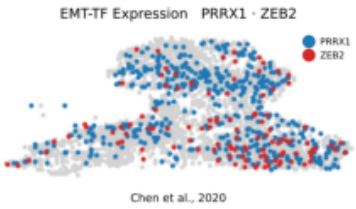
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 3,236 malignant cells (8 samples).

Verdict.
NOT REPRODUCED (embryonic geography absent; Gate 2 NO-GO). Malignant NPC cells form a mixed, continuous, non-branching manifold with a real, cell-cycle-independent Crest-terminal gradient (Crest opposes Gast −0.64 / pSom −0.44 after cc-correction). But the three-module embryonic pattern is not present: Gast/pSom decoupled (r=+0.01, vs Kürten GO +0.33) and pSom does not participate (flat, r(t,pSom)=−0.17). This NO-GO corrects, rather than echoes, the prior attempt's 'generic-carcinoma' rationale — the axis is embryonic-flavoured and NOT mere proliferation; what fails is the coupled Gast/pSom body the gate requires.

Parameter choices

Expression	UMI → normalize_total, log1p (no scale)	**Threshold**	min 107 malignant cells/sample (gap 36→107, ×2.97) → 8 samples, 3,674 cells
Embedding	overdispersed → PCA(19 PCs, >0.5% var) → Harmony(sample, θ=2) → knn15 → UMAP → Leiden 0.3	**Island removal**	438 cells (clusters 2,4,6,7,8) → 3,236 cells, 1 component
Subtypes	Cycling, Protein-maturation, Hypoxia; 3CA meta-programs + Chen calls	**Manifold**	scFates curve on X_umap (non-branching; 0 branch nodes, 12-combo PPT sweep); rooted Gast/pSom-high
EMT-TFs shown	SOX4/PRRX1/TCF4/SNAI2/ZEB2/FOXC1 + 6 more (detect≥10%); canonical SNAI1/ZEB1/TWIST1 dropped		

Chen 2020 NPC — analysis summary

RUN 5 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony on sample (θ=4, n_pcs=19): iLISI 1.08 → 2.33 of a 3.23 ceiling (72%; ceiling <8 because P11+P01=72% of cells). Single connected kNN component over all 3,674 cells — continuous, sample-integrated. No patient islands.

Gate 2 — module geography → GO (non-branching)

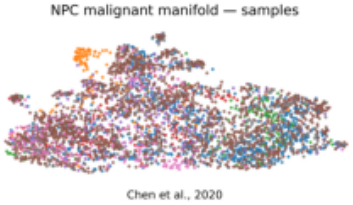
Gast/pSom overlap, Crest inverse terminal gradient. Per-node-cluster corr: Gast~pSom = +0.13 (OVERLAP, not cross-repression); Gast~Crest = −0.82; pSom~Crest = −0.50. Curve pseudotime: corr(t,Gast)=−0.58, corr(t,pSom)=−0.11, corr(t,Crest)=+0.44 — a single Gast/pSom-high → Crest-high axis. Cycling confound cleared (Kürten lesson): Gast~G2M = −0.49, progenitor is quiescent.

Gate 3 — hub cross-repression → NA

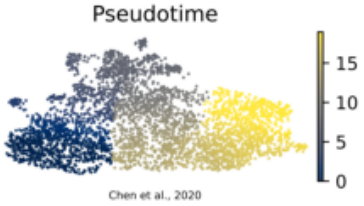
No bifurcation; branching-only correlation / modules notebooks not produced (spec §6).

STEP FIGURES

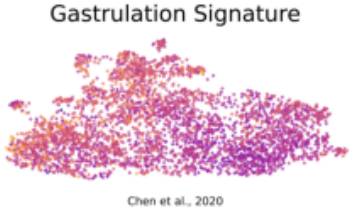
Manifold — samples



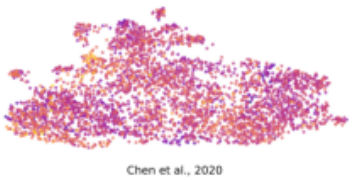
Pseudotime



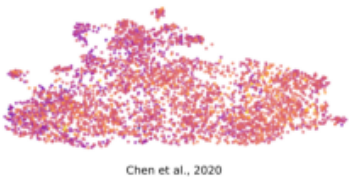
Gastrulation



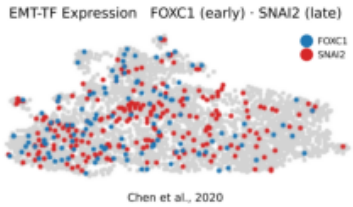
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 14 samples; 3,674 malignant cells retained (of 3,763; 8 samples ≥107 cells).

Verdict.
The embryonic pattern reproduces as the NON-BRANCHING case: a single developmental axis from a Gast/pSom-high progenitor to a Crest-high terminus (structurally the Kürten 2021 outcome, not Puram's branched hub-and-spoke). Gate 1 GO / Gate 2 GO / Gate 3 NA. Detectable EMT-TFs FOXC1, SNAI2, PRRX1, ZEB2 (canonical SNAI1/ZEB1/TWIST below detection).

Parameter choices

Min-cell threshold	107 (count-gap rule, 8/14 samples)	**Scaling**	no sc.pp.scale (log1p, reference-matched)
Genes	1,290 overdispersed	**Embedding**	PCA n_pcs=19; Harmony θ=4
Trajectory	scFates curve Nodes=10; root = Gast/pSom-high tip	**Signatures**	score_genes, union control pool
EMT-TFs	FOXC1 / SNAI2 / PRRX1 / ZEB2 (canonical SNAI1/ZEB1/TWIST below floor)		

Chen 2020 NPC — analysis summary

RUN 7 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/14 samples (≥107 malignant cells; multiplicative count gap 107→36) intermix on one continuous manifold — mean iLISI 2.16 (≈50% of neighbourhoods cross the >2 threshold), a single connected component; the 39-cell detached speck was removed. No per-patient islands.

Gate 2 — module geography → GO (non-branching)

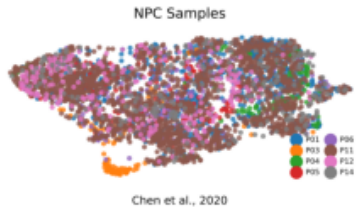
Gast & pSom do not resolve into two competing arms — both localise at the origin (high-decile mean pseudotime 0.22 each; Gast~pSom r=+0.00, no cross-repression). Crest is the inverse/terminal gradient (r=−0.78 vs Gast; high-decile pt 0.97; along pseudotime Gast ρ=−0.79, Crest ρ=+0.82). Non-branching.

Gate 3 — hub mutual exclusivity → NA

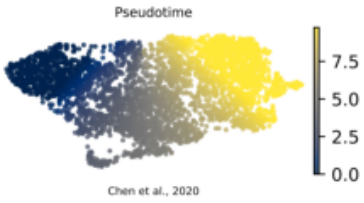
Single monotone Gast/pSom → Crest axis — no bifurcation and no Gast/pSom-overlap hub, so there is no branch pair to test for competitive cross-repression (branching-only gate; §2).

STEP FIGURES

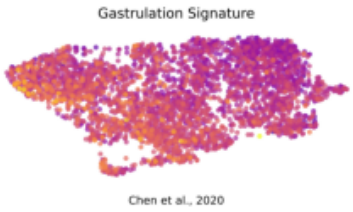
Gate 1: integration



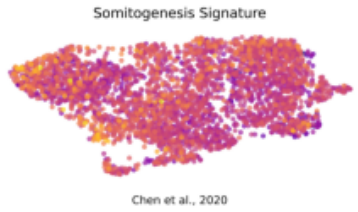
Trajectory / root



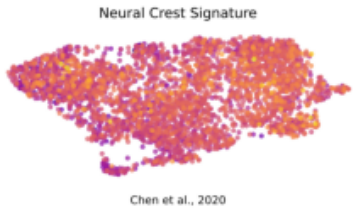
Gast



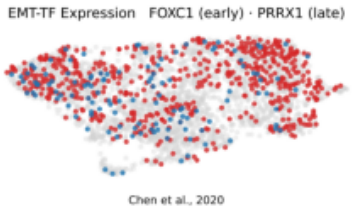
pSom



Crest



EMT-TFs



Dataset.
Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 71,896 cells / 14 samples (3,763 malignant). Malignant only: 3,674 pass the ≥107-cell sample threshold; 3,635 (8 samples) enter the trajectory after the 39-cell detached speck is removed.

Verdict.
✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. The NPC malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. EMT-TFs FOXC1 (early/root) → PRRX1 (late/terminus); prototypical SNAI1/TWIST1 fall below the 3% detection floor and the panel is weakly organised (|ρ|<0.18).

Parameter choices

Min cells/sample	≥107 (keeps 8/14; gap 107→36)	**Integration**	Harmony(sample) θ=5; PCA 20PC
Island removal	radius-graph (−39 detached speck)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Geometry**	non-branching curve (roundish, aspect 1.4)
Subtypes	markers + Chen 2020 (3CA mp_top)	**EMT-TFs**	FOXC1 (early) → PRRX1 (late)

Caron 2020 Pediatric ALL — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Malignant manifold is sample-mixed AND continuous. 8 patients across 3 genetic ALL groups (ETV6-RUNX1 Pre-B, high-hyperdiploid B-ALL, Pre-T) merge on ONE connected component after Harmony (θ=10) on sample. iLISI 1.00→4.71 of 8. Leiden split is proliferation (cycling 0.87 vs 0.06), not B-vs-T lineage; 0 islands removed.

Gate 2 — module geography → GO (non-branching)

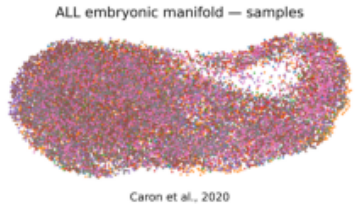
Embryonic pattern present as organized cross-repression, NOT a convergence hub. Gast & pSom sit at opposite poles (smoothed r=−0.45; joint-high enrichment 0.81<1); Crest is an inverse/terminal gradient (pSom–Crest r=−0.72). Root Gast-high (ρ=−0.72 vs pseudotime) → terminus Crest-high (ρ=+0.33). No bifurcation.

Gate 3 — hub cross-repression → NA

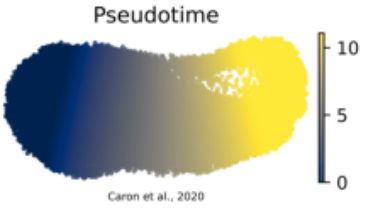
Not evaluated — non-branching topology (principal curve, 2 tips, 0 forks, fold-free). No tree node-clusters, so no mutual Gast–pSom cross-repression test applies. Terminates at the expression step.

STEP FIGURES

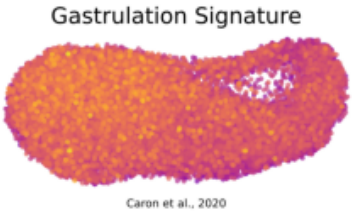
Manifold — samples



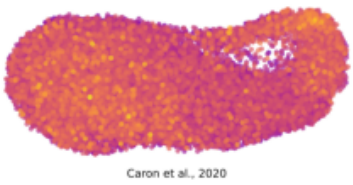
Pseudotime



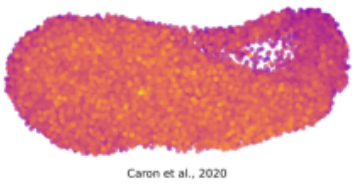
Gastrulation



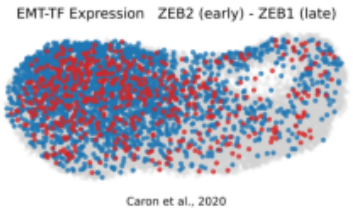
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric acute lymphoblastic leukemia (Caron et al., 2020), 10x — 8 malignant samples / 25,788 cells across 3 genetic ALL groups (ETV6-RUNX1 Pre-B, high-hyperdiploid B-ALL, Pre-T).

Verdict.
Pattern reproduces as the NON-BRANCHING case: a single embryonic axis with organized Gast–pSom cross-repression (r=−0.45) and an inverse terminal neural-crest gradient — a strict linear subset of the Puram hub-and-spoke, with no bifurcation. Root Gast/pSom-high → terminus Crest-high. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (root) → ZEB1 (terminus).

Parameter choices

Min-cell threshold	floor 100 (no dominant cliff; all 8 samples ≥1585) → 8/8 kept, 25,788 cells	**Integration**	Harmony on sample, θ=10 (same-sample nbr frac 0.999→0.272; 3 genetic ALL groups)
Embedding	PCA 50→20 on X_pca_harmony, k=30, spread=1.3, seed=1; radius-graph 0 removed	**Geometry**	principal CURVE (scFates EPG N=20 λ=0.3 μ=100); PPT over-forks a dense blob; curve fold-free
Root	curve tip 0 (Gast/pSom-high, Crest-low) — never Crest (late) or a cycling state	**Modules**	score_genes union control pool (Gast 66/81, pSom 77/98, Crest 93/104); RAW per-cell scores
EMT-TFs (>3%)	ZEB2 (40.6%), ZEB1 (8.9%), SNAI1 (6.8%); overlay pair ZEB2→ZEB1		

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) lifts per-neighbourhood iLISI 1.0→4.7 of 8 samples; the 3 genetic subtypes intermix on one connected manifold (single component after island removal). Read on the embedding + iLISI sweep, not a scalar.

Gate 2 — module geography → GO (non-branching)

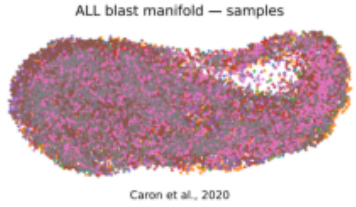
PPT tree = 0 forks → non-branching. Organized Gast↔pSom cross-repression (r=−0.45; −0.37 after removing cell cycle) + Crest inverse gradient (r=−0.72 vs pSom) SURVIVE cell-cycle regression → real developmental axis, not a proliferation artefact. FLAG: Crest leans to the Gast end (inversion of ideal pSom-arm), passes anyway.

Gate 3 — hub cross-repression → NA

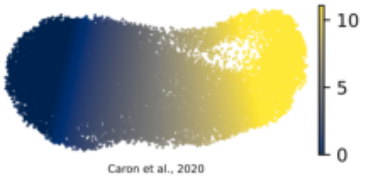
Non-branching topology has no bifurcation to test (branching-only gate).

STEP FIGURES

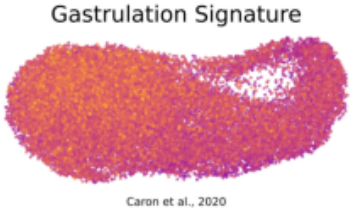
Manifold — samples



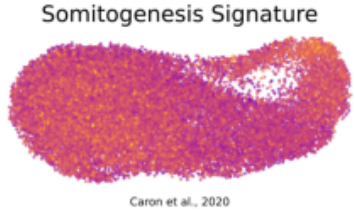
Pseudotime



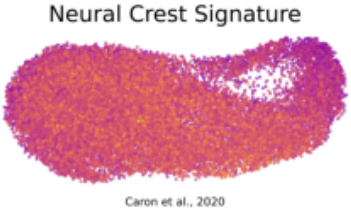
Gastrulation



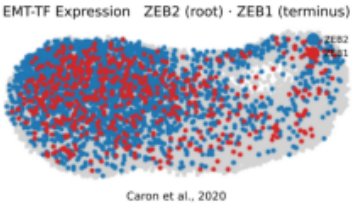
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric B/T-cell ALL (Caron et al., 2020), 10x — 25,788 malignant blasts, 8 samples, 3 genetic subtypes (ETV6-RUNX1, HHD, pre-T).

Verdict.
Pattern reproduces as a non-branching developmental axis — organized Gast↔pSom cross-repression with a terminal Crest gradient, cell-cycle-independent. Gate 1 GO / Gate 2 GO / Gate 3 NA. Crest-arm inversion flagged. Caveat: the dominant axis is the CELL CYCLE (pseudotime ρ=+0.69 with cycling; cc explains R^2=0.37/0.17/0.11 of Gast/pSom/Crest); the developmental cross-repression is the residual structure surviving cell-cycle removal (Zhou 2020 precedent). EMT-TFs ZEB2 (root) → ZEB1 (terminus); other factors <3% detection in blasts.

Parameter choices

Threshold	none — 8 samples span 1,585–5,816 cells with no clean multiplicative cliff; gap rule is a no-op (keep all)	**Embedding**	n_pcs=20, UMAP spread=1.3; Harmony θ=10 (iLISI plateaus by θ=5; subtypes need strong integration)
Geometry	NON-BRANCHING principal curve (ElPiGraph, fold-free self-approach 3.0); PPT tree gave 0 branch nodes	**Root**	quiescent Gast/pSom-high pole (origin; Crest is late)
Cell cycle	dominant axis is cell cycle (pt ρ=+0.69); cc explains R^2=0.37/0.17/0.11 of Gast/pSom/Crest; dev cross-repression survives cc removal	**EMT-TFs**	ZEB2 (root) → ZEB1 (terminus); other factors <3% detection

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) fuses the three genetically distinct subtypes: same-sample neighbour fraction 1.00→0.26, iLISI 1.0→4.2/8 (sample) and 2.5/3 (subtype), one connected kNN component at n_pcs=20/30/40. B-ALL (ETV6-RUNX1, HHD) and T-ALL intermix on one continuous manifold; the only within-manifold structure is a cycling↔quiescent axis (both poles carry all 3 subtypes), not a lineage split.

Gate 2 — module geography → GO (non-branching)

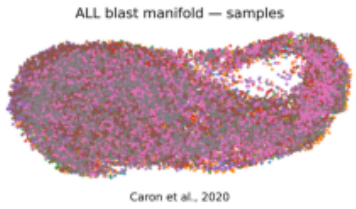
Signatures collapse onto a single (Gast/pSom)→Crest axis: smoothed pSom↔Crest r=−0.74, Gast↔pSom r=−0.46 (organized cross-repression), Gast↔Crest r=−0.20. Gast and pSom sit at OPPOSITE poles — no co-expression hub, no two competing arms. Crest is a terminal gradient (joint pSom&Crest enrichment 0.28 < 1). Along pseudotime Gast falls (ρ=−0.73), Crest rises (decile pt 0.68). Curve fold-free (2 tips / 0 forks). Crest does NOT enrich a Gast arm — no flag.

Gate 3 — hub cross-repression → NA

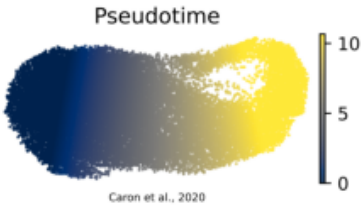
Non-branching topology — no bifurcation and no Gast/pSom hub, so the branching-only cross-repression test does not apply. Correlation & modules notebooks are correctly not produced.

STEP FIGURES

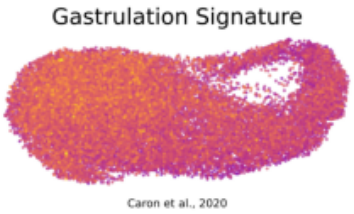
Manifold — samples



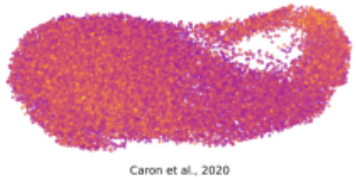
Pseudotime



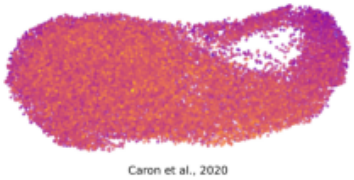
Gastrulation



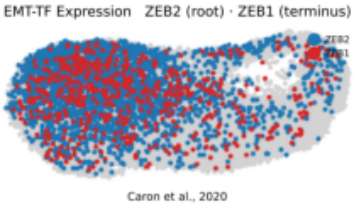
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric ALL (Caron et al., 2020), 10x — 25,784 malignant lymphoblasts over 8 samples spanning B-ALL (ETV6-RUNX1), B-ALL (HHD) and T-ALL.

Verdict.
Pattern reproduces as the NON-BRANCHING case — one gastrulation/somitogenesis → neural-crest developmental axis with organized Gast↔pSom cross-repression; no hub, no bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means (raw) Gast −0.017 / pSom +0.000 / Crest −0.010 — Linear. EMT-TFs ZEB2 (root, 40.6%) → ZEB1 (terminus, 8.9%).

Parameter choices

Min-cell threshold	100 floor — counts fall smoothly (no dominant cliff); all 8 samples kept	**Integration**	Harmony θ=10 (mixing plateaus θ=10–15, θ=20 regresses); n_pcs=30, n_neighbors=30, spread=1.3
Islands removed	4 cells / 3 detached specks (UMAP radius graph, largest component)	**Geometry**	Non-branching scFates curve (ElPiGraph), Nodes=20 λ=0.3 μ=100; 2 tips / 0 forks, fold-free
Root node	curve tip 0 = Gast/pSom-high, Crest-low (origin; Gast→Crest along pseudotime)	**Module means**	Gast −0.017 / pSom +0.000 / Crest −0.010 — Linear
EMT-TFs	ZEB2 (root, 40.6%) → ZEB1 (terminus, 8.9%); SNAI2/TWIST1/FOXC2 <0.1% gated out		

Caron 2020 ALL — analysis summary

RUN 4 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates all 8 samples; sample-iLISI rises 1.0→4.1 of a 6.85 ceiling (plateau θ 5–10). The manifold resolves into a cycling lobe and a low-cycling embryonic lobe; the cycling lobe is SCOPED OUT (25,788→19,800). On the scoped embryonic manifold samples and subtypes intermix on one connected component — GO.

Gate 2 — module geography → GO (non-branching)

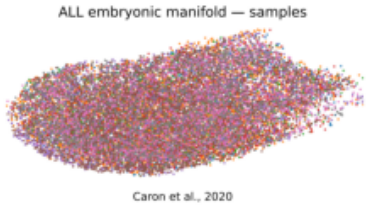
On the scoped manifold gastrulation is high at the root and falls along pseudotime (ρ=−0.65); Crest rises to a sharp inverse terminal gradient (decile t=0.97) opposing Gast (r=−0.74). Gast/pSom cross-repress (r=−0.55); pSom sits mid-axis (decile t=0.45). No resolution into two module-distinct arms — non-branching GO.

Gate 3 — cross-repression → NA

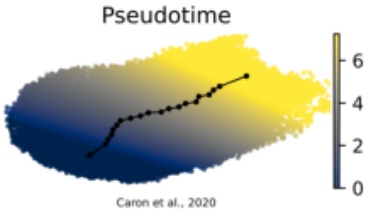
Non-branching topology: on the scoped embryonic manifold the PPT tree yields 2 tips / 0 forks (Gast-high root, Crest-high terminus) — no Gast/pSom co-expression hub. A clean fold-free curve (self-approach 3.0) is fit — no bifurcation to test — NA.

STEP FIGURES

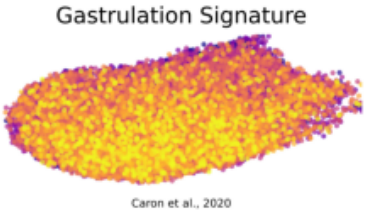
Manifold — samples



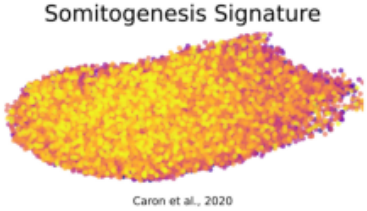
Pseudotime



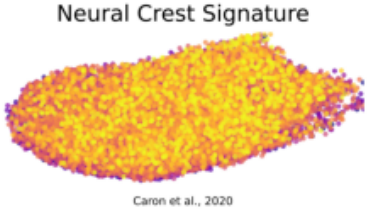
Gastrulation



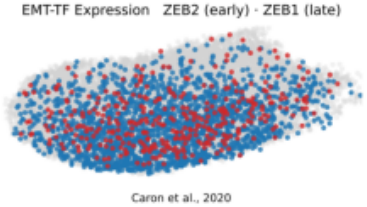
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric B/T-cell acute lymphoblastic leukemia (Caron et al., 2020), 10x — 19,800 embryonic blasts (cycling lobe scoped out) across 8 samples / 3 subtypes (ETV6-RUNX1, HHD, T-ALL). Non-branching Linear.

Verdict.
Pattern reproduces (non-branching / Linear) after scoping out the cycling lobe. A single continuous gastrulation → neural-crest embryonic axis on the 19,800-cell embryonic manifold, rooted at the Gast-high pole, with organised Gast/pSom cross-repression (r=−0.55) and a sharp terminal Crest gradient (decile t=0.97). The proliferation confound is resolved (pt↔proliferation ρ +0.62→+0.21). No bifurcation — Gate 3 NA.

Parameter choices

Min-cell threshold	≥100 (smooth counts) → 8/8 samples, 25,788 malignant cells	**Integration**	Harmony θ=10; n_pcs=20, n_neighbors=30, spread=1.3; 1,332 HVG
Compartment scoping	cycling lobe (86.5% cycling) scoped out → 19,800 embryonic cells	**Geometry**	Linear (non-branching); scFates curve N=20, fold-free (3.0); 2 tips / 0 forks
Root	curve tip at Gast-high, low-cycling origin (Gast/pSom early, Crest late)	**Modules recovered**	Gast 66/81 · pSom 77/98 · Crest 93/104
EMT-TFs	ZEB2 (early/root) → ZEB1 (late/terminus); SNAI1 also expressed	**Confound resolved**	pt↔proliferation ρ +0.62→+0.21; continuity |corr(t,DC1)|=0.91

Galen 2019 AML — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

19/28 samples (≥31 cells) intermix on one continuous manifold with no per-patient islands (iLISI 5.66 of 9.95 ceiling; radius-graph: 1 detached); both lobes (Cycling 9,872 / Low-cycling 6,227) carry the trajectory.

Gate 2 — module geography → GO (non-branching, inversion FLAG)

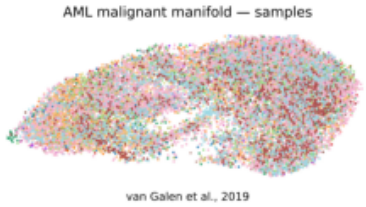
Gast/pSom cross-repress (r=−0.67); along pseudotime pSom declines / Gast rises (ρ=−0.87/+0.44); Crest anti-correlated with pSom (r=−0.90) but positive with Gast (r=+0.39) — the inversion — rising to a terminal readout (ρ=+0.83; crest-high decile mean pt=0.99).

Gate 3 — cross-repression → NA

Single monotone axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for mutual exclusivity.

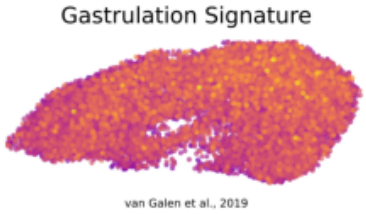
STEP FIGURES

Manifold — samples

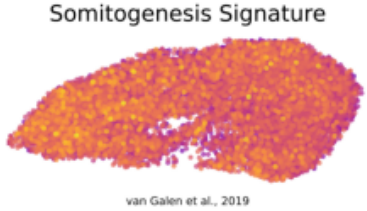


Pseudotime

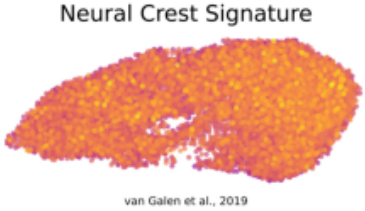
Gastrulation



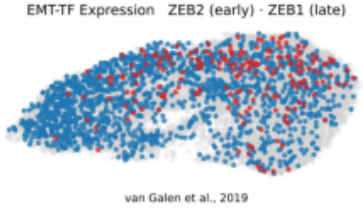
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Acute Myeloid Leukemia (van Galen et al., 2019), SeqWell — 37 samples / 33,392 cells (master scope); malignant only: 16,169 load, 16,099 across 19 samples (≥31) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Galen's AML malignant manifold is a single monotone Gast/pSom (cross-repressing) → terminal-Crest gradient rooted at the primitive, pSom/Gast-high, low-cycling pole; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The developmental axis spans both lobes (pSom pole low-cycling, Crest terminus cycling), so the whole manifold is kept (Bischoff/Sun/Lee no-scoping precedent). FLAG: Crest enriches the Gast end (inversion). EMT-TFs ZEB2 (early/root) → ZEB1 (late/terminus); FOXC1/SNAI1/TWIST below 3% floor.

Parameter choices

Min cells/sample	31 (keeps 19/28; cliff 61→16)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**Compartments**	Cycling 9,872 / Low-cycling 6,227
Geometry	Linear (non-branching); inversion FLAG	**EMT-TFs**	ZEB2 (early/root) · ZEB1 (late/terminus)

Wu 2020 AML — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ=10 on sample (iLISI 9.37 of 23.63 ceiling climbs to θ=10, drops at θ=20). After removing 20 cells in 17 detached specks, all 96,881 cells form ONE connected component: samples intermix per-neighbourhood on a single continuous manifold. Judged on the regenerated embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

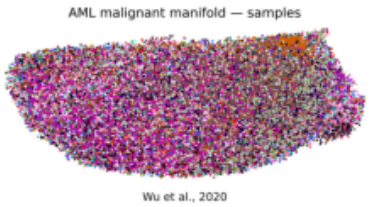
Single Gast/pSom-high → Crest-high developmental axis. Organized Gast↔pSom cross-repression (smoothed r=−0.68); Crest opposes Gast (r=−0.67), co-varies weakly with pSom (r=+0.11) and rises to the terminus (Crest-high decile at pseudotime 0.82; Spearman ρ=+0.43). No Gast/pSom co-expression hub and no opposing Gast arm — non-branching, not a bifurcation. Crest terminal, not on the Gast arm (no inversion flag).

Gate 3 — hub cross-repression → NA

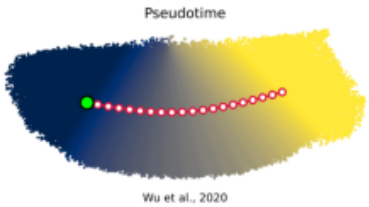
Branching-only. The manifold does not bifurcate — the diagnostic tree fit shows no Gast/pSom co-expression hub and no opposing Gast arm (both non-root tips are pSom-high / Crest-low), and the committed curve is fork-free. There is no hub population to test, so Gate 3 does not apply.

STEP FIGURES

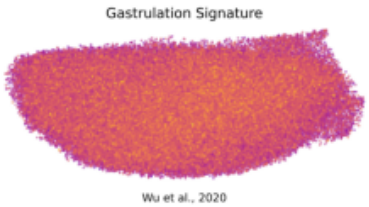
Manifold — samples



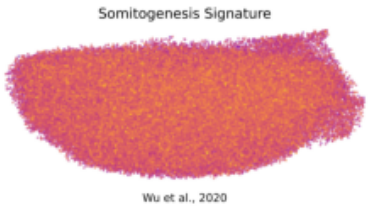
Pseudotime



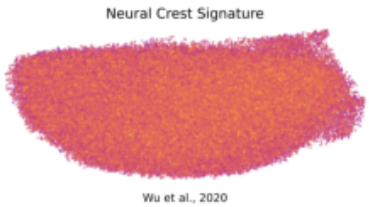
Gastrulation



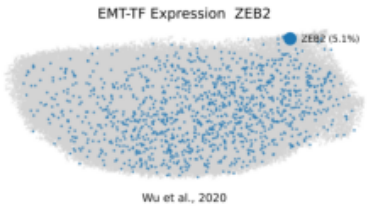
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Acute myeloid leukemia (Wu et al., 2020), Microwell-seq — 96,881 malignant cells (36 samples); single low-cycling body (cycling fraction 0.001).

Verdict.
The embryonic-program pattern reproduces in a NON-BRANCHING form: one continuous, sample-integrated AML manifold carrying a single Gastrulation/Somitogenesis-high → Neural-crest-high developmental axis with organized Gast↔pSom cross-repression (r=−0.68) and a terminal Crest gradient. No bifurcation, so Gate 3 is NA. Gate 1 & 2 GO; geometry Linear-pSom/Crest. EMT-TF ZEB2 (5.1%) is the sole detectable canonical factor.

Parameter choices

Min-cell threshold	100 (floor; no dominant cliff) → 36/38 samples, 96,901 cells (dropped P18, P13)	**Integration**	Harmony θ=10, n_pcs=20, k=30, spread=1.3 (iLISI peaks θ=10, falls θ=20); 443 overdispersed
Island removal	20 cells / 17 specks (UMAP radius graph 6× median NN) → 96,881	**Geometry**	Non-branching curve N=20, fold-free (SA 2.90); |corr(t, DC1)|=0.91
Root	curve tip at Gast/pSom-high, Crest-low pole; pt vs cycling ρ=−0.07	**Module means (raw)**	pSom +0.007 > Gast +0.002 > Crest −0.004 (Linear-pSom/Crest)
EMT-TF readout	ZEB2 (5.1%) — sole detectable canonical EMT-TF; broad panel below floor/absent		

Wu 2020 Acute Myeloid Leukemia — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

36 AML samples intermix across one continuous manifold; iLISI 9.4 of 23.6 ceiling (θ=10 peak, from 1.1 pre-Harmony), single connected component, one low-cycling compartment.

Gate 2 — module geography → GO (non-branching)

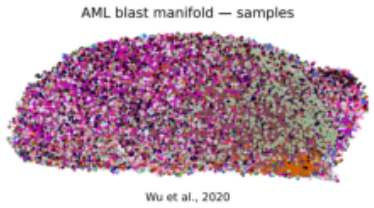
Single continuous Gast/pSom axis with organized cross-repression (smoothed r=−0.68) and a terminal Crest gradient (ρ=+0.43 vs pseudotime). Non-branching case → passes.

Gate 3 — hub cross-repression → NA

No bifurcation: the PPT tree's tips/fork carry near-identical signature means (geometric, not biological), so there is no co-expression hub or opposing arms to test. Branching-only gate not applicable.

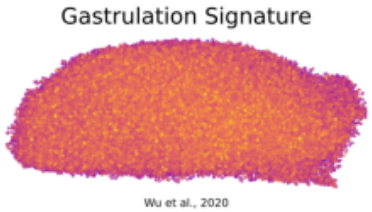
STEP FIGURES

Manifold — samples

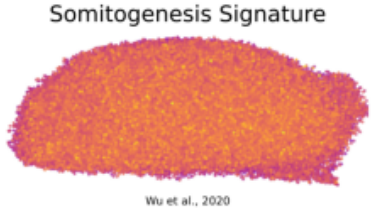


Pseudotime

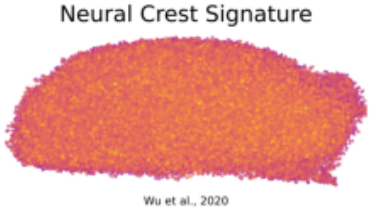
Gastrulation



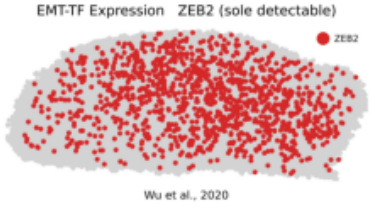
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Acute myeloid leukemia (Wu et al., 2020), Microwell-seq — 96,881 malignant blasts from 36 bone-marrow samples.

Verdict.
The embryonic-development pattern reproduces in its non-branching form — a single continuous gastrulation/somitogenesis axis under organized cross-repression, resolving into a terminal neural-crest gradient. No competing arms (Gate 3 NA). Geometry: Linear-pSom/Crest. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast +0.0019 / pSom +0.0066 / Crest −0.0042. EMT-TF ZEB2 (sole detectable factor; epithelial Snail/Zeb1/Twist silent in blasts).

Parameter choices

Min-cell threshold	202 (viable-range cliff 417→98, geo-mean); 36/38 samples kept	**Integration**	Harmony θ=10 (iLISI sweep peak; drops at θ=20)
Embedding	n_pcs=20, n_neighbors=30, spread=1.3	**Geometry**	principal curve (ElPiGraph); fold-free ratio 2.90; continuity DC1 0.92
Root	curve tip 0 = Gast/pSom-high, Crest-low (origin)	**Module means**	Gast +0.0019 / pSom +0.0066 / Crest −0.0042
EMT-TF	ZEB2 (sole detectably-expressed factor; Snail/Zeb1/Twist silent)		

Wu 2020 Acute Myeloid Leukemia — analysis summary

RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

iLISI 1.00 (fully patient-segregated) → 10.2 of 36 samples after Harmony (θ=10); kNN graph = 1 component. One continuous, uniformly low-cycling dome.

Gate 2 — module geography → GO (non-branching)

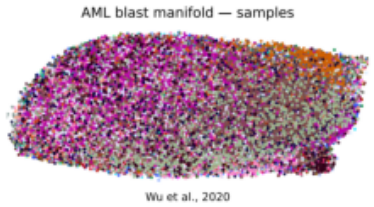
Organized Gast↔pSom cross-repression (r=−0.64) with a single terminal Crest gradient (Crest-high decile at pseudotime 0.86); no diverging arms.

Gate 3 — hub cross-repression → NA

Branching-only. Manifold is a fold-free curve (2 tips, 0 forks); no bifurcation to test.

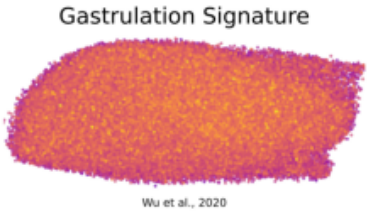
STEP FIGURES

Manifold — samples

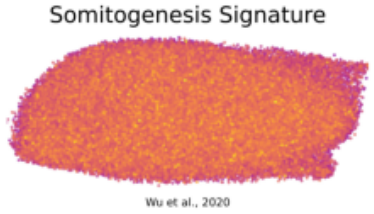


Pseudotime

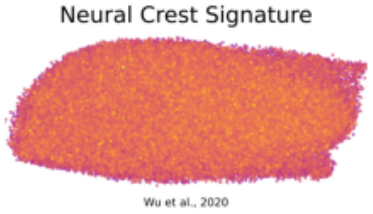
Gastrulation



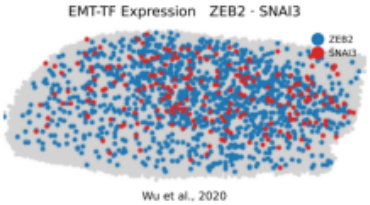
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Acute Myeloid Leukemia (Wu et al., 2020), Microwell-seq — 96,875 malignant cells (36/38 samples); counts (log1p of normalize_total).

Verdict.
Pattern reproduces (non-branching). Single continuous gastrulation → neural-crest axis with organized Gast↔pSom cross-repression; no bifurcation. Crest-arm inversion flag: none (Crest opposes Gast). Geometry: Linear. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (5.1%) + SNAI3 (0.9%) — canonical EMT-TF panel largely silent in this myeloid malignancy.

Parameter choices

Min-cell threshold	≥100 cells (floor) → drop P18/P13; no dominant multiplicative cliff (top/second 1.35)	**Integration**	Harmony θ=10 (iLISI plateau; θ≥20 over-integrates); n_pcs=30, no sc.pp.scale
Geometry	scFates curve N=20, fold-free (2 tips/0 forks); PPT tree overfits 5–10 spurious branches	**Root**	Gast/pSom-high, Crest-low tip (origin; Crest late/terminal); cell-cycle-free axis
Cross-repression	Gast↔pSom r=−0.64; Crest-arm inversion flag: none (Crest opposes Gast)	**EMT-TFs**	ZEB2 (5.1%) + SNAI3 (0.9%) — canonical panel largely silent

Wu 2020 AML — analysis summary

RUN 4 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

One connected kNN component at every θ (2/10/20); 488 cells (0.5%) in tiny detached specks, removed. Harmony θ=10 minimises residual same-sample enrichment (2.7×; θ=20 re-segregates). iLISI 9.5/36 (26% of ceiling). Decisive: between-sample R^2 of the embedding = 0.01 — patient identity explains ~1% of manifold structure, so this is an integrated continuum, NOT a per-patient clonal archipelago (cf. Liu2021 MM).

Gate 2 — module geography → NO-GO

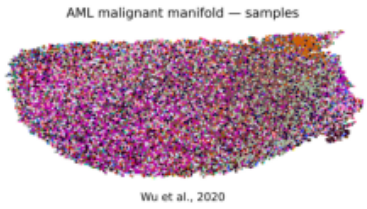
Embryonic module pattern ABSENT. Smoothed signature geography is featureless (score std ~0.007; no Gast/pSom poles, no Crest gradient); module means ~0 (Gast +0.002, pSom +0.007, Crest −0.004). Gast and pSom ANTI-correlate (smoothed r=−0.37) instead of overlapping as the non-branching case requires. No developmental axis: modules unaligned with any diffusion component (|r|≤0.14), flat along pseudotime, and the Crest "terminal" position FLIPS with the root choice (0.59 vs 0.41) — a rooting artifact, not biology.

Gate 3 — cross-repression → NA

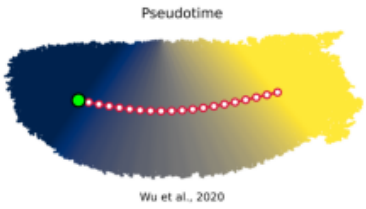
Not applicable — no bifurcation (Gate 2 NO-GO). Branching-only 04_correlation / 05_modules not produced.

STEP FIGURES

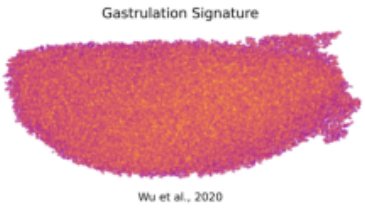
Manifold — samples



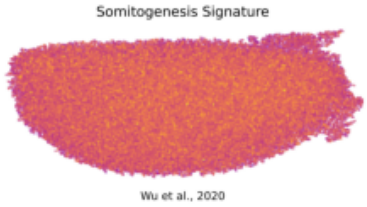
Pseudotime



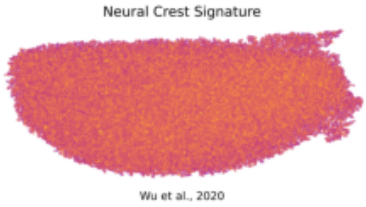
Gastrulation



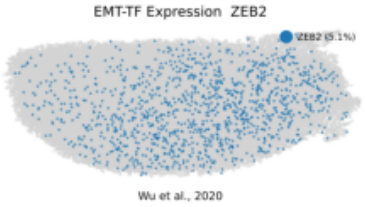
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Bone-marrow acute myeloid leukemia (Wu et al., 2020), Microwell-seq — 96,413 malignant blasts across 36 samples. A trajectory is fit, but the embryonic module pattern is absent (Gate 2 NO-GO; pattern-absent).

Verdict.
The embryonic developmental pattern does NOT reproduce — malignant AML blasts form an integrated, continuous manifold with no Gast/pSom hub, no arms, and no terminal Crest gradient (Gate 2 NO-GO). Geometry: none (pattern absent). Corrects three prior attempts that read a union-control-pool scoring artifact off an arbitrary root on a featureless blob. Gate 1 GO (integrated continuum, between-sample R^2=0.01), Gate 3 NA.

Parameter choices

Min-cell threshold	≥100 (clean gap 98→417, 4.26×) → 36/38 samples, 96,901 cells	**Integration**	Harmony θ=10 (residual-enrichment min); n_pcs=20, k=30, spread=1.3; 443 HVG
Island removal	radius-graph 6× median, keep largest → 488 removed, 96,413 final	**Trajectory**	scFates ElPiGraph curve (N=20, λ=0.3, μ=100); non-branching; root Gast/pSom-high tip
Gate 2 verdict	pattern absent — module means ~0, geography featureless, Gast–pSom anti-correlate (−0.37)	**Rooting artifact**	Crest terminal position flips with root (0.59 tip0 vs 0.41 tip2)
EMT-TFs	panel SNAI/ZEB/TWIST/FOXC/PRRX (VIM excluded); only ZEB2 (5.1%) clears 3% floor		

Roider 2020 B-cell lymphoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates 9 patients despite a severe clonal batch effect: same-sample neighbour fraction 0.995→0.26; iLISI median 1.00→4.58 of 9. One connected component (1 speck removed). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

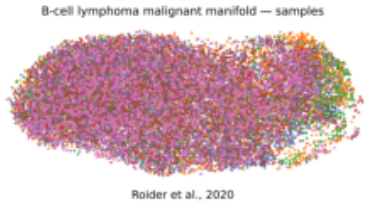
Non-branching case: a single (Gast/pSom)→Crest developmental gradient with organized Gast↔pSom cross-repression (smoothed r=−0.41; Crest↔pSom r=−0.52). Along pseudotime Gast/pSom fall (ρ −0.77/−0.15), Crest rises to a terminal tip (decile pt 0.77). Joint pSom&Crest enrichment 0.50 (<1 = inverse gradient, not a hub).

Gate 3 — cross-repression → NA

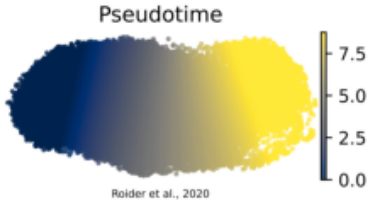
Not applicable — non-branching topology (principal curve, 2 tips / 0 forks, fold self-approach 2.95). No resolved bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

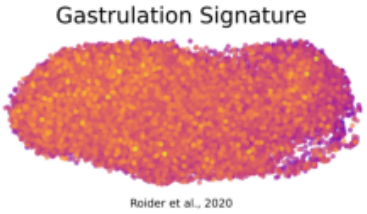
Manifold — samples



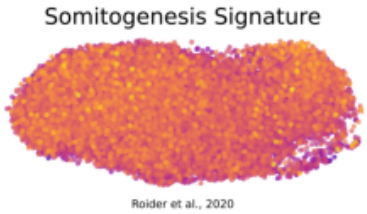
Pseudotime



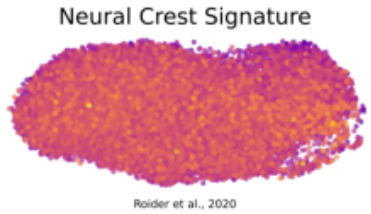
Gastrulation



Somitogenesis



Neural Crest



EMT-TF expression

Dataset.
B-cell lymphoma lymph-node biopsies (Roider et al., 2020), 10x — 22,599 malignant cells over 9 samples (DLBCL/FL/tFL); reactive-lymph-node normals contribute no malignant cells; single low-cycling body (~11% cycling). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs ZEB2 (early/root) → ZEB1 (terminus).

Parameter choices

Sample threshold	≥100 cells (largest gap 1583→327 above floor) → 9/9 samples	**Overdispersed genes**	1,334 (scFates find_overdispersed)
Integration	Harmony on sample θ=10 (mixing plateau; clonal patient batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 1 island removed → 22,599
Geometry	non-branching curve N=20, λ=0.3, μ=100; 2 tips / 0 forks (2.95)	**Root**	tip 0 = Gast/pSom-high, Crest-low, low-cycling origin
Module means (raw)	Gast −0.027 · pSom +0.002 · Crest +0.005 (Linear-pSom/Crest)	**EMT-TFs**	ZEB2 (early/root) → ZEB1 (late/terminus); only ZEB1/ZEB2 clear >3% floor

Gaiti 2019 CLL — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

The 7 SmartSeq2 plates and 4 patients intermix per-neighbourhood on one continuous, low-cycling manifold (iLISI 1.99 pre-Harmony → 5.53 of 7 samples at θ=5; same-sample-neighbour fraction 0.66→0.21). One connected component, 0 detached islands.

Gate 2 — module geography → GO (non-branching)

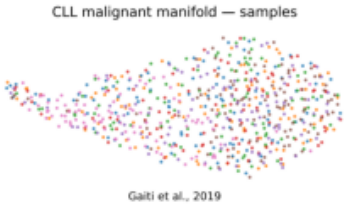
Single (gastrulation+somitogenesis)→neural-crest axis, not a hub with two cross-repressing arms. Gast and pSom co-vary (raw r +0.07, smoothed +0.64) and both oppose Crest (raw −0.57 / −0.59); along pseudotime Gast ρ −0.96 and pSom ρ −0.66 fall while Crest ρ +0.97 rises to a terminal tip. Joint pSom&Crest enrichment 0.43 (<1 → inverse gradient, not a hub).

Gate 3 — cross-repression → NA

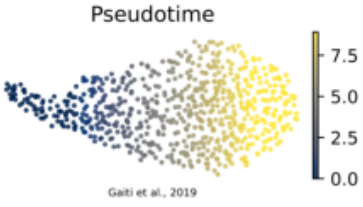
Non-branching topology (principal curve: 2 tips, 0 forks, fold self-approach 3.0, stable across Nodes 15/20/30; a PPT tree finds 0 branch points) — no bifurcation, so the hub cross-repression test does not apply.

STEP FIGURES

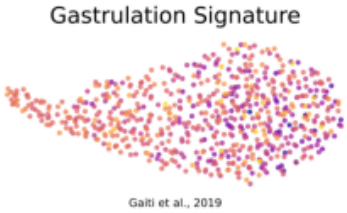
Manifold — samples



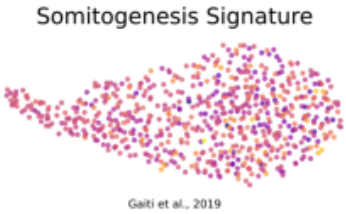
Pseudotime



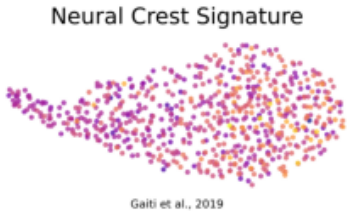
Gastrulation



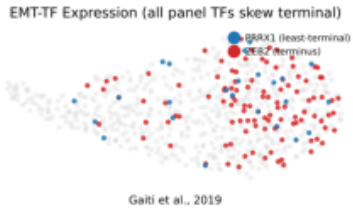
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Chronic lymphocytic leukemia (Gaiti et al., 2019), SmartSeq2 (plate-based) — 672 malignant CD19+ cells, 7 plates / 4 patients. Non-branching Linear-pSom/Crest.

Verdict.
The embryonic pattern reproduces as a non-branching single (gastrulation/somitogenesis)→neural-crest axis (Linear-pSom/Crest) — GO / GO / NA. EMT-TFs ZEB2 (terminus) vs PRRX1 (least terminus-biased); all panel TFs skew terminal, no root-end factor.

Parameter choices

Min-cell threshold	67 (96→48 plate cliff geomean); 7 full plates / 672 cells	**Integration**	Harmony on sample θ=5; n_pcs=20; overdispersed 45 → PCA 44
Geometry / root	principal curve (ElPiGraph), non-branching; root = tip 0 (Gast/pSom-high)	**Hub identity**	NA — non-branching, no convergence hub
EMT-TFs	ZEB2 (terminus) vs PRRX1 (least terminal); no root-end factor; VIM excluded		

Rendeiro 2020 CLL — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) integrates all 7 samples: same-sample neighbour fraction 0.97 → 0.22, iLISI median 1.00 → 4.95 of 7. 205 cells in 3 detached islands removed; one connected component. Read on the regenerated embedding + per-neighbourhood iLISI, not a scalar: samples intermix AND the manifold is continuous.

Gate 2 — module geography → GO (non-branching)

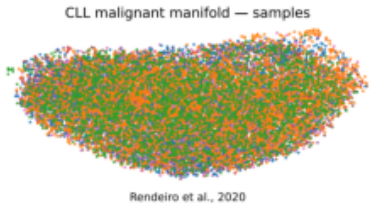
Non-branching case: a single (Gast)→Crest developmental gradient. Gast opposes Crest in space (smoothed r=−0.75; pSom weakly opposes both). Along pseudotime Gast falls (ρ −0.64) while Crest rises to a terminal tip (Crest-high decile pt=0.74) and pSom rises (ρ +0.46). Joint Gast&Crest enrichment 0.27 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks; a PPT tree throws 6–7 spurious forks on this near-round cloud). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

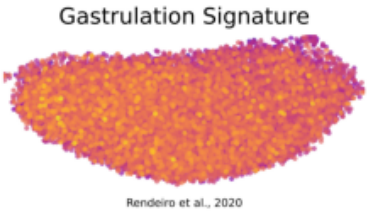
STEP FIGURES

Manifold — samples

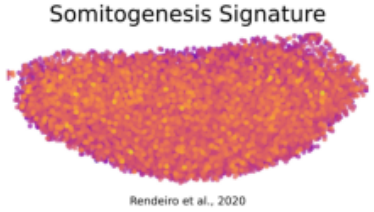


Pseudotime

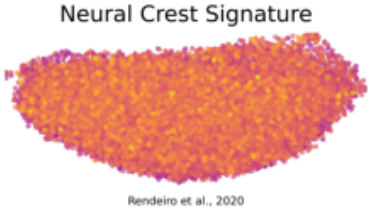
Gastrulation



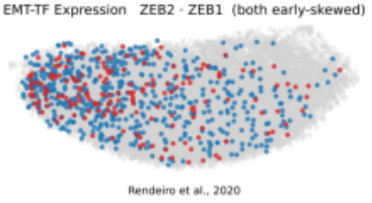
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
CLL peripheral blood (Rendeiro et al., 2020), 10x — 18,452 malignant cells over 4 patients (7 longitudinal samples); single diffuse-cycling body (~44% cycling, cell-cycle-free axis).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation → neural-crest axis, Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. CLL barely expresses EMT-TFs — only ZEB2 and ZEB1 clear the detection floor, both root-skewed (no terminus EMT-TF).

Parameter choices

Sample threshold	geom-mean 284 (cliff 1012→80, ratio 12.6) → 7/11 samples; 18,452 cells	**Overdispersed genes**	528 (scFates find_overdispersed)
Integration	Harmony on sample, θ=10 (mixing plateau; patient-private CLL batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 205 cells / 3 islands removed
Geometry	Non-branching scFates curve N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.58; corr(x_lin,t)=0.92	**Root**	tip 0 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast −0.020 · pSom +0.008 · Crest +0.010 (Linear-pSom/Crest)	**Cell cycle**	43.8% cycling, diffuse single lobe; axis cell-cycle-free
EMT-TFs shown	ZEB2 + ZEB1, both root-skewed (no terminus EMT-TF); >3% gate		

Rendeiro 2020 CLL — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7 samples / 4 patients (CLL1/5/6/8) intermix on one continuous manifold (iLISI 1.0→4.7 of 7; radius-graph: 166 detached cells stripped); one diffuse compartment, no separable cycling lobe.

Gate 2 — module geography → GO (non-branching)

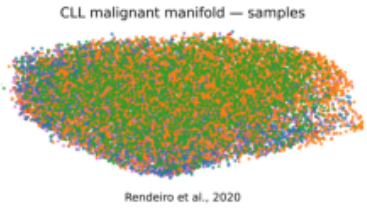
Gast declines / pSom rises on one axis (ρ=−0.62 / +0.49); Crest anti-correlated with both (raw r=−0.55 / −0.55), rising toward a terminal readout (crest-high decile mean pt=0.73).

Gate 3 — hub mutual exclusivity → NA

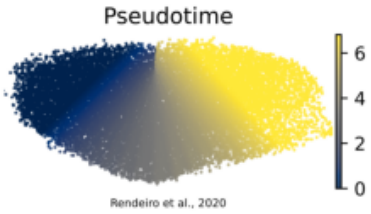
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual cross-repression.

STEP FIGURES

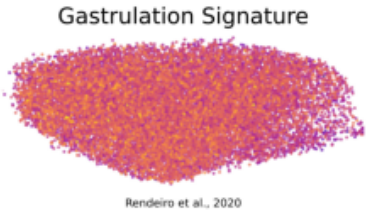
Manifold — samples



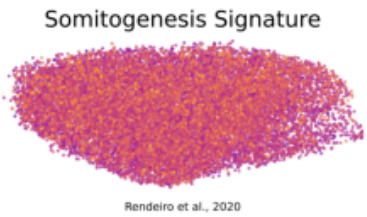
Pseudotime



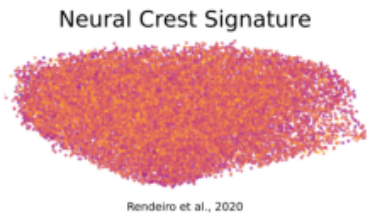
Gastrulation



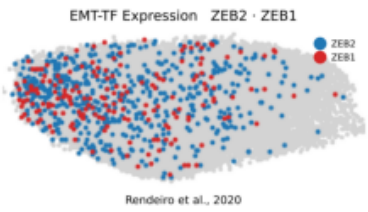
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Chronic Lymphocytic Leukemia (Rendeiro et al., 2020), 10x — peripheral blood, longitudinal ibrutinib cohort. Malignant only: 18,854 load; 18,657 across 7 samples / 4 patients (≥284, multiplicative gap 1012→80) enter preprocessing; 18,491 after island removal enter the trajectory (whole manifold, no lobe scoping — proliferation is diffuse).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Rendeiro's CLL malignant manifold is a single monotone Gast → terminal-Crest gradient (Linear-pSom/Crest), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. A PPT tree fabricates 4–7 forks on this compact near-round cloud, while an ElPiGraph curve gives a clean fold-free 2-tip backbone (self-approach 2.80) — genuinely non-branching, not a collapsed Y. EMT-TFs near-silent in this blood cancer: only ZEB2 (12%) / ZEB1 (4%) clear the 3% floor (SNAI/TWIST/FOXC/PRRX essentially unexpressed), and their pseudotime trend is flat (ρ≈0), reported as-is.

Parameter choices

Min cells/sample	≥284 (keeps 7/11; gap 1012→80)	**Scope**	whole manifold (no scoping)
Root node	tip 2 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=5; 20 PC; spread=1.3
Trajectory	curve N=20 λ=0.3 μ=100	**Geometry**	Linear-pSom/Crest

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) fuses the four patient-private CLL clones: per-neighbourhood iLISI 1.0→~5.1 / 7 across a θ×n_pcs sweep (plateau at θ=10; θ=20 over-collapses), one connected kNN component after island removal. Decisive check: the Gast→Crest gradient reproduces WITHIN each clone (Crest↔Gast −0.47…−0.58 in every patient), so integration aligns a shared intra-clonal gradient, not a fabricated continuum; proliferation diffuse (corr≈0).

Gate 2 — module geography → GO (non-branching)

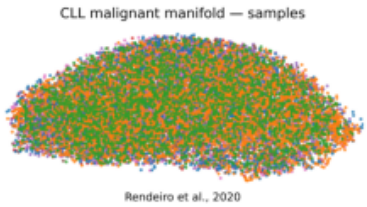
Signatures collapse onto a single (Gast/pSom)→Crest axis: raw per-cell Crest↔Gast r=−0.55, Crest↔pSom r=−0.55, Gast↔pSom r≈+0.05 (no cross-repression — no two competing arms). Crest is an inverse terminal gradient. Principal curve is fold-free (2 tips / 0 forks; self-approach 1.6). Crest is NOT enriched on a Gast arm — no inversion flag.

Gate 3 — hub cross-repression → NA

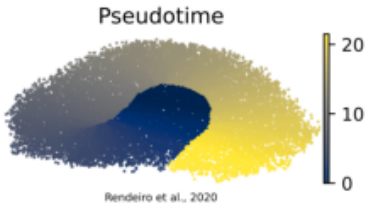
Non-branching topology — no bifurcation, no Gast/pSom hub, so the branching-only cross-repression test does not apply. PPT tree gives only unstable spurious forks (4→6→9 as nodes 100→200) over a blob. Modules notebooks not produced.

STEP FIGURES

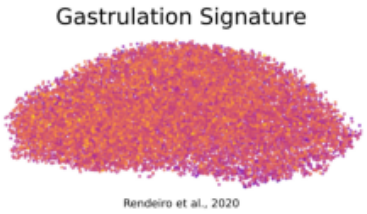
Manifold — samples



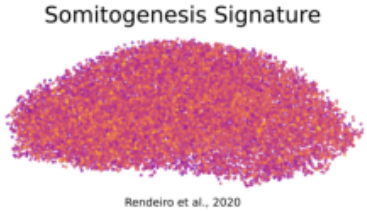
Pseudotime



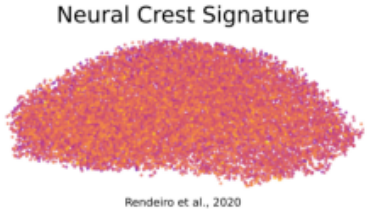
Gastrulation



Somitogenesis



Neural Crest



EMT-TF expression

Dataset.
Chronic lymphocytic leukemia (Rendeiro et al., 2020), 10x — 18,314 malignant B-cells over 7 samples / 4 patients.

Verdict.
Pattern reproduces as the NON-BRANCHING case — a single gastrulation/somitogenesis → neural-crest developmental axis; no hub, no bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means (raw) Gast −0.017 / pSom +0.007 / Crest +0.007 — Linear. EMT-TFs ZEB2 (12.1%) + ZEB1 (3.8%); Snail/Twist/FOXC/PRRX silent.

Parameter choices

Min-cell threshold	285 — geom mean of the 1012→80 spillover cliff; keeps 7/11 samples, 4 patients	**Integration**	Harmony θ=10 (patient-private CLL clones), n_pcs=30, spread=1.3
Islands removed	343 cells / 57 detached specks (UMAP 6× radius graph, largest component)	**Geometry**	Non-branching scFates curve (ElPiGraph), Nodes=20; 2 tips / 0 forks, fold-free (SA 1.6)
Root node	curve tip 2 = Gast-high, Crest-low (origin; Crest terminal along pseudotime)	**Module means**	Gast −0.017 / pSom +0.007 / Crest +0.007 — Linear
EMT-TFs	ZEB2 (12.1%) + ZEB1 (3.8%); Snail/Twist/FOXC/PRRX silent		

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Unintegrated iLISI 1.0 of 7 (complete per-patient segregation); Leiden clusters 99.3% patient-pure; patient identity 99.5% kNN-predictable. Mixing appears only when Harmony is forced (θ≥3), collapsing patient centroids rather than revealing a shared axis. Per-patient archipelago — no continuous, sample-mixed malignant manifold exists to fit.

Gate 2 — module geography → NA

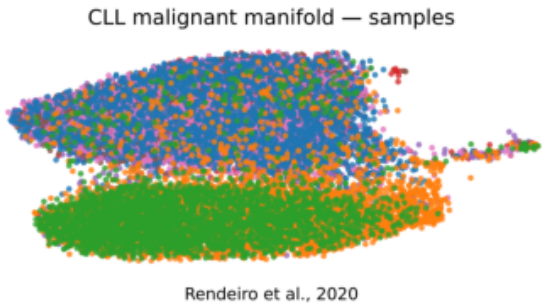
Not evaluated — gate chain stops at the Gate 1 NO-GO (§6).

Gate 3 — hub cross-repression → NA

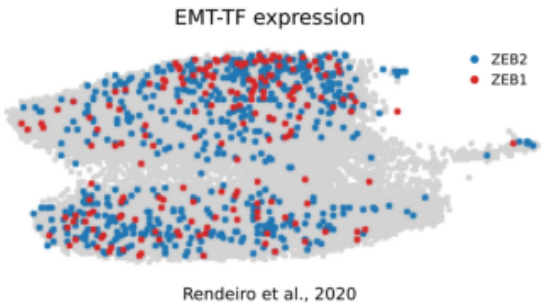
Not evaluated — branching-only; no trajectory fit.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Chronic Lymphocytic Leukemia (Rendeiro et al., 2020), 10x — 18,657 malignant cells · 7 samples / 4 patients.

Verdict.
Pattern does NOT reproduce. The malignant compartment is a set of independent per-patient CLL clones, not a continuous developmental manifold. NO-GO at Gate 1; no trajectory, no bifurcation. Descriptive EMT-TF overlay: ZEB2 (12.4%), ZEB1 (3.8%).

Parameter choices

Sample threshold	≥100 cells → 7 of 11 samples (18,657 cells); largest multiplicative count gap 1012→80 (12.6×)	**n_pcs**	20 (scree elbow at 6, floored to atlas robust range)
Integration	Harmony on sample, swept 0→5; protocol-default θ=2 shown; higher over-integrates	**Island removal**	none (disconnected components are whole patients, not noise)
Geometry	NA (no trajectory); manifold not sample-mixed; Gate 1 NO-GO	**EMT-TFs**	ZEB2 (12.4%), ZEB1 (3.8%); SNAI2/TWIST/FOXC2/PRRX2 absent

Rendeiro 2020 CLL — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4 CLL clones (CLL1/5/6/8, 7 samples) integrate to one continuous mass; sample iLISI 1.0→4.7/7, patient 1.0→2.6/4 at θ=5. Gast→Crest axis reproduces within every patient clone (r=−0.47..−0.59) — shared biology, not a per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

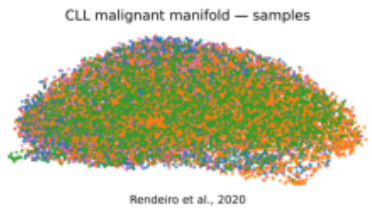
Single Gast→Crest developmental axis: Gast opposes Crest (smoothed r=−0.74), Crest an inverse terminal readout (Crest-high decile pseudotime 0.75 vs Gast 0.29). No Gast/pSom co-expression hub (joint-high enrichment 0.34 < 1). Curve fold-free.

Gate 3 — hub cross-repression → NA

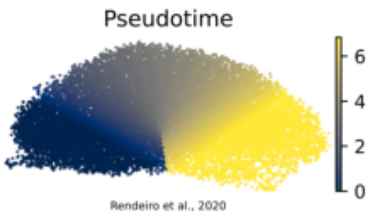
Single collapsed axis, no bifurcation — no two competing arms to test for mutual Gast↔pSom cross-repression.

STEP FIGURES

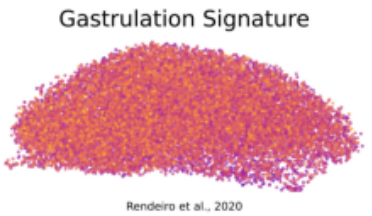
Manifold — samples



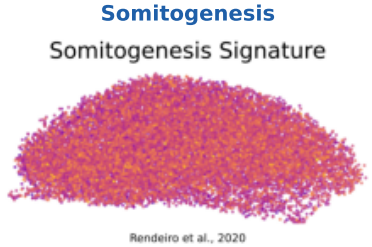
Pseudotime



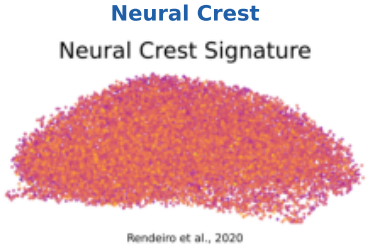
Gastrulation



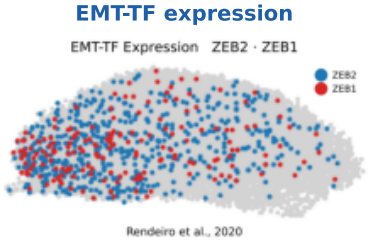
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Chronic lymphocytic leukemia (Rendeiro et al., 2020), 10x, longitudinal ibrutinib time-course — 18,854 malignant raw → 18,657 (7 samples ≥100 cells) → 18,490 after island removal (167 detached-speck cells, 2 rounds). 4 patients; 43.8% cycling (axis-orthogonal).

Verdict.
Pattern reproduces as a non-branching embryonic axis: a Gast/pSom-high origin → Crest-high terminus; developmental order Gast ρ=−0.63, Crest terminal (decile 0.75) along pseudotime, confirmed on a diffusion embedding independent of the fitted UMAP (ρ=0.92). Linear geometry; Gate 3 NA. EMT-TFs near-silent (lymphoid malignancy). Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Min cells/sample	100 (gap 1012→80, 12.65×)	**Integration**	Harmony θ=5; PCA20; nn30; spread=1.3
Scope	all 4 patients, one manifold	**Trajectory**	scFates curve N=20 λ=0.3 μ=100 (2 tips/0 forks)
Root node	Gast/pSom-high tip 0	**EMT-TFs**	ZEB2, ZEB1 (both early; only TFs >3% detect)

Rendeiro 2020 CLL — analysis summary

RUN 6 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Judged on the regenerated embedding across a θ-sweep (1/2/3/5/10 × n_pcs 20/30/40) with per-neighbourhood iLISI. The malignant compartment is a per-patient archipelago: unintegrated kNN patient accuracy 99.7%, Leiden 99.3% patient-pure. Harmony mixes only by forced over-integration — even at θ=10, patient kNN accuracy stays 52% (chance 34%) and iLISI plateaus ~3.0/4. Signatures carry no manifold geography (Moran's I +0.02 = random-signature control). No single connected continuous developmental gradient — NO-GO.

Gate 2 — module geography → not evaluated

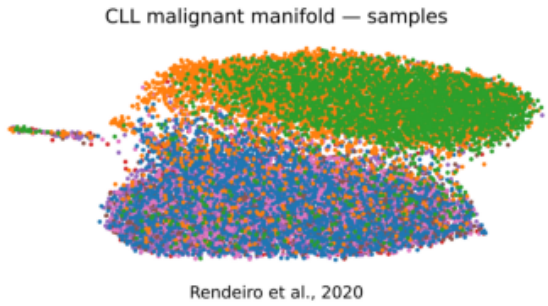
Not evaluated — the gate chain stops at the Gate 1 NO-GO (no integrated continuous manifold on which to read a Gast/pSom hub or arms). For the record, modules are near-zero and cell-cycle-orthogonal, and Gast–pSom co-vary (r=+0.05, no cross-repression).

Gate 3 — cross-repression → not evaluated

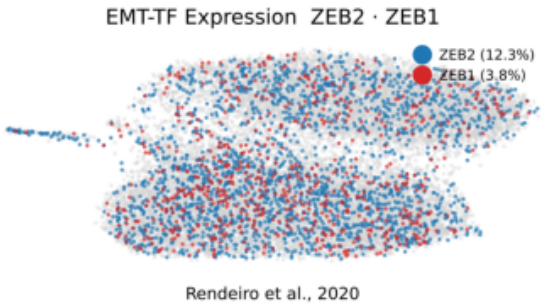
Not evaluated — reached only when a branching manifold clears Gates 1–2. No trajectory is fit (correlation / modules notebooks correctly not produced).

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Chronic lymphocytic leukemia (Rendeiro et al., 2020), 10x — 18,598 malignant B-cells over 7 samples / 4 patients (longitudinal ibrutinib). Each patient is a genetically distinct CLL clone.

Verdict.
Pattern does NOT reproduce. Gate 1 NO-GO — a per-patient clonal archipelago with no shared continuous manifold and no embryonic module geography. No trajectory is warranted; analysis stops at preprocessing + the unconditional EMT-TF step.

Parameter choices

Min-cell threshold	≥100 (multiplicative cliff 1012→80, 12.65×); 7/11 samples	**Malignant cells**	18,854 raw → 18,657 (threshold) → 18,598 (island removal)
Overdispersed genes	528; PCA 50 comps, NO scaling	**Integration**	Harmony on sample θ=2 canonical; swept 1/2/3/5/10 × n_pcs 20/30/40
Embedding	n_pcs=20, n_neighbors=30, spread=1.3; 59 island cells removed	**Cell cycle**	42.2% cycling; modules cell-cycle-orthogonal (r +0.02) — no lobe to scope
Geometry	None — no manifold/trajectory fit (Gate 1 NO-GO); Gast–pSom +0.05 (not branching)	**EMT-TFs**	ZEB2 (12.3%) · ZEB1 (3.8%) — only TFs above 3% floor; VIM excluded

Giustacchini 2017 CML — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=6) integrates all 35 plates: cross-sample neighbour fraction 0.82; one connected component, 0 islands removed. Read on the regenerated embedding across an n_pcs/θ sweep, not a scalar: θ=6 folds the blast-crisis patient triad into the body (vs deleting ~180 cells at θ=2).

Gate 2 — module geography → GO (non-branching)

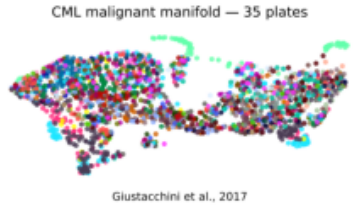
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.70), both oppose Crest (Gast↔Crest r=−0.93, pSom↔Crest r=−0.80). Along pseudotime Gast falls (ρ −0.57), Crest rises to a terminal tip (ρ +0.60, Crest-high decile pt=0.90). Joint pSom&Crest enrichment 0.27 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

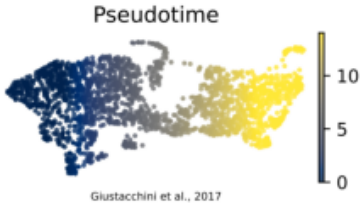
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

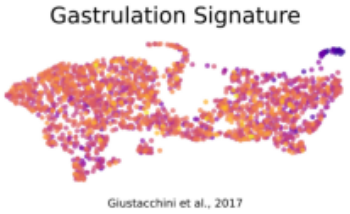
Manifold — samples



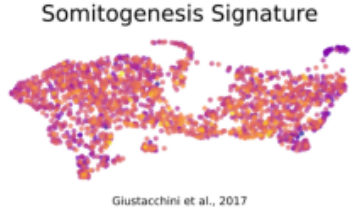
Pseudotime



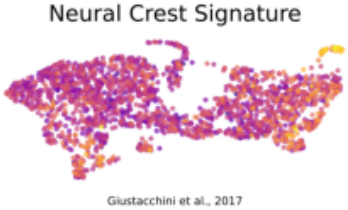
Gastrulation



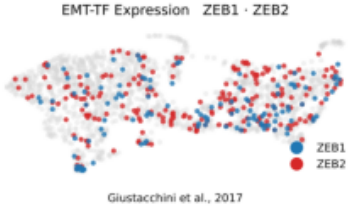
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
CML (Giustacchini et al., 2017), SmartSeq2 — 2,055 malignant BCR-ABL cells over 36 patient plates; single low-cycling body (~16% cycling), one continuous developmental axis.

Verdict.
Pattern reproduces in the non-branching Linear-Gast form: one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB1 & ZEB2 (only factors clearing detection), rising toward the Crest terminus.

Parameter choices

Sample threshold	n≥20 cells (35/36 plates, 2,040); ≥100 floor rejected (SmartSeq2 caps at 154); gap 15→22	**Overdispersed genes**	5,909 (scFates find_overdispersed)
Integration	Harmony on sample, θ=6 (stability sweep; folds blast-crisis patients into body)	**Embedding**	n_pcs=20 (scree elbow; n_pcs=40 fragments a patient speck)
Geometry	Non-branching scFates curve N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.97; corr(x_lin,pt)=+0.99	**Root**	curve tip 0 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast +0.016 ≥ pSom +0.013 > Crest −0.016 (Linear-Gast); cell-cycle-free	**EMT-TFs shown**	ZEB1 (23%) & ZEB2 (42%); FOXC1/PRRX1/TWIST1/SNAI1 <2%, gated out

Giustacchini 2017 CML — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

29/34 primary samples (≥29 cells) intermix on one continuous manifold with no per-patient islands (iLISI 8.67 of 23.39 ceiling, n_pcs=40 θ=10; radius-graph: 0 detached). K562 cell line (91 cells) scoped out — a detached, cycling clone reproducing none of the reference relationships.

Gate 2 — module geography → GO (non-branching)

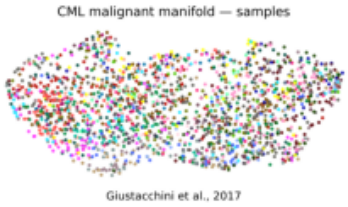
Single developmental axis: Gast/pSom overlap (r=−0.04, no competing arms); Crest an inverse terminal gradient opposing Gast (r=−0.88) and pSom (r=−0.37). Along pseudotime Gast falls (ρ=−0.96), Crest rises to a terminal readout (ρ=+0.79; crest-high decile mean pt=0.98).

Gate 3 — cross-repression → NA

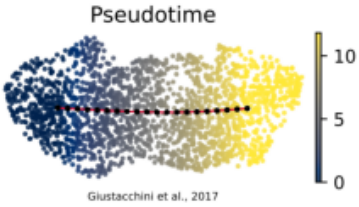
Single monotone axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

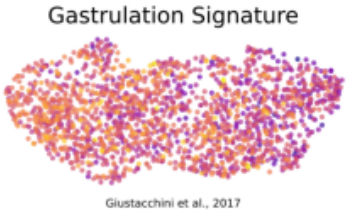
Manifold — samples



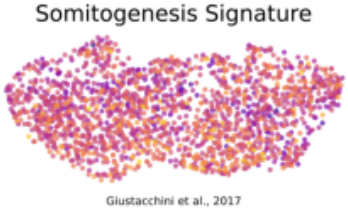
Pseudotime



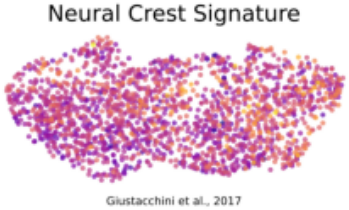
Gastrulation



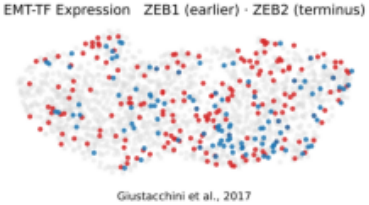
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Chronic myeloid leukemia (Giustacchini et al., 2017), SmartSeq2/HiSeq 2000 — 42 samples / 2,287 cells (master scope); malignant only: 2,055 load, 1,845 primary cells across 29 samples (≥29) enter the trajectory after the K562 cell line is scoped out. Non-branching Linear-pSom/Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The primary CML malignant manifold is a single monotone gastrulation → terminal-neural-crest gradient rooted at the Gast/pSom-high, Crest-low origin; it lacks Puram's hub-and-spoke bifurcation (PPT tree: 2 tips, 0 forks), so Gate 3 has no hub to test. Crest peaks at the low-Gast terminus — no inversion. EMT-TFs ZEB1 (earlier) → ZEB2 (terminus); SNAI1/TWIST1/FOXC1/PRRX1 below the 3% floor.

Parameter choices

Min cells/sample	≥29 (keeps 29/34; smooth, 30→29 step)	**Scope**	primary tumour (K562 cell line dropped)
Root node	tip 0 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=10; n_pcs=40; PCA50; spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100	**Geometry**	Linear-pSom/Crest (non-branching)
EMT-TFs	ZEB1 (earlier) → ZEB2 (terminus); rest below 3% floor		

Steen 2021 DLBCL — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Whole malignant manifold (2,478 cells, 3 tumours) is one connected component (0 islands). Harmony (θ=10) lifts per-neighbourhood mixing from iLISI 1.00→1.88 of 3 samples; the regenerated embedding shows one continuous cloud with samples intermixed — no per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

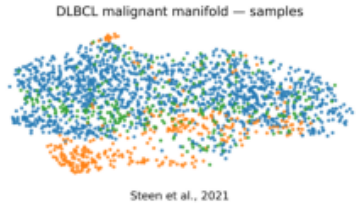
A distinct proliferative lobe (513 cells, 95.5% cycling) confounds Gastrulation on the whole manifold (Gast↔cycling r=−0.74). Scoping to the low-cycling lobe restores the reference relationships: Gast/pSom co-vary (r=+0.6) and both oppose Crest (Crest↔Gast r=−0.9). One (Gast/pSom)→Crest axis — Gast ρ=−0.94, pSom ρ=−0.73 decline; Crest rises to a terminal tip (ρ=+0.87, Crest-high decile pt 0.95; joint pSom&Crest enrichment 0.26<1). Single axis, no bifurcation — GO.

Gate 3 — cross-repression → NA

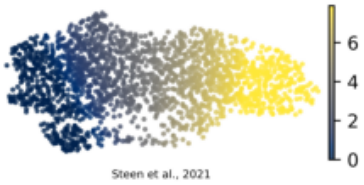
Not applicable: the manifold is non-branching (principal curve, 2 tips / 0 forks, fold-free self-approach 2.9 across Nodes=15/20/30). With no Gast/pSom hub there is no bifurcation to test.

STEP FIGURES

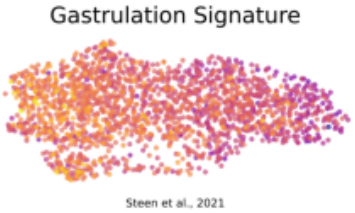
Manifold — samples



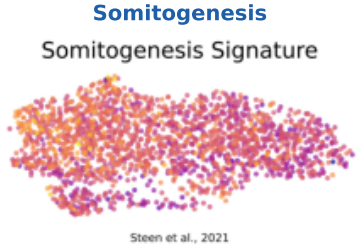
Pseudotime



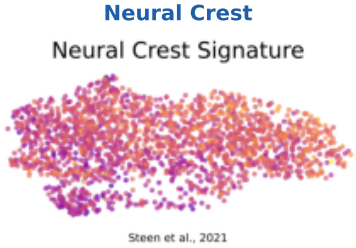
Gastrulation



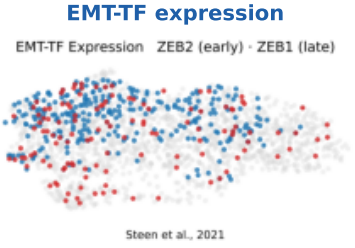
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Diffuse large B-cell lymphoma (Steen et al., 2021), 10x — 2,522 malignant cells → 2,478 after threshold (3 tumours) → 1,963 scoped to the low-cycling developmental lobe. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching). Malignant DLBCL cells recapitulate a single gastrulation/somitogenesis → neural-crest developmental axis once the proliferative compartment (513 cells, 95.5% cycling) is scoped out. Gate 1 GO, Gate 2 GO, Gate 3 NA (no bifurcation). EMT-TFs ZEB2 (early/root) → ZEB1 (late/terminus).

Parameter choices

Min-cell threshold	127 (geo-mean of 8.4× cliff 371→44); drops thin DLBCL111B → 3 tumours, 2,478 cells	**Integration**	Harmony on sample θ=10 (mixing floor by θ=5); n_pcs=20, spread=1.3
Scope	low-cycling Leiden lobe (res=0.15): 1,963 cells; excludes 95.5%-cycling lobe	**Geometry / root**	principal curve (ElPiGraph, non-branching); root Gast/pSom-high tip; Linear-pSom/Crest
EMT-TFs	ZEB2 (61%) & ZEB1 (26%) — only factors above 3% floor; rest near-silent in B-cell tumour		

Zhang 2019 Follicular Lymphoma — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) integrates the clonal FL batch: same-sample neighbour fraction 0.975 → 0.453, iLISI median 1.00 → 2.26 of 4. One connected component (0 islands). θ=2 leaves a per-patient archipelago; judged on the regenerated embedding + per-neighbourhood iLISI, not a scalar: mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

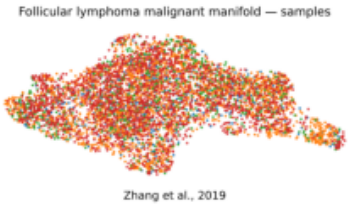
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom overlap at the origin (joint enrichment 1.00, smoothed r=−0.16), both oppose Crest (Gast↔Crest r=−0.55, pSom↔Crest r=−0.58). Along pseudotime Gast falls (ρ −0.48), Crest rises to a terminal tip (decile pt=0.58). Joint pSom&Crest enrichment 0.23 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

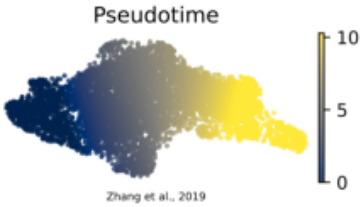
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

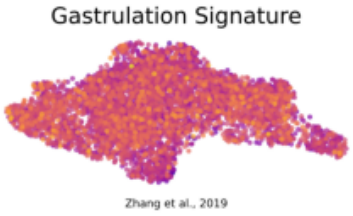
Manifold — samples



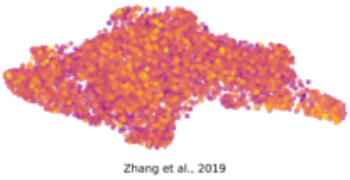
Pseudotime



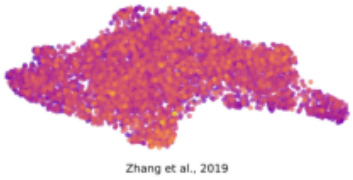
Gastrulation



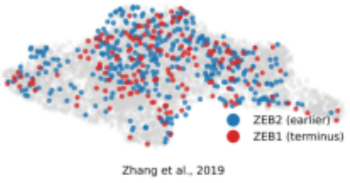
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma malignant B-cell compartment (Zhang et al., 2019), 10x — 5,645 malignant cells over 4 samples / 2 patients (primary + progressed/transformed); single low-cycling body (~14% cycling).

Verdict.
Pattern reproduces in the non-branching form (Linear-Gast): one continuous gastrulation → neural-crest axis, Crest terminal, Gast globally strongest. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 & ZEB1 (both terminus-loaded, ZEB2 earlier).

Parameter choices

Sample threshold	≥100 (floor; largest gap ratio 2.74, no dominant cliff) → 4/4 samples, 5,645 cells	**Overdispersed genes**	636 (scFates find_overdispersed)
Integration	Harmony on sample, θ=10 (mixing improves monotonically; clonal FL patient batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	Non-branching scFates curve N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.96; pt vs cycling ρ=+0.29	**Root**	tip 2 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast +0.012 / pSom +0.006 / Crest +0.009 (Gast strongest → Linear-Gast)	**EMT-TFs shown**	ZEB2 & ZEB1 (>3% detected; both terminus-loaded, ZEB2 earlier; no root-loaded TF)

Zhang 2019 Follicular Lymphoma — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 4 samples (2 patients) intermix on one continuous connected manifold at Harmony θ=10 (θ=2 = per-patient archipelago); 0 detached islands; iLISI 1.04→2.26 (max 4).

Gate 2 — module geography → GO (non-branching)

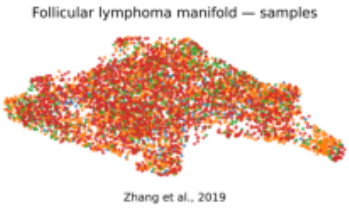
Gast/pSom weakly overlap at one pole (r=−0.16); Crest opposes both (r=−0.54 / −0.57) as an inverse terminal gradient rising along pseudotime (ρ=+0.33). No competing arms; no Crest inversion.

Gate 3 — hub cross-repression → NA

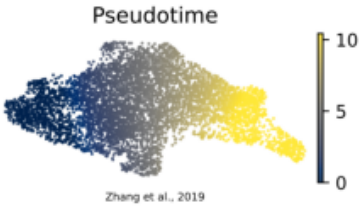
Non-branching manifold — single developmental axis, no bifurcation and no convergence hub, so the branching-only cross-repression test does not apply.

STEP FIGURES

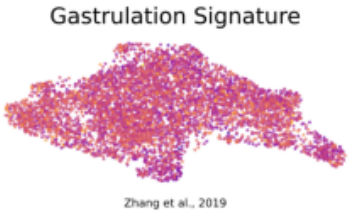
Manifold — samples



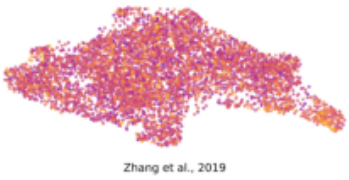
Pseudotime



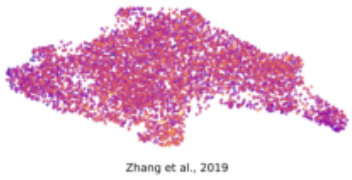
Gastrulation



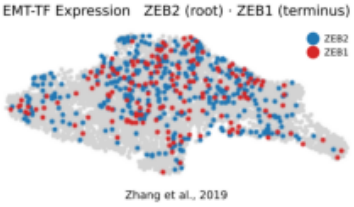
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma (Zhang et al., 2019), 10x — 4 tumour samples / 2 patients; 5,645 malignant B cells analysed.

Verdict.
The embryonic pattern reproduces in its non-branching form: a single continuous gastrulation/somitogenesis → neural-crest axis, rooted at the Gast/pSom-high pole, with EMT transcription factors ZEB2 (early) → ZEB1 (late). No bifurcation, so no hub / Gate 3 NA. Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Sample threshold	all 4 kept (min 680 >> floor; no gap cut — 2× cliff is the 2-large/2-small split)	**Integration**	Harmony θ=10 on sample, PCA(50), n_pcs=15, spread=1.3 (θ from embedding + iLISI sweep)
Island removal	radius graph 6× median-NN on UMAP; 0 cells removed (one connected component)	**Geometry**	non-branching scFates curve (Nodes=20, λ=0.3, μ=100); PPT tree tested = folds, no arm split
Root	curve tip 0 = Gast/pSom-high, Crest-low (origin; Crest late)	**EMT-TFs**	ZEB2 (early/root) ↔ ZEB1 (late/terminus), detection-gated from a 10-factor panel

Zhang 2019 Follicular Lymphoma — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) integrates all 4 samples + both patients (FL1018/FL2001): per-neighbourhood iLISI 1.05→1.88 of a 3.11 perfect-mixing ceiling. One connected component (0 islands removed). Judged on the regenerated embedding + per-neighbourhood iLISI across a θ sweep (θ=1 archipelago → θ=10 mixed): mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

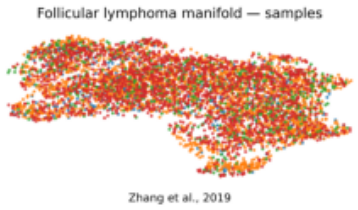
Non-branching case: a single Gast → Crest developmental gradient. Module expression is weak (raw means ~0.01). The Gast→Crest axis is spatially organized and confound-independent: along pseudotime Gast falls (ρ=−0.51, survives regressing out progression+interferon+proliferation) and Crest rises to a terminal gradient (ρ=+0.32). pSom is the exception — ~63% confound-explained, high-decile 54% cycling (vs 14% baseline), so pSom is proliferation-confounded, not an independent axis. Gast↔pSom mild cross-repression (r=−0.18, no hub); Crest opposes both (r=−0.54/−0.60) → inverse terminal gradient, no Gast-arm inversion.

Gate 3 — hub cross-repression → NA

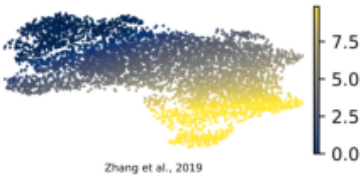
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks robust across PPT Nodes=15/20/30; no Gast/pSom co-expression hub). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

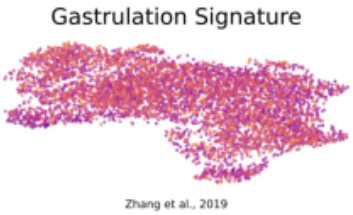
Manifold — samples



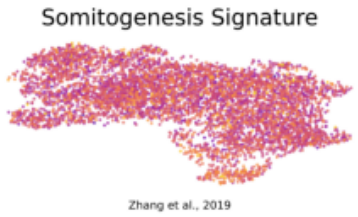
Pseudotime



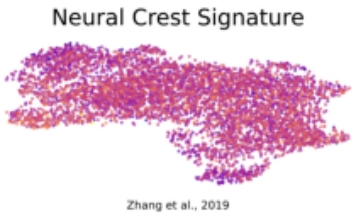
Gastrulation



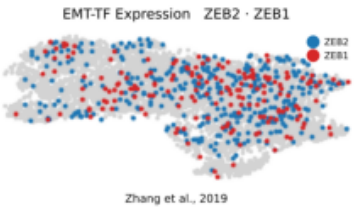
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma / transformed DLBCL (Zhang et al., 2019), 10x — 5,645 malignant B cells over 4 samples (2 patients, each primary + transformed/progressed); Gast-high developmental origin → Crest-high terminal gradient on one continuous manifold.

Verdict.
Pattern reproduces in the non-branching form (Linear): one continuous gastrulation → neural-crest axis, Crest terminal, mild Gast/pSom cross-repression. Gate 1 GO / Gate 2 GO / Gate 3 NA. Gast & Crest confound-independent; pSom proliferation-confounded. EMT-TFs ZEB2/ZEB1 expressed but sparse and not end-polarized.

Parameter choices

Sample threshold	≥100 cells (floor; no dominant cliff, all 4 samples 680–2405) → 4/4 kept, 5,645 cells	**Overdispersed genes**	636 (scFates find_overdispersed); no sc.pp.scale before PCA
Integration	Harmony θ=10 on sample (iLISI peaks at 10, declines at 20/40)	**Embedding**	n_pcs=30, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	Non-branching scFates curve, N=20 λ=0.3 μ=100; 2 tips/0 forks (SA 2.97, fold-free)	**Root**	curve tip = Gast/pSom-high, Crest-low origin
Module means	Gast +0.012 / pSom +0.006 / Crest +0.009 (weak, control-pool level)	**EMT-TFs**	ZEB2 (27%) · ZEB1 (12%); SNAI/TWIST/FOXC/PRRX near-absent

Zhang 2019 Follicular Lymphoma — analysis summary

RUN 4 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ-sweep (5→20) + per-neighbourhood iLISI: mixing plateaus at θ=8 (same-sample nbr frac 0.97→0.46; iLISI sample 1.9/4, patient 1.5/2). Single connected kNN component; 0 detached islands. Patient-private FL clones integrate into one continuous manifold; signature geography is cell-cycle-independent.

Gate 2 — module geography → GO (non-branching)

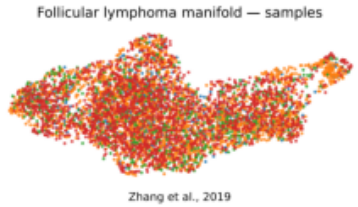
Non-branching case: Gast/pSom overlap on one axis (spatial r=−0.19, no two competing arms); Crest an inverse/terminal gradient (Crest vs Gast −0.52, vs pSom −0.60; joint pSom&Crest-high enrichment 0.23 < 1). Single Gast/pSom → Crest developmental axis present → GO.

Gate 3 — hub cross-repression → NA

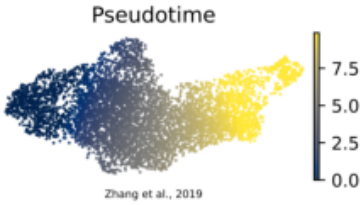
No bifurcation (curve, 0 forks) → branching-only gate not applicable. 04_correlation / 05_modules not produced.

STEP FIGURES

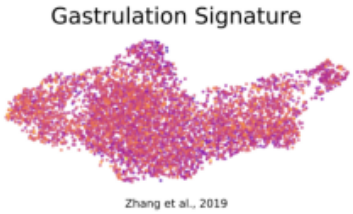
Manifold — samples



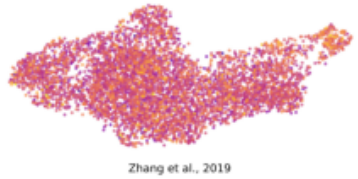
Pseudotime



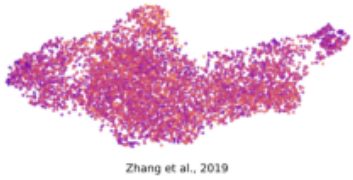
Gastrulation



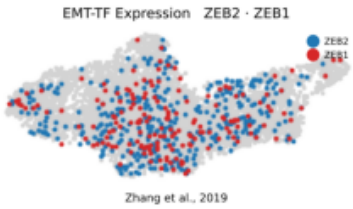
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma malignant B cells (Zhang et al., 2019), 10x — 5,645 malignant cells · 4 samples · 2 patients (FL1018, FL2001; FL1018T2 transformed-DLBCL); one continuous gastrulation → neural-crest axis, no arm split.

Verdict.
Pattern REPRODUCES as the non-branching case — Gate 1 GO / Gate 2 GO / Gate 3 NA (linear). No cross-repression hub. Module means (raw) Gast +0.012 / pSom +0.006 / Crest +0.009. EMT-TFs ZEB2 (27%) · ZEB1 (12%), both terminus-associated; no root-end TF.

Parameter choices

Sample threshold	≥100 cells → all 4 kept (5,645 cells); min sample 680 ≈ 2×floor, no under-populated sample	**Overdispersed genes**	636 (scFates find_overdispersed, GAM residuals)
n_pcs	30 (variance-ratio elbow past PC20; floored to robust 10x range)	**Integration**	Harmony θ=8 on sample (mixing plateau across θ=5–20; FL clones patient-private)
Embedding	k=30, UMAP spread=1.3; radius-graph island removal (0 removed)	**Geometry**	Linear (non-branching); scFates curve N=20, 0 forks, fold-free (SA 2.96); PPT spurious forks
Root	Gast/pSom-high tip (Crest-low); order Gast t≈0.04 → Crest 0.59 → pSom 0.82	**EMT-TFs**	ZEB2 (27%) · ZEB1 (12%) both terminus-associated; no root-end TF; VIM excluded

Zhang 2019 Follicular Lymphoma — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Patients/samples intermix on one continuous manifold (iLISI 1.00→2.26 of 4 at Harmony θ=10; patient 1.67/2); single connected component, 0 detached islands. Strongly clonal FL batch integrates.

Gate 2 — module geography → GO (non-branching)

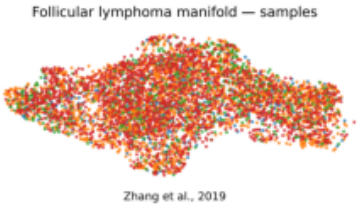
Non-branching: on the cycling-scoped Body, Gast/pSom co-vary (r=+0.33) and both oppose Crest (Crest↔Gast −0.83, Crest↔pSom −0.41); Crest is the terminal readout (high-decile pt=0.58). Single collapsed developmental axis.

Gate 3 — hub cross-repression → NA

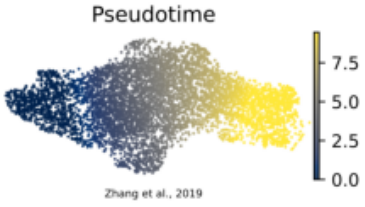
No Gast/pSom two-arm bifurcation (curve: 2 tips, 0 forks) → branching-only correlation/modules steps not produced.

STEP FIGURES

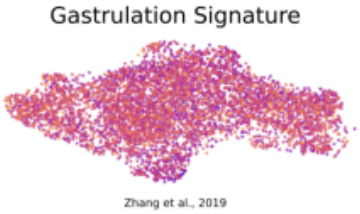
Manifold — samples



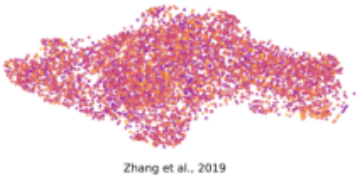
Pseudotime



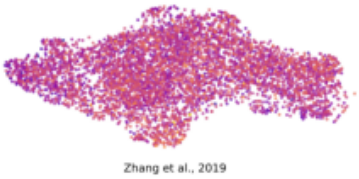
Gastrulation



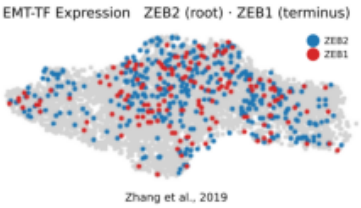
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma (Zhang et al., 2019), 10x — 5,645 malignant B cells (trajectory scoped to 5,449 non-cycling).

Verdict.
Pattern reproduces (non-branching). A single Gast → Crest developmental axis: gastrulation/somitogenesis programs at the root, neural-crest as the terminal readout. No competing arms → Gate 3 not applicable. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means (raw) Gast +0.012 / pSom +0.005 / Crest +0.010. EMT-TFs ZEB2 (28%, earlier) → ZEB1 (12%, most terminal).

Parameter choices

Sample threshold	≥100 floor — kept all 4 samples (largest gap ratio 2.74, not a clean cliff; min 680 a genuine primary FL tumour)	**Integration**	Harmony θ=10 on sample (iLISI sweep + embedding; FL clones patient-private, need strong θ)
Embedding	n_pcs=20, n_neighbors=30, spread=1.3; 636 overdispersed genes	**Cycling scoping**	excluded a 196-cell, 95%-cycling lobe where pSom was proliferation-confounded (pSom↔cycling 0.61→0.18)
Geometry	Linear/non-branching scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips/0 forks; fold-free (SA 2.96, DC1 +0.94)	**Root**	Gast/pSom-high, Crest-low origin (Gast ρ=−0.41, Crest ρ=+0.43 along pt)
Module means	Gast +0.012 / pSom +0.005 / Crest +0.010 (raw)	**EMT-TFs**	ZEB2 (28%, earlier) · ZEB1 (12%, terminal); SNAI/TWIST/FOXC/PRRX <3%; VIM excluded

Zhang 2019 follicular lymphoma — analysis summary

RUN 6 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4/4 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands (iLISI 1.00→2.26 of 4 at Harmony θ=10; same-patient nbr frac 0.60; 1 kNN component, 0 detached). Strongly clonal FL batch integrates into one body.

Gate 2 — module geography → NO-GO

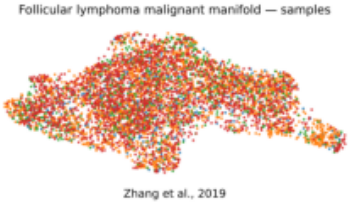
Embryonic programs are at the control-pool noise floor (means Gast +0.012 / pSom +0.006 / Crest +0.009, 20–40× below real-pattern datasets). No Gast/pSom hub (joint-high enrichment 1.0). Gast–pSom r and Crest pseudotime gradient sit INSIDE the random-module 95% null band (48th / 72nd null percentile) — no embryonic-specific signal. Pattern absent.

Gate 3 — cross-repression → NA

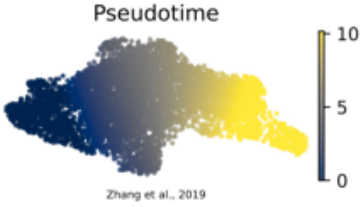
Not reached — Gate 2 NO-GO halts the gate chain. No bifurcation (curve: 2 tips, 0 forks) and no hub, so there is no branch pair to test for cross-repression.

STEP FIGURES

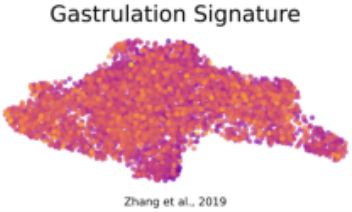
Manifold — samples



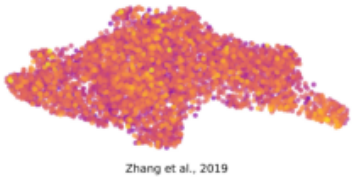
Pseudotime



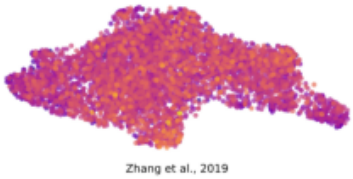
Gastrulation



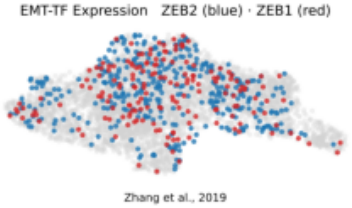
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma (Zhang et al., 2019), 10x — 2 patients (FL1018, FL2001) / 7,002 cells; malignant only: 5,645 B cells across 4 samples (all clear the ≥100 floor); whole malignant manifold (no lobe scoping). Pattern absent — Gate 2 NO-GO.

Verdict.
Pattern does NOT reproduce — Gate 1 GO, Gate 2 NO-GO, Gate 3 NA. The FL malignant manifold is one integrated continuum, but the gastrulation/somitogenesis/neural-crest programs are absent: means at the control-pool floor and, decisively, Gast–pSom co-variation and the Crest pseudotime gradient are indistinguishable from random gene sets of equal size (48th / 72nd null percentile). No Gast/pSom hub (enrichment 1.0). This corrects 5 prior attempts that called Gate 2 GO by reading score_genes shared-pool artifacts (and, in -4, a circularly-scoped 196-cell cycling lobe that flipped Gast–pSom to +0.33). EMT-TFs ZEB2 (28%) · ZEB1 (12%), both terminus-loaded; SNAI/TWIST/FOXC/PRRX below 3% detection.

Parameter choices

Min cells/sample	≥100 floor (4/4; gap 2.74 not a cliff) → 5,645 cells	**Scope**	whole manifold (no scoping; verified robust across scopes)
Root node	tip 2 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=10; PCA20; spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100 (fold-free, 2 tips / 0 forks)	**Decisive test**	200-draw random-gene-set null (same sizes/pool/smoothing)
Gate readout	G1 GO / G2 NO-GO / G3 NA (module means .012/.006/.009)	**EMT-TFs**	ZEB2 (28%) · ZEB1 (12%), both terminus-loaded; rest below 3% detection

Cohen 2021 Multiple Myeloma — analysis summary

RUN 1 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

54 samples intermix per-neighbourhood (iLISI 1.0→14.1 of 54) on one continuous connected component; single Leiden lobe, aspect 1.09, only 2.8% cycling — no islands, no batch archipelago, no cycling compartment.

Gate 2 — module geography → GO (non-branching)

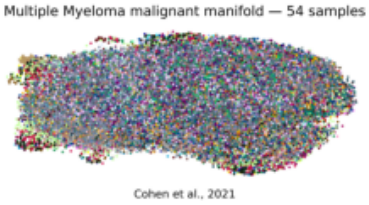
Non-branching: a single Gast/pSom-origin → terminal-Crest axis. Crest inversely opposes Gast (r=−0.69) and pSom (r=−0.61); pSom&Crest joint-high enrichment 0.22 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

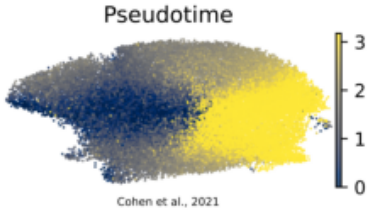
No bifurcation: Gast↔pSom near-orthogonal (r=+0.07), no competing arms — branching-only gate not applicable.

STEP FIGURES

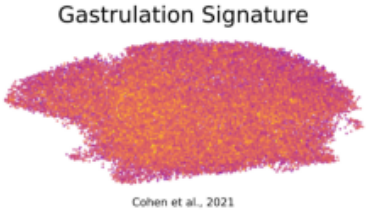
Manifold — samples



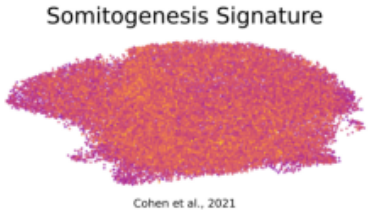
Pseudotime



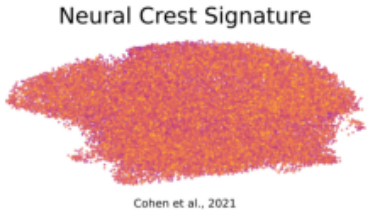
Gastrulation



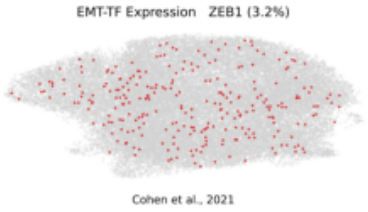
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma bone-marrow plasma cells (Cohen et al., 2021), MARS-seq — 33,076 malignant cells across 54 samples (≥100 cells). Whole malignant manifold; single differentiation continuum.

Verdict.
The embryonic pattern reproduces as a NON-BRANCHING axis: Gast/pSom-high origin → terminal neural-crest, no resolved bifurcation. Linear-pSom/Crest geometry (Gast μ=−0.022, pSom μ=+0.014, Crest μ=+0.001). EMT-TF ZEB1 is the only factor above the detection floor.

Parameter choices

Sample threshold	≥100 cells (floor; smooth decline, no cliff) → 54/73 samples, 33,147 cells	**Overdispersed genes**	482 (scFates); island removal 71 cells / 2 specks → 33,076
Integration	Harmony θ=10 on sample (swept 2/5/10; same-sample nbr frac 0.91→0.11)	**Embedding**	n_pcs=20 (scree plateau), n_neighbors=30, UMAP spread=1.3
Trajectory	scFates curve (ElPiGraph, N=20, λ=0.3, μ=100); non-branching, fold-free (2.5)	**Root**	tip highest Gast+pSom, lowest Crest; Gast ρ=−0.55, Crest ρ=+0.57
EMT-TF readout	ZEB1 (3.2%) — sole factor above floor; others <0.3%	**Annotation**	Malignant plasma cells (author cell_type); single continuum

Cohen 2021 Multiple Myeloma — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

54/73 samples (≥94 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 80 cells / 3 specks removed; iLISI median 1.00→14.2 of 54). Low-θ density lobes were patient batch, resolved at θ=10; one lobe carries the trajectory.

Gate 2 — module geography → GO (non-branching)

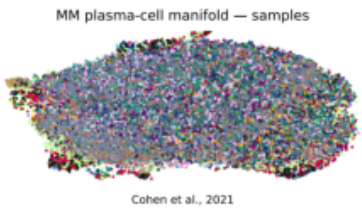
No Gast/pSom divergence into competing arms (spatial r=+0.07 — no bifurcation). A single dominant contrast opposes Crest to both (Crest–Gast r=−0.69, Crest–pSom r=−0.61), Crest rising to a terminal readout (crest-high decile mean pt=0.55). pSom>Gast, Crest≈0 globally.

Gate 3 — hub mutual exclusivity → NA

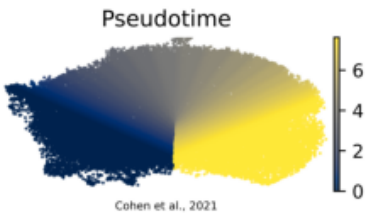
Single continuous axis — no bifurcation and no Gast/pSom convergence hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

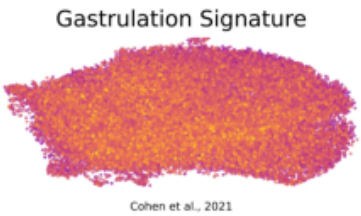
Manifold — samples



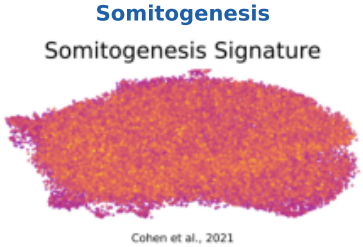
Pseudotime



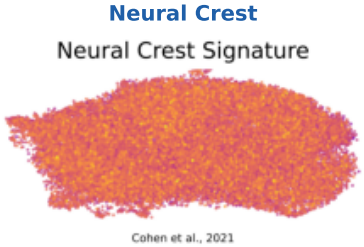
Gastrulation



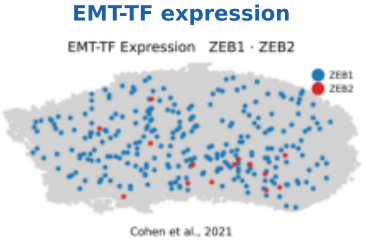
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple Myeloma (Cohen et al., 2021), MARS-seq — 74 samples / 39,157 cells (master scope); malignant only: 33,599 load, 33,147 across 54 samples (≥94) after thresholding, 33,067 after island removal enter the trajectory; whole manifold (no scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Cohen's MM plasma-cell manifold is a single continuous gastrulation/somitogenesis → terminal-neural-crest gradient, not Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Modules are globally weak (Gast −0.022, pSom +0.014, Crest +0.001), consistent with shallow-MARS-seq plasma cells; the axis is orthogonal to (near-absent) proliferation. EMT-TF program essentially absent — only ZEB1 detectable (~3%); Snail/Twist/Foxc/Prrx at the detection floor.

Parameter choices

Min cells/sample	≥94 (54/73; gap 117→77)	**Scope**	whole manifold (no scoping)
Root node	tip 0 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=10; PCA n_pcs=21; spread=1.3
Trajectory	scFates curve N=20 λ=0.3 μ=100	**EMT-TFs**	ZEB1 (~3%); others <0.3% (absent)

Cohen 2021 multiple myeloma — analysis summary

RUN 3 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=5) fuses the patient-private myeloma clones: iLISI median 1.0→12.5 / 54 samples, one connected kNN component. Judged on the regenerated embedding across a θ×n_pcs sweep — samples intermix on one continuous, low-cycling body (no proliferative lobe to scope). θ=5 keeps the manifold axis-bearing; θ=10 over-smooths to an axis-less blob.

Gate 2 — module geography → GO (non-branching)

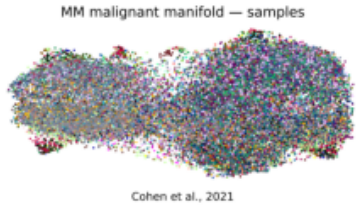
Signatures resolve onto a single (Gast/pSom)→Crest axis: smoothed Gast↔Crest r=−0.73, pSom↔Crest r=−0.65, Gast↔pSom r=+0.28 (overlap, not two competing arms). Crest is an inverse terminal gradient (Crest-high decile pseudotime 0.85 vs Gast 0.31 / pSom 0.15). Principal curve is fold-free (2 tips / 0 forks, self-approach 2.8). No Crest-on-Gast-arm inversion to flag.

Gate 3 — cross-repression → NA

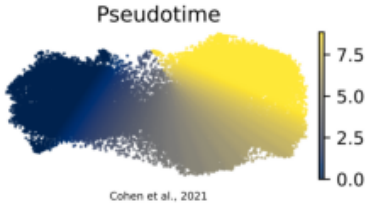
Non-branching topology — no bifurcation and no Gast/pSom hub, so the branching-only cross-repression test does not apply. Correlation & modules notebooks are correctly not produced.

STEP FIGURES

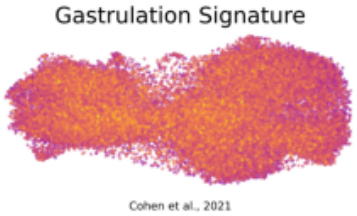
Manifold — samples



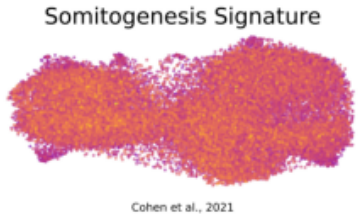
Pseudotime



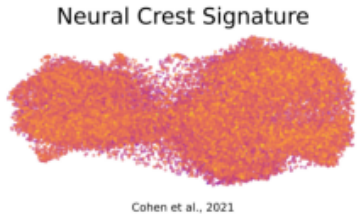
Gastrulation



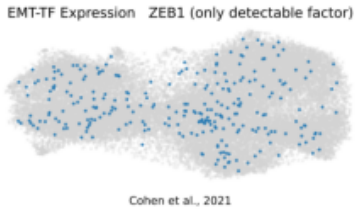
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma (Cohen et al., 2021), MARS-seq, 32,894 malignant plasma cells over 54 samples. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces as the non-branching case — a single gastrulation/somitogenesis → neural-crest developmental axis; no hub, no bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TF ZEB1 is the only factor above the 1% detection floor (terminally-differentiated plasma cells).

Parameter choices

Min-cell threshold	≥100 floor (no dominant cliff); 54/73 samples kept	**Integration**	Harmony on sample θ=5; n_pcs=30, n_neighbors=30, spread=1.3
Islands removed	253 cells / 14 detached specks (largest component)	**Geometry**	Non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips / 0 forks)
Root node	curve tip 2 = Gast/pSom-high, Crest-low (Crest-high at terminus)	**Module means (raw)**	Gast −0.022 · pSom +0.014 · Crest +0.001 (Linear single axis)
EMT-TFs shown	ZEB1 (3.2%) — only EMT-TF above the 1% floor; all others <0.3%		

Liu 2021 Multiple Myeloma — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Clonal MM plasma cells form a per-patient archipelago without integration (pre-Harmony iLISI 1.06). Judged on the regenerated embedding across a θ sweep (not a scalar): at θ=15 the 15 samples share ONE continuous manifold (island strip 42.5→28→11.7→2.4→2.7% as integration rises to a clean-speck level), well mixed per-neighbourhood (iLISI 2.27 of 5.62 ceiling; patient |r|≤0.19 on diffusion axes). Detached specks are unintegrated clones of the same states, not a distinct lineage.

Gate 2 — module geography → GO (non-branching)

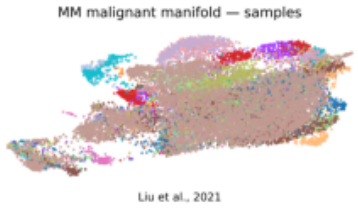
Single developmental axis: Gast & pSom vary along it with organised cross-repression (smoothed r=−0.67), and Crest occupies the opposite terminal end (pSom↔Crest r=−0.95). No Gast/pSom co-expression hub and no opposing Gast arm → non-branching. FLAG: Crest enriches the Gast end (Crest↔Gast r=+0.56, Crest↔pSom r=−0.95) — inversion of the canonical pSom-arm Crest tip; flagged but PASSES.

Gate 3 — hub cross-repression → NA

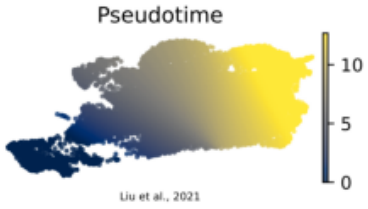
Non-branching manifold has no bifurcation; the tree-cluster cross-repression test does not apply.

STEP FIGURES

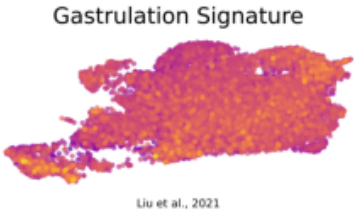
Manifold — samples



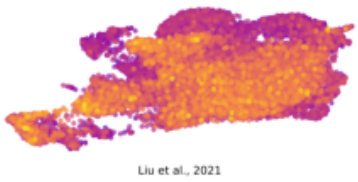
Pseudotime



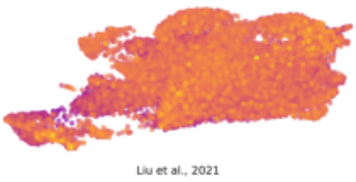
Gastrulation



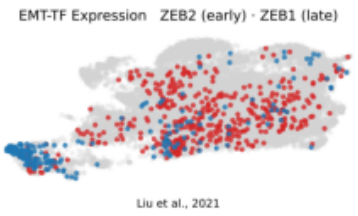
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma plasma cells (Liu et al., 2021), 10x — 24 samples / 74,986 cells (master scope); 17,267 malignant plasma cells; 15 marrows pass the 66→156 count cliff (≥100 cells; 17,003), 16,551 retained after removing detached islands. Global module means Gast −0.024, pSom +0.032, Crest −0.006 → Linear-pSom/Crest class (pSom the strongest module).

Verdict.
The embryonic-development pattern REPRODUCES as an organised non-branching cross-repression axis (Gast/pSom early, Crest terminal), with a Crest-on-Gast-arm inversion flag. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (early) → ZEB1 (late) — the only factors above the 3% detection floor.

Parameter choices

Min-cell threshold	≥100 cells/sample (drops 5 thin marrows; count cliff 66→156, ratio 2.4)	**Integration**	Harmony θ=15 on sample, 20 PCs (iLISI plateaus + island fraction bottoms out)
Embedding	PCA-50, kNN=30, UMAP spread 1.3	**Geometry**	Non-branching principal curve (ElPiGraph, fold-free SA 2.9; no hub/arm)
Scope	Whole manifold (no lobe excised); axis orthogonal to proliferation	**Root**	Gast/pSom-high, Crest-low tip (gastrulation/somitogenesis early, crest late)
EMT-TF panel	ZEB2 (early) & ZEB1 (late) clear the >3% floor; VIM excluded		

Liu 2021 multiple myeloma — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant plasma cells form a per-patient clonal archipelago. Across the full θ×n_pcs Harmony sweep, 9/10 patients show self-neighbour enrichment >2.5× (worst ~63×) — cells sit next to their own clone, not a shared state. iLISI peaks at only ~59% of ceiling (θ=5) then declines; stronger integration does not merge the clones. Inter-patient clonal/CNV heterogeneity, not a sample-integrated continuum. Do not force a trajectory.

Gate 2 — module geography → NA

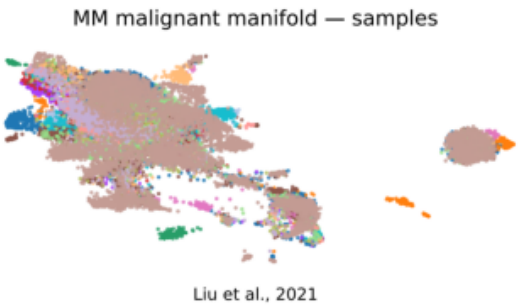
Not evaluated — Gate 1 NO-GO halts the chain before the manifold/signature steps.

Gate 3 — cross-repression → NA

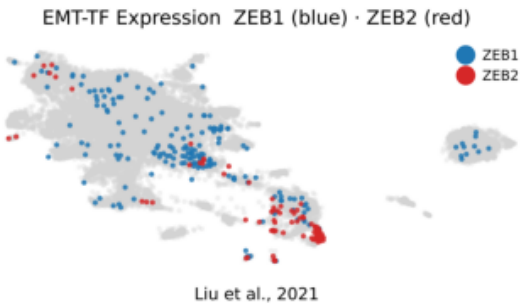
Not evaluated — branching-only, and no trajectory was fit.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Multiple myeloma bone-marrow aspirates (Liu et al., 2021), 10x — 17,003 malignant plasma cells (15 samples ≥101 cells, 10 patients). Gate 1 NO-GO (clonal archipelago).

Verdict.
The developmental hub-and-arm pattern does NOT reproduce. Clonal MM plasma cells are irreducibly patient-segregated (Gate 1 NO-GO); no cell-state manifold or module geography can be honestly assessed. Consistent with the other MM datasets (Cohen 2021) in the atlas. EMT-TFs ZEB1 (11.1%) / ZEB2 (4.7%) — only detectable factors, descriptive only.

Parameter choices

Min-cell threshold	≥101 cells → 15/20 samples (17,003 cells); tail cliff 66→156	**Integration**	Harmony on sample; θ & n_pcs swept → θ=5, n_pcs=40 (iLISI-maximal)
Embedding	n_neighbors=30, spread=1.3	**Geometry**	none fitted (Gate 1 NO-GO); global means Linear-pSom/Crest
EMT-TFs shown	ZEB1 (11.1%) / ZEB2 (4.7%) — only detectable; VIM excluded		

Ledergor 2018 Multiple Myeloma — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=15) fuses the patient-private myeloma clones: same-sample neighbour fraction 0.87→0.26, iLISI 1.0→5.1 / 16, one connected kNN component. Judged on the regenerated embedding across a θ-sweep — samples intermix on one continuous, uniformly low-cycling body (no proliferative lobe to scope).

Gate 2 — module geography → GO (non-branching)

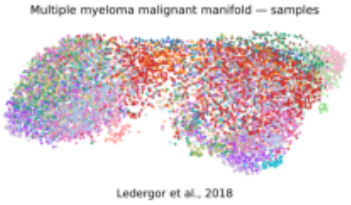
Signatures collapse onto a single (Gast/pSom)→Crest axis: smoothed pSom↔Crest r=−0.97, Gast↔pSom r≈−0.10 (no cross-repression → no two competing arms). Crest is an inverse terminal gradient (crest-high decile pseudotime 0.83; joint pSom&Crest enrichment 0.16). Principal curve is fold-free (2 tips / 0 forks).

Gate 3 — hub cross-repression → NA

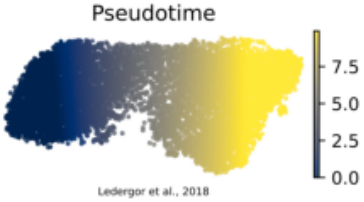
Non-branching topology — no bifurcation and no Gast/pSom hub, so the branching-only cross-repression test does not apply. Correlation & modules notebooks are correctly not produced.

STEP FIGURES

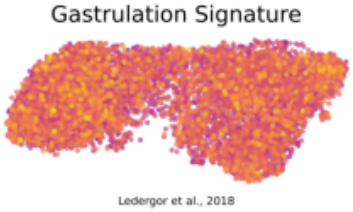
Manifold — samples



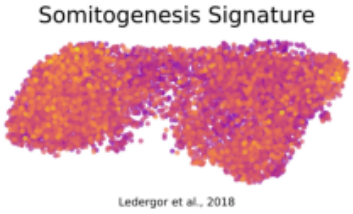
Pseudotime



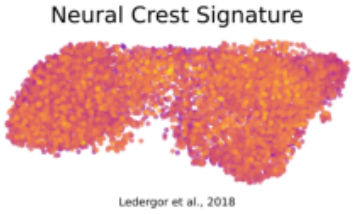
Gastrulation



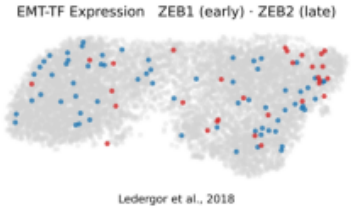
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma (Ledergor et al., 2018), MARS-seq — 8,182 malignant plasma cells over 16 samples. Uniformly low-cycling body (no proliferative lobe to scope).

Verdict.
Pattern reproduces as the NON-BRANCHING case — a single gastrulation/somitogenesis→neural-crest developmental axis; no hub, no bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB1 (root) → ZEB2 (terminus) — the only detectably-expressed factors.

Parameter choices

Min-cell threshold	≥95 (geom-mean of 183→50 cliff; 16/18 samples; drops MM04, MM05)	**Integration**	Harmony on sample, θ=15 (patient-private MM clones), n_pcs=40, k=30, spread=1.3
Islands removed	15 cells / 2 detached specks (UMAP radius graph, largest component)	**Geometry**	Non-branching → scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips / 0 forks, fold-free
Root node	curve tip 0 — Gast/pSom-high, Crest-low (pSom→Crest along pseudotime)	**Module means (raw)**	Gast −0.029 · pSom +0.016 · Crest +0.004 → Linear-pSom/Crest
EMT-TFs shown	ZEB1 (root) → ZEB2 (terminus); detectable 3.3% / 1.5%		

Ledergor 2018 multiple myeloma — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

16/18 samples (≥183 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 13 detached; iLISI median 1.00→5.13 of 16 ≈ 32% of ideal; kNN graph = 1 component at every θ). Self-neighbour enrichment falls with integration — Harmony works, not an archipelago.

Gate 2 — module geography → GO (non-branching)

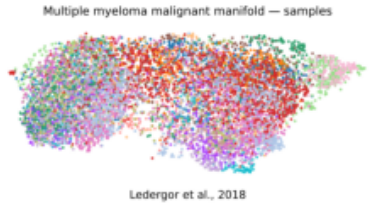
Single dominant pSom→Crest axis: pSom high at the Gast/pSom-high origin declines along pseudotime (ρ=−0.56); Crest anti-correlated (smoothed r=−0.96) and rises to a terminal readout (ρ=+0.57; crest-high decile mean pt=0.81). Gast weak/flat. Axis holds within patients (r=−0.64).

Gate 3 — cross-repression → NA

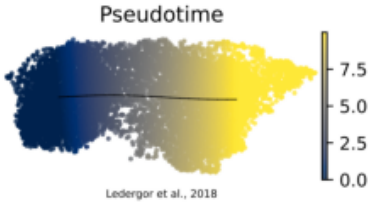
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test.

STEP FIGURES

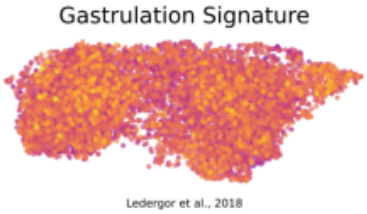
Manifold — samples



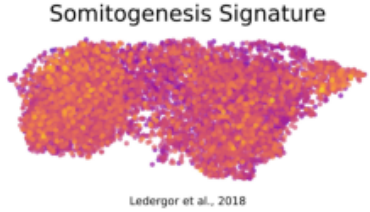
Pseudotime



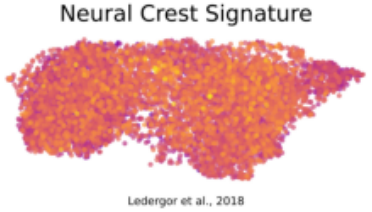
Gastrulation



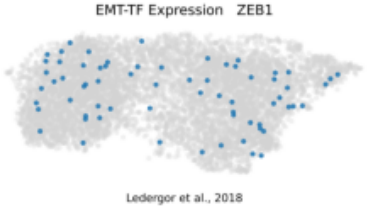
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma (Ledergor et al., 2018), MARS-seq — 8,273 malignant plasma cells load (MM+SMM marrow); 8,197 across 16 samples (≥183) enter preprocessing; 8,184 after island removal; the whole malignant manifold (no lobe scoping) enters the trajectory. Non-branching Linear-pSom/Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The myeloma malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis is a within-patient cell-state gradient (77–86% of module variance within-sample), not clonal/CNV structure, and is orthogonal to cell cycle (0.3% cycling). EMT-TFs: only ZEB1 (3.3%) / ZEB2 (1.4%) clear the detection floor — MARS-seq dropout silences the panel.

Parameter choices

Min cells/sample	≥183 (16/18; gap 183→50)	**Integration**	Harmony θ=20; n_pcs=20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root	tip: Gast/pSom-high, Crest-low	**Geometry**	Linear (pSom/Crest); 0 stable forks
Island removal	13 cells (radius-graph)	**EMT-TFs**	ZEB1 (+ZEB2); rest <3% detection (MARS-seq dropout)

Nam 2019 myeloproliferative neoplasms — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates all 10 samples + both group leaves: per-neighbourhood iLISI 1.43→3.90 of a 4.94 perfect-mixing ceiling. One connected component (1 speck removed). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

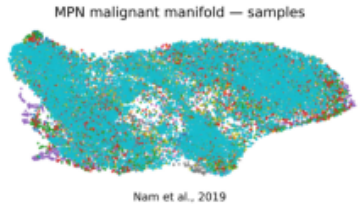
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.23), both oppose Crest (Gast↔Crest r=−0.73, pSom↔Crest r=−0.71). Along pseudotime Gast/pSom fall (ρ −0.63/−0.66), Crest rises to a terminal tip (decile pt=0.86). Inverse terminal gradient, no Gast-arm inversion.

Gate 3 — cross-repression → NA

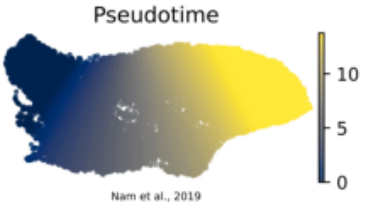
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks across Nodes=15/20/30). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

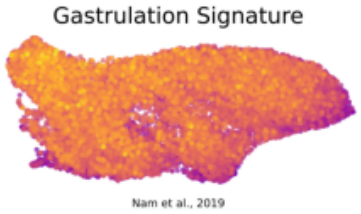
Manifold — samples



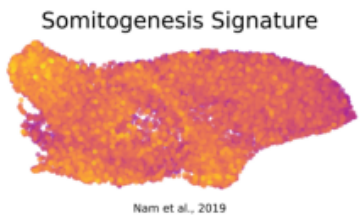
Pseudotime



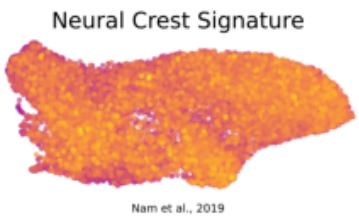
Gastrulation



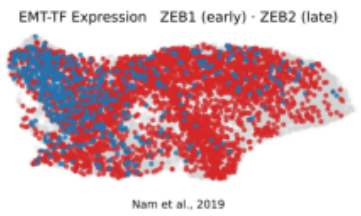
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Myeloproliferative neoplasms (CALR-mut ET/MF) (Nam et al., 2019), 10x — 22,605 malignant cells over 10 samples (2 group leaves); low-cycling Gast/pSom root pole and a cycling Crest terminus, one continuous manifold. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs ZEB1 (early/root) → ZEB2 (terminus).

Parameter choices

Sample threshold	≥100 cells (no cliff) → 10/10 samples, 22,606 cells	**Overdispersed genes**	2,162 (scFates find_overdispersed)
Integration	Harmony on sample θ=10 (iLISI plateau 5–10, declines at 20)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 1 island removed → 22,605
Geometry	non-branching curve (N=20, λ=0.3, μ=100): 2 tips / 0 forks; corr(t,DC1)=0.98	**Root**	tip 0 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast −0.036 · pSom +0.008 · Crest +0.009 (Linear-pSom/Crest)	**EMT-TFs**	ZEB1 (early/root) → ZEB2 (late/terminus); detection-gated, FOXC/PRRX/SNAI in panel

Young 2018 Kidney (Wilms/RCC) — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/25 samples (≥100 cells) intermix on one continuous manifold (kNN = 1 component; 0.91 of the perfect-mixing ceiling), no per-patient islands (radius-graph: 3 specks stripped). No proliferation compartment (most-cycling lobe 1.5%); module means near-identical across all 3 lobes → whole manifold.

Gate 2 — module geography → GO (non-branching)

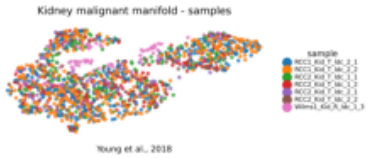
Single dominant pSom→Crest axis: pSom high over the body declines along pseudotime (ρ=−0.75); Crest anti-correlated (r=−0.78), rising to a terminal readout (ρ=+0.72; Crest-high decile mean pt=0.89). Gast weakly structured (ρ=+0.39). Joint pSom&Crest-high depleted (enrichment 0.22) — an inverse gradient, not a co-expression hub.

Gate 3 — hub mutual exclusivity → NA

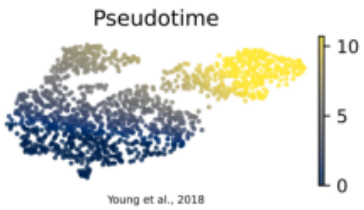
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

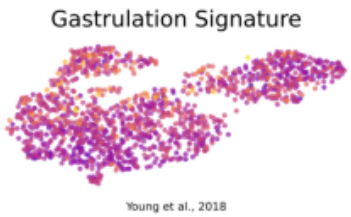
Manifold — samples



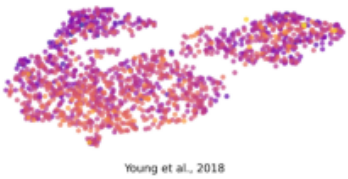
Pseudotime



Gastrulation

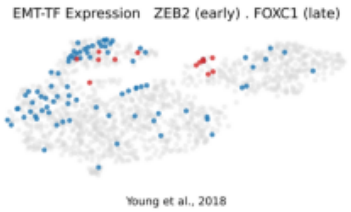


Somitogenesis



Neural Crest

EMT-TF expression



Dataset.
Wilms tumour / RCC (Young et al., 2018), 10x — 66 samples / 125,139 cells (master scope); malignant only: 2,055 cells → 1,438 after ≥100/sample threshold (7/25 samples) → 1,435 after island removal enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Young's kidney malignant manifold is a single monotone somitogenesis→terminal-neural-crest gradient (pSom-high body → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The manifold is continuous and cell-cycle-free (most-cycling lobe 1.5%, Gast vs g2m r=−0.02), so no proliferation caveat; the whole manifold is kept (Bi/Steele no-scoping precedent). Module means Gast −0.008 / pSom +0.042 / Crest −0.029 (pSom-dominant Linear-pSom/Crest). EMT-TFs ZEB2 (early/root, 23.3% expressing, ρ=−0.20) → FOXC1 (late/terminus, 3.6%, ρ=+0.03) — FOXC1 surfaces only via the widened panel; end-separation weak (Δpt=0.27). Curve fold-free (self-approach 2.84, 0 forks).

Parameter choices

Min cells/sample	≥100 floor (7/25; gap non-dominant)	**Scope**	whole manifold (no scoping)
Root node	tip 0 = pSom-high, Crest-low	**Geometry**	Linear-pSom/Crest (non-branching)
Integration	Harmony θ=10; PCA50; spread=1.3	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Overdispersed	262 genes; n_pcs=30	**EMT-TFs**	ZEB2 (early) · FOXC1 (late)

Young 2018 kidney cancer — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

The malignant compartment is a composite of THREE biologically distinct kidney cancers — clear-cell RCC (1,373), Wilms tumour (287), papillary RCC (144). Each sample is (near-)pure for one type (Cramér V(sample,type)=0.93), so integrating "samples" integrates three cancers. Across a θ 0→20 sweep, per-neighbourhood iLISI(tumour-type) stays ~1.0–1.2 vs a 1.64 well-mixed ceiling: the types never intermix at any integration strength. The kNN graph is one component throughout (the misleading scalar), but the embedding shows type-segregated islands.

Gate 2 — module geography → N/A

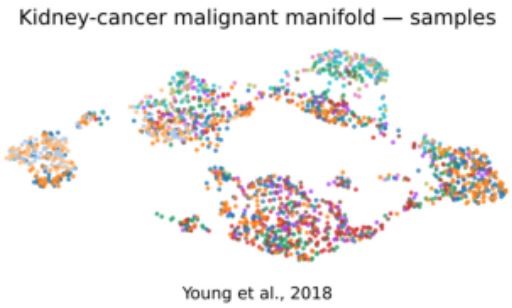
Not evaluated — Gate 1 NO-GO halts the chain before the manifold / signature steps.

Gate 3 — cross-repression → N/A

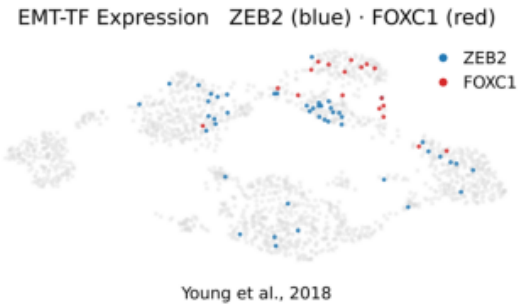
Not evaluated — branching-only, and no trajectory is fit.

STEP FIGURES

Malignant manifold — samples



EMT-TF expression



Dataset.
Kidney cancer (Young et al., 2018), 10x scRNA-seq. 2,055 malignant cells loaded; 1,804 retained after sample thresholding (≥63 cells). A mixture of clear-cell RCC, pediatric Wilms tumour, and papillary RCC.

Verdict.
The embryonic-development cell-state pattern does NOT reproduce: the malignant "manifold" is not one continuous cell-state continuum but three unrelated cancers that refuse to integrate. Gate 1 = NO-GO; no trajectory is fit. Geometry: none (un-integrable composite of three malignancies).

Parameter choices

Sample threshold	≥63 cells (geo-mean of 87→47 gap, 1.85×); 11/25 kept → 1,804 cells	**Overdispersed genes**	531 (scFates find_overdispersed)
Integration	Harmony on sample; θ swept 0–20, read per-neighbourhood; θ=2 for visual	**n_pcs / spread**	20 / 1.3 (10x precedents at ~1.8k cells)
Root / geometry	None — no trajectory fit (Gate 1 NO-GO)	**EMT-TFs shown**	ZEB2 (25.4%) · FOXC1 (9.5%); FOXC/PRRX added, VIM excluded

Bi 2021 Kidney RCC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

5/8 samples (natural-break ≥160 cells) intermix on one continuous manifold; no per-patient islands (41 detached cells removed). Diff-sample kNN 0.46 = 0.67 of the mixing ceiling; 1 component.

Gate 2 — module geography → GO

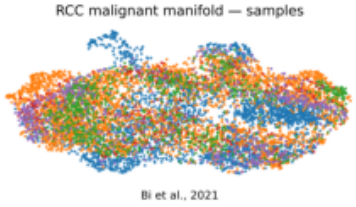
Non-branching: Gast/pSom co-vary from a shared origin (no opposing arms); Crest is the inverse terminal gradient (Crest–pSom r = −0.65; Crest vs pseudotime = +0.71). pSom globally strongest.

Gate 3 — hub mutual exclusivity → NA

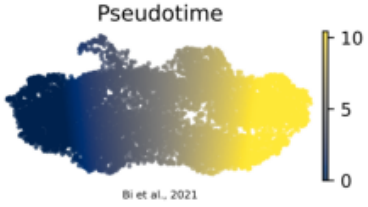
No bifurcation and no Gast/pSom hub — the branching-only cross-repression test does not apply.

STEP FIGURES

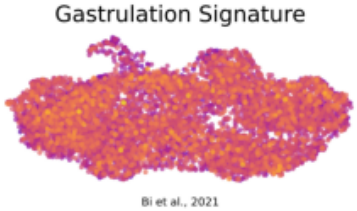
Manifold — samples



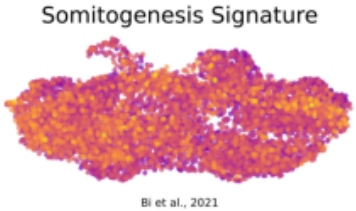
Pseudotime



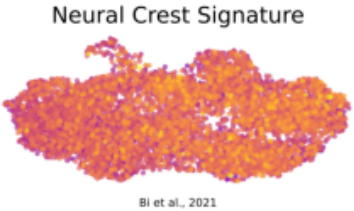
Gastrulation



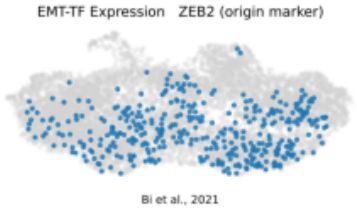
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Kidney renal-cell carcinoma (Bi et al., 2021), 10x — 8 patients / 34,326 cells (master scope); malignant only: 8,040 load, 7,822 across 5 samples enter the trajectory.

Verdict.
Pattern reproduces (non-branching). A single embryonic axis runs from a Gast/pSom-high origin to a Neural-crest-high terminus — the programs are present but not resolved into competing arms, so there is no hub and Gate 3 is NA. EMT-TFs: canonical factors all decline toward the terminus; only ZEB2 clears the detection floor (19%) and marks the mesenchymal/gastrulation origin — no bipolar EMT-TF split.

Parameter choices

Min cells/sample	≥160 (natural break, keeps 5/8)	**Integration**	Harmony on sample, θ=10
Island removal	UMAP radius-graph (−41 cells)	**Trajectory**	scFates curve N=20 (non-branching)
Root node	tip 0 = Gast/pSom origin	**Modules**	score_genes, union control pool
n_pcs / spread	20 / 1.3	**Overdispersion**	scFates find_overdispersed

Obradovic 2021 ccRCC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/8 samples (dominant 522→153 cliff, cut at geo-mean 282) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→4.28 of 6); both low-cycling lobes (main body 8,207 / neural-crest terminus 902) carry the axis.

Gate 2 — module geography → GO (non-branching)

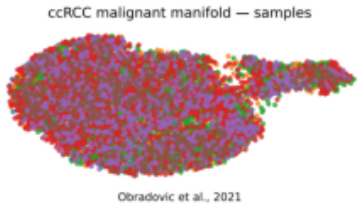
Single dominant pSom→Crest axis: pSom high over the main body declines along pseudotime (ρ=−0.56); Crest anti-correlated (r=−0.95), rising to a terminal readout (ρ=+0.67; Crest-high decile mean pt=0.93). Gast co-varies weakly with pSom (r=+0.47), also declining (early programme, spent); joint pSom&Crest-high depleted (enrichment 0.21) — an inverse gradient, not a co-expression hub.

Gate 3 — hub mutual exclusivity → NA

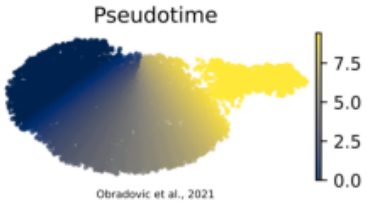
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

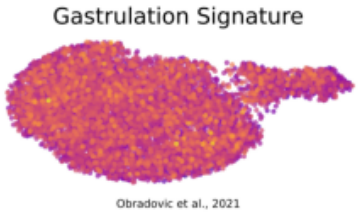
Manifold — samples



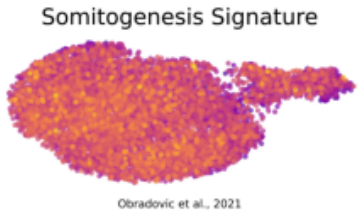
Pseudotime



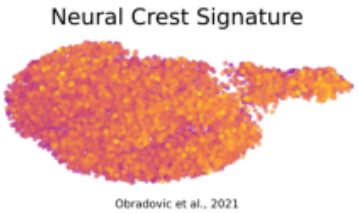
Gastrulation



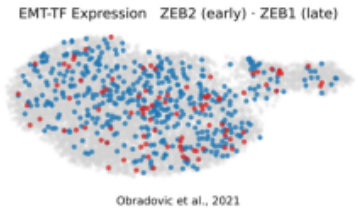
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Clear-cell RCC (Obradovic et al., 2021), 10x — 8 patients / 19,781 cells (master scope); malignant only: 9,369 cells; 9,109 across 6 tumour-bearing samples (drop 2 spillover samples below the 522→153 cliff) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Obradovic's ccRCC malignant manifold is a single monotone somitogenesis→terminal-neural-crest gradient (somitogenesis-high body → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Malignant cells are almost entirely non-cycling (~1%), so unlike Kürten/Lin/Qian there is no proliferative-lobe confound: both lobes are low-cycling with near-identical developmental means, and the whole manifold is kept (Bischoff/Sun/Lee/Steele no-scoping precedent). The clean-count-gap premise is met here (dominant cliff 522→153, ratio 3.41 ≫ runner-up 1.86), so the threshold is cut at the geometric mean (282), not the ≥100 floor. EMT-TFs are root/body-loaded: ZEB2 (early) → ZEB1 (late) chosen by data over Puram's SNAI1/ZEB1 pair, but all expressed factors decline along pt and end-separation is weak (Δpt=0.08); no factor rises to the neural-crest terminus.

Parameter choices

Min cells/sample	282 (geo-mean of 522→153 cliff; 6/8)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = pSom-high, Crest-low	**Lobes**	body 8,207 / terminus 902
EMT-TF axis	ZEB2 (early) → ZEB1 (late)	**Geometry**	Linear / non-branching (fold-free)

Zhang 2021 RCC — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/8 samples (≥369 cells, dominant 710→192 cliff) intermix on one continuous manifold: iLISI 1.00→3.17, single kNN component (0 detached) at every θ; no patient-private island (kNN-purity 4.6× vs 4.5×, no standoff).

Gate 2 — module geography → GO (non-branching)

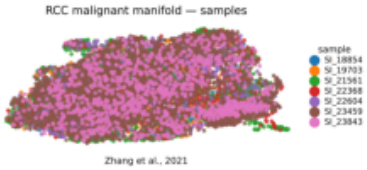
Single developmental axis: pSom-high origin → neural-crest terminus. Crest↔pSom spatially anti-correlated (r=−0.91); Crest rises along pseudotime (ρ=+0.48, high-decile pt 0.72 = terminal); joint pSom&Crest enrichment 0.26 (<1 = inverse gradient, not a co-expression hub). No Gast/pSom arms, no bifurcation. Crest terminal, no inversion flag.

Gate 3 — hub mutual exclusivity → NA

Not tested — no bifurcation and no hub to run 04_correlation/05_modules on (branching-only gate).

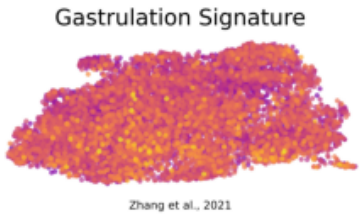
STEP FIGURES

Manifold — samples

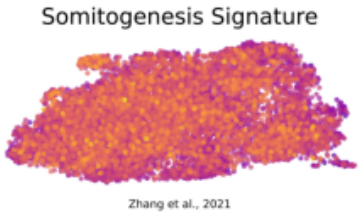


Pseudotime

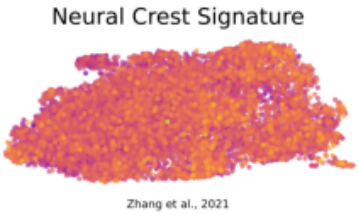
Gastrulation



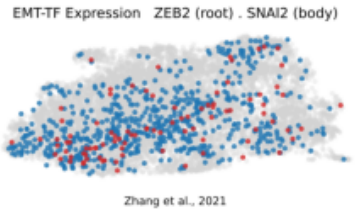
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Clear-cell renal cell carcinoma (Zhang et al., 2021), 10x — 8 patients / 11,410 malignant load; 11,218 across 7 samples (≥369 cells) enter the manifold.

Verdict.
Pattern reproduces (non-branching). Gate 1 GO, Gate 2 GO, Gate 3 NA: RCC malignant cells form one continuous developmental axis — posterior-somitogenesis origin → neural-crest terminus, Crest inverse gradient. EMT programme is root-loaded: ZEB2 (24% det.) concentrates at the pSom origin; no terminus EMT-TF.

Parameter choices

Min cells/sample	≥369 (geom-mean cliff; keeps 7/8)	**Island removal**	none (kNN-purity, no standoff)
Root	tip 0 = pSom-high / Crest-low origin	**Geometry**	non-branching (curve 2 tips 0 forks)
Subtypes	3CA meta-programs (mp_top)	**Integration**	Harmony on sample, θ=10, 30 it
Trajectory	scFates curve N=20, λ=0.3, μ=100	**n_pcs**	20 (10x precedent; scree top-heavy)
Signatures	score_genes, union control pool	**EMT-TFs**	ZEB2 (root) / SNAI2, root-loaded

Zhang 2021 RCC — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/8 samples (≥369 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→3.17 of 7); clear-cell and chromophobe RCC intermix. Both compartments (Cycling/Metabolic 9,545 / Embryonic 1,673) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

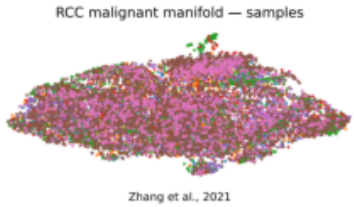
Single dominant pSom→Crest axis: pSom high over the body declines along pseudotime (ρ=−0.27); Crest anti-correlated (r=−0.91), rising to a terminal readout (ρ=+0.48; crest-high decile mean pt=0.72). Gast weakly structured, co-varies with pSom (early, spent).

Gate 3 — hub mutual exclusivity → NA

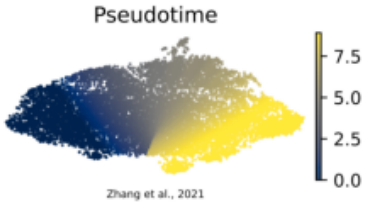
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

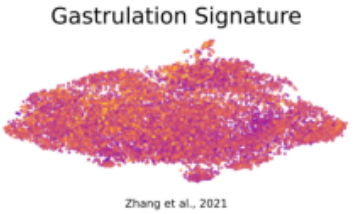
Manifold — samples



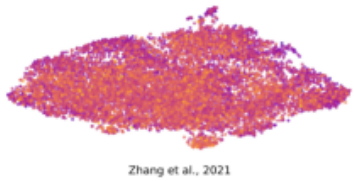
Pseudotime



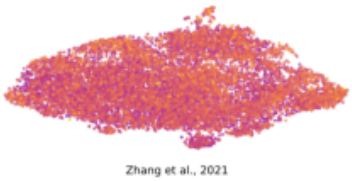
Gastrulation



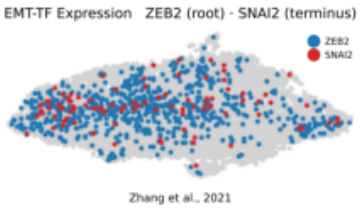
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Renal Cell Carcinoma (Zhang et al., 2021), 10x — 8 malignant samples / 11,410 malignant cells; 11,218 across 7 samples (≥369) enter preprocessing (0 island cells removed); whole malignant manifold (no lobe scoping). Mixed clear-cell + chromophobe RCC.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zhang's RCC malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both proliferation compartments (module·g2m |ρ|≤0.03) and both RCC subtypes carry near-identical developmental means, so the whole manifold is kept (Bischoff/Sun no-scoping precedent). A curve is fold-free (self-approach 2.76); a PPT tree's forks do not resolve a Gast/pSom split (over-fitting of a near-round cloud). EMT-TFs: ZEB2 is the root-end factor (most negative ρ); no well-expressed factor is genuinely terminus-enriched (all ρ≈0), so SNAI2 is only the least-declining — all EMT-TFs sit near the 10x detection floor.

Parameter choices

Min cells/sample	369 (7/8; gap 710→192, ratio 3.7)	**Scope**	whole manifold (no scoping)
Root node	tip 0 = pSom-high, Crest-low	**Integration**	Harmony θ=10; PCA50; n_pcs=20; spread=1.3
Trajectory	scFates curve N=20 λ=0.3 μ=100	**Compartments**	Cycling/Metabolic 9,545 / Embryonic 1,673

Zhang 2021 RCC — analysis summary

RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — integration & continuity →GO

7 samples (≥369-cell threshold) on ONE connected component; clear-cell samples intermix (per-neighbourhood iLISI 3.0/7 at θ=5); the chromophobe histology forms a resolved but ADJOINING region of the same continuous manifold. Cell cycle negligible (3%). No islands.

Gate 2 — module geography →GO (non-branching)

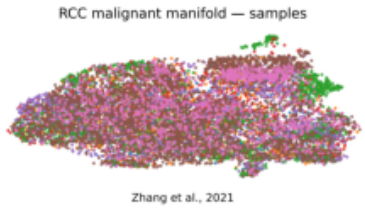
Non-branching case: a shared Gast/pSom gradient with Crest as an inverse TERMINAL gradient (pSom vs Crest r = −0.91; Crest-high decile at pseudotime 0.78). No Gast/pSom hub diverging into two competing arms. Programs present but not resolved into a bifurcation.

Gate 3 — hub cross-repression →NA

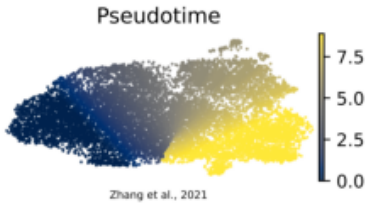
Not applicable — geometry is a single curve (non-branching), so there is no hub to test. An aggressive PPT tree over-fits the compact cloud into 33 spurious forks, not one Y.

STEP FIGURES

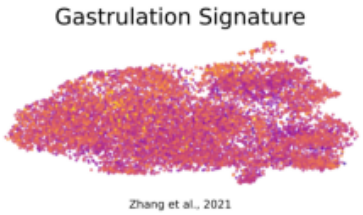
Manifold — samples



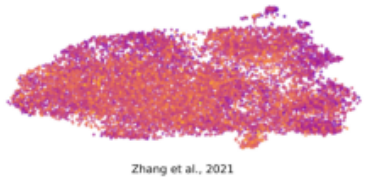
Pseudotime



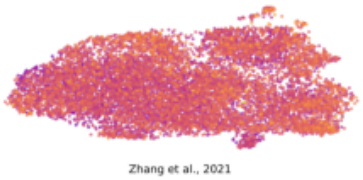
Gastrulation



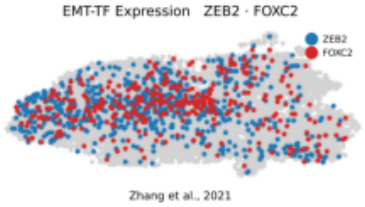
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Renal cell carcinoma (clear-cell + chromophobe) (Zhang et al., 2021), 10x — 11,218 malignant cells, 7 samples.

Verdict.
Pattern reproduces as the NON-BRANCHING case: a single gastrulation/somitogenesis axis terminating in a neural-crest gradient. No Gast/pSom bifurcation (Linear geometry). Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast −0.034 / pSom +0.085 / Crest −0.060. EMT-TFs ZEB2 (23.8%) + FOXC2 (12.2%).

Parameter choices

Min-cell threshold	≥369 (geom-mean of 710→192 gap, 3.7×); drops 1 small sample → 7 kept	**Integration / PCs**	Harmony θ=5, n_pcs=20 (no sc.pp.scale); iLISI-selected, per-neighbourhood
Geometry	Linear — scFates principal curve (2 tips, 0 forks, fold ratio 2.71)	**Root**	Curve tip with max Gast+pSom, min Crest (developmental origin)
Module means	Gast −0.034 / pSom +0.085 / Crest −0.060 (score_genes, union pool)	**EMT-TFs**	ZEB2 (23.8%) + FOXC2 (12.2%) — best-detected, distinct Zeb/Foxc families

Zhang 2021 RCC — analysis summary

RUN 4 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

1 connected component at all θ; iLISI peak 3.12 @ θ=10; the axis recurs within patients (not a between-patient integration artifact).

Gate 2 — module geography → GO (non-branching)

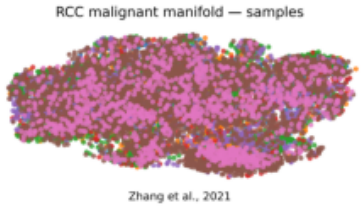
Non-branching: no Gast/pSom co-expression hub (joint co-high 4.9% vs 11.6% expected); pSom↔Crest inverse terminal gradient r=−0.80, Gast/pSom cross-repression r=−0.53. Single organized axis, no divergent arms.

Gate 3 — cross-repression → NA

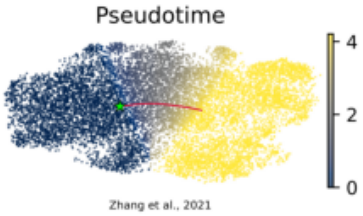
Not applicable — no bifurcation / hub to test for Gast↔pSom node anti-correlation.

STEP FIGURES

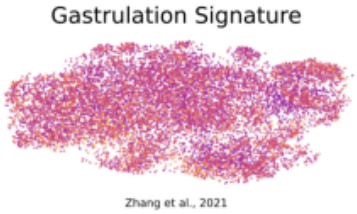
Manifold — samples



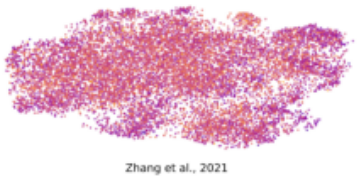
Pseudotime



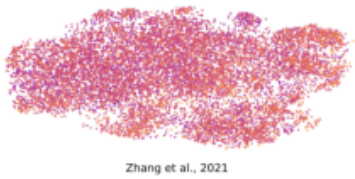
Gastrulation



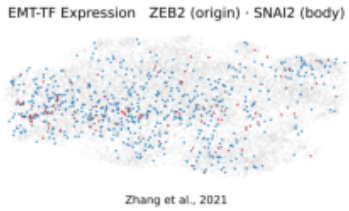
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Clear-cell + chromophobe renal cell carcinoma (Zhang et al., 2021), 10x — 11,410 malignant cells → 11,218 after per-sample threshold → 11,136 after island removal, across 7 samples (6 ccRCC + 1 chromophobe); the whole malignant manifold enters the trajectory (non-branching linear geometry).

Verdict.
The embryonic pattern reproduces as the non-branching case. A single continuous manifold carries an organized posterior-somitogenesis (origin) ↔ neural-crest (terminal) inverse axis (r=−0.80) with Gast/pSom cross-repression (r=−0.53), but no Gast/pSom co-expression hub and no divergent arms — so no bifurcation. Geometry LINEAR; Gate 3 NA. The axis is genuine intra-tumoral biology: it reproduces within every individual patient (mean pSom↔Crest r=−0.57), not a between-patient integration artifact. EMT programme is origin-loaded (all 6 expressed TFs decline from the origin).

Parameter choices

Min-cell threshold	369 (geom-mean of dominant gap 192→710, ratio 3.70); drops 1 sample	**Over-dispersed genes**	1,363 (scFates find_overdispersed); n_pcs=20 (scree knee)
Integration	Harmony on sample, θ=10 (iLISI peak 3.12; θ=20 re-segregates chromophobe)	**Island removal**	radius graph 6× median NN; keep largest comp (99.3%); 82 fringe cells
Geometry	LINEAR — ElPiGraph curve (Nodes 12, λ 2.0); 2 tips, 0 forks, fold-free (2.99)	**Root**	Gast/pSom-high, Crest-low curve tip (developmental origin; Crest is late)
Module means	Gast −0.034 / pSom +0.085 / Crest −0.060 (Crest terminal, no Gast arm)	**EMT-TFs**	ZEB2 (23.9%, origin) / SNAI2 (3.3%, body); all origin-loaded, 0 terminus factor

Krishna 2021 ccRCC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

UT2_Center + UT2_Far intermix on one continuous manifold; 0 detached islands, single Leiden compartment (θ=10, mixing plateau). Judged on the embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

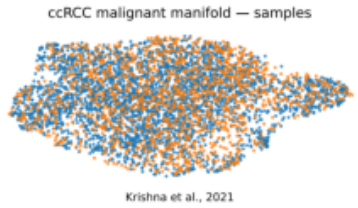
Non-branching: a single pSom→Crest gradient (pSom_sm–Crest_sm r=−0.89); Crest terminal (crest-high decile mean pt=0.97); Gast weak/diffuse.

Gate 3 — cross-repression → NA

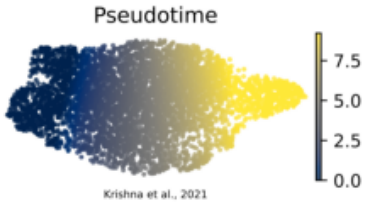
No bifurcation (2 tips, 0 forks) — the competitive Gast↔pSom arm test does not apply. Non-branching case, as in prior Kidney runs (Bi/Zhang).

STEP FIGURES

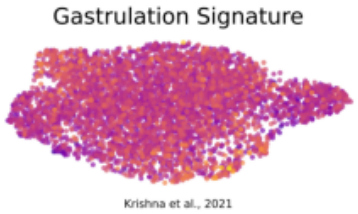
Manifold — samples



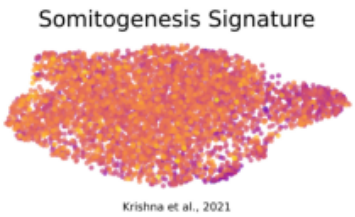
Pseudotime



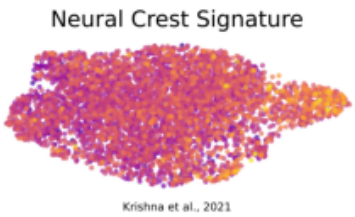
Gastrulation



Somitogenesis



Neural Crest



EMT-TF expression

Dataset.
Clear-cell RCC (Krishna et al., 2021), 10x; 5,715 malignant cells pre-filter, 5,006 after thresholding (all CD45− ccRCC CA9+). One continuous non-branching manifold.

Verdict.
Pattern reproduces (Linear-pSom/Crest). A single embryonic axis rooted at the somitogenesis-high origin flows to a neural-crest terminus (pSom ρ=−0.85, Crest ρ=+0.75 vs pseudotime); no Gast/pSom bifurcation, so Gate 3 is NA. Caveat: the clean 15.9× count gap retains only two regions of one patient (UT2), 5,006/5,715 cells, so integration is region-level; cycling ρ=−0.23 vs pt (not proliferation-driven).

Parameter choices

Min cells/sample	403 (geom-mean 15.9× gap; keeps UT2)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**Geometry**	Linear (non-branching); 2 tips / 0 forks
EMT-TFs	TWIST2 (early) · ZEB2 (late)		

Sun 2021 HCC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 patients (P04, P05, P09, P10, P16, P18) intermix on one connected component; a cycling-enriched lobe (~32%) does not separate, is not gastrulation-high, and leaves the Gast↔Crest gradient intact when removed → full manifold, no scoping.

Gate 2 — module geography → GO (non-branching)

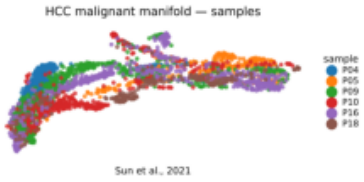
Principal curve fold-free (2 tips, 0 forks); shared developmental gradient — Gast declines (ρ=−0.79) while Crest rises to the terminus (ρ=+0.60); smoothed Gast↔Crest r=−0.71 (inverse terminal gradient).

Gate 3 — hub mutual exclusivity → NA

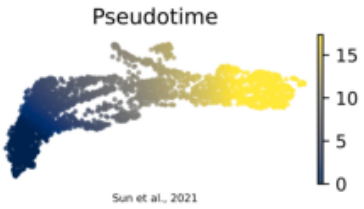
Non-branching manifold — no fork, so the Gast↔pSom cross-repression test does not apply (branching-only gate; §2).

STEP FIGURES

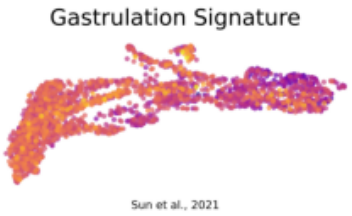
Manifold — samples



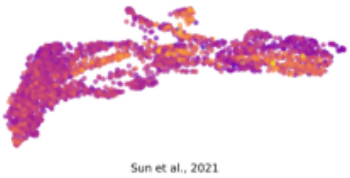
Pseudotime



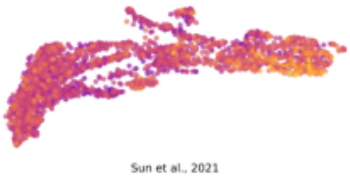
Gastrulation



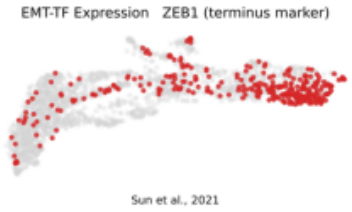
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Hepatocellular carcinoma (Sun et al., 2021), 10x — 16,498 cells / 18 samples (2,737 malignant). Malignant only: 2,338 pass the thin-sample threshold (cut 125, keeps 6 samples); 2,337 enter the trajectory after 1-cell island removal.

Verdict.
Malignant cells recapitulate an embryonic developmental gradient — a non-branching gastrulation→neural-crest axis, no bifurcation. Gast declines while Crest rises to the terminus (smoothed Gast↔Crest r=−0.71). Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. The EMT axis is monotone (no root-specific TF; all candidates rise to terminus); ZEB1 is the strongest terminus marker (53% expressing).

Parameter choices

Malignant cells	2,737 raw (18 samples) → threshold cut 125 keeps 6 samples/2,338 → island removal 1 → 2,337	**Expression / features**	log-normalised 10x; 7,498 overdispersed (scFates find_overdispersed); modules vs union control pool
Embedding	n_pcs=20; Harmony on sample (θ=5); kNN k=15; UMAP spread=1.3; one connected component	**Trajectory**	ElPiGraph principal curve on X_umap (Nodes=20, λ=0.3, μ=100); 2 tips, 0 forks; fold-free (self-approach 2.97)
Root / orientation	root = Gast/pSom-high, Crest-low tip; pseudotime left→right (orient_sx=+1, raw coords)	**EMT-TFs**	panel SNAI/ZEB/TWIST + FOXC/PRRX, 3% floor; axis MONOTONE (all rise to terminus); ZEB1 strongest (53%)

Bischoff 2021 LUAD — analysis summary

RUN 1 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

10/19 samples (≥43 cells) intermix on one continuous manifold with no per-patient islands; the kNN graph is a single connected component (13,382 cells, no island removal). Density is bimodal (a thin transitional waist) but topologically continuous — both poles are multi-patient (terminus: 7 samples >5%).

Gate 2 — module geography → GO (non-branching)

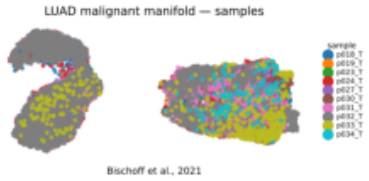
Gast/pSom decline together along pseudotime (ρ=−0.60/−0.23); Crest is spatially anti-correlated (r=−0.40/−0.72) and rises to a terminal readout (ρ=+0.29; crest-high decile mean pt=0.73). The modules ride a dominant alveolar-differentiation axis — present but not resolved into two competing arms.

Gate 3 — cross-repression → NA

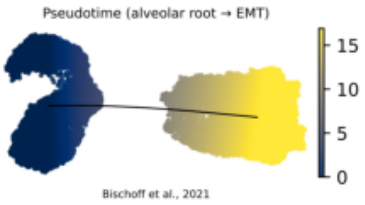
Single monotone differentiation axis — no bifurcation and no Gast/pSom-overlap hub, so there is no branch pair to test for competitive cross-repression.

STEP FIGURES

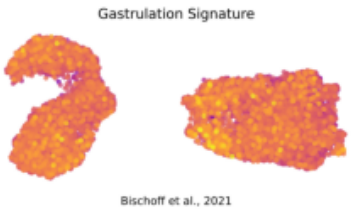
Manifold — samples



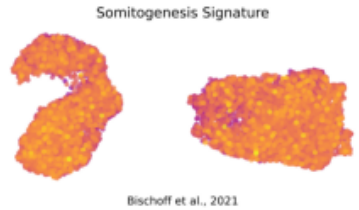
Pseudotime



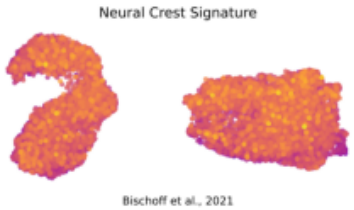
Gastrulation



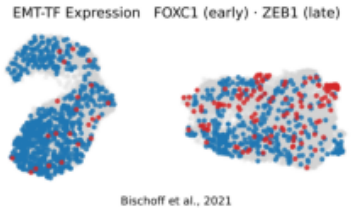
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Lung adenocarcinoma (Bischoff et al., 2021), 10x — early-stage LUAD; malignant only: 13,439 loaded, 13,382 across 10 samples on one continuous manifold; the full compartment carries the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Bischoff's manifold is a single alveolar-differentiation axis (alveolar/AT2 → dedifferentiated); embryonic modules ride it as a secondary Gast/pSom → terminal-Crest gradient, lacking Puram's hub-and-spoke, so Gate 3 has no hub to test. EMT-TFs FOXC1 (early/root, 33% det) → ZEB1 (late/terminus); canonical SNAI1/TWIST1 near the floor.

Parameter choices

Min cells/sample	≥43 (keeps 10/19; gap 127→15, ×8.5)	**Malignant states**	Alv/AT2 4,173 / Secretory 1,761 / Dediff 7,448
Root node	tip 0 = Gast/pSom-high, Crest-low	**Integration**	Harmony(sample) θ=2; 50PC → 12; UMAP spread=1.3
Trajectory	scFates curve N=20 λ=0.3 μ=100 (fold-free, 2 tips)	**Manifold**	full malignant (no lobe split); 1 kNN component
EMT-TFs	FOXC1 (early/root, 33% det) → ZEB1 (late/terminus); SNAI1/TWIST1 near floor		

Bischoff 2021 LUAD — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

MIXED (diff-sample kNN frac 0.02→0.47, ceiling 0.81; both lobes internally sample-mixed, tumours span both) but NOT CONTINUOUS — two islands: PAGA between-lobe 0.01 vs within 0.55; 0.18% of edges & 1.9% of cells bridge; bimodal diffusion axis (peak/valley 387×). Invariant across θ {0,2,5,10}, n_pcs {12,16,20} — not under-integration.

Gate 2 — module geography → NA

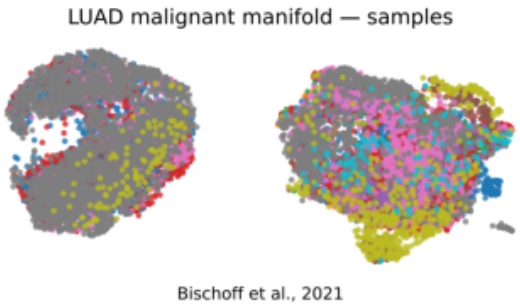
Not evaluated — Gate 1 NO-GO (spec §6). Lobe-identity fork: no lobe is a valid scope target — signatures near union-control baseline everywhere (Gast −0.04 / pSom +0.06 / Crest −0.03); the alveolar lobe shows the Kürten sign pattern but at baseline magnitude and is the DIFFERENTIATED pole (biologically backwards); the EMT/dediff lobe is incoherent (Gast~pSom −0.32).

Gate 3 — hub cross-repression → NA

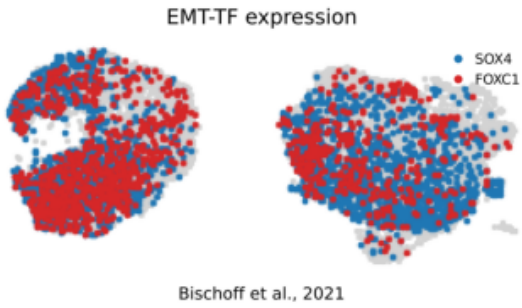
Not evaluated — branching-only, and no trajectory exists.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Lung adenocarcinoma (Bischoff et al., 2021), 10x — 13,382 malignant cells (10 tumour samples, 44-cell threshold).

Verdict.
Embryonic pattern does NOT reproduce — malignant structure is two disconnected EMT/differentiation lobes, not one continuous manifold; no trajectory fit (Gate 1 NO-GO). No lobe is a valid scope target: signatures sit at union-control baseline, and the one lobe with the Kürten sign pattern (alveolar) is the differentiated pole. Descriptive EMT-TF overlay (no trajectory to polarize): SOX4 graded high on the EMT/dediff lobe; SOX9/TEAD1/FOXC1 patchier.

Parameter choices

Min-cell threshold	44 (geom-mean of 127→15 gap, 8.5×) → 10/19 samples	**Over-dispersed genes**	1,976 (scFates GAM)
n_pcs	16 (PCs > 0.5% variance; PC1 = 12.4%)	**Integration**	Harmony on sample, θ=2 (mixing plateaus by θ=2)
Embedding	k=15, UMAP spread=1.3, Leiden res=0.5 (12 clusters)	**Lobe split**	diffusion-axis valley: Alveolar 6,100 / EMT-dediff 7,282
Signatures	score_genes vs union pool (Gast 80 / pSom 95 / Crest 103 genes)	**EMT-TFs shown**	SOX4 83%, SOX9 44%, TEAD1 44%, FOXC1 33% (≥3% floor; VIM excluded)

Bischoff 2021 LUAD — analysis summary

RUN 3 OF 3

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

Malignant structure is mixed and continuous on one manifold. Judged on the embedding + mixing metrics across an n_pcs×θ sweep: θ=0 gives per-sample islands (iLISI 1.05, diff-sample kNN 0.02) — integration mixes them strongly, saturating by θ=2 (iLISI ~3.3–3.7 across θ=2/5/10). At the chosen θ=10 (Kürten precedent) the kNN graph is a single connected component (comp=1, as at every θ) and strongly mixed (iLISI 3.53, diff-sample kNN 0.43 of 0.805 ceiling). The two density lobes are poles of one connected manifold — not disconnected islands (comp=1 throughout) and not sample-driven (both multi-sample); the thin transitional waist is expected differentiation density, not a break. Cell cycle does not structure the manifold (≤6% cycling per lobe) — no compartment to strip.

Gate 2 — embryonic module pattern → GO (non-branching, flagged)

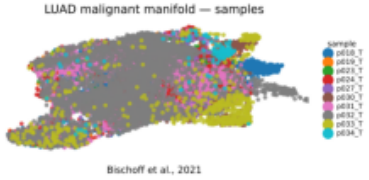
Non-branching case (§1): Crest is a clean inverse/terminal gradient — anti-correlates Gast (−0.42) and pSom (−0.50), and the Gast&pSom co-high hub is strongly Crest-low (z −0.82 vs +0.20). Flag (pass anyway): pSom is near-ubiquitous (score>0 in 94% of cells), Gast sparse (14%), Gast/pSom co-variation weak (−0.15) — an attenuated signal versus the Kürten precedent. The axis is alveolar-differentiation ↔ EMT.

Gate 3 — bifurcation cross-repression → NA

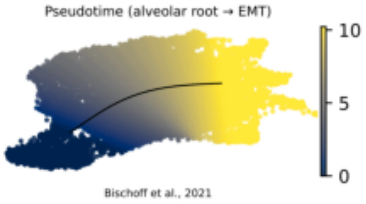
No bifurcation. The curve is fold-free (2 tips, 0 forks, self-approach 2.9); the PPT tree's forks separate no biology (all tips Gast≈−0.035 / pSom≈+0.06 / Crest≈−0.035). Branching-only gate → NA.

STEP FIGURES

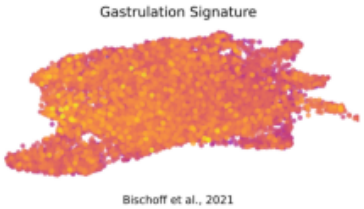
Manifold — samples



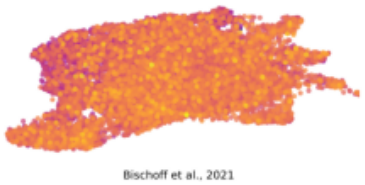
Pseudotime + curve



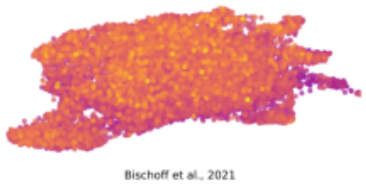
Gastrulation



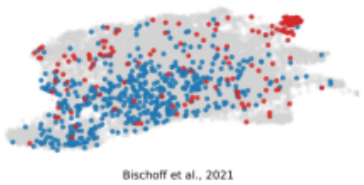
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Lung adenocarcinoma (Bischoff et al. 2021), 10x scRNA-seq. 13,382 malignant cells across 10 tumor samples (of 20; normal-tissue samples carry 2–15 malignant cells and drop out at threshold).

Verdict.
The embryonic pattern reproduces in the non-branching form — a single continuous alveolar→EMT manifold with Crest as an inverse terminal gradient and no Gast/pSom bifurcation. The signal is weaker than the Kürten HNSCC precedent (attenuated; flagged), but present and continuous.

Parameter choices

Min-cell threshold	43 (geo-mean across the 127→15 gap, ratio 8.5) → 10 samples, 13,382 cells	**Islands removed**	0 (no detached components at n_neighbors=30) → 13,382 on the manifold
Overdispersed genes	1,976 (no pre-PCA scaling)	**Integration / embedding**	Harmony on sample θ=10; n_pcs=20, n_neighbors=30, UMAP spread=1.3 (seed 1)
Geometry	2-D continuum (UMAP aspect ≈1.3); scFates curve (non-branching)	**Root**	Alveolar / Crest-low tip (Gast/pSom-high pole; Crest is developmentally late)
EMT-TFs shown	FOXC1, TCF12, TCF3, SNAI1, PRRX1/2, ZEB1/2, TCF4 (≥3% detected; VIM/TGFBI not TFs)	**Gast-pSom r (whole manifold)**	−0.15 raw — attenuated vs Puram hub ref ≈ −0.49 (no windowed hub step; Gate 3 NA)

Kim 2020 LUAD — analysis summary

RUN 1 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

13/14 samples (≥42 cells) intermix on one continuous manifold; radius-graph finds 0 detached islands. Harmony θ=5 (mixing plateaus by θ=5); both compartments span multiple patients.

Gate 2 — module geography → GO (non-branching)

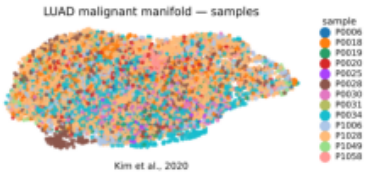
Gast/pSom near-orthogonal (r=−0.11), pSom broadly dominant over the body; Crest anti-correlated (r=−0.55 / −0.41), rising to a terminal readout (ρ=+0.48; crest-high decile mean pt=0.81). Gast declines along pseudotime (ρ=−0.55) — one monotone Gast/pSom → neural-crest axis, no second arm.

Gate 3 — hub mutual exclusivity → NA

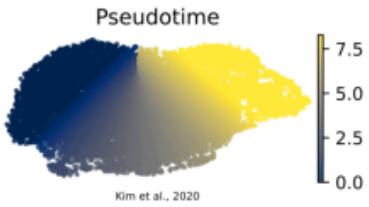
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test.

STEP FIGURES

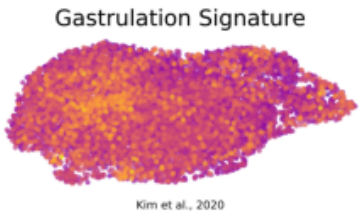
Manifold — samples



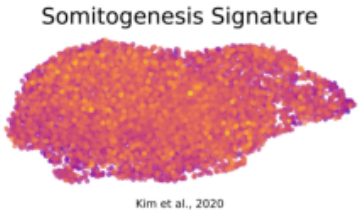
Pseudotime



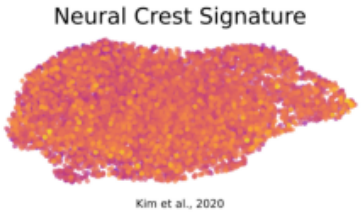
Gastrulation



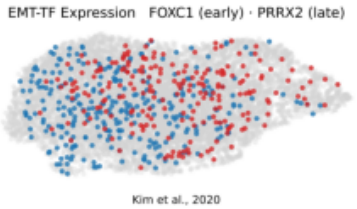
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Lung Adenocarcinoma (Kim et al., 2020), 10x — 14 patients / 32,493 cells (master scope); malignant only: 10,538 load, 10,512 across 13 samples (≥42) enter preprocessing. The whole malignant manifold (no lobe scoping) enters the trajectory: Embryonic 9,315 (6% cycling) + Cycling 1,197 (61%).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Kim's LUAD malignant manifold is a single monotone Gast/pSom → terminal-neural-crest gradient, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The developmental axis spans both proliferation lobes (orthogonal to cell cycle; per-lobe module means near-identical, Crest-high decile split 52/48), so the whole manifold is kept (Bischoff/Sun no-scoping precedent). Caveat: pseudotime co-varies with cycling (ρ=+0.41). EMT-TFs FOXC1 (early/root) → PRRX2 (late/terminus); canonical SNAI1/ZEB1/TWIST1 near the detection floor, end-separation weak (Δpt=0.21).

Parameter choices

Min cells/sample	≥42 (13/14 kept; gap 70→26, ratio 2.7)	**Scope**	whole manifold (no lobe scoping)
Root node	tip 0 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=5 on 50-PC; UMAP spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100	**Compartments**	Embryonic 9,315 / Cycling 1,197

Kim 2020 Lung Adenocarcinoma — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

13/13 malignant-bearing samples intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→2.75 of 13); both compartments (Embryonic 9,315 / Cycling 1,197) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

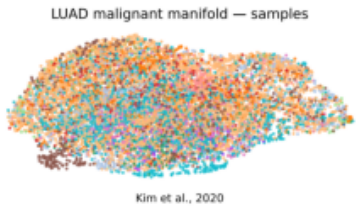
Single Gast→Crest developmental axis: Gast declines along pseudotime (ρ=−0.55) from the Gast/pSom-high origin; Crest anti-correlated with Gast (r=−0.55), rising to a terminal readout (ρ=+0.48; Crest-high decile mean pt=0.81). pSom overlaps the Gast body (ρ=+0.09, flat) — an inverse gradient, not a Gast-arm/pSom-arm co-expression hub. Crest terminal — no inversion flag.

Gate 3 — hub mutual exclusivity → NA

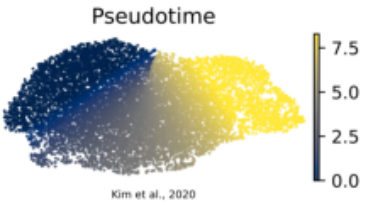
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

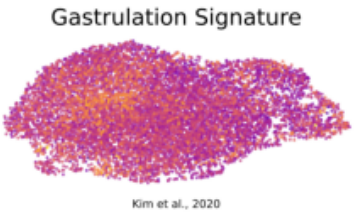
Manifold — samples



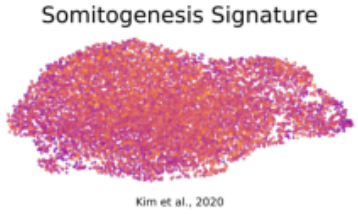
Pseudotime



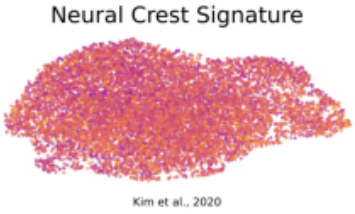
Gastrulation



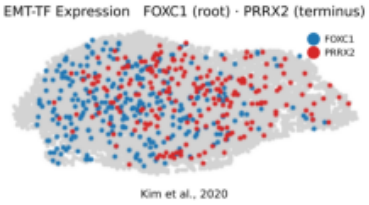
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
LUAD (Kim et al., 2020), 10x — 14 patients / 32,493 cells (full atlas); malignant only: 10,538 cells → 10,512 across 13 samples enter preprocessing (drop 1 sub-42 spillover sample, P0008/26 cells); whole malignant manifold (no lobe scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Kim's LUAD malignant manifold is a single monotone gastrulation→terminal-neural-crest gradient rooted at the Gast/pSom-high origin, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The Embryonic/Cycling split is partly proliferative, but the developmental modules are cell-cycle-orthogonal (|ρ vs g2m|<0.15; both compartments share identical module means), so the whole manifold is kept (Bischoff/Sun no-scoping precedent), carrying pseudotime↔cycling co-variation (ρ=+0.41) as a declared caveat. EMT-TFs FOXC1 (early/root) → PRRX2 (late/terminus) — the only two factors that clear the detection floor, chosen by data over Puram's pair.

Parameter choices

Min cells/sample	42 (gap 70→26, r=2.69)	**Integration**	Harmony θ=5; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Embryonic 9,315 / Cycling 1,197

Kim 2020 LUAD — analysis summary

RUN 3 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

One connected manifold (0 detached islands). Dominant sample P1028 (40%) spread throughout, not an archipelago; median 5 of 11 samples per 30-cell neighbourhood. iLISI 1.00→2.61 across the θ sweep (θ=10 on the mixing plateau). Read on the embedding + iLISI, not a scalar.

Gate 2 — module geography → GO (non-branching)

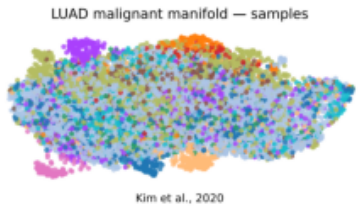
Non-branching: single developmental axis. Gast/pSom overlap (spatial r=−0.11); both oppose Crest (Gast-Crest r=−0.53, pSom-Crest r=−0.38). Crest is a terminal inverse gradient (Crest-high decile pseudotime 0.75; joint-high enrichment 0.31/0.44 < 1). Crest terminal to both modules — no Crest-arm inversion flag. PPT tree over-tiles the blob (no clean Y).

Gate 3 — hub cross-repression → NA

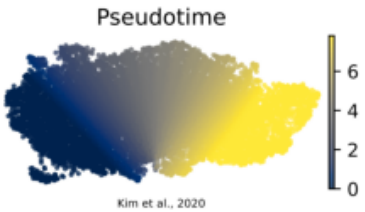
Not applicable — non-branching manifold has no Gast/pSom bifurcation hub to test.

STEP FIGURES

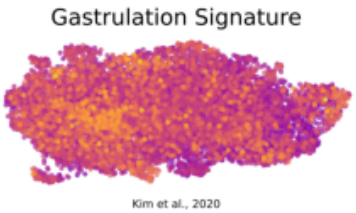
Manifold — samples



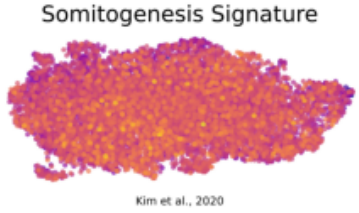
Pseudotime



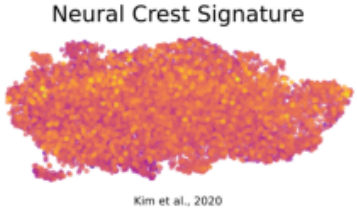
Gastrulation



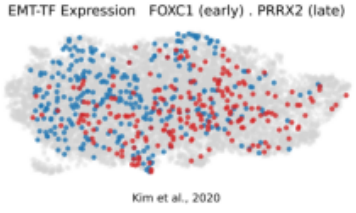
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Lung Adenocarcinoma (Kim et al., 2020), 10x — 11 patients; 10,538 malignant cells (10,368 after thresholding) enter the analysis; the whole malignant manifold enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA (LINEAR geometry). The embryonic-development programme reproduces as a single continuous gastrulation/somitogenesis → neural-crest axis: Gast/pSom co-vary at the origin, Crest rises to the terminus (inverse gradient), no competing-arm bifurcation — so Gate 3 has no hub to test. EMT-TFs FOXC1 (root) → PRRX2 (terminus).

Parameter choices

Min-cell threshold	≥100 cells/sample → 11/14 kept (no dominant cliff, top/2nd 1.2 → floor)	**Integration**	Harmony on sample, θ=10
Geometry	n_pcs=30; no scale before PCA	**Trajectory**	ElPiGraph curve N=20 (fold-free 2.89)
Root	Gast/pSom-high tip 0 (Crest-low origin)	**Module means**	Gast −0.040 · pSom +0.052 · Crest −0.045
EMT-TF axis	FOXC1 (root) → PRRX2 (terminus)		

Qian 2020 NSCLC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/8 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→2.39 of 6); both lobes (Body 3,848 / Cycling 1,400) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

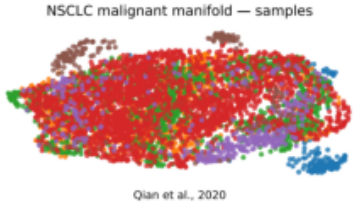
Single dominant pSom→Crest axis: pSom high over the low-cycling body declines along pseudotime (ρ=−0.68); Crest anti-correlated (r=−0.89), rising to a terminal readout (ρ=+0.82; Crest-high decile mean pt=0.98). Gast weakly structured (early programme, spent); joint pSom&Crest-high depleted (enrichment 0.18) — an inverse gradient, not a co-expression hub.

Gate 3 — hub mutual exclusivity → NA

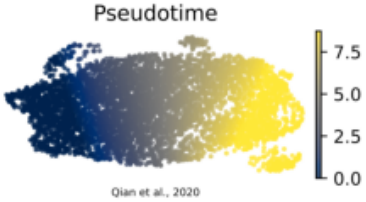
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

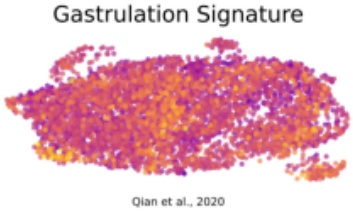
Manifold — samples



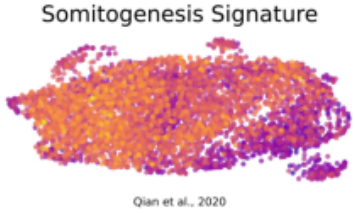
Pseudotime



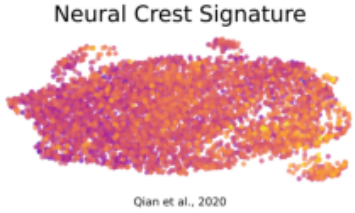
Gastrulation



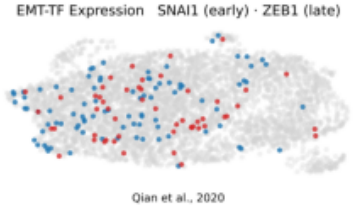
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Non-Small-Cell Lung Cancer (Qian et al., 2020), 10x — 8 patients / 27,262 cells (master scope); malignant only: 5,248 cells across 6 malignant-bearing samples (drop 2 sub-100 spillover samples) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Qian's NSCLC malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient (low-cycling body → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The Body/Cycling split is partly proliferative and the Crest terminus enriches the cycling side, but the developmental modules are cell-cycle-free (Gast 0/81, Crest 0/104 cell-cycle genes) and the pSom↔Crest anti-correlation survives within non-cycling cells alone (r=−0.89→−0.90), so the axis is developmental, not proliferative — the whole manifold is kept (Lin/Steele no-scoping precedent), carrying pseudotime↔cycling co-variation (ρ=+0.45) as a caveat. EMT-TFs are root/body-loaded: SNAI1 (early) → ZEB1 (late) chosen by data over Puram's pair, but all expressed factors decline along pt and end-separation is weak (Δpt=0.08); the only terminus-rising factor (SNAI2, ρ=+0.18) sits just below the 3% detection floor.

Parameter choices

Min cells/sample	≥100 floor (6/8; gap non-dominant)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = pSom-high, Crest-low	**Compartments**	Body 3,848 / Cycling 1,400

Laughney 2020 LUAD — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

θ derived by connectivity (θ=5 fragments an 11% blob; θ=10 = one component). One patient (RU680, 125 cells) stays a same-patient clump at every θ (kNN-purity 16× vs ≤7× for the rest) — an unintegrable island, removed. Remaining 7/13 samples (2,734 cells) share one continuous manifold, max purity 6.6×, iLISI 1.00→3.24 — mixed & continuous.

Gate 2 — module geography → GO (non-branching)

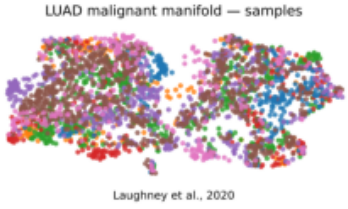
Single developmental axis driven by posterior-somitogenesis (pSom): pSom high at the origin declines along pseudotime (ρ=−0.87) opposed to a terminal neural-crest end that rises (ρ=+0.26; decile pt=0.68); Crest anti-correlated in space (Crest↔Gast −0.72, Crest↔pSom −0.45); Gast weak (ρ=+0.26, mid-axis) and ≈independent of pSom (r=−0.11). Joint pSom&Crest-high depleted (0.36) — inverse gradient, not a hub.

Gate 3 — cross-repression → NA

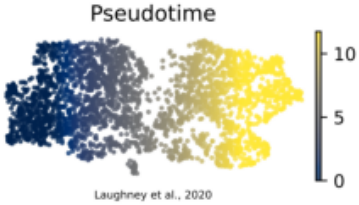
Single monotone axis — no bifurcation (curve: 2 tips, 0 forks) and no Mixed-type hub, so there is no branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

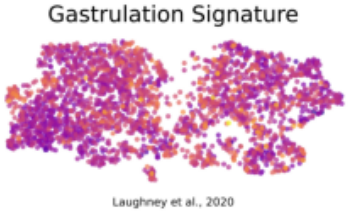
Manifold — samples



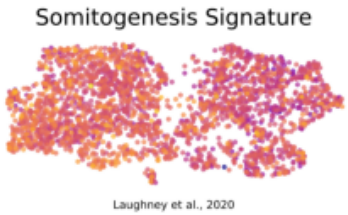
Pseudotime



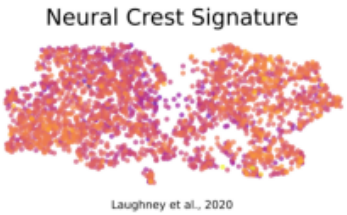
Gastrulation



Somitogenesis



Neural Crest



EMT-TF expression

Dataset.
Metastatic LUAD (Laughney et al., 2020), 10x — 13 patients / 40,505 cells (master scope); malignant only: 3,128 cells; ≥100-cell floor → 2,859 (8 samples), minus one unintegrable patient-private island (RU680, 125) → 2,734 cells / 7 patients on the trajectory (no scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Laughney's LUAD malignant manifold is a single monotone posterior-somitogenesis → terminal-neural-crest gradient (pSom ρ=−0.87, Crest ρ=+0.26); it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test (Linear-pSom/Crest class, shared with Qian / Kürten / Steele). Two decisive derived calls: integration strength θ=10 by connectivity, and removal of RU680 — a patient-private island that stays 16× kNN-pure at every θ, which the connectivity test alone had let pass. No cell-cycle annotation / flat proliferation, so no cycling lobe is scoped (contrast Qian). EMT terminus-loaded: expressed EMT-TFs lean late, ZEB2 marks the neural-crest terminus, no root-end factor (contrast Puram's SNAI1/ZEB1 pair).

Parameter choices

Cells / patients	2,734 / 7 (≥100 floor, −1 island)	**Integration**	Harmony θ=10 (connectivity); PCA50
Island removed	RU680 (125 cells; 16× pure, θ-inv.)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	pSom-high, Crest-low tip	**EMT-TFs**	terminus-loaded (ZEB2); no root factor

Maynard 2020 LUAD — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/38 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands: an integration sweep (θ 5→40) read on the embedding gives one connected component at θ=20 (iLISI median 1.00→2.00 of 7); θ=40 over-integrates into detached islands. Both poles (Body / Terminus) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

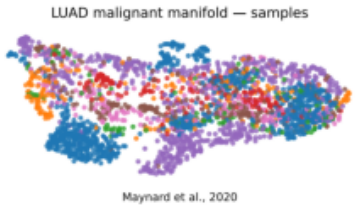
Gast and pSom co-vary (r=+0.43) at one origin pole; both oppose Crest (r=−0.74 / −0.64). Along pseudotime Gast/pSom decline (ρ=−0.64 / −0.63) and Crest rises to a terminal readout (ρ=+0.79; Crest-high decile mean pt=0.83). Joint pSom&Crest-high depleted (enrichment 0.29) — an inverse gradient, not a co-expression hub; Crest enriches the pSom/Gast-low pole, so no inversion flag.

Gate 3 — cross-repression → NA

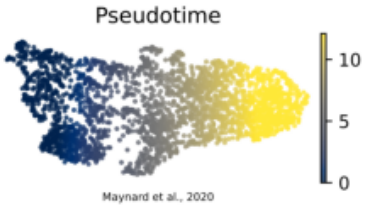
Single monotone axis — 0 branch nodes (curve & PPT tree), fold-free (self-approach 2.96), and no Gast/pSom-overlap hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

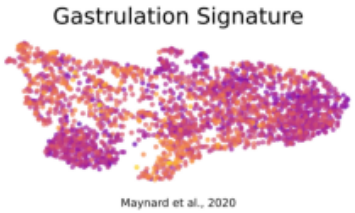
Manifold — samples



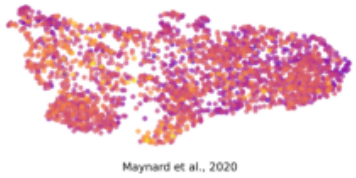
Pseudotime



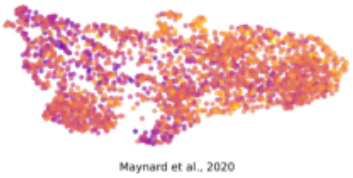
Gastrulation



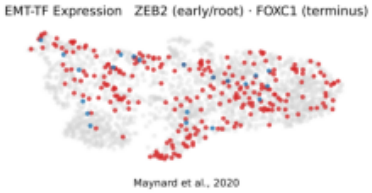
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
LUAD (Maynard et al., 2020), SmartSeq2 — 50 samples / 19,777 cells (master scope); malignant only: 3,184 cells; after ≥100-cell thresholding 2,486 cells across 7 tumour-bearing samples and island removal 2,396 cells enter the trajectory (whole manifold, no scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Maynard's metastatic-LUAD malignant manifold is a single monotone Gast/pSom → terminal-neural-crest gradient (somitogenesis-high origin → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The published cell-cycle annotation is NaN for malignant cells, so cycling was scored de novo; it is diffuse (22%), the modules are cell-cycle-free (Gast 0/81, Crest 0/104 cell-cycle genes), pSom↔Crest anti-correlation survives in non-cycling cells (r=−0.64→−0.67) and pseudotime↔cycling co-variation is negligible (ρ=−0.04) — the axis is developmental, not proliferative. FOXC1 (42% expressing, ρ=+0.15) is the robust neural-crest terminus factor; the root end is weakly marked — only ZEB2 (~3%, at the detection floor) surfaces as an early factor and its call is run-sensitive.

Parameter choices

Min cells/sample	≥100 floor (7/38; gap non-dominant)	**Integration**	Harmony θ=20; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Poles**	Body 1,198 / Terminus 1,198
Geometry	Linear (non-branching); 0 branch nodes	**EMT-TFs**	ZEB2 (early/root) · FOXC1 (terminus, 42%)

Xing 2021 lung cancer — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 18 samples (≥100 malignant cells) intermix per-neighbourhood on one continuous manifold: iLISI 1.4→4.5 across the θ 0→10 sweep, diff-sample kNN fraction 0.72 (of a 0.91 ceiling), a single connected kNN component, 0 detached islands. Cell cycle (~4.5%) is diffuse and does not fragment the cloud. Read on the regenerated embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

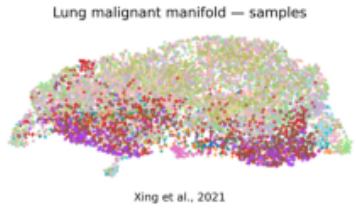
The embryonic axis is present: a Gast/pSom-high origin pole giving way to an inverse terminal Neural-crest gradient (Crest ρ vs pseudotime +0.69, Crest-high decile at pseudotime 0.92; joint pSom&Crest-high depleted, enrichment 0.28). Gast and pSom cross-repress (smoothed r=−0.52, −0.78 at the origin) but do NOT resolve into two competing arms — a branching-capable PPT fit gives no stable fork (forks grow 2→5 with node density on one roundish mass). Non-branching case, GO; no inversion flag.

Gate 3 — cross-repression → NA

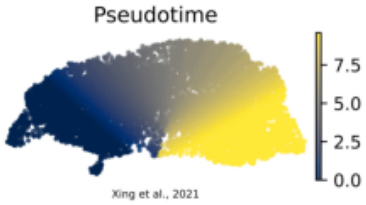
Not applicable: the manifold is non-branching (single continuous developmental gradient, no stable bifurcation), so there is no hub convergence population to test for competitive Gast↔pSom cross-repression. The branching-only correlation and modules notebooks are therefore not produced.

STEP FIGURES

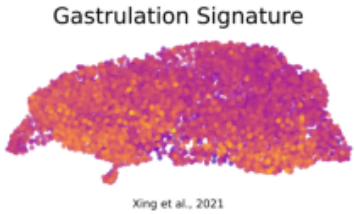
Manifold — samples



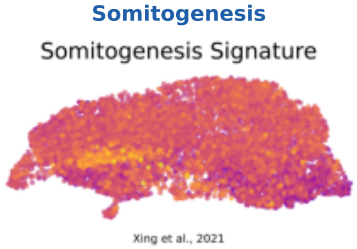
Pseudotime



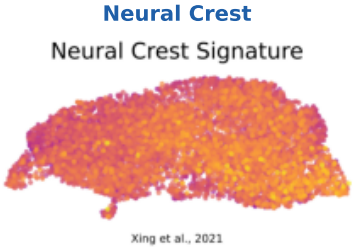
Gastrulation



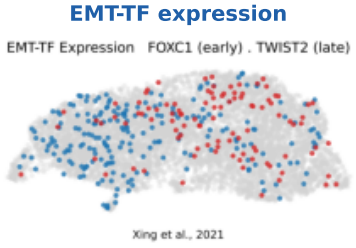
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Lung cancer (subsolid nodules + adjacent margins), 10x UMI counts — 9,268 malignant cells across 25 samples → 8,920 cells / 18 samples after the ≥100-cell floor and island removal.

Verdict.
The embryonic pattern REPRODUCES in the non-branching form: a single continuous gastrulation/somitogenesis → neural-crest axis, rooted at the Gast/pSom-high developmental origin, with organised Gast↔pSom cross-repression that does not resolve into competing arms. EMT-TFs FOXC1 (root) → TWIST2 (terminus). Global module strength pSom>Gast~Crest (Linear-pSom/Crest).

Parameter choices

Malignant load	9,268 cells / 25 samples (publication calls)	**After thresh + islands**	8,920 cells / 18 samples
Integration	Harmony θ=10 (sweep-derived plateau)	**Embedding**	n_pcs=20, n_neighbors=30, UMAP spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100	**Root**	curve tip: Gast/pSom-high, Crest-low
Topology	non-branching (2 tips, 0 forks, self-approach 2.7)	**EMT-TFs**	FOXC1 (early, 9.6%) · TWIST2 (late, 4.5%)

Chan 2021 SCLC — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

24/34 samples pass the mult-gap threshold (≥37 cells); 50,886 malignant cells share one continuous, isotropic manifold (UMAP aspect 1.04). Samples & SCLC-A/N/P subtypes intermix (θ sweep 0.83→0.22); cell-cycle confound regressed out (mean |corr| 0.32→0.01).

Gate 2 — module geography → GO (non-branching)

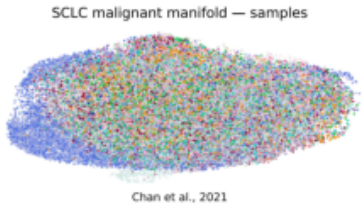
Single monotone axis: Gast & pSom co-vary (r=+0.73) and decline from the root (ρ −0.24/−0.45); Crest opposes both (r=−0.71/−0.88) and rises to a terminal readout (ρ +0.24). No bifurcation.

Gate 3 — hub cross-repression → NA

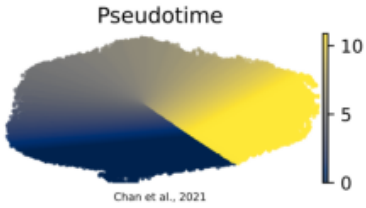
Non-branching manifold has no bifurcation/hub, so the cross-repression test does not apply.

STEP FIGURES

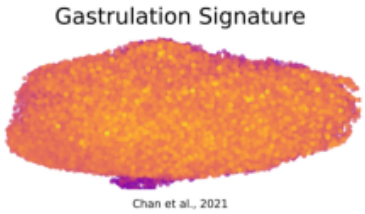
Manifold — samples



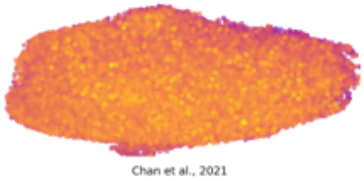
Pseudotime



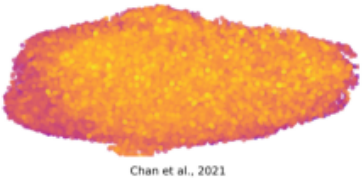
Gastrulation



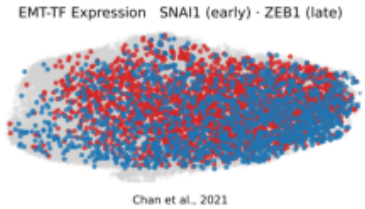
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Small-cell lung cancer (Chan et al., 2021), 10x. Master scope 37 patients / 86,662 cells; malignant compartment 51,052 cells (SCLC-A/N/P + minor NSCLC); 50,886 across 24 samples enter the trajectory.

Verdict.
Pattern reproduces as the non-branching case (Gate 1 GO / Gate 2 GO / Gate 3 NA): a single monotone gastrulation/somitogenesis → terminal neural-crest gradient, not resolved into Puram's two competing arms. SCLC is intensely proliferative, so a raw-embedding curve tracks cell cycle (ρ −0.79); regressing the cell-cycle scores out (post ρ=0.00) recovers developmental order. EMT SNAI1 → ZEB1.

Parameter choices

Min cells/sample	≥37 (mult-gap 92→15; keeps 24/34)	**Cell-cycle**	G1S+G2M regressed out (0.32→0.01)
Root node	tip 0 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=5; PCA50→20; spread=1.3
Trajectory	scFates curve N=30 λ=0.3 μ=100 (SA 2.06)	**EMT-TFs**	SNAI1 (root) → ZEB1 (terminus)

Chan 2021 SCLC/NSCLC — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

24/34 samples (≥37 cells) share one continuous manifold, no per-patient islands (radius-graph: one component; iLISI median 1.00→8.8 of 24 samples at θ=10).

Gate 2 — module geography → GO (non-branching)

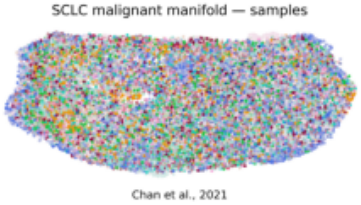
Single dominant pSom→Crest axis: pSom high over the neuroendocrine body, Crest anti-correlated (r=−0.89), rising to a terminal readout (crest-high decile mean pt=0.61). Gast and pSom co-vary (r=+0.28) — overlap, no competitive cross-repression.

Gate 3 — hub mutual exclusivity → NA

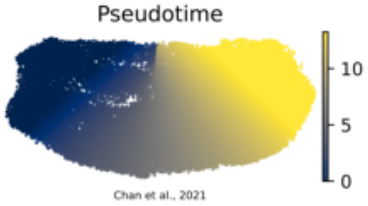
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

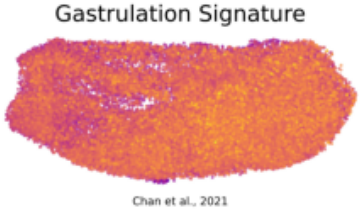
Manifold — samples



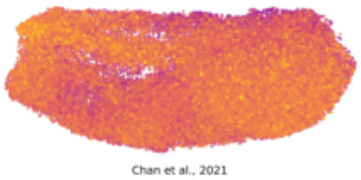
Pseudotime



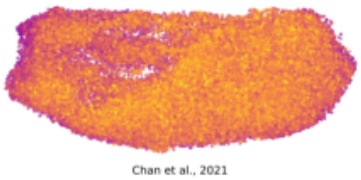
Gastrulation



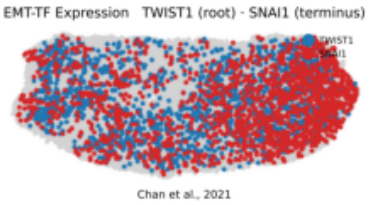
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Small-cell lung cancer (Chan et al., 2021), 10x — 34 samples / 51,052 malignant cells; predominantly SCLC (ASCL1/NEUROD1/POU2F3 neuroendocrine subtypes) with minor NSCLC. 24 samples (≥37 cells) → 50,997 cells enter the trajectory; whole malignant manifold (no lobe scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The SCLC malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. A PPT tree over-forks the compact cloud (8–15 forks, no single dominant Y), while an elastic curve is fold-free (self-approach 2.78); Gast and pSom co-vary rather than competing. The developmental axis is orthogonal to proliferation (module·g2m |ρ|≤0.15), so the whole manifold is kept. Caveat: pseudotime co-varies negatively with cycling (ρ−0.65), as the pSom-high root pole is the more proliferative one (Zhou/Bischoff precedent). EMT-TFs TWIST1 (early/root) → SNAI1 (late/terminus); ZEB1 most-expressed but pseudotime-flat; end-separation modest (Δpt=0.15).

Parameter choices

Min cells/sample	37 (24/34; gap 92→15)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=30 λ=0.3 μ=100
Root node	tip 2 = pSom/Gast-high, Crest-low	**EMT-TFs**	TWIST1 (early) → SNAI1 (late)
Geometry	Linear-pSom/Crest (non-branching)	**n_pcs**	20 (neighbours + iLISI)

Chan 2021 Small-Cell Lung Cancer — analysis summary

RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

24/34 samples (≥37 cells; dominant 92→15 spillover cliff) intermix on one continuous manifold — no per-patient islands (radius-graph: 1 detached cell; iLISI median 1.00→8.79 of 24); both cell-cycle compartments (Body 38,882 / Cycling 12,116) carry multiple samples.

Gate 2 — module geography → GO (non-branching)

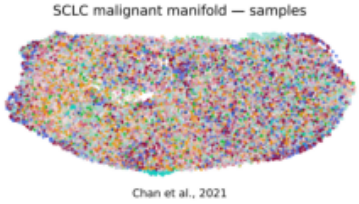
Single embryonic axis, no hub-into-arms. Gast & pSom co-vary (smoothed r=+0.28; within-patient median r=+0.06, 15/23 samples positive; joint-high enrichment 1.42) — overlap, not competition. Crest is an inverse terminal gradient (pSom↔Crest r=−0.89; joint pSom&Crest-high enrichment 0.20). No Crest-on-Gast-arm inversion to flag (there are no arms).

Gate 3 — hub mutual exclusivity → NA

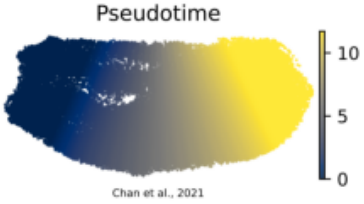
Non-branching: the PPT tree over-forks the compact cloud (9–17 spurious forks across the grid, no single dominant Y) and Gast/pSom co-vary rather than compete — there is no hub or branch pair to test.

STEP FIGURES

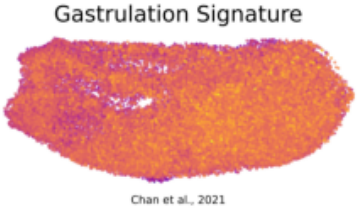
Manifold — samples



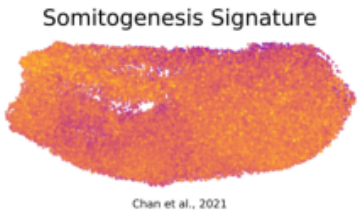
Pseudotime



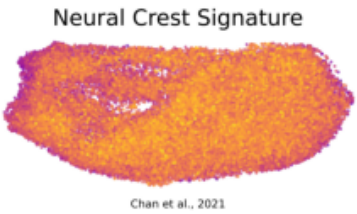
Gastrulation



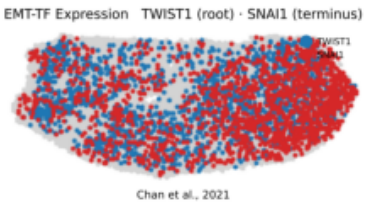
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
SCLC (Chan et al., 2021), 10x — 34 patients / 51,052 malignant cells (86,662 atlas). Threshold ≥37 keeps 24 samples (50,999 cells); whole malignant manifold (50,998 after island removal, no lobe scoping). Reported subtypes: SCLC-A 29,099 / SCLC-N 18,176 / SCLC-P 2,989 / NSCLC 788.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The SCLC malignant manifold is a single continuous axis: Gast and pSom overlap (co-vary within and across patients) while Crest is the inverse/terminal readout — the embryonic programmes are present but not resolved into two competing arms, so there is no Puram-style hub-and-spoke bifurcation and no hub to test. Subtype is fully patient-confounded (24/24 samples >80% one SCLC subtype), so Harmony-on-sample intermixes SCLC-A/N as expected. Modules are cell-cycle-free (|r| with cycling <0.14) and the pSom↔Crest axis strengthens in non-cycling cells. Caveat: this near-isotropic cloud (axis ratio ~1.6) has no strong 1-D trajectory — the curve tracks the dominant Gast/proliferation long-axis while the pSom→Crest gradient runs oblique, so Gate-2 geography is read off the signature embeddings, not the pseudotime ordering (Crest ρ vs pt only +0.07). EMT-TFs TWIST1 (early/root) → SNAI1 (late/terminus); ZEB1 most-expressed but pseudotime-flat.

Parameter choices

Min cells/sample	≥37 (24/34; dominant 92→15 cliff)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**n_pcs**	20 (swept 20/40; robust range)
Geometry	Linear (non-branching)	**EMT-TFs**	TWIST1 (early) → SNAI1 (late)

Chan 2021 SCLC — analysis summary

RUN 4 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

24 of 34 malignant-bearing samples (≥37 cells each) intermix on one continuous manifold with no per-patient archipelago (radius-graph: 118/50,999 island cells removed; iLISI median 6.17 of 24); proliferation lobe kept — modules are cell-cycle-free.

Gate 2 — module geography → GO (non-branching)

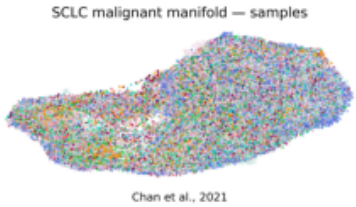
Single embryonic axis: Gast and pSom co-vary (smoothed r=+0.48 — overlap, not competitive cross-repression); Crest is an inverse terminal gradient rising where pSom falls (pSom↔Crest r=−0.80). No hub diverging into two arms; the Gast↔pSom co-variation strengthens in non-cycling cells (+0.48→+0.59).

Gate 3 — cross-repression → NA

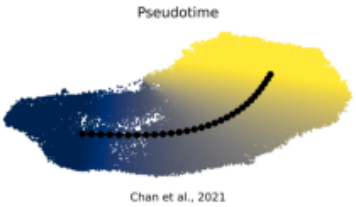
Non-branching manifold — a curve with 2 tips / 0 forks (PPT tree also 0 forks), so there is no branch pair and no convergence hub to test for cross-repression.

STEP FIGURES

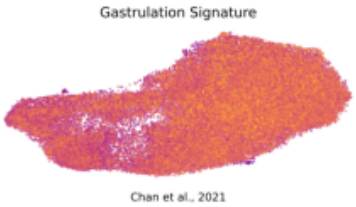
Manifold — samples



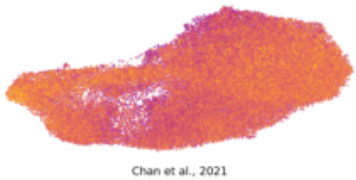
Pseudotime



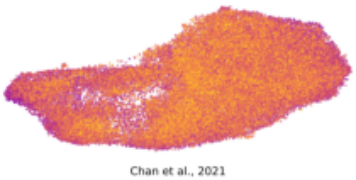
Gastrulation



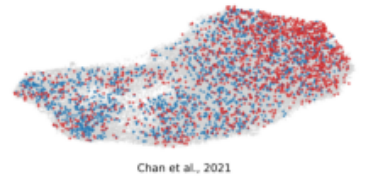
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
SCLC (Chan et al., 2021), 10x — 37 samples / 86,662 cells (CCCA master scope). Malignant only: 51,052 cells across 34 samples → min-cell threshold (≥37 cells/sample) keeps 50,999 cells across 24 samples → island removal → 50,881 cells enter the trajectory; whole manifold (no lobe scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Chan's SCLC malignant manifold is a single continuous gastrulation/somitogenesis → terminal-neural-crest gradient rooted at the Gast/pSom-high pole, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. SCLC-A/SCLC-N subtypes do not form two module-opposed arms, and the embedding carries a proliferation axis, but the embryonic modules are cell-cycle-free (|r| with cycling ≤0.16) and the Gast↔pSom co-variation strengthens in non-cycling cells (+0.48→+0.59; the pSom↔Crest anti-correlation persists, −0.80→−0.77), so the whole manifold is kept. The curve is fold-free (far-node approach / edge = 24.7). EMT-TFs (widened panel incl. FOXC/PRRX): TWIST1 (root-leaning) → SNAI1 (terminus-leaning), all weakly pseudotime-associated (|ρ|≤0.11) — consistent with a diffuse linear axis.

Parameter choices

Min cells/sample	≥37 (dominant 92→15 cliff)	**Integration**	Harmony θ=5; n_pcs=20
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=30, λ=0.3, μ=100
Root node	tip 2 = Gast/pSom-high	**Overdispersed HVG**	3,228
Geometry	Linear (non-branching); 2 tips / 0 forks	**EMT-TFs**	TWIST1 (root) · SNAI1 (terminus); |ρ|≤0.11

Zilionis 2019 NSCLC — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/7 samples (≥131 cells) intermix on one continuous manifold with no per-patient / per-histology islands (radius-graph: 0 detached; iLISI median 1.00→3.35 of 6); both Leiden lobes carry all 6 samples and both histologies (LUAD+LUSC). Read on the embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching, Crest-flagged)

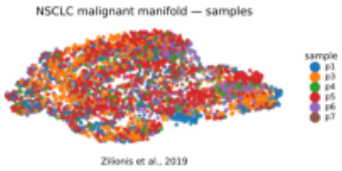
Single dominant Gast→pSom axis: gastrulation high at the origin declines along pseudotime (ρ=−0.58) while somitogenesis rises to a terminal readout (ρ=+0.77; pSom-high decile mean pt 0.99). Strong Gast↔pSom cross-repression (smoothed r=−0.53, near Puram −0.49) but joint-high enrichment 0.73 < 1 — mutually exclusive, one axis not a co-expression hub. FLAG: Crest is globally depleted and weakly U-shaped (high at BOTH tips, lowest mid; $t^2>0$, $p<0.001$) — not the reference terminal readout.

Gate 3 — hub mutual exclusivity → NA

Single monotone axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for mutual exclusivity (the Wu 2021 organised-cross-repression non-branching case).

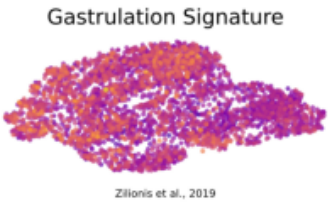
STEP FIGURES

Manifold — samples

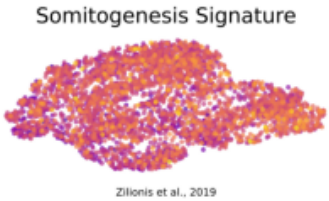


Pseudotime

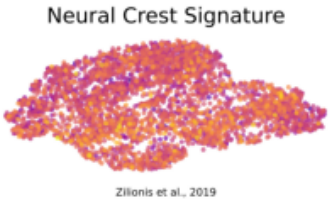
Gastrulation



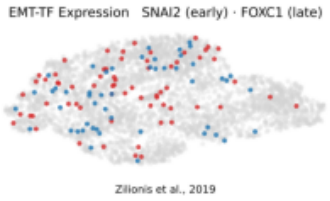
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
NSCLC (Zilionis et al. 2019), inDrop — 7 patients / 31,179 cells (master scope); malignant only: 3,920 cells across 6 malignant-bearing samples (drop 1 sub-100 spillover sample, p2) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zilionis's NSCLC malignant manifold is a single monotone gastrulation → somitogenesis axis with strong Gast↔pSom cross-repression (smoothed r=−0.53), but the decisive hub test shows the two are mutually exclusive (joint-high enrichment 0.73 < 1), so it is one axis, not Puram's co-expression hub — no bifurcation, and Gate 3 has no hub to test. The two density lobes are NOT a proliferative split (6.4% cycling; both lobes low) nor histological nor single-patient; the axis is cell-cycle-free (survives non-cycling cells, r=−0.53) so the whole manifold is kept. EMT-TFs are weak & root/early-loaded: SNAI2 (root) → FOXC1 (terminus) chosen by data over Puram's pair, though end-separation is small (Δpt=0.02). Crest is the weak link: globally depleted and only shallowly U-shaped (high at both tips, not terminal) — flagged.

Parameter choices

Min cells/sample	≥131 (dominant cliff 232→75, ratio 3.09; drops p2; 6/7)	**Integration**	Harmony θ=10; PCA50; n_pcs=20; spread=1.3
Island removal	radius-graph 6× median NN — 0 removed	**Root node**	Gast-high / pSom-low curve tip 2
Geometry	non-branching single axis (curve 2 tips, 0 forks, fold-free)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100 (ElPiGraph)
Scope	whole manifold (no lobe scoping)	**EMT-TFs**	SNAI2 (early) → FOXC1 (late), 3% floor

Zilionis 2019 NSCLC — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/7 samples (≥131 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 2.85 of 6); proliferation globally low (6.4% cycling, <12%/lobe) — no cycling compartment to scope out.

Gate 2 — module geography → GO (non-branching)

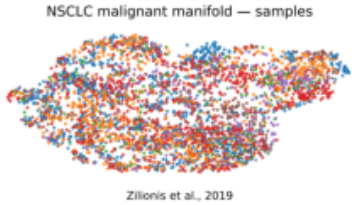
Single Gast(early)→pSom(late) cross-repression axis: Gast falls along pseudotime (ρ=−0.58), pSom rises (ρ=+0.74); smoothed Gast↔pSom r=−0.53. Crest a mid gradient (leans pSom side, r=−0.36 vs −0.51). No Gast/pSom co-expression hub — collapsed bifurcation.

Gate 3 — hub mutual exclusivity → NA

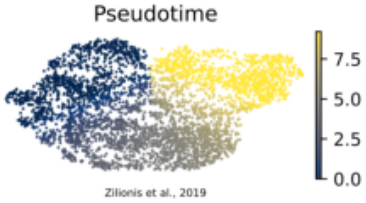
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

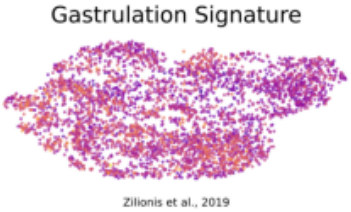
Manifold — samples



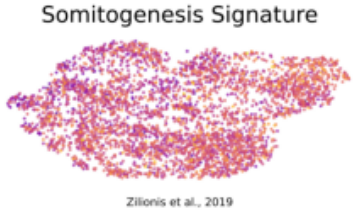
Pseudotime



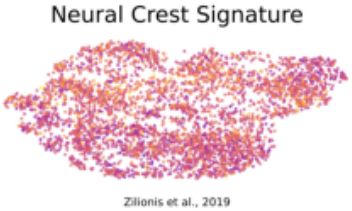
Gastrulation



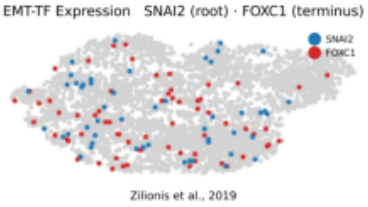
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Non-Small Cell Lung Cancer (Zilionis et al., 2019), inDrop — 7 patients / 31,179 cells (master scope); malignant only: 3,995 load, 3,920 across 6 samples (≥131) enter preprocessing; whole malignant manifold (no lobe scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zilionis NSCLC malignant manifold is a single monotone gastrulation → posterior-somitogenesis axis (Gast-high origin → pSom-high terminus), matching the developmental order; it lacks Puram's hub-and-spoke bifurcation (PPT tree hairballs, no Gast/pSom co-expression hub), so Gate 3 has no hub to test. The developmental axis spans low-cycling lobes, so the whole manifold is kept (Bischoff/Sun no-scoping precedent). EMT-TFs SNAI2 (early/root) → FOXC1 (late/terminus); canonical ZEB1/TWIST1 near the detection floor (1.5%/0.2%); EMT-TF signal weak, end-separation negligible (Δpt 0.03).

Parameter choices

Min cells/sample	131 (keeps 6/7; gap 232→75)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = Gast-high, Crest-low	**Geometry**	Linear-pSom/Crest (non-branching)

Zilionis 2019 NSCLC — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 tumour samples (75-cell p2 dropped at the geometric-mean threshold 131 over the 232→75 spillover cliff) intermix on one continuous connected component with no per-patient islands (radius-graph: 0 detached); iLISI median 1.00→3.26 of 6. Cycling (32%) is interspersed, not a detached proliferative lobe. Read on the regenerated embedding across a θ×n_pcs sweep.

Gate 2 — module geography → GO (non-branching)

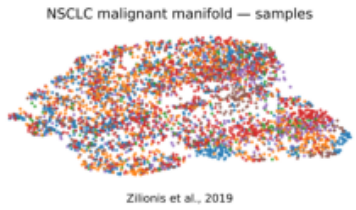
Organized Gast↔pSom cross-repression along one axis (smoothed spatial r=−0.58; Gast-high origin opposing a pSom-high end) with NO co-expression hub (Gast&pSom co-high enrichment 0.73<1). Inverse neural-crest gradient opposing both (Crest↔Gast r=−0.47, Crest↔pSom −0.37). Along pseudotime Gast falls (ρ=−0.60), pSom rises (ρ=+0.78). Crest not on a Gast arm — no flag.

Gate 3 — hub mutual exclusivity → NA

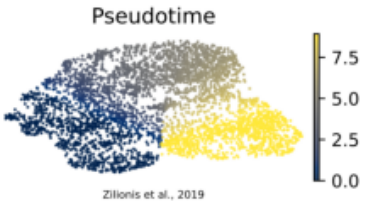
No bifurcation: the principal curve is robustly 2-tip/0-fork and fold-free across Nodes 15/20/30, while a PPT tree throws 10–20 spurious forks (a bush, not a Y), and there is no co-expression hub — so there is no branch pair to test for mutual Gast↔pSom exclusivity.

STEP FIGURES

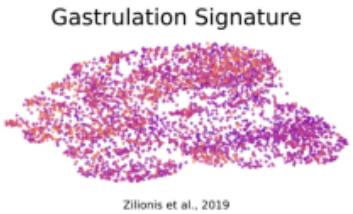
Manifold — samples



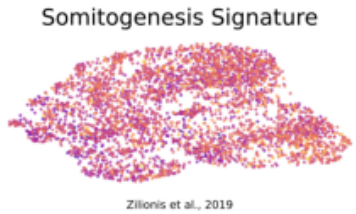
Pseudotime



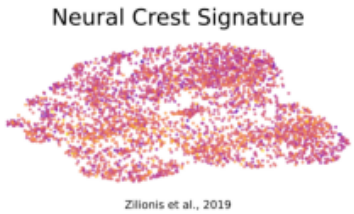
Gastrulation



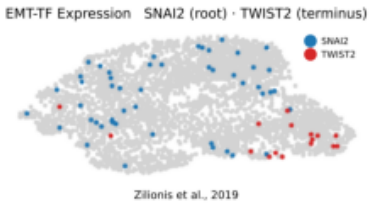
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
NSCLC (Zilionis et al., 2019), inDrop — 7 samples / 31,179 cells (master scope); malignant only: 3,995 cells, of which 3,920 across 6 samples enter preprocessing (75-cell p2 dropped as spillover); whole malignant manifold (single body, no scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zilionis's NSCLC malignant manifold is a single continuous axis of organized Gast↔pSom cross-repression (spatial r=−0.58) rooted at the Gast/pSom-high developmental pole, with an inverse neural-crest gradient; it lacks Puram's co-expression hub-and-spoke bifurcation (Gast&pSom co-high enrichment 0.73<1; PPT bush vs a fold-free 2-tip curve), so Gate 3 has no hub to test. The compartment is 32% cycling but forms one continuous component; the axis is developmental, not proliferative — the modules are cell-cycle-free (Gast 0/81, Crest 0/104 cell-cycle genes) and the Gast↔pSom cross-repression survives within non-cycling cells alone (r=−0.58→−0.59), pseudotime↔cycling near-zero (ρ=−0.14). EMT-TFs SNAI2 (root, 5.1%) → TWIST2 (terminus, 1.7%), sparsity disclosed.

Parameter choices

Min cells/sample	131 (232→75 cliff; drops p2)	**Integration**	Harmony θ=10; n_pcs=30; spread=1.3
Scope	whole manifold (single body)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high	**Geometry**	Linear-pSom/Crest (non-branching)

Zilionis 2019 NSCLC — analysis summary

RUN 4 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/7 samples (≥132 cells) intermix on one continuous manifold with no per-patient islands. An integration sweep (θ 0→10) read on the embedding gives one connected kNN component at every θ (iLISI 1.6→3.0 of 6); mixing plateaus at θ=10. Mixed histology (LUSC p1 + LUAD p3–p7) is interspersed (LUSC kNN self-enrichment 2.3×, far below an island); manifold roundish (aspect 1.1).

Gate 2 — module geography → GO (non-branching)

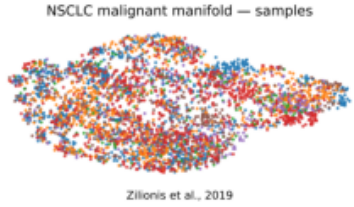
Three embryonic programs present and MUTUALLY EXCLUSIVE at single-cell resolution (raw Gast~Crest −0.47, pSom~Crest −0.48; all smoothed cross-repressions p<0.001 vs a permutation null — REAL, not a smoothing artifact). No Gast/pSom co-expression hub (co-high enrichment 0.39) — no bifurcation. Along pseudotime Gast declines (ρ=−0.44) and pSom rises (ρ=+0.48). Crest FLAGGED: a real anti-correlated territory but U-shaped/non-terminal (ρ=+0.01, p=0.86; quadratic-U p<0.001).

Gate 3 — hub mutual exclusivity → NA

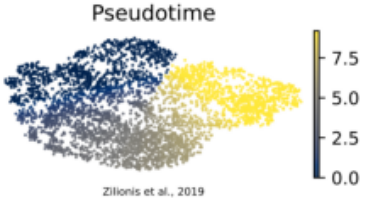
Single non-branching axis — the curve is 2 tips / 0 forks and fold-free (self-approach 1.68), tree fits over-fragment (5 tips/3 forks, unstable across θ). No Gast/pSom-overlap hub, so no branch pair to test.

STEP FIGURES

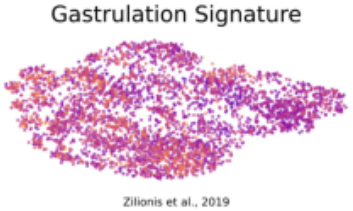
Manifold — samples



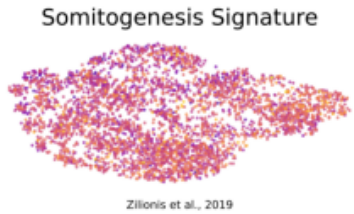
Pseudotime



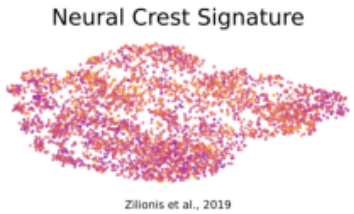
Gastrulation



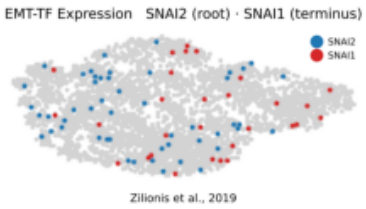
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
NSCLC (Zilionis et al., 2019), inDrop — 7 samples / 31,179 cells (master scope); malignant only: 3,995 cells; after 132-cell thresholding 3,920 across 6 tumour-bearing samples; island removal 3,920 (one connected component, 0 removed) enter the trajectory (whole manifold, no scoping).

Verdict.
Non-branching pattern reproduces with a Crest flag — Gate 1 GO, Gate 2 GO, Gate 3 NA. A single monotone Gast → pSom developmental axis (gastrulation-high origin → somitogenesis-high terminus); Crest is a real but non-terminal, off-axis third territory, so the manifold lacks Puram's hub-and-spoke bifurcation and Gate 3 has no hub to test. The three modules are cell-cycle-free (Gast 0/81, pSom 2/98, Crest 0/104 cell-cycle genes), cycling is low and diffuse (6.4%), and the Gast~pSom anti-correlation survives in non-cycling cells (−0.33→−0.33) — the axis is developmental, not proliferative. EMT-TFs weak/early-loaded: SNAI2 (root) / SNAI1 (terminus) by pseudotime, FOXC1 the best-expressed factor.

Parameter choices

Min cells/sample	132 geom-mean (cliff 232→p2 75; 6/7)	**Integration**	Harmony θ=10; 20 PCs; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Poles**	Origin ~2,070 / Terminus ~1,850

Zilionis 2019 NSCLC — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

All 6 samples and both histologies intermixed on ONE connected manifold (sample iLISI 3.4/6 at θ=10; 0 detached islands). Mixed + continuous — judged on the embedding, not a scalar.

Gate 2 — module geography → GO (non-branching)

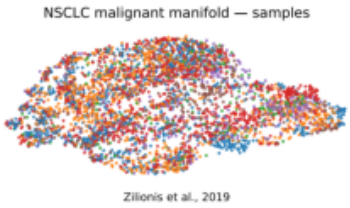
Shared developmental gradient: monotone Gast(origin, ρ=−0.59) → pSom(terminus, ρ=+0.76) axis; Crest an inverse off-axis readout. Present but collapsed onto one axis, not two competing arms.

Gate 3 — hub cross-repression → NA

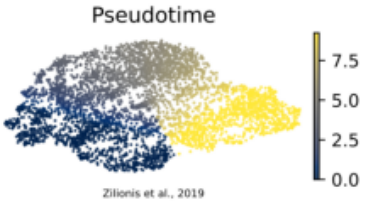
No Gast/pSom co-expression HUB (co-occurrence enrichment 0.07 smoothed / 0.70 raw, both <1) and no stable branch point (transcriptome PPT = 2 tips/0 forks across nodes × seeds). Non-branching → NA.

STEP FIGURES

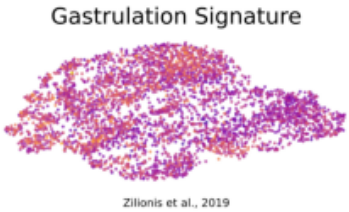
Manifold — samples



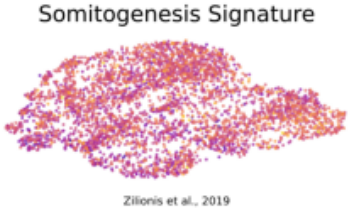
Pseudotime



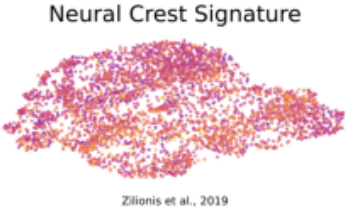
Gastrulation



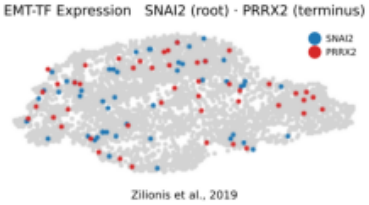
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
NSCLC (Zilionis et al., 2019), inDrop — 3,920 malignant cells (6/7 samples; mixed LUSC + LUAD histology). Geometry: LINEAR (non-branching).

Verdict.
The embryonic pattern reproduces on a single collapsed axis — a gastrulation-high origin resolving to a somitogenesis-high terminus, with neural crest as an inverse off-axis gradient (flagged: non-terminal). Not the Puram hub-and-spoke bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs SNAI2 (early/root) ↔ PRRX2 (late/terminus); weak, root-loaded.

Parameter choices

Threshold	≥131 malignant (geom-mean across 232→75 count cliff, ratio 3.09) → drops p2 (75) → 3,920 cells	**Overdispersed genes**	248 (scFates GAM); PCA 50 PC, n_pcs=20 (~50% variance)
Integration	Harmony θ=10 (iLISI plateau 1.4→3.4/6); n_neighbors=30; UMAP spread=1.3; 0 islands	**Trajectory**	scFates CURVE (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips/0 forks, fold-free (SA 2.45); corr(pt,DC1)=−0.85
Root	Gast-high tip (Crest is late, never the root)	**EMT-TFs**	SNAI2 (early/root) ↔ PRRX2 (late/terminus); weak, root-loaded

Zilionis 2019 NSCLC — analysis summary

RUN 6 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 samples (≥131-cell floor) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI 3.42/6; same-sample neighbour fraction 0.91→0.40 across the θ sweep). Roundish body (UMAP aspect 1.13), single connected component.

Gate 2 — module geography → NO-GO

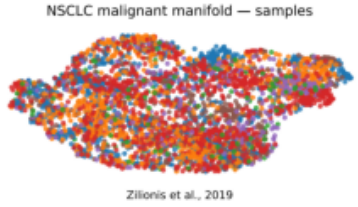
Embryonic pattern absent. In score space the modules look anti-correlated (Gast/pSom −0.53, ≈ Puram's branching −0.49) — but this is a shared-control-pool artifact: on RAW mean-expression all three POSITIVELY co-activate (Gast/pSom +0.58, Gast/Crest +0.45, pSom/Crest +0.51; survives library-size regression). One shared developmental field, not competing arms. No convergence hub (all joint-high enrichments depleted 0.07–0.62); Crest not terminal (high-decile pt 0.56, mid-axis); every principal-tree tip is pSom-dominant (no Gast/Crest arm).

Gate 3 — cross-repression → NA

Not branching and Gate 2 NO-GO — no branch pair / hub to test for mutual exclusivity.

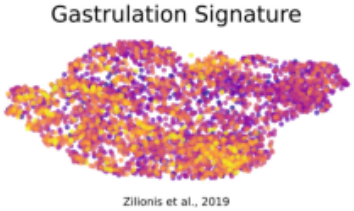
STEP FIGURES

Manifold — samples

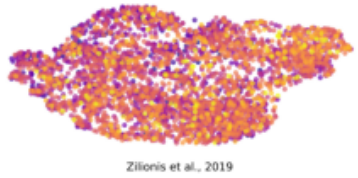


Pseudotime

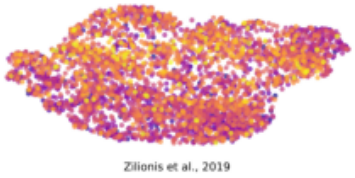
Gastrulation



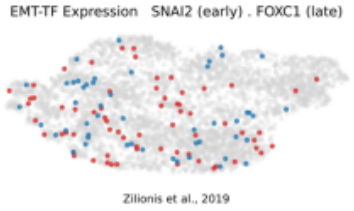
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
NSCLC (Zilionis et al., 2019), inDrop (droplet UMI) — 7 patients / 3,995 malignant cells; 3,920 across 6 samples (all clear the ≥131 floor) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Embryonic pattern does NOT reproduce — Gate 1 GO / Gate 2 NO-GO / Gate 3 NA. The malignant manifold is a single continuous, sample-integrated cloud organized by carcinoma metaprograms (Alveolar / cell-cycle / Stress / IFN; Moran's I 0.54 / 0.24 / 0.22 vs the embryonic modules' 0.10–0.26), not by an embryonic hub-and-arms axis. The three modules form one shared low-amplitude activation field (raw co-activation +0.45–+0.58): no Gast/pSom co-expression hub, no Gast or Crest arm (all tree tips pSom-dominant), and Crest sits mid-pseudotime (0.56), not terminal — the Peng2019 / Hwang2022 NO-GO signature, opposite Steele 2020's terminal-Crest GO. The score-space −0.53 that read as branching in five prior attempts is a union-control-pool subtraction artifact, not cross-repression. EMT-TF step run unconditionally: SNAI2 (early) → FOXC1 (late) chosen by data over Puram's SNAI1/ZEB1, all near the detection floor with negligible end-separation (Δpt ≈ 0) — consistent with no EMT axis.

Parameter choices

Min cells/sample	≥131 (gap 232→75, r3.09; drops p2)	**Integration**	Harmony θ=10; 20 PCs; spread 1.3
Scope	whole manifold (no scoping)	**Trajectory**	curve N=20, λ=0.3, μ=100; fold-free 2.46
Root node	Gast-high, Crest-low tip (origin)	**Overdispersed**	248 genes; PPT tree rejected
Geometry	Non-branching; embryonic pattern absent → NO-GO	**EMT-TFs**	SNAI2 (early) → FOXC1 (late); near floor, Δpt≈0

Song 2019 NSCLC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

3 tumour samples (P4/P3/P1, ≥100-cell floor after dropping the 3-cell P4_Normal spillover) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→1.65 of 3); a single Leiden lobe carries the whole trajectory.

Gate 2 — module geography → GO (non-branching)

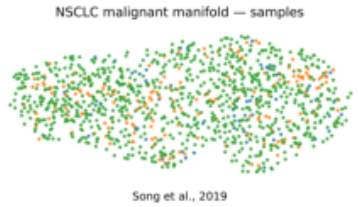
Single dominant (Gast/pSom)→Crest axis: Gast/pSom high over the origin decline along pseudotime (Gast ρ=−0.72, pSom ρ=−0.37); Crest anti-correlated in space (Gast↔Crest r=−0.89), rising to a terminal readout (ρ=+0.74; Crest-high decile mean pt=0.84). Joint pSom&Crest-high depleted (enrichment 0.54) — an inverse gradient, not a co-expression hub.

Gate 3 — cross-repression → NA

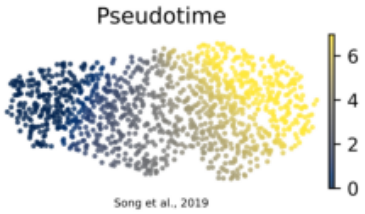
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

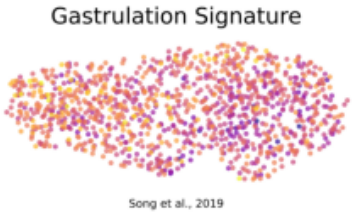
Manifold — samples



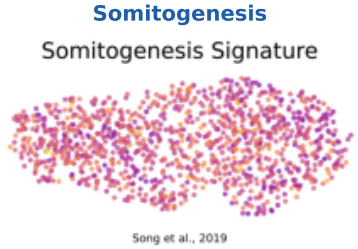
Pseudotime



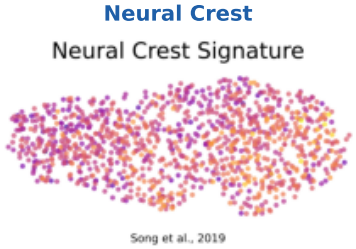
Gastrulation



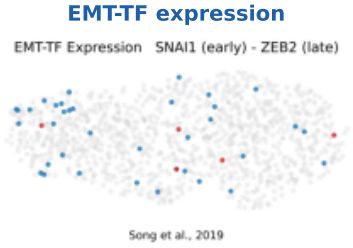
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
NSCLC / lung adenocarcinoma (Song et al., 2019), 10x — 6 samples / 8,772 cells (master scope); malignant only: 1,144 cells across 3 tumour samples (drop the 3-cell P4_Normal spillover); whole malignant manifold (single lobe, no scoping). Non-branching Linear-pSom/Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Song's NSCLC malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient rooted at the Gast/pSom-high pole; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The compartment is 57% cycling but forms one Leiden lobe — the axis is developmental, not proliferative (Gast/Crest cell-cycle-free; Gast↔Crest anti-correlation survives within non-cycling cells, r=−0.87; pseudotime↔cycling ρ=−0.19). EMT-TFs are detection-limited in this sparse 1.1k-cell compartment: only SNAI1 (root, 12.5%) and ZEB2 (terminus, 1.6%) survive the gene filter.

Parameter choices

Min cells/sample	√(62·3)≈13 (62→3 cliff) → 3 tumour samples, 1,144 cells	**Scope**	whole manifold (single lobe, no scoping)
Root node	tip 2 = Gast/pSom-high	**Integration**	Harmony θ=10; n_pcs=20; spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100	**Geometry**	Linear-pSom/Crest (non-branching)
EMT-TFs	SNAI1 (root, 12.5%) → ZEB2 (terminus, 1.6%); rest below detection		

Jansky 2021 Neuroblastoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

13/14 samples (≥269 cells) intermix on one continuous manifold with no per-patient islands; the kNN graph is a single connected component (27,987 cells; one 129-cell spillover sample dropped). Same-sample-neighbour fraction 0.233 vs mixing null 0.111 — well-integrated across 13 patients.

Gate 2 — module geography → GO (non-branching)

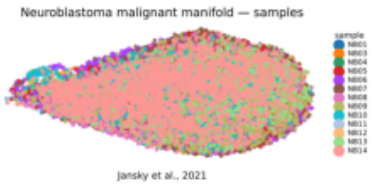
Gast/pSom decline together along pseudotime (ρ=−0.44/−0.14) and co-vary positively (r=+0.55); Crest is spatially anti-correlated (r=−0.72/−0.80) and rises to a terminal readout (ρ=+0.36; crest-high decile mean pt=0.72). Modules ride a dominant noradrenergic-differentiation axis — present, but not resolved into two competing arms.

Gate 3 — hub mutual exclusivity → NA

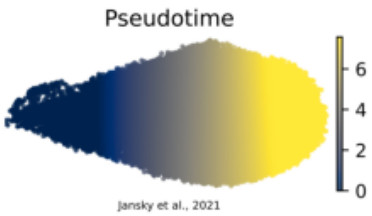
Single monotone differentiation axis — no bifurcation and no Gast/pSom-overlap hub, so there is no branch pair to test for competitive cross-repression.

STEP FIGURES

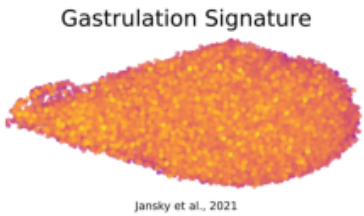
Manifold — samples



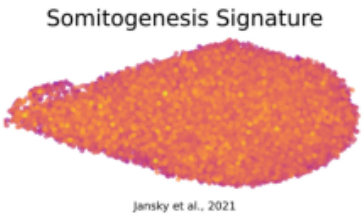
Pseudotime



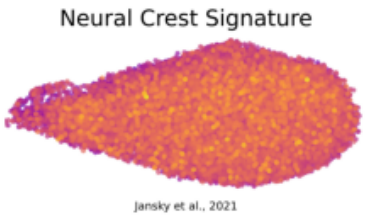
Gastrulation



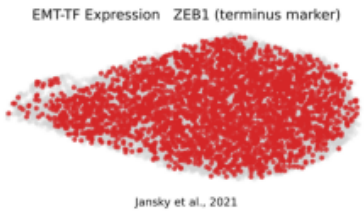
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Neuroblastoma (Jansky et al., 2021), 10x — childhood adrenal NB; malignant only: 28,116 load, 27,987 across 13 samples on one continuous manifold; the full compartment carries the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Jansky's manifold is a single noradrenergic-differentiation axis (progenitor → differentiating neuronal); embryonic modules ride it as a secondary Gast/pSom → terminal-Crest gradient, lacking Puram's hub-and-spoke, so Gate 3 has no hub to test. EMT-TFs monotone: ZEB1 (42% det) rises weakly to the terminus; canonical SNAI1/2, TWIST1, PRRX1, FOXC1 sit near the detection floor (no mesenchymal arm).

Parameter choices

Min cells/sample	≥269 (keeps 13/14; gap 561→129, r4.35)	**Integration**	Harmony(sample) θ=5; 50PC→20; UMAP spread=1.3
Malignant states	Noradren. progenitor 17,333 / Differentiating 10,654	**Trajectory**	scFates curve N=20 λ=0.3 μ=100 (fold-free, 2 tips)
Root node	tip 0 = Gast/pSom-high, Crest-low	**Manifold**	full malignant (no lobe split); 1 kNN component

Kildisiute 2021 NB — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

2 populated samples (≥101 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached). Manifold resolves into two lobes; trajectory is scoped to the Embryonic lobe (1,504) — the Cycling lobe (216) confounds Gast with proliferation.

Gate 2 — module geography → GO (non-branching)

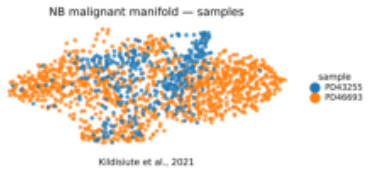
Gast/pSom co-vary (r=+0.91); along pseudotime Gast is flat (ρ=−0.01) and pSom declines (ρ=−0.26). Crest is anti-correlated with both (r=−0.66/−0.84), rising to a terminal readout (ρ=+0.40; neural-crest-high decile mean pt=0.70).

Gate 3 — hub mutual exclusivity → NA

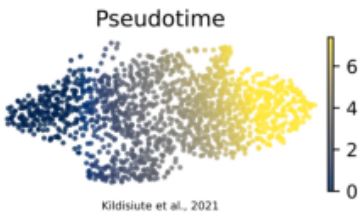
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

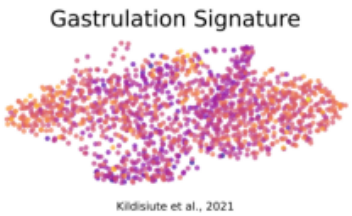
Manifold — samples



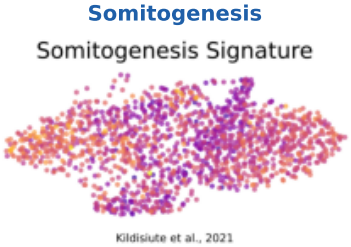
Pseudotime



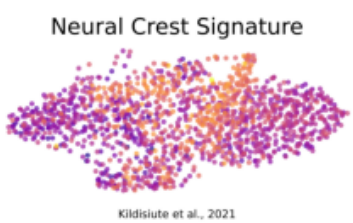
Gastrulation



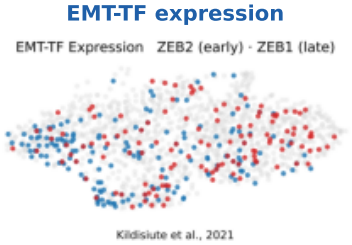
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Neuroblastoma (Kildisiute et al., 2021), 10x + CEL-seq2 — 21 patients / 19,723 cells (master scope). Multi-platform with platform-exclusive patients, so the 10x (UMIcounts) leaf is resolved; malignant only: 1,766 load, 1,720 across 2 samples (≥101) enter preprocessing; Embryonic compartment (1,504).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The neuroblastoma malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Unlike Lee, the axis is confined to the non-cycling lobe (Gast/pSom anti-correlate under proliferation), so the trajectory is scoped to it (Kürten lobe-identity precedent). EMT-TFs ZEB2 (early/root) → ZEB1 (late/terminus); FOXC/PRRX/SNAI near the detection floor; end-separation weak (Δpt=0.25).

Parameter choices

Min cells/sample	≥101 (keeps 2/5; gap 425→24)	**Scope**	10x leaf + Embryonic lobe
Root node	tip 2 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=10; PCA50; spread=1.3
Trajectory	scFates curve N=20 λ=0.3 μ=100	**Compartments**	Embryonic 1,504 / Cycling 216

Kildisiute 2021 neuroblastoma — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant cells form a per-platform / per-patient archipelago. At the best-mixing integration (θ=10) the kNN graph is a single component — the scalar a naïve test reads as "integrated" — yet the embedding is organised BY SAMPLE: platforms NEVER share a neighbourhood (iLISI(technology)=1.00/2, pinned across the entire θ sweep 2–50), and at low θ the manifold is two disconnected blobs split cleanly by platform. The platform batch is perfectly confounded with patient (each patient on one platform only), so Harmony-on-sample has no anchor. Even within one platform the two patients barely mix (iLISI ≈1.8/2) — inter-patient / inter-platform (technical + CNV/clonal) heterogeneity, not a shared cell-state continuum.

Gate 2 — module geography → N/A

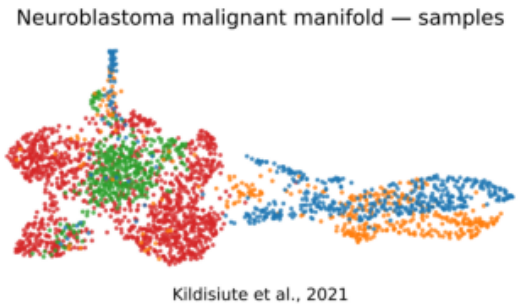
Not reached. Gate 2 tests a Gast/pSom hub and Crest geography on a fitted manifold; with Gate 1 NO-GO no manifold / trajectory is fit, so the module-geography gate does not apply.

Gate 3 — cross-repression → N/A

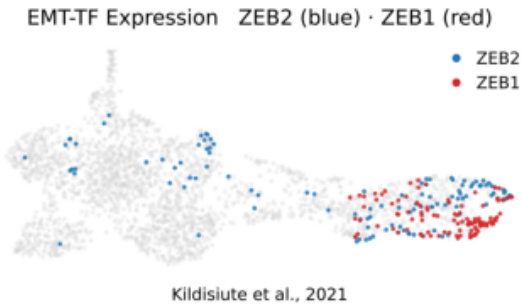
Not reached (branching-only, and downstream of Gate 2). No hub, no bifurcation, no hub r.

STEP FIGURES

Malignant manifold — samples



EMT-TF expression



Dataset.
Neuroblastoma (Kildisiute et al., 2021), Neuroendocrine — 10X (UMI) + CEL-seq2 (TPM) merged (inner join, 32,118 shared genes). ≥166-cell threshold keeps 4/14 samples (2,632 cells); a sparse 10-sample tail is dropped.

Verdict.
The embryonic-program pattern does NOT reproduce in Kildisiute 2021 neuroblastoma: the malignant compartment is a per-platform / per-patient archipelago with no sample-integrated, continuous cell-state manifold. Gate 1 NO-GO — analysis stops before the trajectory. ZEB2 / ZEB1 are the broadly-expressed EMT-TFs, reported unconditionally.

Parameter choices

Data merge	10X UMI + CEL-seq2 TPM, inner join (32,118 genes)	**Sample threshold**	≥166 cells (gap 412→67, 6.15×); keeps 4/14 (2,632 cells)
Gene / PCA	1,513 overdispersed; n_pcs=20 (PC1 ≈57% variance = CNV axis)	**Integration**	Harmony on sample, θ 2/5/10/20/50; θ=10 best; iLISI(tech)=1.00/2 throughout
Root / geometry	None — no trajectory fit (Gate 1 NO-GO)	**EMT-TFs shown**	ZEB2 (45.2%) · ZEB1 (44.4%); FOXC/PRRX added, VIM excluded

Dong 2020 neuroblastoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

13/13 tumours (≥100 cells) intermix on one continuous manifold (1 kNN component; iLISI source 1.00→1.38, sample 3.38 of 13). The two capture groups mix through the body; the small near-pure pockets are the two largest MYCN-amplified tumours — genuine CNV states, not a per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

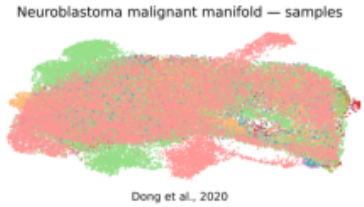
Single pSom→Crest axis: pSom high across the body, Crest anti-correlated (r=−0.65) as an inverse gradient (joint pSom&Crest-high enrichment 0.38); Gast marks the origin, declining along pseudotime (ρ=−0.63). No hub-and-spoke divergence — gradient, not hub.

Gate 3 — cross-repression → NA

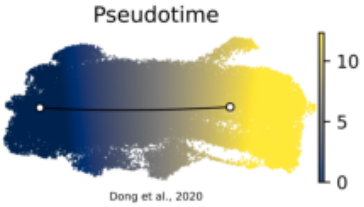
Non-branching: PPT forks scale with node count and the curve is 2-tip / 0-fork, so there is no bifurcation and no Mixed-type hub to test.

STEP FIGURES

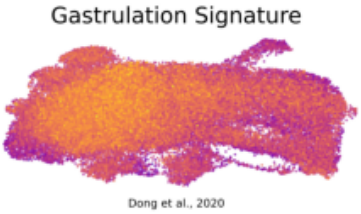
Manifold — samples



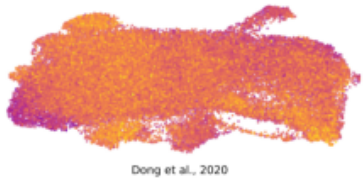
Pseudotime



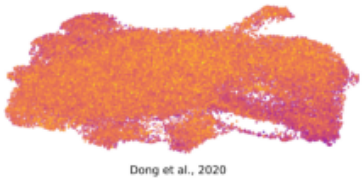
Gastrulation



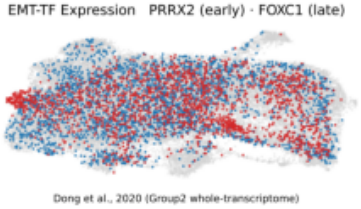
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Neuroblastoma (Dong et al., 2020), 10x — 13 tumours / 55,190 cells; malignant only: 47,384 cells (all 13 tumours ≥100) enter preprocessing. Captured in two panels; the joint manifold is built on the 12,113-gene shared panel so all tumours share one feature space.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The neuroblastoma malignant manifold is one continuous gastrulation/somitogenesis → neural-crest-inverse axis (low-cycling origin → proliferative terminus), matching developmental order; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub. The curve tracks proliferation (pt↔cycling ρ=+0.80), but the modules are cell-cycle-free (Gast 0/81, Crest 0/104 cc-genes) and pSom↔Crest anti-correlation survives in non-cycling cells (r=−0.60) — developmental, so the whole manifold is kept (cycling co-variation a caveat). EMT-TFs PRRX2 (early) → FOXC1 (late, ρ=+0.15).

Parameter choices

Min cells	≥100 floor (13/13; gap 1.06×)	**Integration**	Harmony θ=10; 20 PCs; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Body 35,196 / Cycling 12,188 (modules cc-free)
Geometry	Linear (non-branching); 2 tips / 0 forks	**EMT-TFs**	PRRX2 (early) → FOXC1 (late); Group2 whole-transcriptome

Zhang 2022 HGSOC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

20/22 samples (≥41 cells) / 11 patients intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); both lobes (Body 8,002 / Terminal cap 756) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

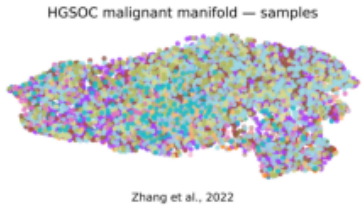
Gast/pSom co-vary (r=+0.55) and decline along pseudotime (ρ=−0.65/−0.80); Crest anti-correlated (r=−0.83/−0.82), rising to a terminal readout (ρ=+0.87; crest-high decile mean pt=0.94).

Gate 3 — hub mutual exclusivity → NA

Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

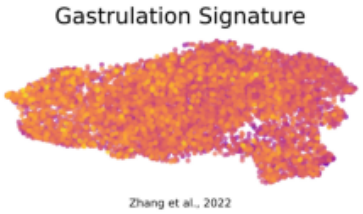
STEP FIGURES

Manifold — samples

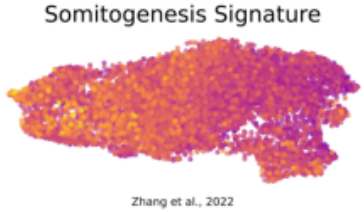


Pseudotime

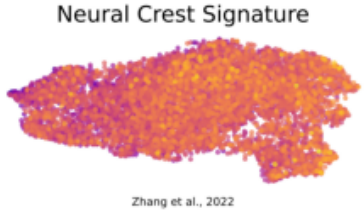
Gastrulation



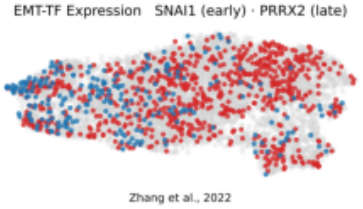
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-Grade Serous Ovarian Carcinoma (Zhang et al., 2022), 10x — 22 samples / 11 patients / 51,786 cells (master scope); malignant only: 8,806 load, 8,758 across 20 samples (≥41) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zhang's HGSOC malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The developmental axis is orthogonal to cycling (ρ=−0.003) — no proliferation confound, and both lobes are low-cycling, so the whole manifold is kept (Bischoff/Sun no-scoping precedent). EMT-TFs SNAI1 (early/root) → PRRX2 (late/terminus); canonical ZEB1/TWIST near the detection floor (6.5%/3.1%); EMT-TF end-separation weak (Δpt=0.29).

Parameter choices

Min cells/sample	≥41 (20/22; gap 57→30)	**Integration**	Harmony θ=5; PCA 50PC; spread=1.3
Scope	whole manifold (no lobe scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Body 8,002 / Terminal cap 756

Geistlinger 2020 HGSOC — analysis summary RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

5 patients share ONE continuous manifold (single kNN component; iLISI 1.87/5, limited by 54% T59 dominance). θ=5 resolves a T59-only under-integration lobe seen at θ=2. Judged on the embedding.

Gate 2 — module geography → GO (non-branching)

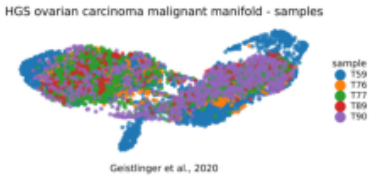
Embryonic pattern present, non-branching: pSom↔Crest single inverse axis (r=−0.95 smoothed); Gast co-varies with pSom (+0.61), both oppose Crest. Crest is the terminal gradient (crest-high decile at pseudotime 0.98). Not a proliferation artifact — pSom↔Crest holds in non-cycling cells (r=−0.62). Crest at the terminus, not the Gast arm — no flag.

Gate 3 — hub cross-repression → NA

Branching-only. No bifurcation (fold-free 2-tip curve; PPT branch-node counts unstable 3–8, no reproducible fork; Gast⊥pSom, r≈0) — no hub cross-repression test. Non-branching case, per §1–2.

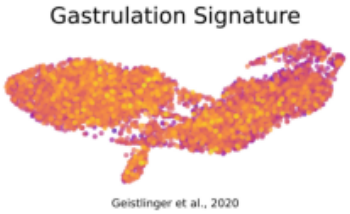
STEP FIGURES

Manifold — samples

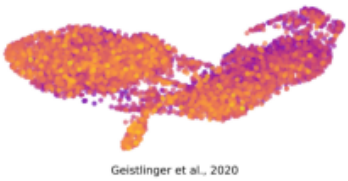


Pseudotime

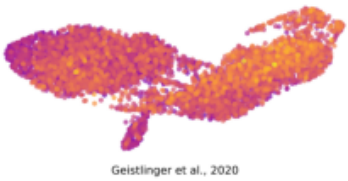
Gastrulation



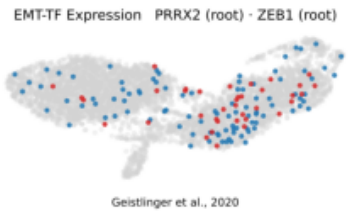
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-Grade Serous Ovarian Carcinoma (Geistlinger et al., 2020), 10x — 5,914 malignant cells (5 patients). Harmony integration on sample (θ=5), principal-curve trajectory (scFates) on the whole manifold.

Verdict.
The embryonic-program pattern REPRODUCES as the non-branching case: one continuous gastrulation/somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high origin, with Crest an inverse terminal gradient — a strict linear subset of the Puram branching hub. Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. No hub / no cross-repression (branching-only).

Parameter choices

Min-cell threshold	≥100 floor → 5/5 kept (gap at head 3180→1298; precedent floor)	**n_pcs / integration**	20 PCs · Harmony θ=5 (merges T59 under-integration lobe)
Island removal	radius-CC, 17-cell speck (740-cell θ=2 lobe was under-integration, kept)	**Geometry**	Non-branching · scFates curve (fold-free 2-tip, self-approach 2.97)
Root	Gast/pSom-high tip (Crest-low); Crest developmentally late	**Subtype annotation**	TCGA + 3CA MPs + Leiden (~85% DIF, continuous gradient)
EMT-TFs shown	PRRX2 · ZEB1 (both root/mesenchymal-enriched, opposite Crest)		

Geistlinger 2020 HGSOC — analysis summary RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

5/5 samples (≥259 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 1 speck, 25 cells removed; iLISI median 1.00→2.23 of 5); both compartments (Embryonic 2,857 / Cycling-Metabolic 3,049) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

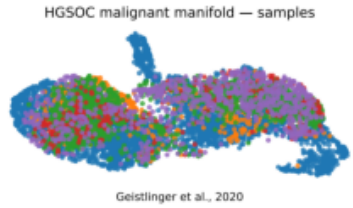
Single Gast/pSom → Crest axis: Gast & pSom co-vary (r=+0.63) and decline along pseudotime (ρ=−0.79/−0.63); Crest anti-correlated (r=−0.79/−0.95), rising to a terminal readout (ρ=+0.79; Crest-high decile mean pt=0.98). Global strength pSom>Gast>Crest-neg (Linear-pSom/Crest).

Gate 3 — cross-repression → NA

All four PPT-tree tips are pSom-leaning (no Gast arm) — the “forks” are density + cycling structure inside one co-varying programme, not a bifurcation. No Mixed hub, no branch pair to test.

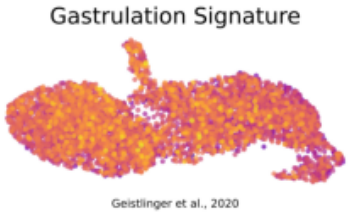
STEP FIGURES

Manifold — samples

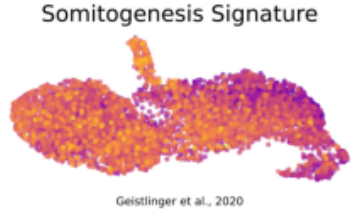


Pseudotime

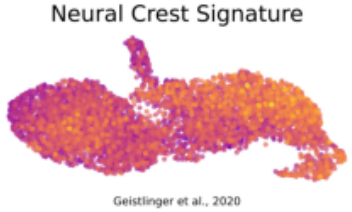
Gastrulation



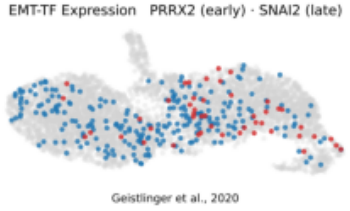
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-grade serous ovarian carcinoma (Geistlinger et al., 2020), 10x — 5 patients / 41,031 cells (master scope); malignant only: 5,931 load, 5,906 across 5 samples (≥259) after a 25-cell speck; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Geistlinger’s HGSOC malignant manifold is a single monotone Gast/pSom → terminal-neural-crest gradient, matching the developmental axis; it lacks Puram’s hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both density lobes (cycling is a weaker secondary gradient, ρ(cyc,pt)=−0.47), so the whole manifold is kept (Bischoff/Sun/Zhou no-scoping precedent). EMT-TFs are a root/mesenchymal readout, all declining from the Gast/pSom-high root: PRRX2 (root-end, steepest) → SNAI2 (terminus-end, least declining); canonical ZEB1/TWIST near the detection floor (6.2%/2.0%).

Parameter choices

Min cells/sample	≥100 (keeps 5/5; top-of-distribution gap, no thin tail)	**Scope**	whole manifold (no scoping)
Integration	Harmony θ=10; n_pcs=20; spread=1.3	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Embryonic 2,857 / Cycling 3,049
EMT-TFs	PRRX2 (root) → SNAI2 (terminus)	**Geometry**	Linear-pSom/Crest (non-branching; fold-free 2.99)

Izar 2020 HGSOC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Both platforms and all 11 retained samples intermix on one connected manifold; the platform batch that sample-only Harmony leaves (same-platform frac 1.00) is removed by joint platform+sample Harmony (→0.65). No islands.

Gate 2 — module geography → GO (non-branching)

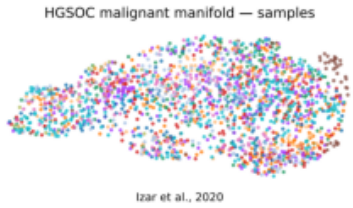
Shared Gast/pSom gradient (geography r=+0.53; reproduced within each platform) with Crest the inverse, terminal end (Crest~Gast −0.68, Crest~pSom −0.92; Crest-high decile at pt 0.81).

Gate 3 — cross-repression → NA

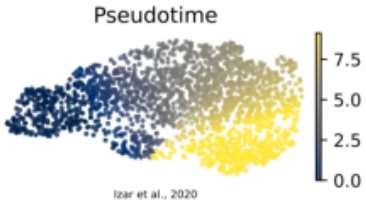
Non-branching principal curve (0 forks across Nodes 15–30), so there is no hub to test.

STEP FIGURES

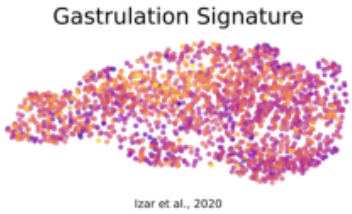
Manifold — samples



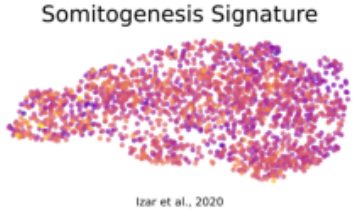
Pseudotime



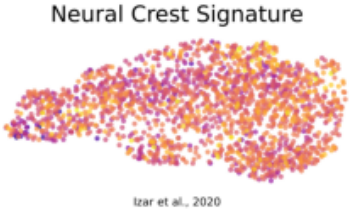
Gastrulation



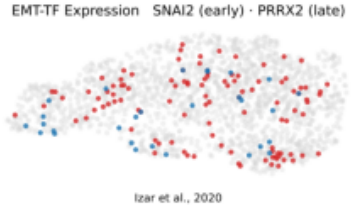
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
HGSOC malignant ascites (Izar et al., 2020), two platforms (SmartSeq2 1,158 + 10x 751 malignant); 1,909 loaded → 1,904 after thresholding. Both leaves concatenated on the shared gene space; platform treated as a batch covariate alongside sample.

Verdict.
The embryonic-program pattern reproduces as a single non-branching axis — a Gast/pSom-high developmental origin flowing to a neural-crest-high terminus, with proliferation orthogonal to (not aligned with) the axis.
Consistent with the Linear-pSom/Crest outcome seen across most of the atlas.

Parameter choices

Min cells/sample	10 (mult-gap 21→5; keeps 11/12)	**Integration**	Harmony [platform, sample] θ=10; PCA20
Scope	whole manifold (no lobe scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip Gast/pSom-high, Crest-low	**Geometry**	Linear (non-branching); 0 forks
EMT-TFs	SNAI2 (root) → PRRX2 (terminus)		

Nath 2021 HGSOC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on 15 retained patients integrates the manifold: iLISI median 1.00 → 5.50 (of 15). Read on the regenerated embedding + per-neighbourhood iLISI — patients intermix across one connected component (a 3-cell speck removed). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

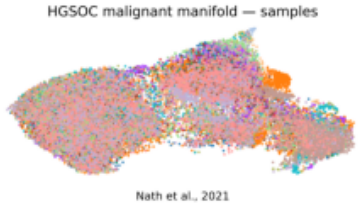
Over the low-cycling body a single (Gast/pSom) → Crest gradient: along pseudotime Gast/pSom fall (ρ −0.22 / −0.60) and Crest rises to a terminal end (ρ +0.21, Crest-high decile late). Crest opposes Gast in space (r=−0.41); joint pSom&Crest-high enrichment 0.42 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — cross-repression → NA

Not applicable — the manifold is non-branching (principal curve: 2 tips, 0 forks, fold-free). With no bifurcation there are no competing arms to test for mutual Gast↔pSom anti-correlation.

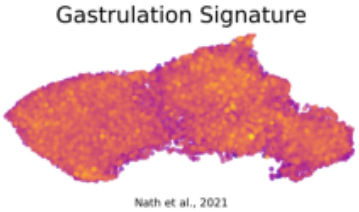
STEP FIGURES

Manifold — samples

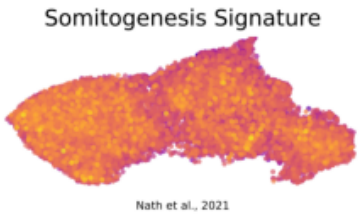


Pseudotime

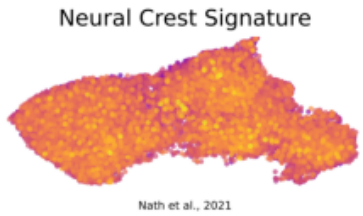
Gastrulation



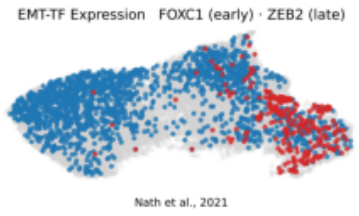
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-grade serous ovarian carcinoma (Nath et al., 2021), 10x — 32,484 malignant cells, 15 patients; trajectory scoped to the 26,703-cell low-cycling body. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching). A single gastrulation/somitogenesis → neural-crest developmental axis over the low-cycling HGSOC body; no bifurcation. Geometry Linear-pSom/Crest. Gates G1 GO / G2 GO / G3 NA. EMT-TFs FOXC1 (early/root, 27%) → ZEB2 (terminus, 4%).

Parameter choices

Sample threshold	cliff 767→149 (ratio 5.15) → geom-mean 338; keeps 15/18 (drops iCell8 + P10)	**Integration**	Harmony on sample θ=10; n_pcs=20, n_neighbors=30, spread=1.3
Scope (cell-cycle fork)	proliferative lobe (5,781, 74% cyc) scoped out; body 26,703; modules cc-free	**Geometry / root**	non-branching curve (N=20, μ=100): 2 tips/0 forks; corr(t,x)=+0.98; root tip 0
Module means (raw)	Gast −0.004 · pSom +0.085 · Crest −0.010 (Linear-pSom/Crest)	**EMT-TFs**	FOXC1 (early/root, 27%) → ZEB2 (terminus, 4%); VIM excluded

Zhang 2019 HGSOC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates both bilateral omental sites of patient VOA11543: same-sample neighbour fraction 0.997→0.641 (perfect-mix floor 0.559), iLISI median 1.00→1.78 of 2. One connected component after removing a 20-cell island. Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

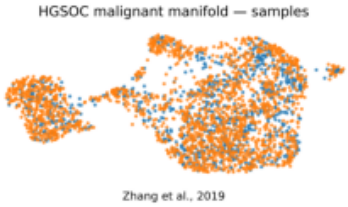
Single (Gast/pSom)→Crest developmental gradient. Gast & pSom do NOT compete out of a hub (smoothed r=−0.10), both oppose Crest (pSom↔Crest r=−0.64, Gast↔Crest r=−0.44). Along pseudotime Gast falls (ρ −0.50), Crest rises to a terminal tip (ρ +0.34, decile pt=0.80). Joint pSom&Crest enrichment 0.35 (<1 = inverse gradient).

Gate 3 — cross-repression → NA

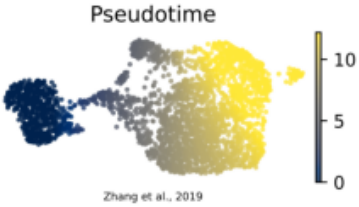
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced).

STEP FIGURES

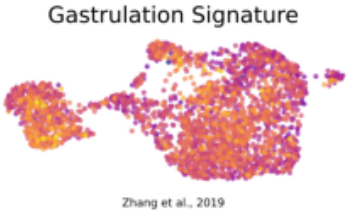
Manifold — samples



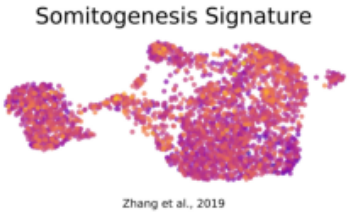
Pseudotime



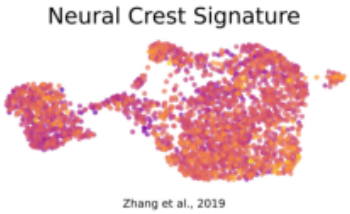
Gastrulation



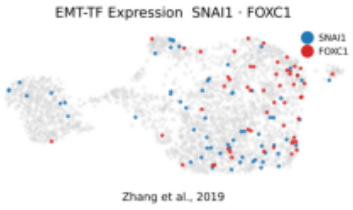
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

HGSOC bilateral ovarian tumour (Zhang et al., 2019), 10x — 2,782 malignant cells over 2 omental sites (1 patient, VOA11543 left+right ovary); predominantly low-cycling (~9% cycling). Non-branching Linear-pSom/Crest.

Verdict.

Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs SNAI1 (root-side) → FOXC1 (terminus).

Parameter choices

Sample threshold	≥100-cell floor; 2 bilateral sites (no cliff) → 2/2, 2,802 cells	**Overdispersed genes**	1,247 (scFates find_overdispersed)
Integration	Harmony on sample θ=10 (bilateral-site batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 20-cell island removed
Geometry	non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips / 0 forks)	**Root**	tip 0 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast −0.011 · pSom +0.042 · Crest −0.034 (Linear-pSom/Crest)	**EMT-TFs**	SNAI1 (root-side) → FOXC1 (terminus); FOXC/PRRX in panel

Olbrecht 2021 HGSOC — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 samples / 5 patients share ONE continuous manifold (single kNN component, no detached islands). Per-neighbourhood iLISI rises 1.00→3.33 under Harmony (θ=10): sample-integrated, not a per-patient archipelago. Judged on the regenerated embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

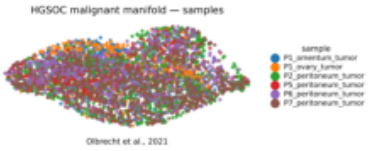
Gast/pSom co-vary (r=+0.44; +0.71 among non-cycling) and BOTH oppose Crest (pSom↔Crest r=−0.68); one developmental axis, Crest an inverse terminal gradient. Cell-cycle confound tested & rejected: anti-correlation survives excluding cycling cells; modules cc-free.

Gate 3 — hub cross-repression → NA

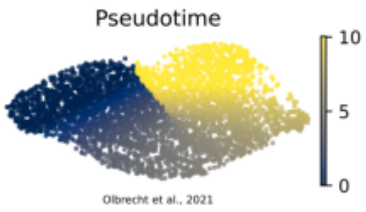
Branching-only. No bifurcation (curve: 2 tips / 0 forks, fold-free) — not tested. The non-branching case passes Gate 2 and reports Gate 3 as NA (distinct from a NO-GO where the pattern is absent).

STEP FIGURES

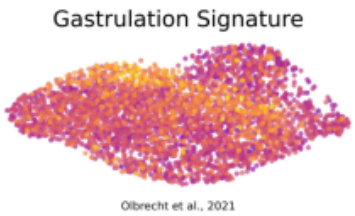
Manifold — samples



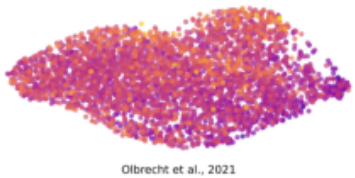
Pseudotime



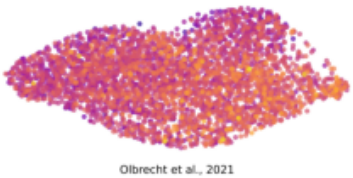
Gastrulation



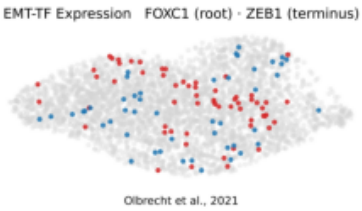
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-Grade Serous Tubo-Ovarian Carcinoma (Olbrecht et al., 2021), 10x — 3,936 malignant cells across 6 samples / 5 patients (≥155-cell threshold keeps 6/10 samples). Harmony integration on sample (θ=10), principal-curve trajectory (scFates) on the whole manifold.

Verdict.
Pattern REPRODUCES (non-branching). The embryonic programme recapitulates on a single continuous axis: a Gast/pSom-high origin resolving into an inverse, terminal Neural-Crest gradient (pSom↔Crest r=−0.68; along pseudotime Gast −0.33, pSom −0.06, Crest +0.40 terminal). Linear-pSom/Crest geometry (means Gast +0.029 / pSom +0.077 / Crest −0.020), not the Puram hub-and-spoke bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT: FOXC1 (root) → ZEB1 (terminus).

Parameter choices

Sample threshold	≥155 → 6/10 (gap 384→63, ratio 6.10; geom-mean cut)	**n_pcs**	20 (scree elbow; Kürten(20)/Puram(40) precedent)
Integration	Harmony θ=10 (iLISI 1.00→3.33, monotone)	**Island removal**	0 cells (UMAP radius-graph, single component)
Geometry	Non-branching curve (2 tips / 0 forks, fold-free)	**Root**	tip 0 = Gast/pSom-high; corr(x_lin,pt)=+0.84
Cell-cycle	whole manifold kept (pSom↔Crest survives non-cycling −0.63; cc-free)	**EMT-TF readout**	FOXC1 (root, ρ=−0.06) → ZEB1 (terminus, ρ=+0.11)

Olbrecht 2021 HGSTOC — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(sample) swept over θ×n_pcs with per-neighbourhood iLISI; at θ=10 iLISI plateaus at 3.34/6 and the 6 samples form one intermixed, connected manifold (single component; 0 islands removed).

Gate 2 — module geography → GO (non-branching)

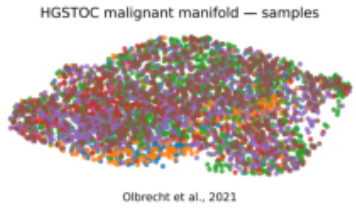
Gast & pSom co-vary (smoothed r=+0.44) at the root pole; Crest is an inverse terminal gradient (Crest~pSom r=−0.68, Crest-high decile mean pseudotime 0.87). A single embryonic axis, not two competing arms.

Gate 3 — cross-repression → NA

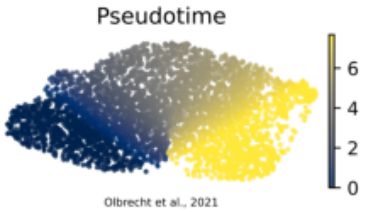
Branching-only. No bifurcation (PPT forks scale with node count 6→13 = noise; curve fits 0 forks, fold-free ratio 2.70), so there is no hub to test for Gast↔pSom mutual exclusivity.

STEP FIGURES

Manifold — samples

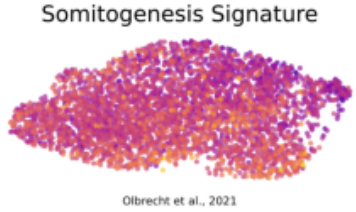


Pseudotime

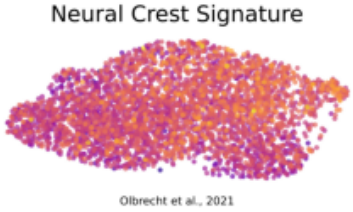


Gastrulation

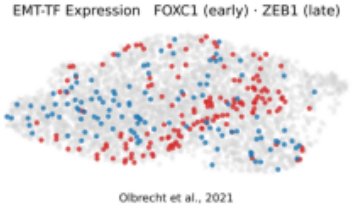
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-grade serous tubo-ovarian carcinoma (Olbrecht et al., 2021), 10x scRNA-seq — 3,936 malignant cells across 6/10 samples (≥155-cell threshold) enter the analysis; the whole malignant manifold enters the trajectory (non-branching linear geometry).

Verdict.
The embryonic pattern reproduces as a single collapsed axis: a Gast/pSom co-expressing developmental origin resolving into a neural-crest-high terminus (Gate 1 GO / Gate 2 GO / Gate 3 NA). Pseudotime correlations Gast ρ=−0.41, pSom −0.18, Crest +0.47 (smoothed). Linear-pSom/Crest geometry; Gast & pSom co-vary rather than diverge, so there is no Gast–pSom bifurcation (Gate 3 NA). EMT-TFs FOXC1 (root) → ZEB1 (terminus).

Parameter choices

Min-cell threshold	155 (geo-mean of 384→63 count gap, ratio 6.1) → 6/10 samples	**Integration**	Harmony(sample), θ=10, n_pcs=20 (iLISI plateau 3.34/6)
Island removal	radius graph 6× median NN → 0 cells removed (single component)	**Geometry**	Linear-pSom/Crest — curve; 2 tips, 0 forks, fold-free 2.70
Root	Gast/pSom-high tip (developmental origin; Crest is late)	**Pseudotime ρ**	Gast −0.41, pSom −0.18, Crest +0.47 (smoothed)
Overdispersed genes	1,413 (scFates find_overdispersed)	**EMT-TFs**	FOXC1 (early/root) → ZEB1 (late/terminus)

Olalekan 2021 HGSOC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates all 6 patients: same-sample neighbour fraction 0.96→0.38, iLISI median 1.00→3.99 of 6. One connected component (0 islands). Read on the regenerated embedding + per-neighbourhood iLISI, not a scalar: mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

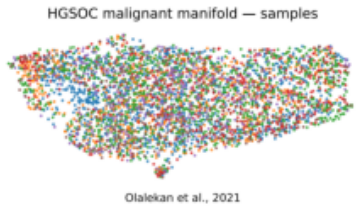
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.80), both oppose Crest (pSom↔Crest r=−0.85, Gast↔Crest r=−0.72). Along pseudotime Gast/pSom fall (ρ −0.77/−0.64), Crest rises to a terminal tip (decile pt=0.99). Joint pSom&Crest enrichment 0.35 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — cross-repression → NA

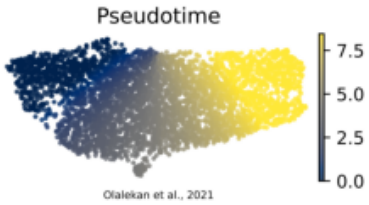
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced, per the deliverable spec).

STEP FIGURES

Manifold — samples

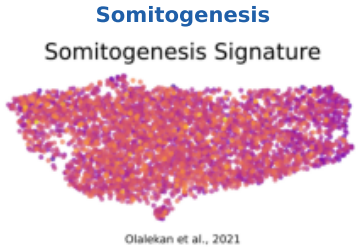


Pseudotime

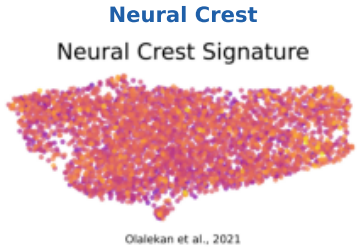


Gastrulation

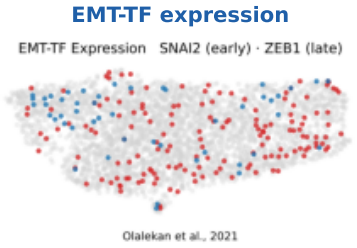
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
HGSOC omental metastases (Olalekan 2021), Drop-seq, 4,125 malignant cells over 6 patients; single low-cycling body (~5% cycling).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs SNAI2 (early) → ZEB1 (terminus).

Parameter choices

Sample threshold	≥100 cells (no dominant cliff, ratio 1.13); 6/6 samples, 4,125 cells	**Overdispersed genes**	381 (scFates find_overdispersed)
Integration	Harmony on sample, θ=10 (mixing plateau)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	Non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips / 0 forks)	**Root**	tip 0 = Gast/pSom-high, Crest-low (developmental origin)
Module means (raw)	Gast +0.018 · pSom +0.027 · Crest −0.028 (Linear-pSom/Crest)	**EMT-TFs**	SNAI2 (early/root) ↔ ZEB1 (late/terminus); FOXC/PRRX in panel

Qian 2020 Ovarian — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=20) interleaves all 4 patients: same-sample neighbour fraction 1.00 → 0.63 (chance 0.35), iLISI median 1.00 → 2.13 of 4. θ=30 REVERSES mixing to 0.88 (over-integration fragments into per-sample lobes), so θ=20 is the strongest integration on the plateau. 97 cells / 1 detached island removed → one connected component. Read on the regenerated embedding + iLISI: mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

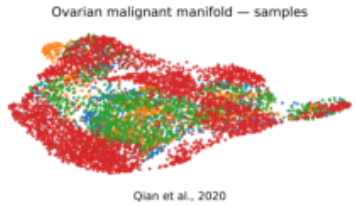
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-express (joint enrichment 1.20 > 1; smoothed r=+0.14), and both oppose Crest (Gast↔Crest r=−0.28, pSom↔Crest r=−0.26; joint enrichment 0.58 / 0.63 < 1 = inverse, not a co-expression hub). Along pseudotime Gast/pSom fall (ρ −0.46 / −0.55), Crest rises to a terminal tip (high-decile pt = 0.83).

Gate 3 — hub cross-repression → NA

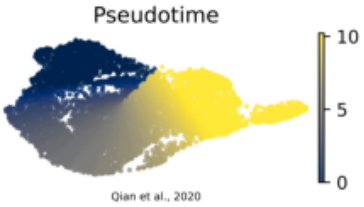
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks, fold self-approach 2.41). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

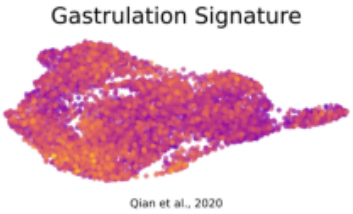
Manifold — samples



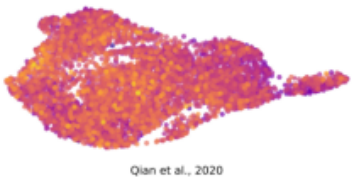
Pseudotime



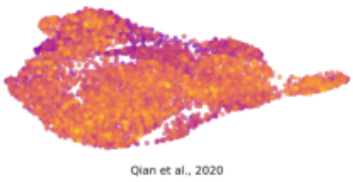
Gastrulation



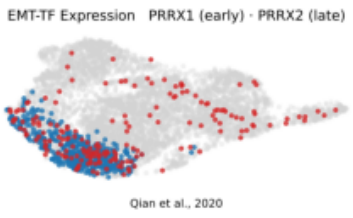
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Ovarian cancer (Qian et al., 2020), 10x — 6,393 malignant cells over 4 treatment-naive patients; one continuous low-confound body (embryonic modules proliferation-independent).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs PRRX1 (early/root) → PRRX2 (late/terminus).

Parameter choices

Sample threshold	≥100 cells (floor; no dominant cliff, ratio 1.89) → 4/4 samples, 6,490 cells	**Overdispersed genes**	1,013 (scFates find_overdispersed)
Integration	Harmony on sample, θ=20 (mixing plateau; θ=30 over-integrates and fragments)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 97 cells / 1 island removed → 6,393
Geometry	Non-branching → scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.41	**Root**	tip 0 = Gast/pSom-high, Crest-low origin; corr(x_lin,pt)=+0.83
Module means (raw)	Gast +0.042 · pSom +0.065 · Crest −0.019 (Linear-pSom/Crest)	**EMT-TFs shown**	PRRX1 (early/root, ρ −0.19) ↔ PRRX2 (late/terminus, ρ +0.00); detection-gated

Qian 2020 ovarian cancer — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4/4 tumours (920–3345 malignant cells) intermix on one continuous manifold at Harmony θ=10 (per-neighbourhood iLISI 1.87 of a 2.89 ceiling); radius-graph: 0 detached islands.

Gate 2 — module geography → GO (non-branching)

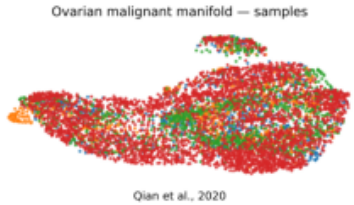
Gast/pSom co-vary (r=+0.36) and decline along pseudotime (ρ=−0.48/−0.54); Crest anti-correlated (r=−0.28/−0.43), rising to a terminal readout (ρ=+0.48; crest-high decile pt=0.68). Single (Gast/pSom)→Crest developmental axis.

Gate 3 — cross-repression → NA

Single monotone axis — a PPT tree's arms co-vary in Gast/pSom (r=+0.68), so there is no bifurcation and no Gast/pSom hub; no branch pair exists to test for cross-repression.

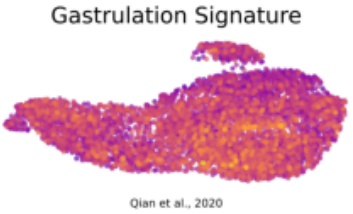
STEP FIGURES

Manifold — samples

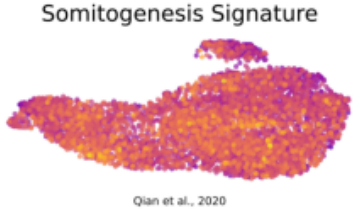


Pseudotime

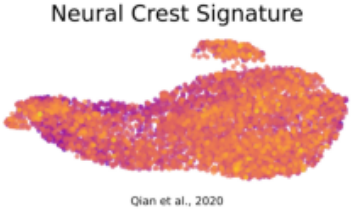
Gastrulation



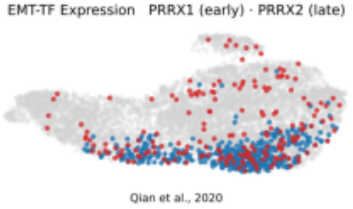
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Ovarian cancer (Qian et al., 2020), 10x — 4 tumours / 16,951 cells (master scope); malignant only: 6,490 load across all 4 samples (no thin tail to threshold; 0 islands removed). Whole malignant manifold (no lobe scoping). Non-branching Linear-pSom/Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Qian's ovarian malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (Linear-pSom/Crest), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The developmental axis is orthogonal to the proliferation split (holds among non-cycling cells alone), so the whole manifold is kept (Lee/Bischoff no-scoping precedent). EMT-TFs PRRX1 (early/root) → PRRX2 (late); all TFs decline from the mesenchymal origin.

Parameter choices

Min cells/sample	≥100 (keeps 4/4; no thin tail)	**Integration**	Harmony θ=10; PCA 20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=15, λ=0.3, μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Compartments**	non-cycling 4,971 / cycling 1,519 (axis orthogonal to split)
EMT-TFs	PRRX1 (early/root) → PRRX2 (late); SNAI1/FOXC2 near detection floor		

Shih 2018 ovarian cancer — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=20) collapses the strong patient batch effect: same-sample neighbour fraction 0.73→0.20 (perfect-mixing floor 0.16), iLISI median 2.98→5.82 of 7 samples. One connected component (0 islands). Patient islands (pre-Harmony) fuse into one continuous mixed cloud.

Gate 2 — module geography → GO (non-branching)

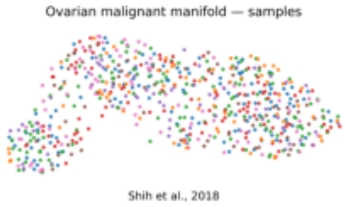
Single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.23; joint enrichment 0.95 ~ origin overlap), both oppose Crest (Gast↔Crest r=−0.80, pSom↔Crest r=−0.74). Along pseudotime Gast/pSom fall (ρ −0.86/−0.62), Crest rises to a terminal tip (decile pt=0.99); joint pSom&Crest enrichment 0.26 (<1 = inverse gradient, not a hub).

Gate 3 — cross-repression → NA

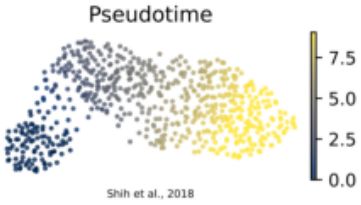
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks; PPT tree 0 branch nodes). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

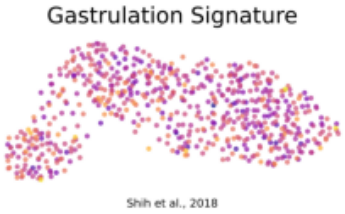
Manifold — samples



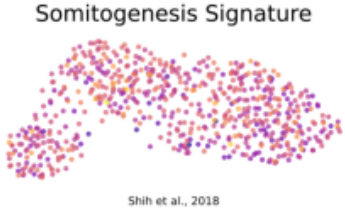
Pseudotime



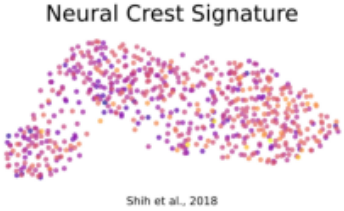
Gastrulation



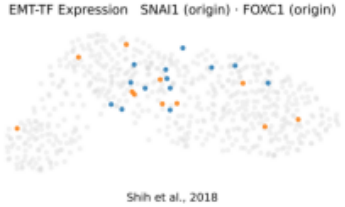
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
HGSOC/LGSOC ovarian tumours (Shih et al., 2018), SmartSeq2 — 633 malignant cells over 7 samples; sparse (~1250 genes/cell), a single low-cycling body with a cycling spur (~43% cycling). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs SNAI1/FOXC1/PRRX2 all origin/root-enriched (no terminal EMT-TF).

Parameter choices

Sample threshold	floor ≥100 rejected (3/16); cliff 60→37 → geo-mean 47 → 7/16, 633 cells	**Feature selection**	scFates HVG underpowered on 633 sparse cells (33) → Seurat HVG 2000
Integration	Harmony on sample θ=20 (mixing collapses to floor; strong ovarian batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3 (SS2 precedent); 0 islands removed
Geometry	non-branching curve N=20, λ=0.3, μ=100; 2 tips / 0 forks (2.84)	**Root**	tip 2 = Gast/pSom-high, Crest-low origin; axis cell-cycle-free (ρ=−0.03)
Module means (raw)	Gast −0.013 · pSom +0.065 · Crest −0.034 (Linear-pSom/Crest)	**EMT-TFs**	SNAI1 (8.4%), FOXC1 (6.8%), PRRX2 (3.5%) — all origin/root-enriched; overlay SNAI1 & FOXC1

Peng 2019 PDAC — analysis summary

RUN 1 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

16/16 samples (≥100 cells) intermix on one continuous manifold; no per-sample islands (0 detached cells removed). Median iLISI 4.33 = 57% of the proportional-mixing maximum (7.53); neighbour graph is a single connected component.

Gate 2 — module geography → NO-GO

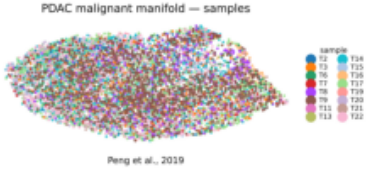
The three module-high territories sit at three separate UMAP corners (triangle, angles 44/95/42° — not a Gast/pSom body flowing to a Crest terminus). Gast & pSom are spatially mutually exclusive: joint-high 0.8% vs 6.2% expected if independent (enrichment 0.13, <1), so there is no co-expression hub and no co-varying body. Along the rooted axis Gast (ρ=−0.56) and pSom (ρ=+0.11) move in opposite directions and Crest is flat (ρ=+0.10; high-decile pseudotime 0.70, mid-axis, not terminal). The anti-correlation holds at every clustering scale (sub-region sweep). Neither GO pattern — branching hub nor non-branching Gast/pSom overlap — is present.

Gate 3 — hub cross-repression → NA

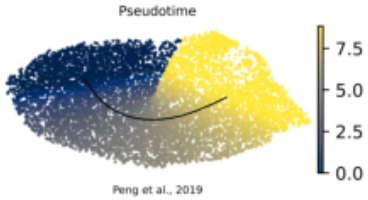
No bifurcation (single non-branching axis); Gate 3 is not evaluated.

STEP FIGURES

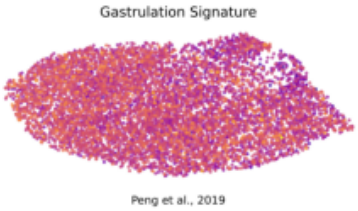
Manifold — samples



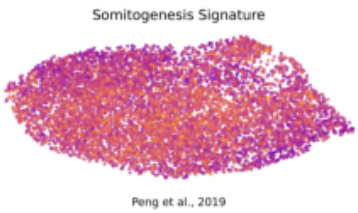
Pseudotime



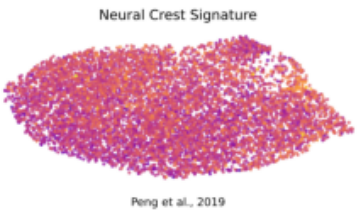
Gastrulation



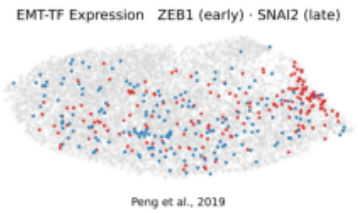
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic ductal adenocarcinoma (Peng et al., 2019), 10x — 16 patients / 27,953 cells (master scope); malignant only: 7,945 load, all 7,945 enter the trajectory (one Leiden lobe, no subtypes).

Verdict.
Pattern does not reproduce — Gate 2 NO-GO. The embryonic modules are present (pSom mean +0.10, strongest in the atlas) but resolved into three mutually-exclusive territories, not a Gast/pSom hub or co-varying body; no bifurcation, no terminal Crest. The smoothed Gast~pSom = −0.41 reflects two separate corners, not cross-repression within a shared hub.

Parameter choices

Min cells/sample	≥ 100 (keeps 16/16)	**Island removal**	UMAP radius-graph (−0 cells)
Root	tip max(Gast+pSom−Crest)	**Integration**	Harmony on sample, θ=10, 30 it
Trajectory	ElPiGraph curve N=20 μ=50 λ=0.3	**Geometry**	non-branching blob; 1 lobe, no subtypes

Peng 2019 PDAC — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 16 samples intermix on one continuous connected component (0 detached islands; UMAP aspect 1.17; iLISI median 1.0→5.3 of 16 at θ=10) — judged on the embedding, not a scalar. Cell cycle (6.6% cycling) is not a module confound (|ρ|≤0.24; the pSom–Crest spatial anti-correlation is unchanged when cyclers are excluded), so the whole malignant manifold is carried forward.

Gate 2 — module geography → NO-GO

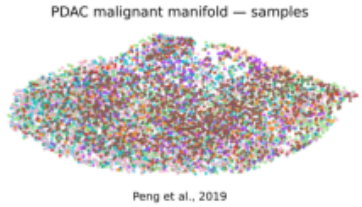
Embryonic module pattern absent. The three smoothed modules occupy a 2-D triangle (PCA variance 0.55/0.40; Gast/pSom/Crest top-decile peaks 110–134° apart) — no Gast/pSom co-expression hub. They do not order along pseudotime (pSom ρ=+0.11, Crest ρ=−0.10) and Crest is non-terminal (Crest-high decile at pseudotime 0.60; joint pSom&Crest enrichment 0.41). Neither the branching hub nor the non-branching pSom→Crest axis reproduces.

Gate 3 — hub cross-repression → NA

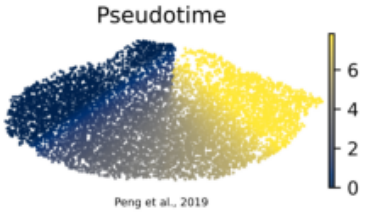
Branching-only gate. The manifold is non-branching (curve: 2 tips, 0 forks) and Gate 2 did not pass. No hub — Gast and pSom are anti-correlated and sit at opposite triangle corners (spatial r=−0.41).

STEP FIGURES

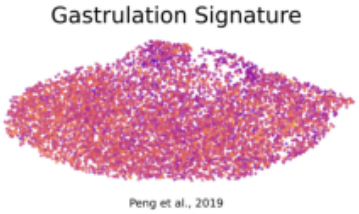
Manifold — samples



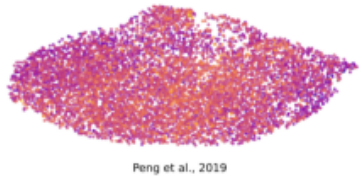
Pseudotime



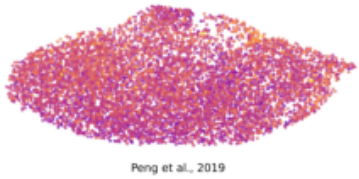
Gastrulation



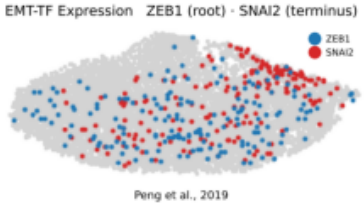
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic ductal adenocarcinoma (Peng et al., 2019), 10x — 7,945 malignant cells, 16 samples.

Verdict.
Pattern does NOT reproduce. Gate 1 GO / Gate 2 NO-GO / Gate 3 NA — developmental programs are present as scores but geographically unresolved (a triangle, not an axis): a triangular, non-axial module geography, not the Linear-pSom/Crest developmental axis seen in the same-tissue Steele/Lin/Moncada PDAC GO runs. EMT-TFs ZEB1 (root end) / SNAI2 (terminus end).

Parameter choices

Threshold	≥100-cell floor (gap-finding premise unmet) → all 16 samples kept	**Integration**	Harmony θ=10; iLISI median 1.0→5.3 of 16
Cell cycle	6.6% cycling; not a module confound (|ρ|≤0.24); whole manifold carried	**Module geometry**	2-D triangle (PCA var 0.55/0.40; peaks 110–134° apart); no hub
Trajectory	scFates curve (2 tips, 0 forks); root = Gast/pSom-high tip	**EMT-TFs**	ZEB1 (root) / SNAI2 (terminus); Snail/Zeb/Twist + FOXC/PRRX, 3% floor

Peng 2019 PDAC — analysis summary

RUN 3 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Pre-Harmony the 16 tumours form a per-patient archipelago (iLISI median 1.0 of 16, many detached islands). With Harmony on sample they intermix into one continuous mass — iLISI median 4.05, a single kNN-graph component, 0 detached cells. Both mixed and continuous. θ=2 from a mixing+geometry sweep (mixing saturates by θ=2; higher θ only collapses the developmental axis, aspect 1.37→1.05).

Gate 2 — module geography → GO (non-branching)

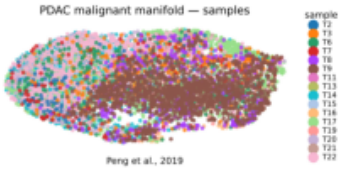
Decided on geography, not means. Not a Puram hub-and-spoke: no Gast/pSom co-expression hub (interior Gast-XOR vs pSom-XOR r=+0.29, not −0.49) and a PPT tree over-fits the continuous cloud (26–43 forks, no clean Y). Non-branching case (§1): organized Gast↔pSom cross-repression (spatial r=−0.53) with a terminal neural-crest gradient — pSom↔Crest r=−0.57 (GO range: Steele −0.68, Moncada −0.56), and on the origin-rooted pseudotime Crest is the latest/terminal module (ρ=+0.40). Crest opposes pSom more than Gast — no Gast-arm inversion (no flag).

Gate 3 — hub cross-repression → NA

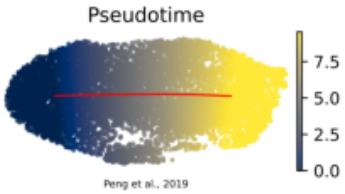
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks, fold-free). With no bifurcation there is no Mixed-type hub and no branch pair to test for mutual Gast↔pSom exclusivity, so the branching-only 04_correlation / 05_modules notebooks are not produced (per the deliverable spec).

STEP FIGURES

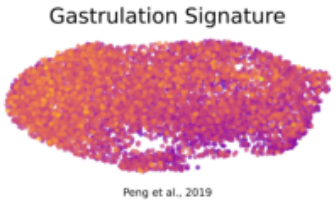
Manifold — samples



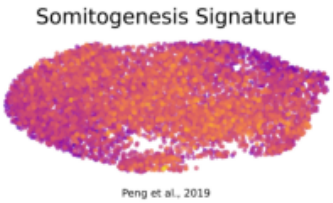
Pseudotime



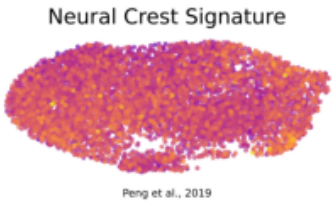
Gastrulation



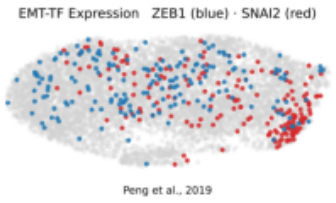
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
PDAC (Peng et al., 2019), 10x — 16 tumours / 27,953 cells (master scope); malignant only: 7,945 cells across all 16 samples (≥100-cell floor; the multiplicative count gap is only 2.31×, mid-distribution, so no clean spillover cliff). Whole malignant manifold (two density lobes, near-identical developmental means, not scoped).

Verdict.
Pattern reproduces in the non-branching form (Linear). One continuous, sample-integrated manifold (G1 GO) carrying organized Gast↔pSom cross-repression with a terminal neural-crest gradient (G2 GO): rooted at the gastrulation/somitogenesis-high origin, the developmental ordering is Gast → pSom → Crest-terminal, matching the passing same-tissue Steele/Moncada PDAC runs. It lacks Puram's hub-and-spoke bifurcation, so there is no hub to test (G3 NA). EMT-TFs ZEB1 (root-side) → SNAI2 (far end); canonical SNAI1/TWIST near the detection floor.

Parameter choices

Sample threshold	≥100-cell floor (mult. gap 2.31× < 4× degenerate); 16/16 kept → 7,945 cells	**Overdispersed genes**	1,288 (scFates find_overdispersed); no sc.pp.scale before PCA
Integration	Harmony on sample, θ=2 (mixing saturates by θ=2; higher collapses axis 1.37→1.05)	**Embedding**	n_pcs=20, n_neighbors=30, UMAP spread=1.3; 0 islands removed; 1 component
Geometry	Linear (non-branching) principal curve; no hub, organized cross-repression + terminal Crest	**Root**	tip 0 = Gast/pSom-high, Crest-low origin; curve fold-free (ratio 2.98); Crest terminal on DPT
Module means (raw)	Gast −0.047 · pSom +0.098 · Crest −0.055 (means match Steele PASS; geography decides)	**EMT-TFs shown**	ZEB1 (8.8%) & SNAI2 (9.3%) — most end-separated expressed factors; VIM excluded (not a TF)

Steele 2020 PDAC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

13/13 samples (≥117 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→5.15 of 13); both lobes (Ductal-classical 6,782 / Stress-UPR 1,092) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

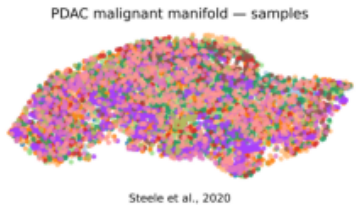
Single dominant pSom→Crest axis: pSom high over the classical-ductal body declines along pseudotime (ρ=−0.70); Crest anti-correlated (r=−0.69), rising to a terminal readout (ρ=+0.74; crest-high decile mean pt=0.87). Gast weakly structured (early programme, spent); joint pSom&Crest-high depleted (enrichment 0.44) — an inverse gradient, not a co-expression hub.

Gate 3 — hub mutual exclusivity → NA

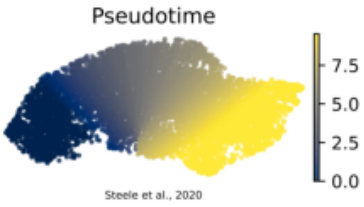
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

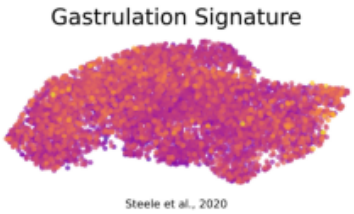
Manifold — samples



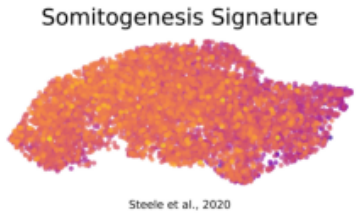
Pseudotime



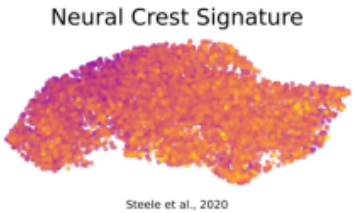
Gastrulation



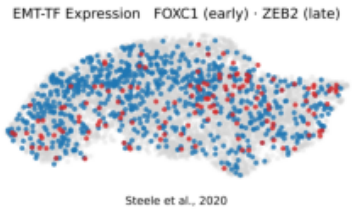
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
PDAC (Steele et al., 2020), 10x — 16 patients / 48,570 cells (master scope); malignant only: 7,874 cells across 13 malignant-bearing samples (all clear the ≥100 floor) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Steele's PDAC malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient (classical-ductal body → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both transcriptional lobes (orthogonal to cell cycle; module–cyc |ρ|≤0.08), so the whole manifold is kept (Bischoff/Sun/Lee no-scoping precedent). Unlike Zhou, cell cycle does not define the lobes (PDAC ~94% non-cycling) and pseudotime is uncorrelated with cycling (ρ=+0.02), so no proliferation caveat applies. The raw module means (Gast −0.04 / pSom +0.08 / Crest −0.05) nearly match the failed Peng 2019 PDAC run, but the geography differs: Peng showed three mutually-exclusive territories with no terminal readout (NO-GO), whereas Steele shows a coherent terminal-crest gradient (GO) — the gate turns on geometry, not means. EMT-TFs FOXC1 (early/root) → ZEB2 (late/terminus); canonical SNAI1/ZEB1 near the detection floor; EMT-TF end-separation weak (Δpt=0.12).

Parameter choices

Min cells/sample	≥100 floor (13/13; gap degenerate)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = pSom-high, Crest-low	**Compartments**	Ductal-cl. 6,782 / Stress-UPR 1,092

Lin 2020 PDAC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates 15 samples (iLISI 1.0→4.1, same-sample frac 0.94→0.45); one continuous connected component after removing a justified detached island (652 cells, 14 samples, developmentally flat).

Gate 2 — module geography → GO (non-branching)

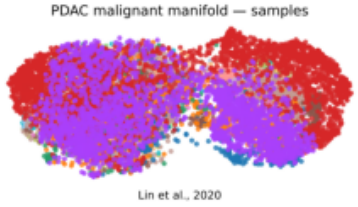
Single pSom→Crest axis: pSom high over ductal body falls along pseudotime (ρ=−0.59), Crest rises to a terminal end (ρ=+0.32, decile pt 0.82); anti-correlated in space (r=−0.38, joint-high enrichment 0.52 = no hub). Gast weak (ρ=−0.24). Crest terminal, not on the Gast end → no inversion flag.

Gate 3 — cross-repression → NA

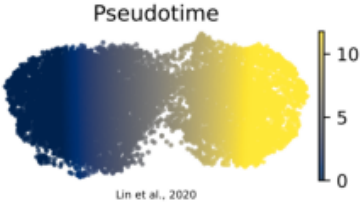
No bifurcation (branching-only gate). Geometry is linear, so the tree cross-repression test does not apply.

STEP FIGURES

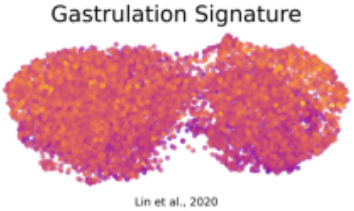
Manifold — samples



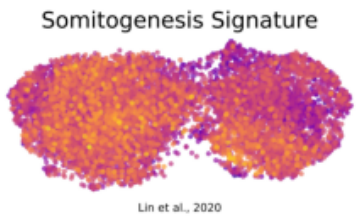
Pseudotime



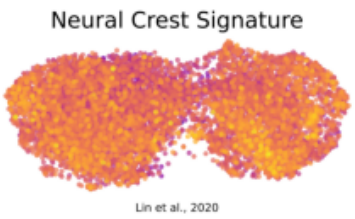
Gastrulation



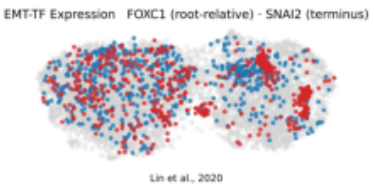
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic Ductal Adenocarcinoma (Lin et al., 2020), 10x — 16 samples; 10,993 malignant cells (10,304 after threshold + island removal) enter the analysis; the whole malignant manifold enters the trajectory (non-branching linear geometry).

Verdict.
The embryonic pattern reproduces as a single somitogenesis→neural-crest developmental axis (the non-branching / collapsed-bifurcation case) — no Gast/pSom hub, no competing arms; Linear pSom/Crest. Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. EMT-TFs FOXC1 (root-relative) → SNAI2 (terminus).

Parameter choices

Threshold	53 cells (geom-mean 76→37, gap ratio 2.05) → 15/16 kept	**Island**	652 cells removed (radius-graph 6× med-NN; multi-sample, dev-flat)
Integration	Harmony θ=10; n_pcs=20; spread=1.3	**Trajectory**	scFates curve (ElPiGraph) Nodes=20, μ=100 (2 tips, 0 forks; self-approach 2.98)
Root	pSom-high / Crest-low tip	**Scope**	whole manifold kept (modules cycle-free; axis holds in non-cycling lobe)
EMT-TF axis	FOXC1 (root-relative) → SNAI2 (terminus)	**Caveat**	pt vs cycling ρ=+0.42 (cycle-split lobes)

Hwang 2022 PDAC — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

38 retained patients intermix on one continuous manifold (same-sample neighbour fraction 0.96→0.20 across the θ sweep, near the mixing floor); radius-graph leaves one connected component.

Gate 2 — module geography → NO-GO

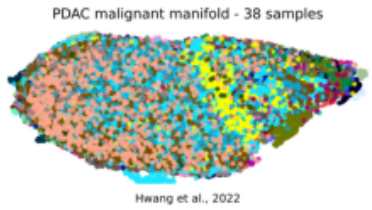
Pattern ABSENT. Three mutually-exclusive territories, not a hub→arms axis: all modules pairwise anti-correlated (Gast/pSom −0.33, pSom/Crest −0.39, Gast/Crest −0.30) and no two co-occur (all joint-high enrichments depleted → no convergence hub). Crest mid-axis (decile pt 0.50), NOT terminal.

Gate 3 — hub cross-repression → NA

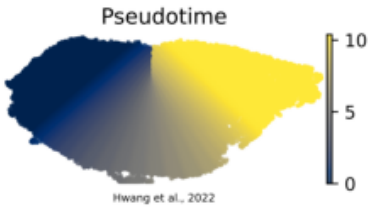
Not branching and Gate 2 NO-GO — cross-repression test not applicable.

STEP FIGURES

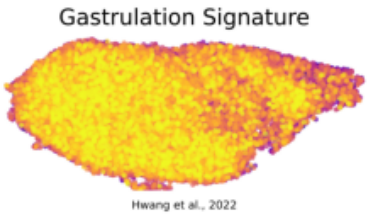
Manifold — samples



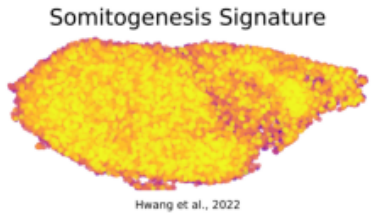
Pseudotime



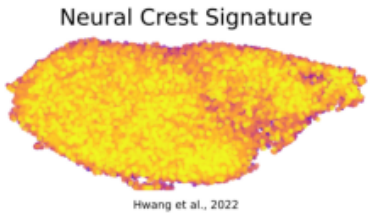
Gastrulation



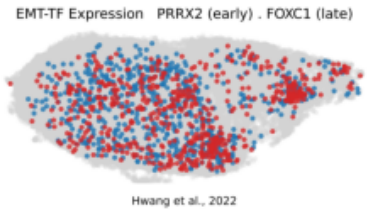
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic ductal adenocarcinoma (Hwang et al., 2022), 10x; 64,538 malignant cells across 38/43 samples (≥14-cell threshold). Whole malignant manifold; cell cycle flat across lobes.

Verdict.
Embryonic pattern does NOT reproduce (Gate 1 GO / Gate 2 NO-GO / Gate 3 NA). Modules resolve into three exclusive non-developmental territories with no terminal Crest readout — the Peng2019 same-PDAC NO-GO signature; territories track EMT-III / G2-M / classical states, not one developmental axis.

Parameter choices

Min cells/sample	14 (mult-gap 24→9, ratio 2.67; 38/43)	**Scope**	whole manifold (cell cycle flat)
Root node	Gast-high tip (developmental origin)	**Integration**	Harmony θ=10 on sample; n_pcs=20
Trajectory	scFates curve N=20 (PPT rejected)	**EMT-TFs**	PRRX2 (early) · FOXC1 (late)

Hwang 2022 PDAC — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Samples mixed (iLISI 9.3/38 vs 1.0 raw; falls to 6.7 at θ=20) on one continuous connected manifold; density lobes are poles of one axis, not batch islands nor a cycling compartment.

Gate 2 — module geography → GO (non-branching)

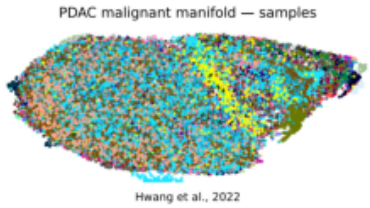
Organized single-axis geography: Gast high at root, opposes pSom/Crest spatially (Gast↔pSom r=−0.33); Crest an inverse/terminal gradient. Non-branching case → GO.

Gate 3 — hub cross-repression → NA

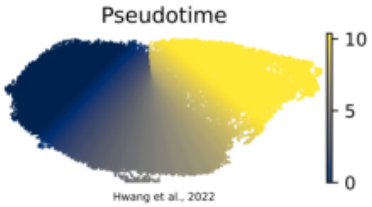
No bifurcation (scFates curve, not tree) → cross-repression scatter N/A.

STEP FIGURES

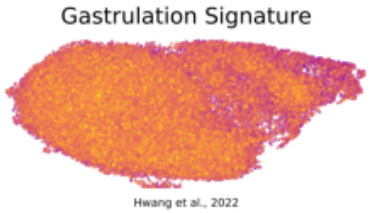
Manifold — samples



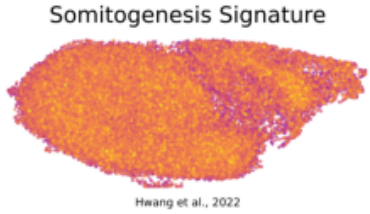
Pseudotime



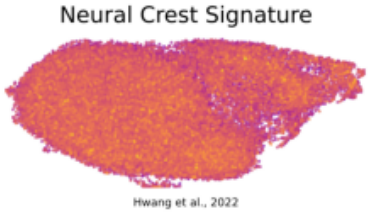
Gastrulation



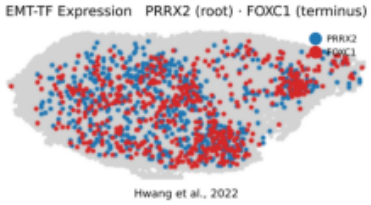
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic ductal adenocarcinoma (Hwang et al., 2022), snRNA-seq / 10x — 64,401 malignant cells across 38 tumours after threshold + island removal.

Verdict.
Pattern reproduces as the NON-BRANCHING case: one continuous gastrulation → somitogenesis / neural-crest axis, rooted at the Gast/pSom-high developmental pole. No competing arms and no cross-repression bifurcation. Geometry = Linear-pSom/Crest. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast −0.005 / pSom −0.002 / Crest +0.003 (weak, spatially organized). EMT-TFs PRRX2 (root) · FOXC1 (terminus); ZEB1 most expressed (~16%).

Parameter choices

Malignant cells	64,538 raw → 64,507 (thresh) → 64,401 (islands)	**Min-cell threshold**	14 (gap-finding; smooth tail, only 24→9 gap); 5 tiny samples dropped
Integration	Harmony θ=10, n_pcs=20, spread=1.3	**Overdispersed genes**	1,664 (of 19,457 retained)
Geometry	Linear-pSom/Crest (curve: 2 tips, 0 forks, fold-free 2.72)	**Root**	Gast/pSom-high tip (cycling flat, ρ=−0.03)
Module means	Gast −0.005 / pSom −0.002 / Crest +0.003	**EMT-TFs**	PRRX2 (root) · FOXC1 (terminus); ZEB1 most expressed (~16%)

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

29/29 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 98 cells / 6 specks detached; iLISI median 1.00→9.31 of 29 samples); UMAP axis ratio 1.37 (compact mass, not a strip of islands). Proliferation is not a confound (module·cycling |ρ|≤0.05).

Gate 2 — module geography → GO (non-branching)

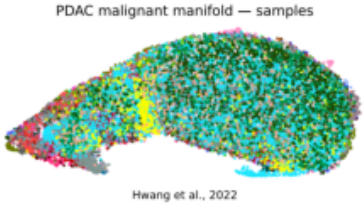
Single organized cross-repression axis, not a bifurcation: Gast↔pSom mutually exclusive (smoothed r=−0.32; joint Gast&pSom-high enrichment 0.44 — no co-expression hub). pSom high at the origin declines along pseudotime (ρ=−0.34), Gast rises (ρ=+0.29); neural crest is a terminal / inverse readout (crest-high decile mean pt=0.63), anti-correlated with both modules (vs pSom r=−0.39, vs Gast r=−0.30 — not a crest-arm inversion).

Gate 3 — hub mutual exclusivity → NA

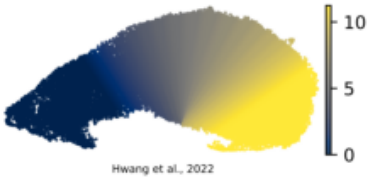
Single monotone axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for competitive cross-repression. Branching-only correlation / modules steps are not produced.

STEP FIGURES

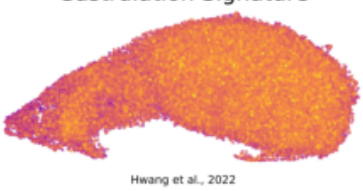
Manifold — samples



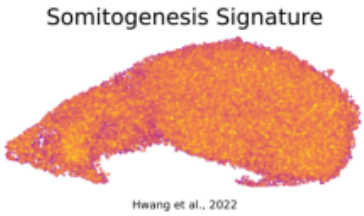
Pseudotime



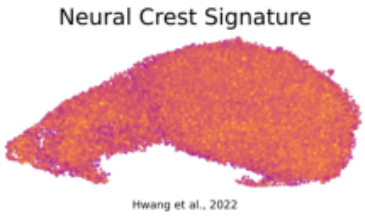
Gastrulation



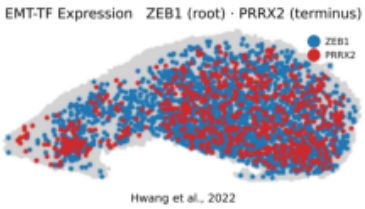
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
PDAC (Hwang et al., 2022), snRNA-seq TP10K — 43 samples / 225k nuclei across 3 group matrices; malignant only: 64,538 nuclei → 64,104 across 29 samples ≥100 cells → 64,006 after island removal enter the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Hwang's PDAC malignant manifold is a single continuous axis rooted at a somitogenesis-high / neural-crest-low origin, with organized Gast↔pSom cross-repression and neural crest as a terminal inverse gradient — the developmental axis without Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The curve is fold-free (2 tips, 0 forks; self-approach 2.81) and continuous on an independent Harmony-PCA geometry (ρ=0.73; centroid-path ratio 2.27). The axis spans both transcriptional compartments (Basal-like 54,795 / Classical-like 9,211) and is orthogonal to cell cycle (93% non-cycling), so the whole manifold is kept. Raw module means are low-amplitude (Gast −0.004 / pSom −0.002 / Crest +0.003) but the geography is decisive. EMT-TFs ZEB1 (early/root) → PRRX2 (late/terminus); ZEB1 dominant (15.7%).

Parameter choices

Min cells/sample	100 floor (29/29 kept)	**Integration**	Harmony θ=10; n_pcs=20
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 2 = pSom-high, Crest-low	**Compartments**	Basal-like 54,795 / Classical-like 9,211
Geometry	Linear (non-branching)	**EMT-TFs**	ZEB1 (root) → PRRX2 (terminus); ZEB1 dominant 15.7%

Hwang 2022 PDAC — analysis summary

RUN 4 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Pre-Harmony the 29 tumours form a per-patient archipelago (iLISI 1.0 of 29; same-sample neighbour frac 0.95). Harmony on sample intermixes them into one continuous mass — iLISI 8.8, same-sample 0.17, a single kNN-graph component after complete island removal (107 detached-speck nuclei → 63,997). θ=2 from the sweep (mixing saturates by θ=2; geography θ-invariant). Three density lobes, all 2–8% cycling — no proliferation compartment.

Gate 2 — module geography → NO-GO

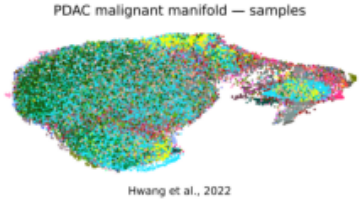
Judged on geography vs an explicit null, not the means (flattest in the atlas: Gast −0.004 / pSom −0.002 / Crest +0.003). The embryonic modules carry NO coherent territory: each module's Moran's I sits at/below its own size-matched union-pool null (Gast I=0.06 z=−0.4 p=0.65; pSom z=−0.2 p=0.55; Crest z=−0.5 p=0.59). Gast↔pSom spatial r=−0.15, far short of the passing same-tissue PDAC bar (Peng/Steele/Moncada/Lin, −0.53 to −0.68); no terminal Crest (module-high deciles mid-pseudotime). The manifold is organized by PDAC meta-programs (EMT/Hypoxia/Chromatin), not the developmental axis → the embryonic pattern is absent.

Gate 3 — cross-repression → NA

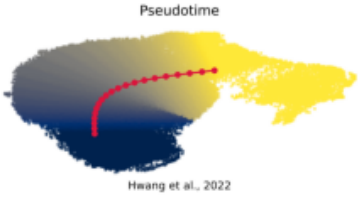
Not applicable — Gate 2 did not pass, so there is no Gast/pSom hub and no bifurcation to test. The branching-only 04_correlation / 05_modules notebooks are not produced (§6); the EMT-TF step is the final notebook (04).

STEP FIGURES

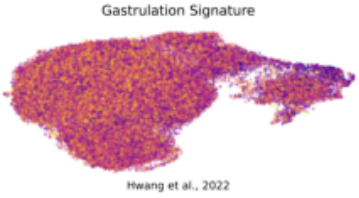
Manifold — samples



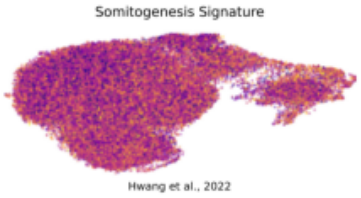
Pseudotime / curve



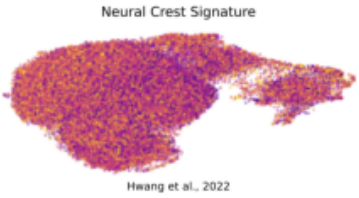
Gastrulation



Somitogenesis



Neural Crest



EMT-TF expression

Dataset.
PDAC (Hwang et al., 2022), 10x snRNA-seq — 43 tumours / 224,988 cells (master scope); malignant only: 64,538 nuclei → 63,997 across 29 samples (≥100-cell floor; the multiplicative count gap is degenerate, sitting in the speck tail). Whole malignant manifold; three density lobes, not scoped.

Verdict.
The embryonic pattern does NOT reproduce. Gate 1 passes — one continuous, sample-integrated malignant manifold — but the developmental modules have no coherent geography (all at/below random-null level), no Gast↔pSom cross-repression axis (spatial r=−0.15 vs passing PDAC −0.53..−0.68), and no terminal Crest. The manifold is structured by PDAC meta-programs, not gastrulation/somitogenesis. EMT-TFs FOXC1 (blue) and ZEB1 (red) mark the manifold descriptively; canonical SNAI1/TWIST are below the detection floor. G1 GO / G2 NO-GO / G3 NA. Stopped at Gate 2 per §6; 01–03 + expression(04) produced.

Parameter choices

Sample threshold	≥100-cell floor (gap 2.67× in speck tail → degenerate); 29/43 kept	**Overdispersed genes**	1,645 (scFates find_overdispersed); no sc.pp.scale
Integration	Harmony on sample, θ=2 (iLISI 1.0→8.8; geography θ-invariant)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 107 specks removed; 1 kNN comp
Geometry	Non-trajectory (Gate 2 NO-GO); probe curve 2 tips/0 forks, no hub	**Module means (raw)**	Gast −0.004 · pSom −0.002 · Crest +0.003 (geography decides)
Root	curve tip 0 (max Gast+pSom−Crest); ordering does not reproduce	**EMT-TFs shown**	FOXC1 (3.6%) · ZEB1 (15.7%); SNAI1/TWIST <0.1%; VIM excluded

Raghavan 2021 PDAC — analysis summary

RUN 1 OF 5

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

20/20 PDAC samples (≥120 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 50 cells / 1 speck removed; iLISI median 1.00→7.22 of 20); both lobes (Ductal-body 6,494 / Cycling 674) carry the trajectory. One PANET sample (662 cells, a distinct neuroendocrine malignancy) was excluded to match the declared PDAC scope.

Gate 2 — module geography → GO (non-branching)

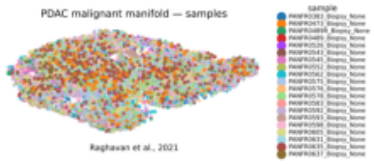
Single dominant pSom→Crest axis: pSom high over the classical-ductal body declines along pseudotime (ρ=−0.74); Crest anti-correlated (r=−0.68), depleted at the root and rising to a distal plateau (ρ=+0.61; crest-sm peaks mid-to-distal, decile pt=0.59). Gast co-varies with pSom (joint-high enrichment 1.15); joint pSom&Crest-high depleted (0.23) — an inverse gradient, not a hub.

Gate 3 — hub mutual exclusivity → NA

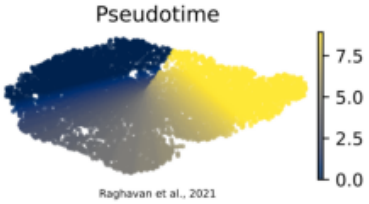
Single monotone axis — no bifurcation and no Mixed-type hub (curve: 2 tips, 0 forks, self-approach 2.57), so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

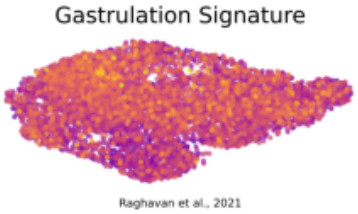
Manifold — samples



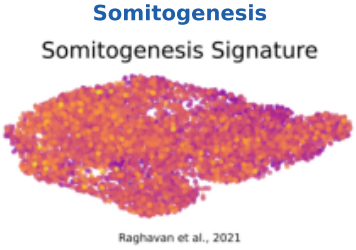
Pseudotime



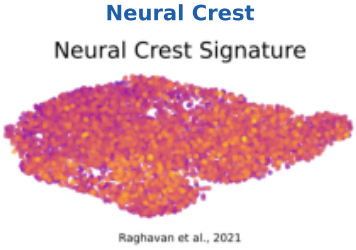
Gastrulation



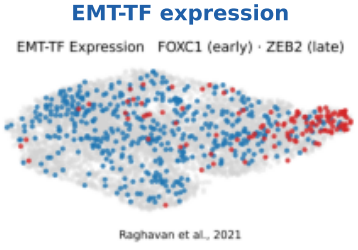
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
PDAC (Raghavan et al., 2021), Seq-Well — 24 patients / 24,412 cells (master scope); malignant only: 7,904 cells, of which 7,242 are PDAC (662 PANET neuroendocrine cells dropped). After the ≥120-cell floor, 7,168 cells across 20 samples enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Raghavan's metastatic PDAC malignant manifold is a single monotone somitogenesis → distal-neural-crest gradient (classical-ductal body → neural-crest-like distal pole); it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both lobes and is orthogonal to cell cycle (module–cyc |ρ|≤0.20; pt–cyc ρ=+0.11), so the whole manifold is kept. Raw module means are pSom-dominant and geography is a coherent inverse pSom→Crest gradient (Crest depleted at root, rising to a distal plateau) — not the mutually-exclusive territories of the failed Peng 2019 PDAC run. EMT-TFs FOXC1 (early/root) → ZEB2 (late) on the widened FOXC/PRRX panel.

Parameter choices

Min cells/sample	≥73 (geom-mean 120→45)	**Scope**	PDAC only; whole manifold
Root node	tip 0 = pSom-high, Crest-low	**Integration**	Harmony θ=10; PCA50; spread=1.3
Trajectory	scFates curve N=20 λ=0.3 μ=100	**Compartments**	Ductal-body 6,494 / Cycling 674

Raghavan 2021 PDAC — analysis summary

RUN 2 OF 5

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

21 samples intermix per-neighbourhood (iLISI 1.0→7.0 of 21) on one continuous component (0 islands). Developmental axis orthogonal to proliferation (|ρ|≤0.15 vs cell cycle) → whole manifold kept.

Gate 2 — module geography → GO (non-branching)

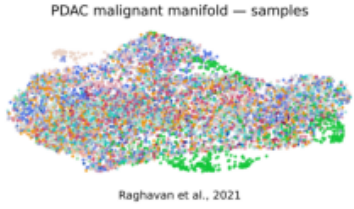
Single Gast/pSom→Crest axis: Gast & pSom co-vary (r=+0.25), both oppose Crest (r=−0.75 / −0.65); Crest an inverse terminal gradient. No Gast-vs-pSom arm split. Crest peaks at the terminus, not the Gast arm → no inversion flag.

Gate 3 — hub cross-repression → NA

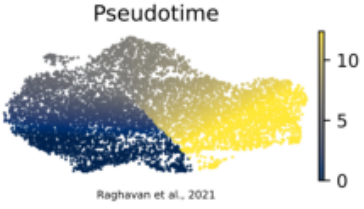
No bifurcation (PPT tree tips share one pSom-high identity; branch-node count unstable 2–4). Branching-only gate does not apply; 04_correlation / 05_modules not produced.

STEP FIGURES

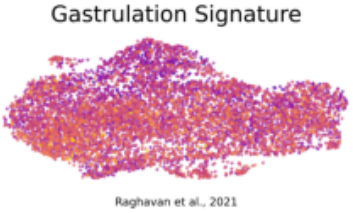
Manifold — samples



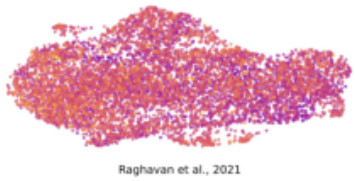
Pseudotime



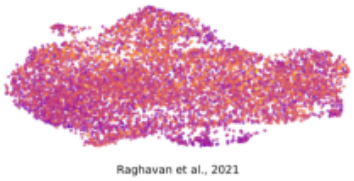
Gastrulation



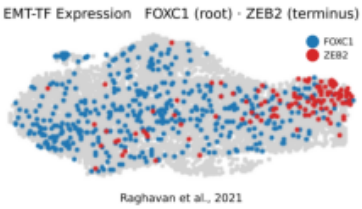
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Metastatic pancreatic ductal adenocarcinoma (Raghavan et al., 2021), Seq-Well — 7,830 malignant cells, 21 samples (mostly liver mets).

Verdict.
The embryonic pattern reproduces in collapsed form: one continuous gastrulation/somitogenesis → neural-crest axis (global means Gast −0.011 / pSom +0.072 / Crest −0.024; Linear-pSom/Crest), rooted at the Gast/pSom-high origin, EMT-TFs FOXC1 (root) → ZEB2 (terminus). Non-branching — no competing arms, so Gate 3 is NA. Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Sample threshold	geom-mean across largest count gap (120→45) = 73 → 21/23 samples	**Genes / PCA**	overdispersed selection; PCA 50, n_pcs = 20 (43% var)
Integration	Harmony θ = 10 (mixing plateaus; strongest without collapse)	**UMAP / neighbours**	n_neighbors = 30, spread = 1.3; island removal = radius-graph (0 removed)
Geometry	principal curve (ElPiGraph), Nodes 20, λ 0.01, μ 50; fold-free (0 forks)	**Root**	derived: Gast/pSom-high, Crest-low tip (Crest is late)
EMT-TFs	FOXC1 (early) · ZEB2 (late), detection-gated wide panel		

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

21/23 samples (≥120-cell gap threshold) intermix on one continuous manifold — iLISI median 1.00→7.04 of 21 (θ sweep 2/5/10, monotone), 0 detached islands, UMAP aspect 1.10 (one blob); both compartments (Classical-ductal 3,092 / Cycling 4,738) span all 21 samples.

Gate 2 — module geography → GO (non-branching)

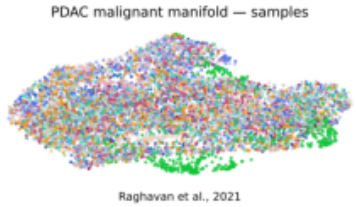
Single pSom→Crest axis: somitogenesis high at the origin declines along pseudotime (ρ=−0.64) while neural crest rises to a terminal readout (ρ=+0.48). Gast/pSom co-express (joint-high enrichment 1.13 >1 — no competing arms); Crest anti-correlated with both (r=−0.75 Gast, −0.65 pSom); joint pSom&Crest-high depleted (0.23) — an inverse gradient, not a hub.

Gate 3 — hub cross-repression → NA

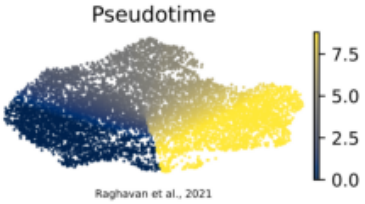
Non-branching manifold (principal curve, 2 tips / 0 forks; PPT forks are blob over-fit, not dev-signature arms). No bifurcation and no Gast/pSom co-expression hub — no branch pair to test; the correlation / modules notebooks are not produced.

STEP FIGURES

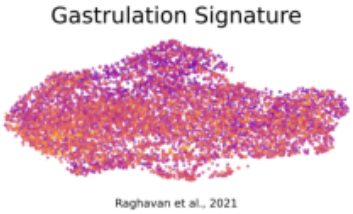
Manifold — samples



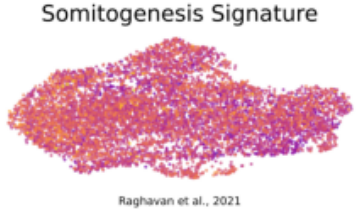
Pseudotime



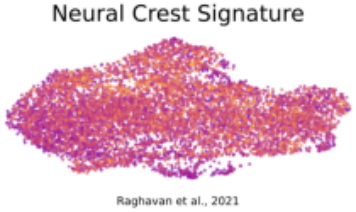
Gastrulation



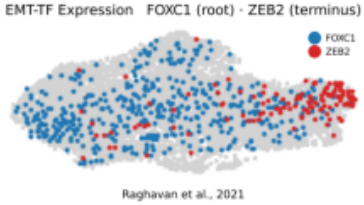
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
PDAC (Raghavan et al., 2021), Seq-Well metastatic biopsies — 24 samples / 24,412 malignant cells (master scope). After the 120-cell gap threshold, 21 samples / 7,830 cells enter preprocessing; whole malignant manifold (no lobe scoping).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous somitogenesis → neural-crest axis rooted at the Gast/pSom-high origin. Gast and pSom co-express (no bifurcation), so Gate 3 has no hub to test (NA). Crest is the terminal readout, not Gast-enriched (no inversion flag). Raw module means (Gast −0.011 / pSom +0.072 / Crest −0.024) match the failed prior run, but the call turns on geometry: a coherent single pSom→Crest gradient with a terminal crest readout → GO. Proliferation is orthogonal to the axis (module·cycling |ρ|≤0.19). EMT-TFs FOXC1 (early/root, 23.6%) → ZEB2 (late/terminus, 6.4%); canonical SNAI1/TWIST1 near the detection floor. Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Min cells/sample	120 (gap 120→45, r=2.67) → 21/23	**Integration**	Harmony θ=10; n_pcs=20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = pSom-high, Crest-low	**Geometry**	Linear (non-branching); 2 tips / 0 forks
Annotation	Moffitt classical/basal + Raghavan mp	**EMT-TFs**	FOXC1 (early) · ZEB2 (late)

Raghavan 2021 PDAC — analysis summary

RUN 4 OF 5

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ-sweep resolves a per-patient archipelago (θ=0, iLISI 1) into one continuous well-mixed mass (θ=3, iLISI 4.8/21). Island-free (largest component 99.1%; 72-cell speck removed).

Gate 2 — module geography → GO (non-branching)

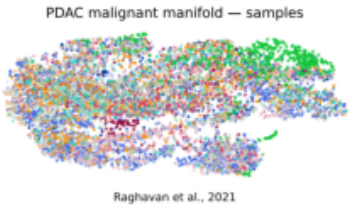
Single embryonic axis: Gast/pSom co-vary at the origin (r=+0.12), both oppose Crest toward the terminus (Gast↔Crest −0.50, pSom↔Crest −0.61; pSom&Crest joint-high enrichment 0.16). No two-arm hub.

Gate 3 — hub cross-repression → NA

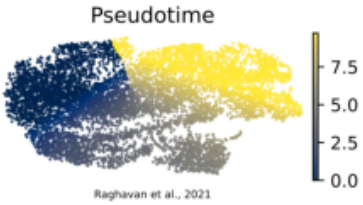
Curve has 0 forks (PPT tree forks unstable 2→3 across N — round-blob overfit). No branch to test.

STEP FIGURES

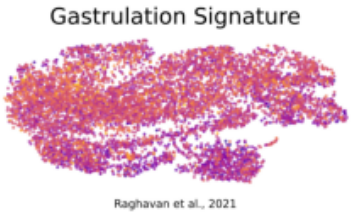
Manifold — samples



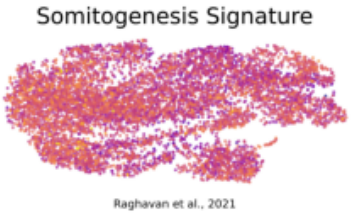
Pseudotime



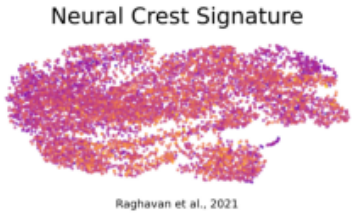
Gastrulation



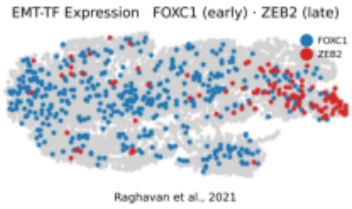
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Metastatic PDAC (Raghavan et al., 2021), Seq-Well — 7,904 malignant cells → 7,830 after 100/sample threshold (21/23 samples) → 7,758 after island removal. Subtypes: Classical 3,314 · Mixed 3,341 · Basal 1,103.

Verdict.
The embryonic pattern reproduces as the non-branching case — one continuous Gast/pSom-origin → Crest-terminus axis with pSom/Crest mutual exclusivity; developmental EMT-TF order FOXC1 → ZEB2. No bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast −0.011 / pSom +0.072 / Crest −0.024.

Parameter choices

Min cells/sample	100 (gap 45→120; 21/23 kept)	**Integration**	Harmony θ=3 · n_pcs=30 · spread=1.3
Gene selection	1,497 overdispersed (find_overdispersed)	**Scope**	whole manifold (cycling |ρ|≤0.19, orthog.)
Trajectory	scFates curve Nodes=20, λ=0.3, μ=100; fold-free (SA 2.56); 0 forks	**Root**	tip 0 = Gast/pSom-high, Crest-low (dev. origin)
Subtype source	Raghavan 2021 · Moffitt basal/classical	**EMT-TFs**	FOXC1 (early) · ZEB2 (late)

Raghavan 2021 PDAC — analysis summary

RUN 5 OF 5

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (sample, θ=5): same-sample neighbour fraction 0.94→0.26 (plateau by θ=2); samples intermix per-neighbourhood (iLISI) on one continuous connected manifold; 0 detached islands removed.

Gate 2 — module geography → GO (non-branching)

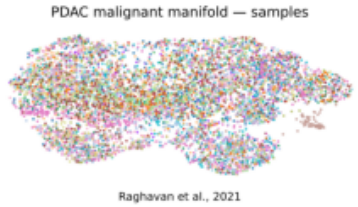
Non-branching: Gast & pSom co-vary (smoothed r=+0.32, joint-high enrichment 1.15 = overlap) and both oppose Crest (r=−0.80 / −0.69); single Gast/pSom→Crest gradient with Crest terminal (decile pseudotime 0.68). Classical/basal plasticity axis is real (r=−0.70) but ORTHOGONAL to the embryonic modules (pSom vs classical/basal +0.04/+0.05) — not the developmental bifurcation.

Gate 3 — cross-repression → NA

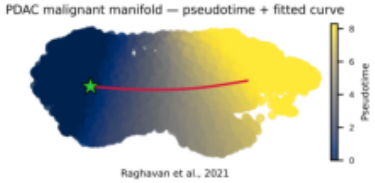
No Gast/pSom bifurcation (they overlap rather than cross-repress) → no hub to test. Gate 3 not applicable.

STEP FIGURES

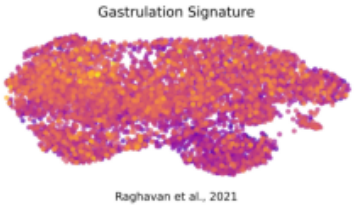
Manifold — samples



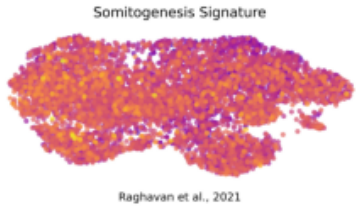
Pseudotime



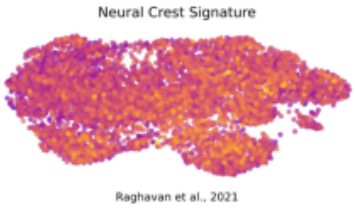
Gastrulation



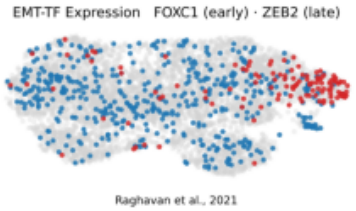
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic ductal adenocarcinoma (Raghavan et al., 2021), Seq-Well — 7,168 malignant cells (20 samples); disease-restricted to PDAC (662 PANET cells excluded). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching): a single Gast/pSom→Crest developmental axis with Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. The classical/basal plasticity axis is real (r=−0.70) but orthogonal to the embryonic modules — not the developmental bifurcation. EMT-TFs FOXC1 (early/root) → ZEB2 (late/terminus).

Parameter choices

Sample threshold	≥100 cells; 20/22 samples kept (45 & 29-cell samples dropped)	**Disease restriction**	PDAC only (662 PANET cells, patient PANFR0580, excluded)
Overdispersed genes	1,308; PCA 50PC; Harmony(sample, θ=5)	**Embedding**	knn n_pcs=20, n_neighbors=30; UMAP spread=1.3
Trajectory	scFates curve N=15, λ=0.3, μ=100 (2 tips, 0 forks); corr(x,pt)=+0.97	**Root**	Gast/pSom-high tip
Geometry	Linear (non-branching); classical/basal axis orthogonal	**EMT-TFs**	FOXC1 (early/root) → ZEB2 (late/terminus); 3% floor, no VIM

Moncada 2020 PDAC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Both patients intermix on one continuous manifold — no per-patient islands. Single connected component (radius-graph 0 detached; kNN-graph in Harmony-PCA also 1 component); iLISI median 1.00→1.97 of 2. Density dips mid-axis but no empty bins — one manifold, not two islands.

Gate 2 — module geography → GO (non-branching)

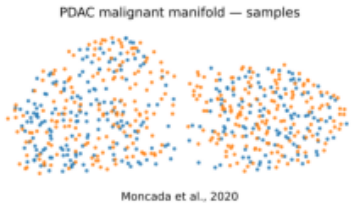
Single pSom→Gast/Crest axis: pSom high at the origin falls along pseudotime (ρ=−0.89) while Gast (ρ=+0.77) and Crest (ρ=+0.27) rise to the terminus. Gast↔pSom strongly anti-correlated in space (r=−0.89) — organised cross-repression — but the manifold does not bifurcate (0 forks). Joint pSom&Crest enrichment 0.29 (<1: inverse gradient). FLAG (passes anyway): Crest enriches the Gast end (r(Crest,Gast)=+0.38 vs r(Crest,pSom)=−0.56) — inversion of the canonical arm assignment.

Gate 3 — cross-repression → NA

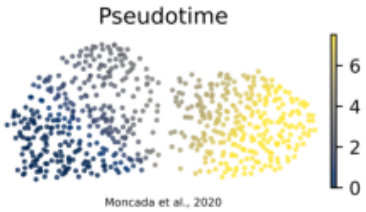
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

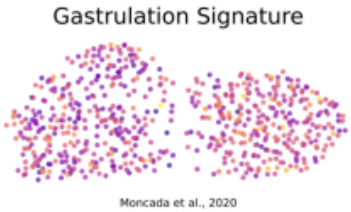
Manifold — samples



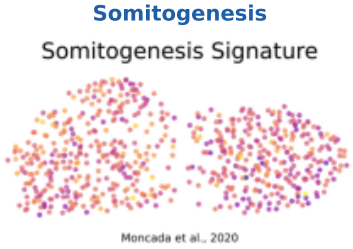
Pseudotime



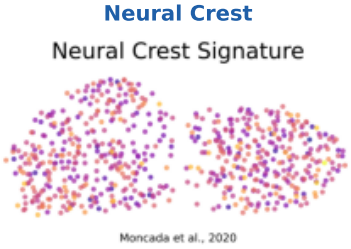
Gastrulation



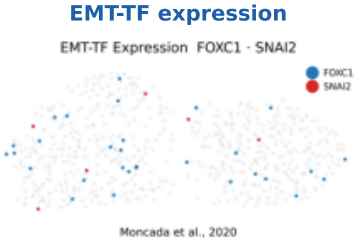
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
PDAC (Moncada et al., 2020), inDrop — 2 patients / 3,659 cells; malignant only: 635 cells (publication cell_type=="Malignant"), both samples clear the ≥100-cell floor, whole malignant manifold (no lobe scoping). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous somitogenesis → gastrulation/neural-crest axis, rooted at the pSom-high origin. Gast/pSom show organised cross-repression (r=−0.89) without bifurcation, so Gate 3 has no hub to test (NA). Crest inverts onto the Gast end (flagged). EMT-TFs are root-enriched (FOXC1 dominant at 46%, SNAI2 next); no distinct terminus factor. G1 GO / G2 GO / G3 NA.

Parameter choices

Sample threshold	≥100-cell floor (only 2 samples → gap-finding degenerate); both kept → 635 cells	**Overdispersed genes**	68 (scFates find_overdispersed)
Integration	Harmony on sample θ=10 (mixing reaches 2-sample chance floor)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3 (inDrop scale); 0 islands removed
Geometry	non-branching curve N=20, λ=0.3, μ=100; 2 tips / 0 forks (2.84)	**Root**	tip 2 = Gast/pSom-high, Crest-low developmental origin
Module means (raw)	Gast −0.011 · pSom +0.115 · Crest −0.064 (Linear-pSom/Crest)	**EMT-TFs**	FOXC1 (root, 46%) & SNAI2 (root, 9%); all expressed EMT-TFs root-enriched

He 2021 mCRPC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

11/12 samples (≥10 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 2.11→5.56 of 11 samples). All three transcriptional lobes (213 / 298 / 322 cells) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

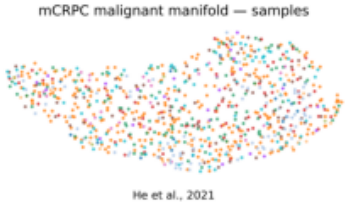
Single dominant Gast/pSom→Crest axis: gastrulation high at the origin declines steeply along pseudotime (ρ=−0.99); neural-crest anti-correlated (r=−0.82 vs Gast, −0.70 vs pSom), rising to a terminal readout (ρ=+0.85; crest-high decile mean pt=0.91). Gast↔pSom weakly positive (+0.18 — co-vary, not compete); joint pSom&Crest-high depleted (enrichment 0.15) — an inverse gradient, not a co-expression hub.

Gate 3 — cross-repression → NA

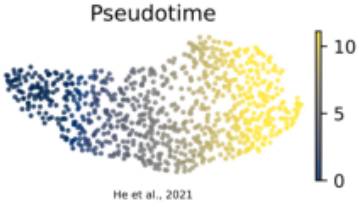
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

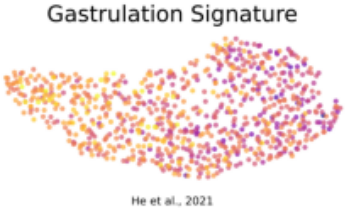
Manifold — samples



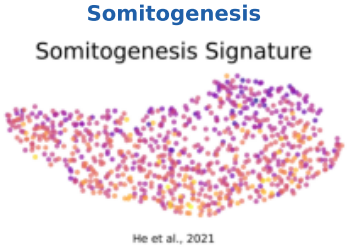
Pseudotime



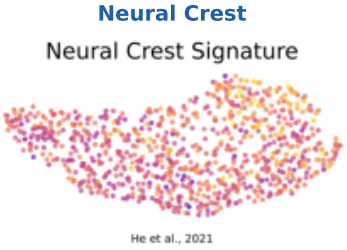
Gastrulation



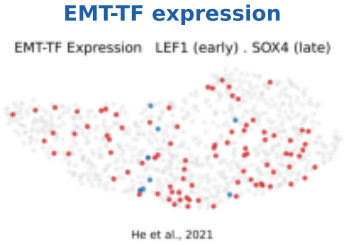
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Metastatic prostate cancer (He et al., 2021), SmartSeq2 — 15 samples / 2,170 cells (master scope); malignant only: 836 cells across 12 samples; 833 across 11 samples enter preprocessing (≥5-cell floor drops one 3-cell sample); the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. He's mCRPC malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans all three transcriptional lobes (orthogonal to cell cycle; Crest–cyc ρ≈0.00, Gast–cyc ρ=−0.31), so the whole manifold is kept. Raw module means (Gast −0.30 / pSom +0.05 / Crest +0.05) place it in the Linear-pSom/Crest class; the gate turns on geography — a resolved single axis with a terminal Crest readout — not the means. EMT-TFs: canonical SNAI1/ZEB1/ZEB2/TWIST1 all below the 3% floor (FOXC/PRRX absent from the SmartSeq2 matrix), so the detectable EMT signal is the neural-crest/mesenchymal TF program rising to the terminus (SOX4 42%, ρ=+0.31); overlay LEF1 (early) → SOX4 (late), end-separation weak (Δpt=0.12).

Parameter choices

Min cells/sample	≥5 (geo-mean of 10→3 gap; 11/12)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Embedding**	n_pcs=15; lobes 213/298/322
Geometry	Linear (non-branching); no hub	**EMT-TFs**	LEF1 (early) → SOX4 (late); canonical set below 3% floor

Chen 2021 Prostate — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

12/13 samples (≥55 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); the whole low-cycling malignant manifold (14,266 cells, cycling 2.1%) carries the trajectory — no separable compartment.

Gate 2 — module geography → GO (non-branching)

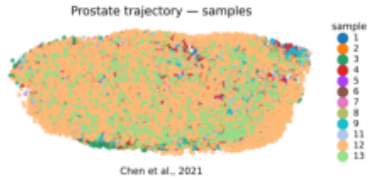
Somitogenesis & Gastrulation decline along pseudotime (ρ=−0.32 / −0.18); Neural-crest anti-correlated with pSom (r=−0.76), rising to a terminal readout (ρ=+0.41; crest-high decile mean pt=0.92). Gast/pSom do not co-vary (r=−0.11) — an inverse gradient, no hub.

Gate 3 — hub mutual exclusivity → NA

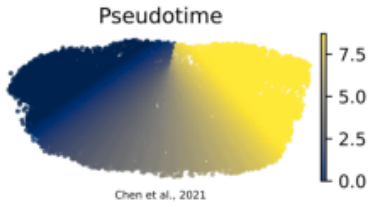
Single monotone pSom→Crest axis — no bifurcation and no Gast/pSom co-expressing hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

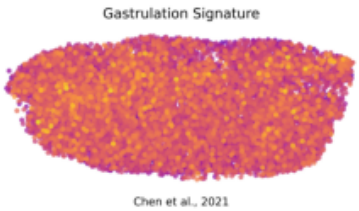
Manifold — samples



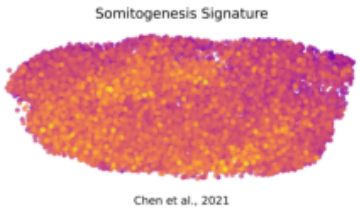
Pseudotime



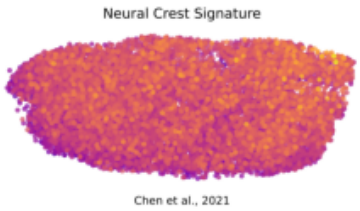
Gastrulation



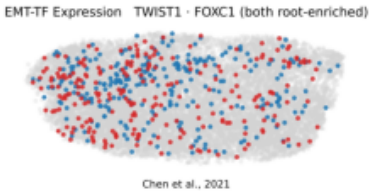
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Prostate cancer (Chen et al., 2021), 10x — 13 samples / 36,424 cells (master scope); malignant only: 14,291 load, 14,266 across 12 samples (≥55) enter preprocessing; all 14,266 (one continuous manifold, 0 detached islands) enter the trajectory.

Verdict.
Non-branching pattern — Gate 1 GO, Gate 2 GO, Gate 3 NA. Chen's prostate malignant manifold is a single monotone Somitogenesis → terminal Neural-crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Expressed EMT-TFs (TWIST1, FOXC1) are root-enriched and decline toward the terminus — no canonical EMT-TF is induced at the neural-crest end; SNAI1/ZEB1 near the detection floor (3.0% / 0.4%).

Parameter choices

Min cells/sample	≥ 55 (keeps 12/13; gap 125→25)	**Compartment**	whole manifold (2.1% cycling, no split)
Root node	tip = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=10; PCA 50PC; spread=1.3
Trajectory	scFates curve N=20 λ=0.3 μ=100	**Orientation**	x-flip, no swap (raw-x ∥ pt, r=+0.94)

Chen 2021 Prostate Cancer — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Integration + continuity: iLISI rises from a sample-segregated baseline (median 1.0) and plateaus by θ=10 (median 3.6, 12 samples intermixed); 0 detached islands — one continuous manifold.

Gate 2 — module geography → GO (non-branching)

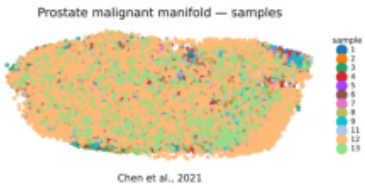
Shared Gast/pSom developmental gradient with Crest as an inverse terminal gradient (pSom↔Crest r=−0.76; Crest-high decile at pseudotime 0.92). No two competing arms.

Gate 3 — hub cross-repression → NA

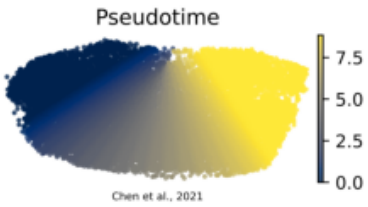
No bifurcation: a PPT tree over-forks a continuous cloud — all tips share one profile (low Gast, moderate pSom, low Crest), no opposing Gast-high vs pSom-high arms. Curve fit: 2 tips, fold-free.

STEP FIGURES

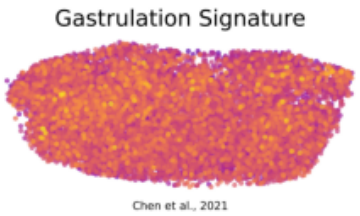
Manifold — samples



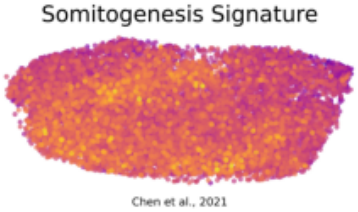
Pseudotime



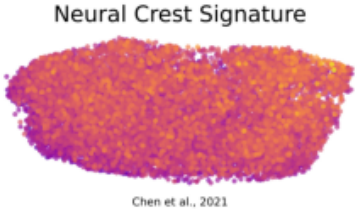
Gastrulation



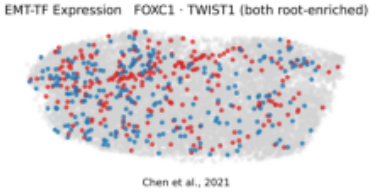
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Prostate adenocarcinoma (Chen et al., 2021), 10x — 14,266 malignant cells (12/13 samples above threshold).

Verdict.
Pattern reproduces as the non-branching case — one continuous gastrulation/somitogenesis → neural-crest developmental axis rooted at the Gast/pSom-high pole; no hub or bifurcation, so no cross-repression to test. Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. EMT-TFs FOXC1 / TWIST1 are both root-enriched — no expressed EMT-TF marks the neural-crest terminus.

Parameter choices

Min-cell threshold	55 (geom. mean, 125→25 gap; 12/13)	**Integration**	Harmony θ=10, n_pcs=20
Overdispersed genes	1,583	**Geometry**	Linear — non-branching curve
Root	Gast/pSom-high tip (origin)	**Module means**	Gast +0.004 / pSom +0.036 / Crest −0.069
Hub identity / r	NA (non-branching)	**EMT-TFs shown**	FOXC1 / TWIST1 (both root-enriched)

Song 2022 Prostate Cancer — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

14/15 samples clear the ≥100-cell floor and intermix per-neighbourhood (iLISI median 1.95 → 7.12 of 14 after Harmony θ=10); one continuous manifold, 0 detached islands.

Gate 2 — module geography → GO (non-branching)

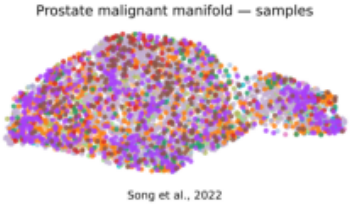
Dominant inverse gradient pSom↔Crest (r=−0.85); Gast↔pSom weak (r=−0.19). Tree tips all share one program → spurious forks; curve fold-free (self-approach 2.93). Single pSom→crest axis.

Gate 3 — hub mutual exclusivity → NA

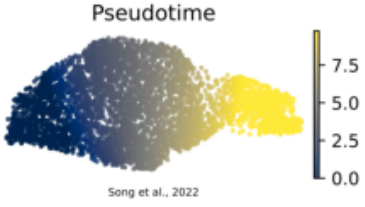
No bifurcation, so no convergence hub to test. Joint pSom&Crest-high enrichment 0.27 (≪1) confirms an inverse gradient, not a co-expression hub.

STEP FIGURES

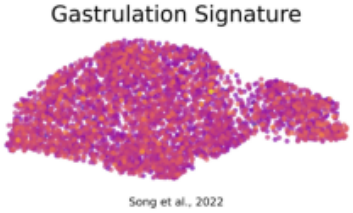
Manifold — samples



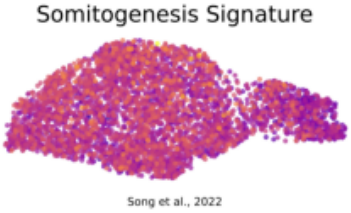
Pseudotime



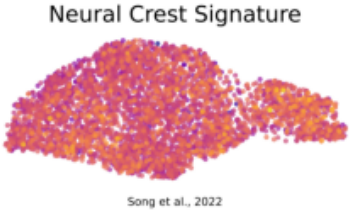
Gastrulation



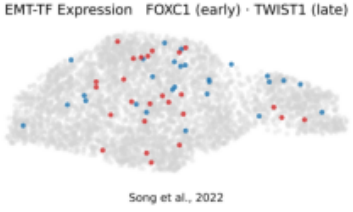
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Prostate Cancer (Song et al., 2022), Seq-Well S3 — 15 samples / 21,743 cells (master atlas); 4,658 malignant cells, of which 14 samples clear the ≥100-cell floor → 4,574 cells on the whole-manifold trajectory. Uniformly luminal AR-driven adenocarcinoma (KLK3 in 97% of cells). Two AR-output compartments: Luminal-AR-active (3,916) and Luminal-AR-low (658). Cycling 0.7%, orthogonal to all developmental programs.

Verdict.
Non-branching linear trajectory on a somitogenesis→neural-crest embryonic axis. Root at the Gast/pSom-high, Crest-low tip; pseudotime runs Gast/pSom-high → Crest-high (Crest decile mean pseudotime 0.95, terminal). Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. Classical mesenchymal EMT-TFs are largely below detection (max FOXC1 2.4%) — the honest low-EMT readout for a luminal epithelial adenocarcinoma; FOXC1 (early) / TWIST1 (late) shown.

Parameter choices

Min-cell threshold	≥100 floor (mult. gap degenerate at rank 0); 14/15 kept	**Integration**	Harmony on sample θ=10 (plateaus by θ=5); PCA 50→20
Overdispersed genes	199; neighbours k=30; UMAP spread=1.3, seed=1	**Islands / lobes**	UMAP radius graph 6× median-NN (0 removed); Leiden res=0.15
Trajectory	curve 20 nodes, epg λ=0.3 μ=100 (fold-free 2.93)	**Root**	Gast/pSom-high tip (Crest-low origin)
EMT-TF axis	FOXC1 (early) → TWIST1 (late)	**Geometry**	Linear — pSom/Crest (non-branching)

Dong 2020 Prostate — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Malignant cells from all 4 patients (9,698; ≥100-cell floor, no sample dropped) form one continuous manifold — 0 detached islands, single connected component; every transcriptional lobe contains all 4 patients. Judged on the embedding, not the composition-capped iLISI (1.78/4, one patient ≈49% of cells).

Gate 2 — module geography → GO (non-branching)

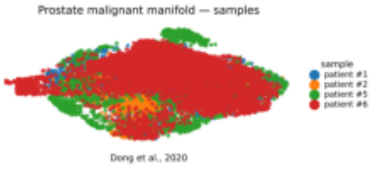
Single developmental axis: Gast & pSom co-vary (smoothed r=+0.90; joint-high enrichment 2.4 — an overlap, not competing arms) and jointly oppose neural-crest (pSom↔Crest r=−0.53). Along pseudotime Gast/pSom decline (ρ −0.58/−0.71), Crest rises to a terminal end (ρ +0.79, Crest-high decile pt 0.98). Non-branching — passes.

Gate 3 — hub cross-repression → NA

No bifurcation (fold-free single curve, 2 tips / 0 forks). The branching-only cross-repression test does not apply; correlation & modules notebooks are not produced.

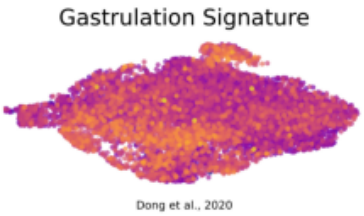
STEP FIGURES

Manifold — samples

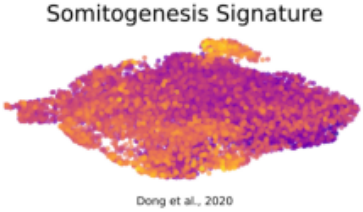


Pseudotime

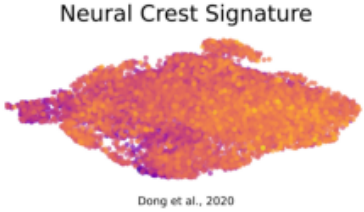
Gastrulation



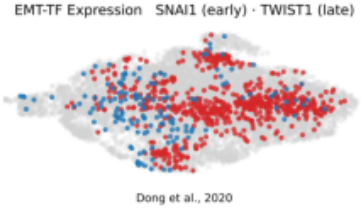
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Prostate adenocarcinoma (Dong et al. 2020), 10x — 9,698 malignant cells across 4 patients (≥100-cell floor, all samples kept; publication cell_type calls). Non-branching embryonic axis.

Verdict.
The embryonic program reproduces as a single monotone gradient — a Gast/pSom-high (overlap) origin flowing to an inverse neural-crest terminus — the collapsed-bifurcation (Linear-pSom/Crest) case, distinct from Puram's hub-and-spoke. Gate 1 GO, Gate 2 GO, Gate 3 NA.

Parameter choices

Malignant / threshold	9,698 cells, 4 patients; ≥100 floor (no cliff; all kept)	**Overdispersion / PCA**	1,670 overdispersed; 50 PC → 20 for kNN
Integration	Harmony on sample, θ=10 (one patient ≈49%); spread=1.3	**Island removal**	radius graph 6× median NN → 0 removed (one component)
Compartments	Leiden res=0.15 → Body 8,696 + Cycling 1,002; whole kept	**Cell-cycle**	cc-free (anti-corr survives 7,988 non-cycling; pt↔cyc ρ=−0.24)
Trajectory	scFates curve N=20 λ=0.3 μ=100; 2 tips 0 forks (self-approach 2.59)	**Root / Orientation**	tip 2 (Gast/pSom-high); rigid rotation −170.4°, corr(x_lin,pt)=+0.85
EMT-TFs	SNAI1 (root, 6.0%) → TWIST1 (terminus, 26.8%); Δpt=0.37		

Dong 2020 Prostate Cancer — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

The malignant manifold clusters BY SAMPLE. After Harmony(sample) the kNN graph is a single component and median iLISI rises 1.00→1.31 — the scalars a naive test reads as 'integrated'. But on the regenerated embedding each patient holds its own territory: self-neighbour enrichment 3.3×/7.3×/3.7×/1.7× (patients #1/#2/#5/#6); 3 of 4 cross the 2.5× segregation line, and patient #6 (small-cell NEPC, ~49% of cells) forms its own pole. Integration does not fix it — enrichment barely moves across θ 2→20 and iLISI plateaus at 1.31 (θ=50 collapses); holds across n_pcs 20/30/40. The geography is inter-sample clonal/CNV + histologic (adeno vs NEPC) heterogeneity, not a shared cell-state continuum.

Gate 2 — module geography → NA

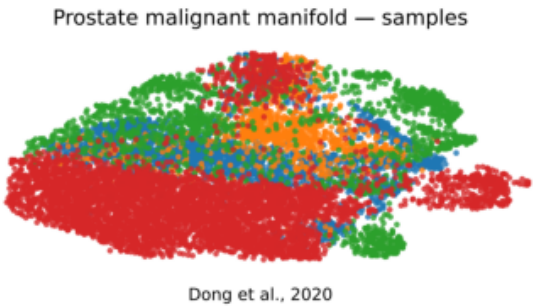
Not evaluated — Gate 1 NO-GO (chain stops). Every sample carries the same embryonic modules, but there is no shared sample-integrated manifold on which to fit a trajectory.

Gate 3 — hub cross-repression → NA

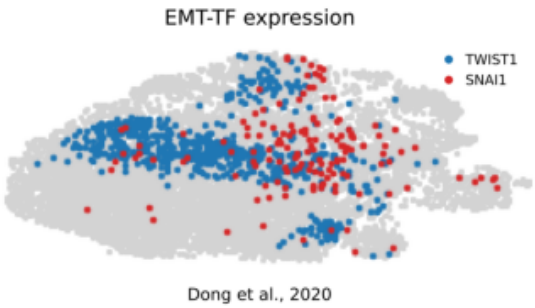
Not evaluated — requires a branching trajectory, which Gate 1 precludes.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Prostate adenocarcinoma + small-cell NEPC (Dong et al., 2020), 10x — 9,698 malignant cells (4 patients).

Verdict.
The embryonic-program pattern does NOT reproduce — the malignant compartment does not form a shared, sample-integrated cell-state manifold. It fragments into per-patient islands (clonal/CNV heterogeneity) plus a distinct small-cell NEPC pole (patient #6), so no trajectory is fit. Gate 1 NO-GO / Gates 2 & 3 NA. Descriptive EMT-TF overlay: TWIST1 (26.8%) · SNAI1 (6.0%).

Parameter choices

Min-cell threshold	floor 100 (all 4 samples kept, 915→4,765 cells); gap-finding rule REJECTED (premise unmet)	**Overdispersed genes**	1,670 (scFates GAM)
PCA / n_pcs	n_pcs=30 (Kürten 20 / Puram 40); Gate-1 call robust at 20/30/40	**Integration**	Harmony(sample); θ swept 2/5/10/20(+50); θ=10 best (max iLISI); none integrates
Geometry	None — no trajectory fit (Gate 1 NO-GO); per-sample archipelago	**EMT-TFs**	TWIST1 (26.8%) · SNAI1 (6.0%); 10-TF panel widened to FOXC/PRRX; VIM excluded

Dong 2020 Prostate Cancer — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

One connected kNN component; adeno (#1/#2/#5) intermix and are continuous with the NEPC pole (#6). iLISI rises 1.00 → 1.70 (ceiling 4) under Harmony θ=10, plateauing by θ=20; n_pcs=30 (robust band). Overturns the prior scalar-led NO-GO — the embedding is one continuum.

Gate 2 — module geography → GO (non-branching)

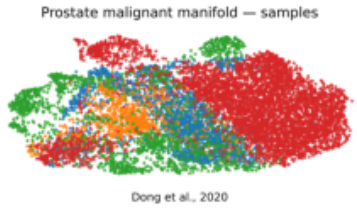
Single developmental axis: Gast & pSom co-vary (r=+0.91) at one pole; Crest inverse-gradient terminus. Crest opposes Gast (−0.34) and pSom (−0.48); joint pSom&Crest enrichment 0.66 (<1 → inverse, not a hub). No Gast arm exists to invert onto → no Crest-arm flag. Passes as the non-branching case.

Gate 3 — hub cross-repression → NA

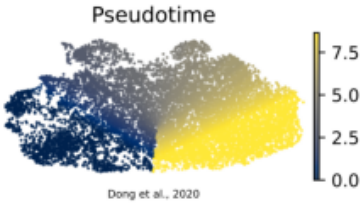
Committed principal curve is 2 tips / 0 forks. Branching check (shown, not asserted): a forced PPT tree (Nodes=200) gives 7 tips / 5 branch nodes — but node-density over-segmentation, not a bifurcation: branch count collapses 5→4→3→2→1 as PPT nodes drop 200→100→50→30→20, and the 7 tips co-vary in Gast & pSom (r=+0.90), not competing arms. No bifurcation → Gate 3 not applicable.

STEP FIGURES

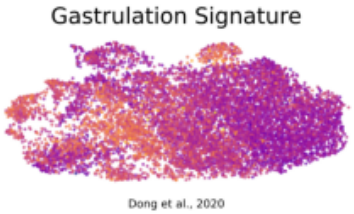
Manifold — samples



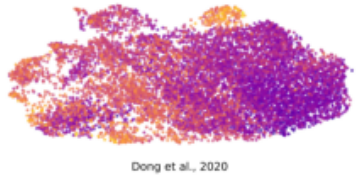
Pseudotime



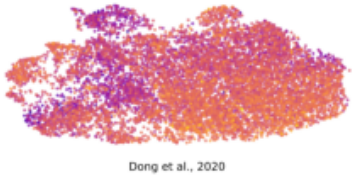
Gastrulation



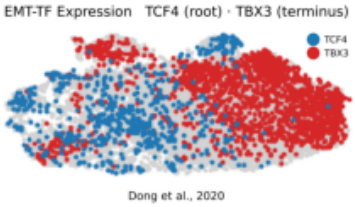
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Primary prostate tumours (Dong et al., 2020), 10x — 5 adenocarcinoma + 1 small-cell NEPC; malignant compartment, 9,698 cells across 4 malignant-bearing samples.

Verdict.
The pattern reproduces as the NON-BRANCHING case — one gastrulation/somitogenesis-high origin (adeno) → neural-crest-high terminus (NEPC) axis, cross-repression collapsed. Geometry: Linear. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast +0.035 / pSom +0.100 / Crest −0.041. EMT-TFs TCF4 (early/root) → TBX3 (late/terminus); Crest-high decile mean pseudotime 0.92 (terminal).

Parameter choices

Min-cell threshold	floor ≥100 (gap premise fails: counts 915–4765, gap at top)	**Samples kept**	4/4 malignant-bearing (of 6 total patients)
n_pcs / θ	30 (robust 20–40 band) / Harmony θ=10 (iLISI plateau)	**Geometry**	Linear — principal curve (ElPiGraph, 20 nodes, fold-free 2.64)
Root	Gast/pSom-high curve tip (developmental origin, Crest-low)	**Module means**	Gast +0.035 / pSom +0.100 / Crest −0.041 (raw score_genes)
EMT-TFs	TCF4 (early/root) → TBX3 (late/terminus), detection-gated		

Dong 2020 Prostate Cancer — analysis summary

RUN 4 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant cells split into two disconnected lineage lobes — adenocarcinoma (patients #1/#2/#5) and small-cell neuroendocrine NEPC (patient #6) — and never sample-integrate. Judged on the regenerated embedding across a θ sweep with per-neighborhood iLISI (max 4): iLISI stays ~1.2 and each lobe is a per-patient archipelago; n_pcs-invariant (20/30/40). Mixing appears only at θ=8 by destroying the embedding. Not mixed, not continuous → do not force a trajectory.

Gate 2 — module geography → NA

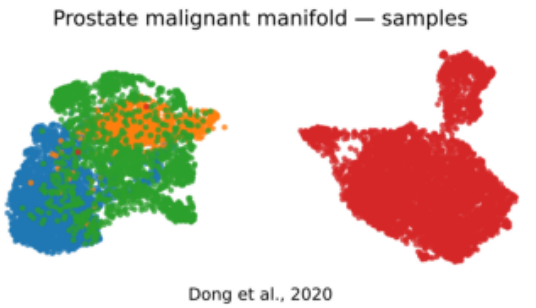
Not evaluated: the gate chain stops at the Gate 1 NO-GO (no manifold/trajectory is fit).

Gate 3 — hub cross-repression → NA

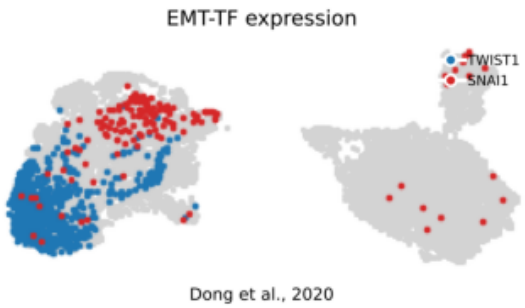
Not evaluated (branching-only; requires a fitted tree, which Gate 1 precludes).

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Prostate cancer (Dong et al., 2020), 10x — 6 patients, 21,292 cells total; 9,698 malignant, of which 4,933 adenocarcinoma and 4,765 small-cell NEPC. Patients #3/#4 contribute no malignant cells.

Verdict.
The embryonic-program pattern does NOT reproduce. The malignant compartment is two distinct prostate-cancer lineages (adeno vs NEPC) that neither integrate nor form one continuous manifold; the adeno lobe alone is a per-patient archipelago. Gate 1 NO-GO — no trajectory fit. The EMT-TF program (TWIST1 dominant) is confined to the adeno lobe.

Parameter choices

Min-cell threshold	≥100 malignant/sample → keeps all 4 present samples (#3/#4 have 0)	**Overdispersed genes**	1,670 (scFates GAM)
Integration	Harmony(sample); θ sweep 0/2/5/8; n_pcs=20 (grid-invariant 20/30/40)	**Gate-1 metric**	per-neighborhood iLISI (k=90) + embedding read; mean iLISI 1.0/1.2/1.6/1.9 by θ (max 4)
Lobe biology	adeno: Gast–pSom r=+0.55, both oppose Crest; NEPC: distinct AR+/CHGA+ lineage	**Geometry**	none fit (Gate 1 NO-GO)
EMT-TFs	TWIST1 (dominant), SNAI1, FOXC1; VIM excluded		

Dong 2020 Prostate — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Samples intermix per-neighbourhood (iLISI 1.93 of 4 at θ=10, up from 1.0 pre-Harmony) on ONE connected component (kNN + independent UMAP radius graph); 9,698 malignant cells, no islands removed.

Gate 2 — module geography → GO (non-branching)

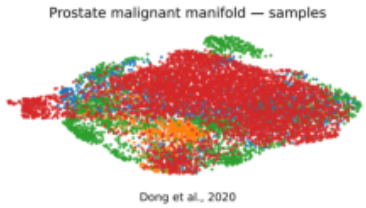
Shared Gast/pSom gradient (smoothed r=+0.90, co-vary) with Crest an inverse terminal readout (r=−0.39 / −0.53). Curve fold-free (0 forks); Crest-high decile at pseudotime 0.98.

Gate 3 — hub cross-repression → NA

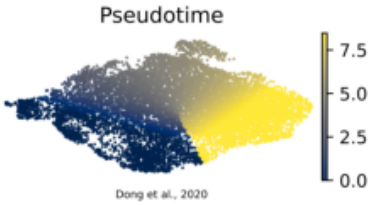
Gast/pSom co-vary (not compete) → single collapsed axis, no bifurcation to test.

STEP FIGURES

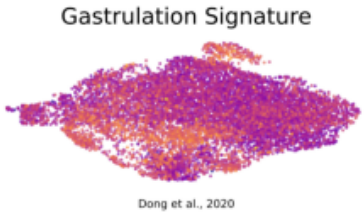
Manifold — samples



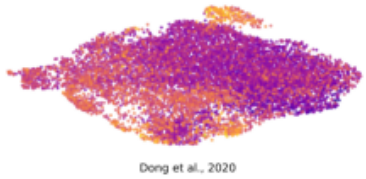
Pseudotime



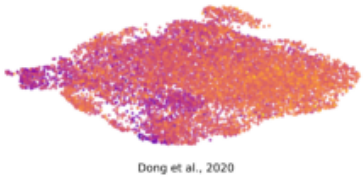
Gastrulation



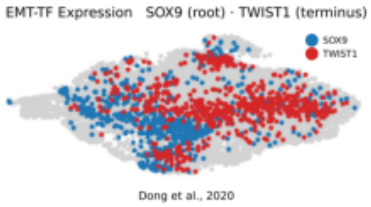
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Prostate cancer (Dong et al., 2020), 10x — 4 malignant-bearing patients (5 adeno + 1 small-cell NEPC by clinical histology). 9,698 malignant cells; luminal & neuroendocrine states are poles of one continuous manifold.

Verdict.
Pattern reproduces as a non-branching embryonic axis: Gast/pSom-high origin → Crest-high terminus; developmental order pSom ρ=−0.71, Crest ρ=+0.79 along pseudotime, checked off a diffusion embedding independent of the fitted UMAP. Linear geometry (pSom globally strongest module). Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs SOX9 (root) → TWIST1 (terminus).

Parameter choices

Min cells/sample	≥100 (no low-count gap; 4/4 kept)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Scope	whole manifold (luminal+NE)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	Gast/pSom-high tip	**EMT-TFs**	SOX9 (root) → TWIST1 (terminus)

Dong 2020 prostate cancer — analysis summary

RUN 6 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 4 malignant-bearing samples (≥915 cells each) intermix in one connected manifold — single kNN component, 0 detached islands; iLISI median rises 1.00→1.28/4 under integration (one patient is ~49% of cells, capping the scalar). Judged on the embedding across a θ×n_pcs sweep, not a scalar.

Gate 2 — module geography → GO (non-branching)

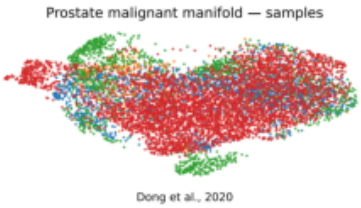
Single Gast/pSom → Crest axis. Gast & pSom spatially OVERLAP and co-vary (r=+0.47) — not a pSom-arm vs Gast-arm split; both oppose Crest (Gast–Crest −0.38, pSom–Crest −0.40). Along pseudotime Gast/pSom fall (pSom ρ=−0.41) and Crest rises (ρ=+0.39) — an inverse terminal gradient, not a hub-and-spoke bifurcation.

Gate 3 — cross-repression → NA

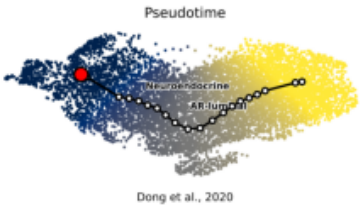
Single monotone axis — no bifurcation and no Gast/pSom-overlap "Mixed" hub with divergent arms, so there is no branch pair to test. (Spatially-defined hub Gast↔pSom r≈−0.04, i.e. no cross-repression; the apparent −0.38 under sum-thresholding is a collider/selection artifact, not competition.)

STEP FIGURES

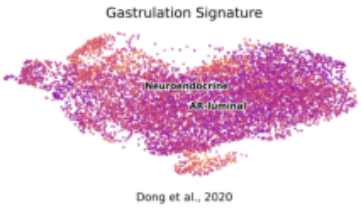
Manifold — samples



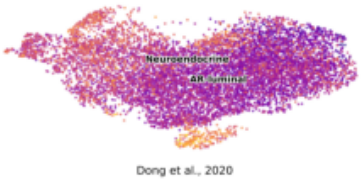
Pseudotime



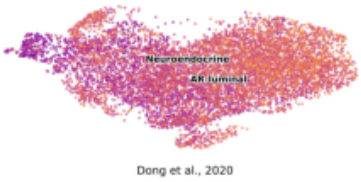
Gastrulation



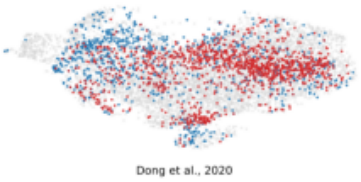
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Prostate cancer (Dong et al., 2020), 10x — 9,698 malignant cells across 4 samples (≥10x floor keeps all; no clean count gap). 8,281 enter the trajectory after a cross-patient cycling compartment (1,417 cells) is scoped out; the developmental manifold is a single connected component.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient, a strict linear subset of Puram's hub-and-spoke: Gast & pSom co-vary and overlap rather than diverging into competing arms, so there is no Mixed hub and no bifurcation to test. Raw module means Gast +0.035 / pSom +0.100 / Crest −0.041. An orthogonal AR-luminal ↔ neuroendocrine lineage axis (decoupled from clinical histology) is continuous and multi-patient — not a second trajectory. Curve is fold-free (self-approach 1.74; independent-diffusion continuity −0.79). EMT-TFs: SOX9 (early/root) → TWIST1 (terminus), detection-gated, VIM excluded.

Parameter choices

Min cells/sample	≥100 (no count gap; all 4 samples 915–4765)	**Integration**	Harmony θ=10; PCA 50; spread=1.3
Scope	cycling lobe excluded (1,417) → 8,281 dev	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	Gast/pSom-high tip (Crest-low origin)	**Embedding**	n_pcs=20; overdispersed=1,670
Geometry	Linear (non-branching); 2 tips / 0 forks	**EMT-TFs**	SOX9 (early/root) → TWIST1 (terminus); hub r NA

Zhou 2020 Osteosarcoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9/11 samples (≥599 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→4.51 of 9); both lobes (Embryonic 22,156 / Cycling 10,650) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

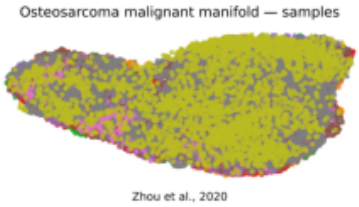
Single dominant pSom→Crest axis: pSom high over the osteoblastic body declines along pseudotime (ρ=−0.66); Crest anti-correlated (r=−0.88), rising to a terminal readout (ρ=+0.55; crest-high decile mean pt=0.93). Gast weakly structured (early programme, spent).

Gate 3 — hub mutual exclusivity → NA

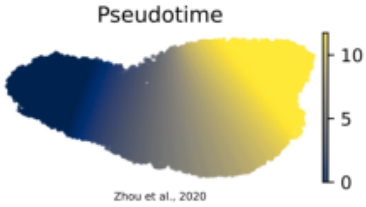
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

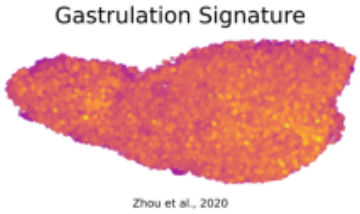
Manifold — samples



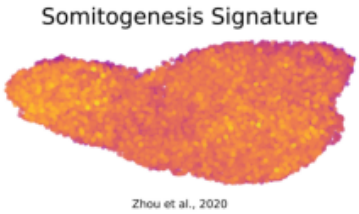
Pseudotime



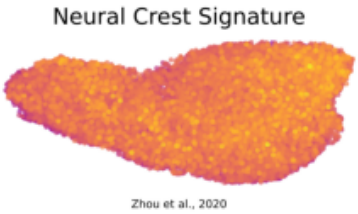
Gastrulation



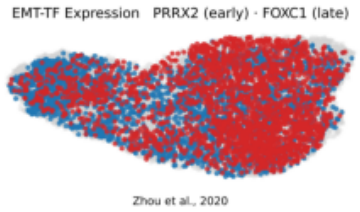
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Osteosarcoma (Zhou et al., 2020), 10x — 11 patients / 64,557 cells (master scope); malignant only: 33,459 load, 32,806 across 9 samples (≥599) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zhou's osteosarcoma malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient (osteoblastic body → chondroblastic tip), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both proliferation lobes (orthogonal to cell cycle; module–g2m |ρ|≤0.14), so the whole manifold is kept (Bischoff/Sun no-scoping precedent). Caveat: pseudotime co-varies negatively with cycling (ρ≈−0.55) as the somitogenesis-high root lobe is the more proliferative one. EMT-TFs PRRX2 (early/root) → FOXC1 (late/terminus); canonical SNAI1/ZEB1 near the detection floor; EMT-TF end-separation weak (Δpt=0.21).

Parameter choices

Min cells/sample	≥599 (9/11; gap 1007→357)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = pSom-high, Crest-low	**Compartments**	Embryonic 22,156 / Cycling 10,650

Jerby-Arnon 2021 Synovial Sarcoma — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

3 patients (SyS12pt/13/14) intermix on one continuous manifold; the kNN graph is a single component across the θ sweep (same-sample frac 1.00→0.72 at θ=5). Cross-platform concat fails (10x/SmartSeq2 stay platform-segregated, ~0.87 at every θ), so the run is scoped to the 10x leaf.

Gate 2 — module geography → GO (non-branching)

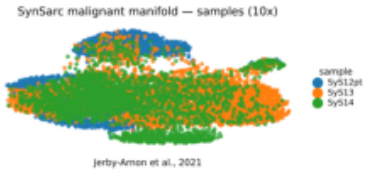
Gast & pSom co-vary (smoothed r=+0.76) while Crest opposes both (−0.75 / −0.97); the Crest-high decile sits at the trajectory terminus (mean pt 1.00). pSom-dominant → Linear-pSom/Crest.

Gate 3 — hub cross-repression → NA

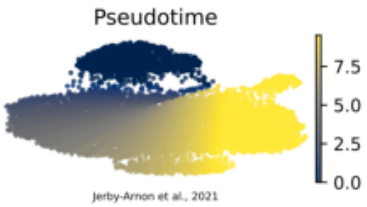
Single monotone curve (0 forks), so the branching-only cross-repression test does not apply.

STEP FIGURES

Manifold — samples

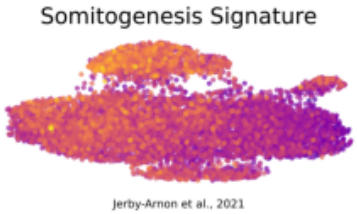


Pseudotime

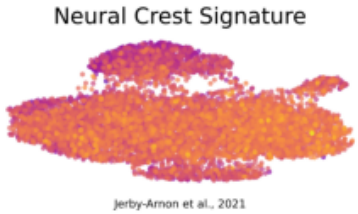


Gastrulation

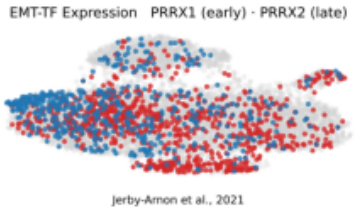
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Synovial sarcoma (Jerby-Arnon et al., 2021); 10x scRNA-seq (UMI counts); 8,322 malignant cells, 3 patients, after island removal. The SmartSeq2 leaf (4,371 cells, 11 patients) reproduces the same axis direction as an independent check.

Verdict.
Pattern reproduces (non-branching). A single gastrulation/somitogenesis-high origin flows to a neural-crest terminus; the EMT axis runs PRRX1 (root) → PRRX2 (terminus). Gate 1 GO / Gate 2 GO / Gate 3 NA — the Linear-pSom/Crest outcome shared by most CCCA datasets.

Parameter choices

Min cells/sample	none (3 balanced samples; ratio 1.59)	**Scope**	10x leaf (platform un-integratable)
Root node	tip 2 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=5 (θ=10 over-integrates)
Trajectory	scFates curve N=20 λ=0.3 μ=100 (SA 2.58)	**EMT-TFs**	PRRX1 (root) → PRRX2 (terminus)

Jerby-Arnon 2021 synovial sarcoma — analysis summary RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO (scoped to SmartSeq2)

Two platforms (SmartSeq2 TPM, 12 pt; 10x UMI, 3 pt) are an irreducible batch — technology iLISI stays 1.0/2 under every Harmony strength and batch key. The SmartSeq2 compartment is one continuous, patient-integrated manifold (iLISI 1.0→6.2 of 11 patients, single component); 10x has only 3 patients and fragments. Per the biology-before-geometry rule the trajectory is scoped to SmartSeq2.

Gate 2 — module geography → GO (non-branching)

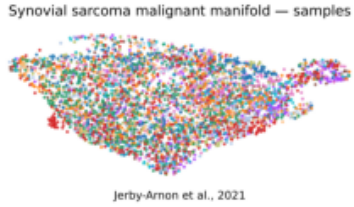
Signatures collapse onto one axis: smoothed Gast & pSom co-vary (r=+0.23) and both oppose Crest (−0.37 / −0.35). Along pseudotime Gast declines (ρ −0.47) and Crest rises to a terminal pole (ρ +0.48; Crest-high decile at t=0.71). No Gast-vs-pSom split → non-branching (still a GO). Crest enriches the terminus, not the Gast origin → no inversion flag.

Gate 3 — cross-repression → NA

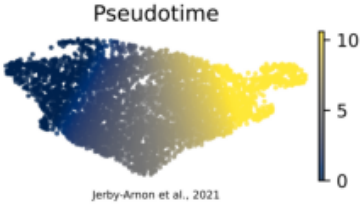
No bifurcation: a PPT tree only wallpapers the compact cloud (forks grow 6→9→15 with node count), while the fold-free principal curve has 0 forks. With no two arms there is no hub to test for mutual Gast↔pSom repression.

STEP FIGURES

Manifold — samples

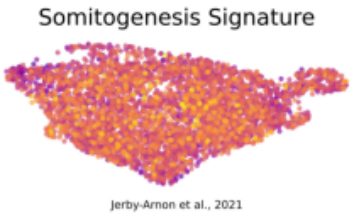


Pseudotime

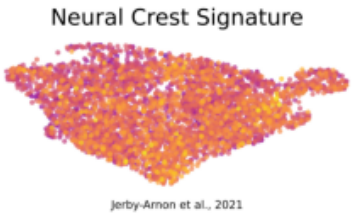


Gastrulation

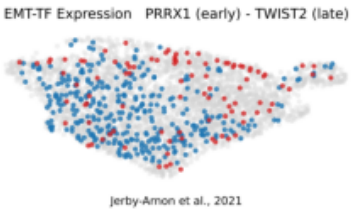
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Synovial sarcoma (Jerby-Arnon 2021); SmartSeq2 + 10x. 12,694 malignant cells merged; trajectory on the 4,338-cell SmartSeq2 manifold (11 patients after thin-sample removal).

Verdict.
The embryonic programme reproduces as a single non-branching developmental axis — a Gast/pSom-high origin flowing to a neural-crest-high terminus, with cell cycle orthogonal (ρ −0.01). This is the Linear-pSom/Crest geometry (pSom globally dominant, mean +0.32). No hub, no cross-repressing arms → Gate 3 NA. EMT axis: PRRX1 (early/root) → TWIST2 (late/terminus).

Parameter choices

Integration scope	SmartSeq2 (technology irreducible: iLISI_tech=1.0/2)	**Sample threshold**	≥98 cells (mult-gap 296→33, 9.0×); drop SyS10
Malignant cells	12,694 merged → 4,338 SmartSeq2 (11 patients)	**Integration**	Harmony θ=10 on sample (iLISI 1.0→6.2/11); PCA20, spread 1.3
Geometry	Linear-pSom/Crest (0 forks, fold ratio 2.95)	**Root**	curve tip 0 = Gast/pSom-high, Crest-low
Gene recovery	Gast 78/81 · pSom 95/98 · Crest 102/104	**EMT-TFs**	PRRX1 (root) → TWIST2 (terminus)

Tirosh 2016 Melanoma — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 samples (≥182 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); mean cross-sample kNN fraction 0.47. All cells MITF-high — a single melanocytic compartment carries the trajectory (no lobe split).

Gate 2 — module geography → GO (non-branching)

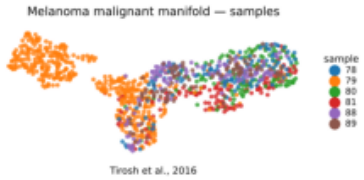
Gast/pSom co-vary (r=+0.93) and decline along pseudotime (ρ=−0.73/−0.71); Crest anti-correlated (r=−0.81/−0.83), rising to a terminal readout (ρ=+0.41; crest-high decile mean pt=0.92).

Gate 3 — hub mutual exclusivity → NA

Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

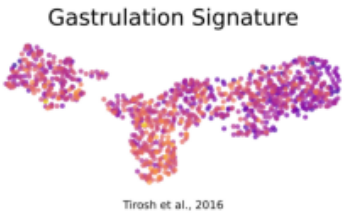
STEP FIGURES

Manifold — samples

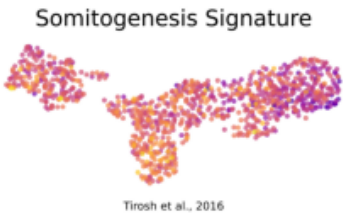


Pseudotime

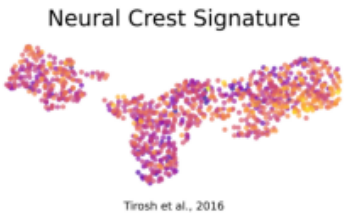
Gastrulation



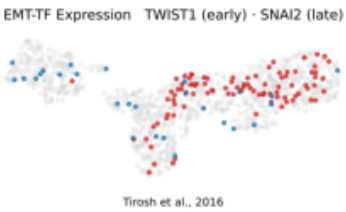
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Melanoma (Tirosh et al., 2016), SmartSeq2 — 19 patients / 4,645 cells (master scope); malignant only: 1,257 load across 15 samples, 1,061 across 6 samples (≥182 cells) enter preprocessing. All cells form one continuous MITF-high melanocytic compartment that carries the trajectory (no lobe split).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Tirosh's melanoma manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Data-derived EMT-TFs are TWIST1 (early/root, ρ=−0.05) → SNAI2 (late/terminus, ρ=+0.44); Puram's canonical SNAI1/ZEB1 are expressed (10%/11%) but not the most end-specific factors here.

Parameter choices

Min cells/sample	≥182 (keeps 6/15; gap 98→54)	**Compartment**	single melanocytic (all MITF-high)
Root node	tip = Gast/pSom-high, Crest-low	**Integration**	Harmony(sample) θ=2; PCA 50PC; UMAP spread=1.0
Trajectory	scFates curve N=20 λ=0.3 μ=100	**Subtypes**	Cycling/Hypoxic, Melanocytic-quiescent, Pigmented-Stress

Tirosh 2016 Melanoma — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/15 samples (≥72 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→4.00 of 6). Each of the 6 patients traverses the shared gastrulation→neural-crest axis end-to-end (per-patient ρ(axis,Crest) −0.97..−0.80).

Gate 2 — module geography → GO (non-branching)

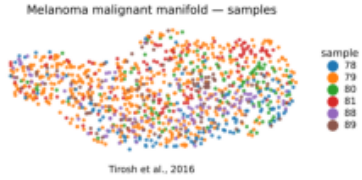
Single dominant Gast/pSom → Crest axis. Gast & pSom overlap at the origin (joint enrichment 1.37; spatial r=+0.97) and decline along pseudotime (ρ=−0.98/−0.98); Crest is the inverse terminal gradient (pSom&Crest enrichment 0.23, spatial r=−0.94), peaking at the terminus (Crest-high decile mean pt=0.98). Crest enriches the terminus, not a Gast arm — no inversion flag.

Gate 3 — hub mutual exclusivity → NA

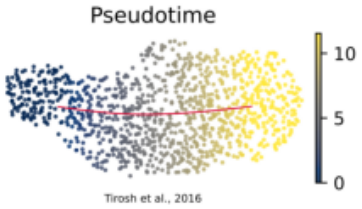
Single monotone axis — no bifurcation and no Gast/pSom-overlap hub, so there is no branch pair to test for cross-repression.

STEP FIGURES

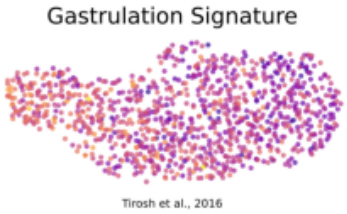
Manifold — samples



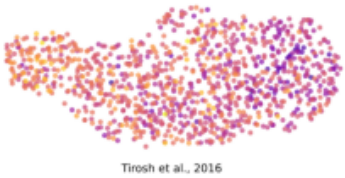
Pseudotime



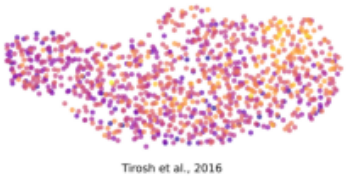
Gastrulation



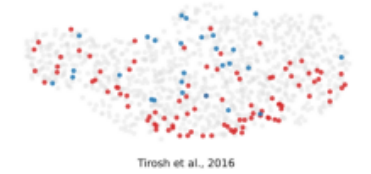
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Melanoma (Tirosh et al., 2016), SmartSeq2 — 19 patients / 4,645 cells (full atlas); malignant only: 1,257 load, 1,061 across 6 well-sampled tumours (≥72 cells) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Tirosh's melanoma malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient. Modules are globally Crest-dominant (Gast −0.51, pSom −0.07, Crest +0.15) — the canonical neural-crest melanoma signature; Gast and pSom co-vary and both oppose Crest, so the axis is developmental, not a competing hub-and-spoke bifurcation — Gate 3 has no hub to test. The axis is cell-cycle-decoupled (module–prolif |r|≤0.11; 9.5% cycling), so the whole manifold is kept. EMT-TFs TWIST1 (body/origin) → SNAI2 (terminus); ZEB2/SNAI2 broadly expressed, canonical TWIST1 near the detection floor.

Parameter choices

Min cells/sample	≥72 (6/15; cliff 98→54, ratio 1.81)	**Integration**	Harmony θ=10; PCA50; n_pcs=20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Geometry**	Linear — non-branching (Crest terminus)

Jerby-Arnon 2018 melanoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

θ derived by connectivity (θ=2 fragments the manifold into 2 components / 185 detached; θ=5 = one component, 0 detached, stable through θ=10/20). Per-patient CNV texture is θ-invariant (kNN-purity 2–8× at every θ) but no patient forms a disconnected island. Decisive: each of the 8 patients TRAVERSES the shared embryonic axis (ρ(axis,Crest) −0.66..−0.95, spanning 59–96%) — one continuous gastrulation→neural-crest gradient, not a per-patient archipelago. iLISI 1.00→3.71 — mixed & continuous; no island removed.

Gate 2 — module geography → GO (non-branching)

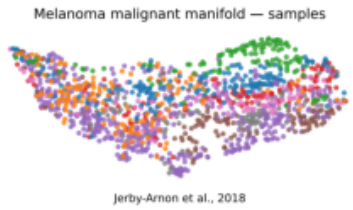
LOW-pSom, Crest-dominant melanoma (raw means Gast −0.49, pSom −0.14, Crest +0.20). Gast & pSom OVERLAP at the origin (spatial r=+0.93; joint-high enrichment 1.44) and decline together along pseudotime (Gast ρ=−0.98, pSom ρ=−0.92), opposed to a terminal neural-crest end that rises (Crest ρ=+0.93; decile pt=0.97). Crest anti-correlated in space (Crest↔Gast −0.91, Crest↔pSom −0.82); joint pSom&Crest-high depleted (0.28) — inverse gradient, not a hub. Crest peaks at the terminus, not a Gast arm — no inversion flag.

Gate 3 — cross-repression → NA

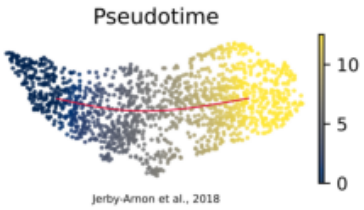
Single monotone axis — no bifurcation (curve: 2 tips, 0 forks; PPT tree yields only spurious forks) and no Gast/pSom-overlap Mixed hub, so there is no branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

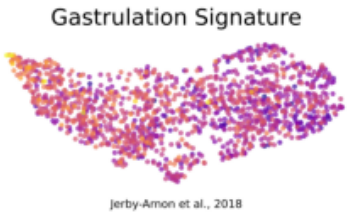
Manifold — samples



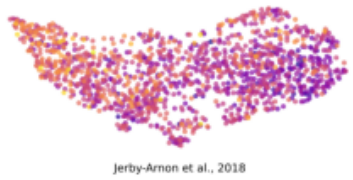
Pseudotime



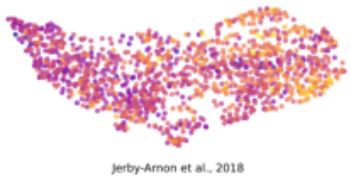
Gastrulation



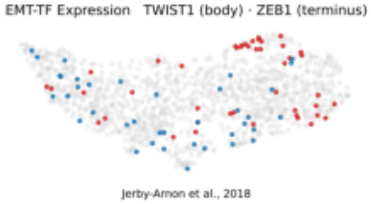
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous/metastatic melanoma (Jerby-Arnon et al., 2018), SmartSeq2 — 23 samples / 7,186 cells (master scope); malignant only: 2,018 cells; ≥100-cell floor → 1,379 cells (8 samples) on the trajectory (no island removal, no scoping).

Verdict.
Non-branching pattern reproduces — G1 GO, G2 GO, G3 NA. The melanoma malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient (Gast ρ=−0.98, pSom ρ=−0.92, Crest ρ=+0.93); it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test (Linear class). The atlas's LOW-pSom, Crest-dominant case — Crest the only positive module (+0.20), Gast (−0.49) and pSom (−0.14) globally weak, the neural-crest identity of a neural-crest-derived tumour. Two decisive derived calls: θ=5 by connectivity, and a patient-SHARED continuous-axis test clearing Gate 1 despite θ-invariant per-patient CNV texture. EMT terminus-loaded: ZEB1 marks the neural-crest terminus, TWIST1 the body factor (contrast Puram's SNAI1/ZEB1).

Parameter choices

Cells / patients	1,379 / 8 (≥100 floor, 0 islands)	**Integration**	Harmony θ=5 (connectivity); PCA50
Island removed	none (patient-shared axis)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	Gast/pSom-high, Crest-low tip	**Geometry**	Linear (non-branching); 2 tips / 0 forks
EMT-TFs	TWIST1 (body) · ZEB1 (terminus); terminus-loaded, no root TF		

Li 2019 Melanoma — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Mixed & continuous. At Harmony θ=40 the 3 samples share ONE connected component (0 detached cells), sample- and cell-cycle-mixed throughout. The 2-island split at θ=20 is a technical MARS-seq low-complexity axis (minor island 11/12 markers are MT-pseudogenes / lincRNAs) merged by stronger integration, not deleted.

Gate 2 — module geography → GO (non-branching)

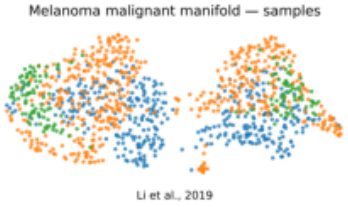
Embryonic pattern present (non-branching). pSom/Gast co-localize at the origin with organized Gast↔pSom anti-correlation (r=−0.61); Crest is an inverse terminal gradient (Crest↔pSom r=−0.59) peaking at the terminus (decile pseudotime 0.80), NOT on a Gast arm — no inversion flag. No bifurcation.

Gate 3 — hub cross-repression → NA

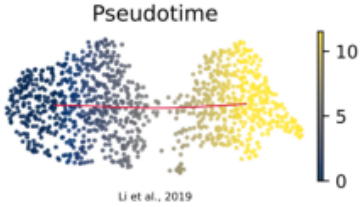
Not applicable — the manifold is non-branching (scFates curve: 2 tips, 0 forks, fold self-approach 2.98; a PPT tree adds no stable fork). With a single axis and no bifurcation there is no hub to test for mutual Gast↔pSom cross-repression, so Gate 3 is NA (branching-only).

STEP FIGURES

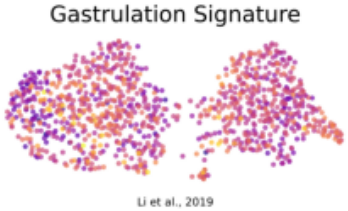
Manifold — samples



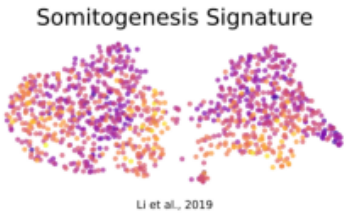
Pseudotime



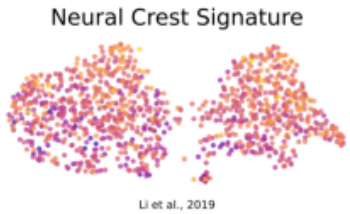
Gastrulation



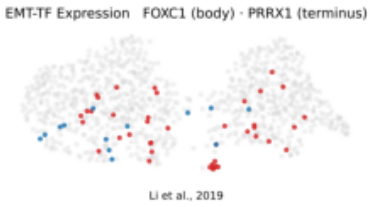
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Melanoma (Li et al., 2019), MARS-seq — 1,082 malignant cells (3 samples ≥ threshold); low-pSom, Crest-terminal melanoma (neural-crest lineage).

Verdict.
The embryonic program REPRODUCES as a non-branching axis — a single gastrulation/somitogenesis → neural-crest developmental gradient (Gate 1 GO / Gate 2 GO / Gate 3 NA), the expected melanoma (neural-crest lineage) pattern. EMT-TFs FOXC1 (body/origin) → PRRX1 (terminus).

Parameter choices

Min-cell threshold	1160→1082 cells; dominant cliff 154→21 (ratio 7.3); geom-mean 56 keeps p25/p21/p26, drops 17 spillover	**Integration**	Harmony θ=40 on sample (first strength giving 1 continuous component; θ=20 leaves 2 islands, θ=80 over-fragments); n_pcs=20
Geometry / root	Non-branching scFates curve; root = Gast/pSom-high, Crest-low tip; fold-free (SA 2.98); corr 0.99	**Module means (raw)**	Gast −0.015 · pSom +0.016 · Crest −0.002 (Linear-pSom/Crest)
EMT-TFs	FOXC1 (body/origin) · PRRX1 (terminus); wide panel (Snail/Zeb/Twist + FOXC/PRRX); VIM excluded		

Li 2019 Melanoma — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Mixed + continuous. Harmony θ-sweep read on the embedding + per-neighbourhood iLISI (1.00→2.00 of 3 samples at θ=5); all 3 samples populate both density lobes (no patient archipelago). One connected component; lobes bridged. Sub-region biology settled: lobes near-equal cycling (0.58/0.64) and no developmental module tracks proliferation → one developmental axis, whole manifold kept.

Gate 2 — module geography → GO (non-branching)

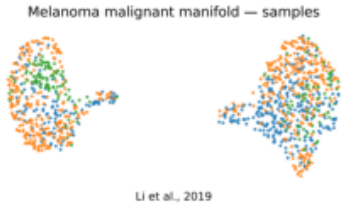
PPT tree over-branches within each lobe at every setting (13–20 forks) = noise-fitting, not a hub→two-arms bifurcation; the manifold is one elongated axis, fit by a fold-free curve (2 tips, 0 forks). Single Gast/pSom-high origin → pSom-leaning axis → inverse terminal Crest gradient (Crest-high decile pseudotime = 1.00). Crest is terminal, not on a Gast arm → no inversion flag.

Gate 3 — hub cross-repression → NA

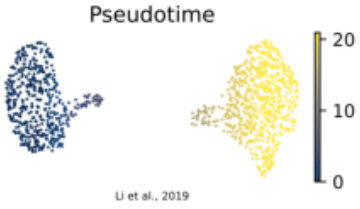
Branching-only. The manifold is non-branching (single continuous curve), so there is no bifurcation hub to test for Gast↔pSom cross-repression. Gate 3 not applicable.

STEP FIGURES

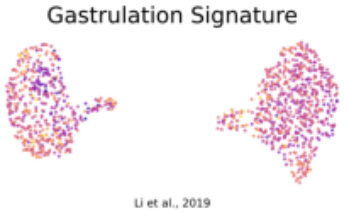
Manifold — samples



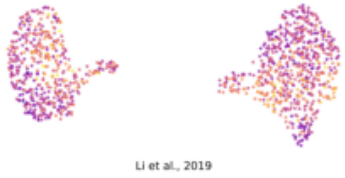
Pseudotime



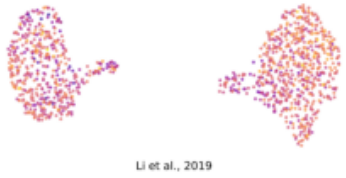
Gastrulation



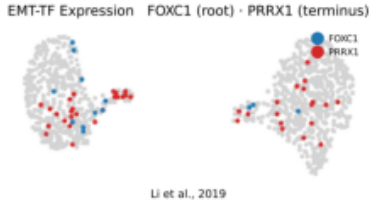
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Melanoma, MARS-seq (Li et al., 2019) — 1,160 malignant cells → 1,082 after per-sample threshold (3 samples).

Verdict.
Pattern reproduces as the non-branching case: one embryonic developmental axis from a Gast/pSom-high origin to a Crest-high terminus, sample-integrated and continuous. Gate 1 GO / Gate 2 GO / Gate 3 NA. Geometry Linear-pSom/Crest; global means Gast −0.015 / pSom +0.016 / Crest −0.002. EMT-TFs FOXC1 (early/root) → PRRX1 (late/terminus).

Parameter choices

Threshold	56 (gap-finding: 154→21, 7.3× drop; derived)	**Integration**	Harmony θ=5 (near-max iLISI mixing; θ=10 fragments the layout); n_pcs=20
Root	Gast/pSom-high curve tip (origin; Crest late) — not a proliferation state	**Geometry**	Non-branching single continuous curve (Linear-pSom/Crest); means Gast −0.015 / pSom +0.016 / Crest −0.002
Cell cycle	scored (no publication phase calls); no module tracks proliferation	**EMT-TFs**	FOXC1 (early/root) → PRRX1 (late/terminus), detection-gated

Li 2019 Melanoma — analysis summary

RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ-sweep read on the regenerated embedding + per-neighbourhood iLISI (never a scalar): the kNN graph is ONE connected component at every θ∈{2,5,10,20} × n_pcs∈{20,30,40}; iLISI rises 1.30→1.80/3 and plateaus (θ=10 settled). Each of the 3 patients spans BOTH density lobes (p21 43/57, p25 59/41, p26 58/42) — a patient-shared axis, not an archipelago. The two visual poles are biologically identical (signatures + cell cycle equal), so they are density poles of one manifold.

Gate 2 — module geography → GO (non-branching)

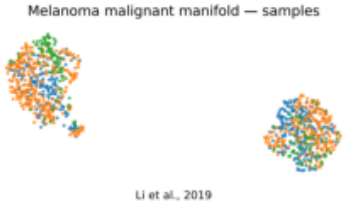
Low-signal melanoma (raw means Gast −0.015, pSom +0.016, Crest −0.002). Gast & pSom co-localise at the origin and decline along pseudotime (Gast ρ=−0.47; pSom flat ρ=+0.05) opposed to a terminal neural-crest end that rises (Crest ρ=+0.59; Crest-high decile pt=1.00). Crest anti-correlated with both in space (Crest↔Gast −0.37, Crest↔pSom −0.43); joint Gast&pSom-high enrichment 0.20 (<1) — an inverse terminal gradient, not a co-expression hub. Crest peaks at the terminus, not on a Gast arm — no inversion flag.

Gate 3 — hub cross-repression → NA

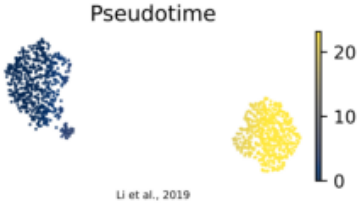
Single monotone axis — no bifurcation (curve: 2 tips, 0 forks, fold-free self-approach 2.99; a PPT tree on the same manifold also yields 0 forks) and no Gast/pSom co-expression hub, so there is no branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

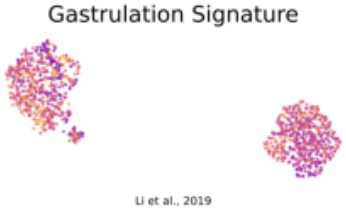
Manifold — samples



Pseudotime



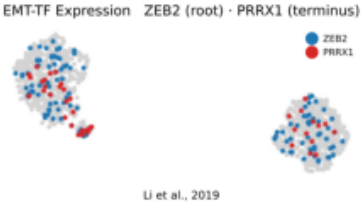
Gastrulation



Somitogenesis

Neural Crest

EMT-TF expression



Dataset.
Melanoma malignant cells (Li et al., 2019), MARS-seq — an immune-focused atlas: 1,160 of 78,336 cells are malignant; ≥56-cell dominant-gap floor keeps 3/20 samples → 1,082 cells on the trajectory.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous, sample-integrated gastrulation → terminal-neural-crest axis rooted at the Gast/pSom-high origin. Gate 1 GO / Gate 2 GO / Gate 3 NA; no Puram-style hub-and-spoke bifurcation (curve 0 forks), so Gate 3 has no hub to test. EMT-TFs ZEB2 (body) / PRRX1 (sparse), detection-gated on this shallow dataset.

Parameter choices

Sample threshold	56 cells (dominant multiplicative gap 154→21, ratio 7.3×); 3/20 samples, 1,082 cells	**Overdispersed genes**	217 (scFates find_overdispersed); no sc.pp.scale before PCA
Integration	Harmony θ=10 (iLISI plateau; strongly patient-private melanoma programs)	**Embedding**	n_pcs=30 (mid robust 20–40), n_neighbors=15; single elongated axis (aspect 7.5)
Geometry	Non-branching scFates curve, N=20 λ=0.3 μ=100; 2 tips/0 forks, fold-free (SA 2.99); PPT 0 forks	**Root**	tip 0 = Gast/pSom-high, Crest-low origin
Module means	Gast −0.015 / pSom +0.016 / Crest −0.002 (low-signal melanoma)	**EMT-TFs**	ZEB2 (body) ↔ PRRX1 (sparse terminus); VIM excluded

Li 2019 melanoma — analysis summary

RUN 4 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Harmony reaches near-maximal sample mixing (iLISI 1.00→1.78 of 3), but the malignant structure resolves into TWO density lobes that are transcriptomically disconnected — inter-cluster PAGA on the PCA-space graph ≈ 0.001 (~0), invariant across n_pcs 20/30 × θ 10/15, and stronger integration never merges them. Neither lobe reproduces the embryonic axis (in both, Gast/pSom ANTI-correlate rather than co-vary). A broken manifold with no cell-state continuum — a trajectory is not forced.

Gate 2 — module geography → N/A

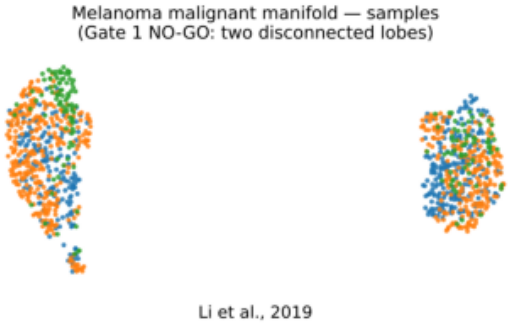
Not evaluated — Gate 1 NO-GO halts the chain before the signature step.

Gate 3 — cross-repression → N/A

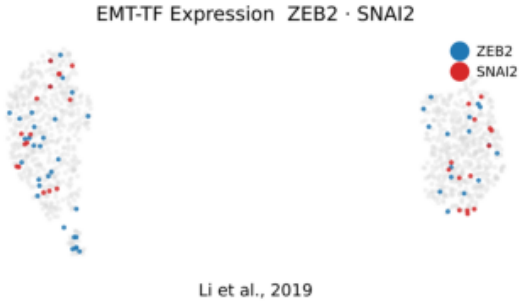
Not evaluated — branching-only; no trajectory fit.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Melanoma (Li et al., 2019), MARS-seq immunotherapy atlas — 1,082 malignant cells across 3 tumours (3/20 clear the ≥56-cell threshold). Gate 1 NO-GO: two disconnected lobes.

Verdict.
The embryonic-program pattern does NOT reproduce in Li et al. 2019. The small malignant compartment (1,082 cells, 3 tumours) forms two transcriptomically disconnected lobes rather than one integrated, continuous cell-state manifold, and the Gast/pSom/Crest modules are salt-and-pepper with no coherent geography — Gast and pSom anti-correlate in each lobe instead of co-varying. Gate 1 is a NO-GO; Gates 2–3 are not evaluated. EMT-TFs (ZEB2, SNAI2) are reported unconditionally on the best-attempt embedding, without a pseudotime claim.

Parameter choices

Malignant load	1,160 cells of 78,336 (MARS-seq atlas); Malignant subset only	**Sample threshold**	≥56 cells (largest gap 154→21, ratio 7.3) → p25/p21/p26, 1,082 cells
Overdispersed genes	217 (scFates GAM); PCA n_pcs=20	**Integration**	Harmony on sample θ=10 (near-maximal iLISI; higher over-corrects, no merge)
Continuity test	UMAP radius-graph lobes + PAGA on the PCA graph (layout-independent)	**EMT-TFs**	ZEB2 (43%) + SNAI2 (27%), top-decile expressers; no pseudotime claim
Geometry	none — trajectory not fit (Gate 1 NO-GO)		

Paulson 2020 Merkel cell carcinoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (sample, θ=10) collapses the 3 tumour samples onto ONE continuous component (100%, 0 detached); per-neighbourhood iLISI 1.0→2.25 (cross-patient 1.75/2); every sample traverses the shared embryonic axis end-to-end (|ρ(axis,Crest)|≈0.87, span 100%) — mixed & continuous, not a per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

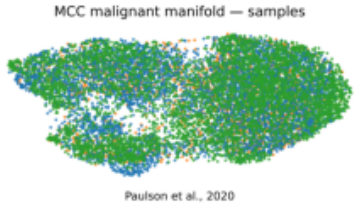
Gast & pSom co-vary along one axis (smoothed r=+0.38; joint-high enrichment 1.22 — no two-arm divergence); Crest is an inverse TERMINAL gradient (pSom↔Crest r=−0.70; Crest-high decile mean pt=0.96) peaking at the terminus, not a Gast arm — no inversion flag.

Gate 3 — cross-repression → NA

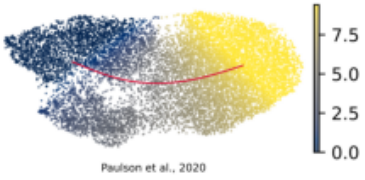
Branching-only. No diverging Gast/pSom hub (single fold-free curve, 0 forks) — no bifurcation to test, so Gate 3 does not apply.

STEP FIGURES

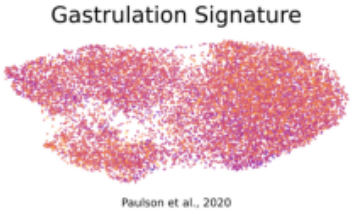
Manifold — samples



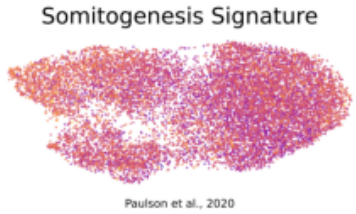
Pseudotime



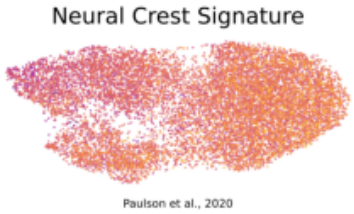
Gastrulation



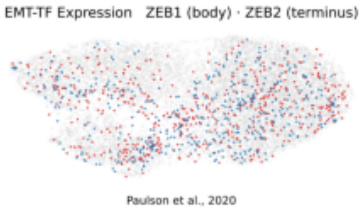
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Merkel cell carcinoma (neuroendocrine skin) (Paulson et al., 2020), 10x — 12,124 malignant cells (3 tumours, 2 patients); non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching): a single continuous gastrulation/somitogenesis → terminal neural-crest axis, rooted at the Gast/pSom-high origin. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB1 (body) · ZEB2 (terminus), both origin/body-loaded; only factors >3% floor.

Parameter choices

Min-cell threshold	≥100 floor (smallest sample 2,031); 3/3 kept, 12,124 cells	**Integration**	Harmony on sample θ=10 (swept 2/5/10); one continuous component
n_pcs / spread	20 / 1.3 (PC1=0.23 CNV-dominated)	**Overdispersed genes**	2,075 (scFates find_overdispersed)
Trajectory	scFates curve N=20, λ=0.3, μ=100 (2 tips, 0 forks, SA 2.83)	**Root**	tip 2 = Gast/pSom-high, Crest-low (Crest is late)
Module means (raw)	Gast +0.037 · pSom +0.051 · Crest −0.006	**EMT-TFs shown**	ZEB1 (14.4%) · ZEB2 (19.1%); both origin/body-loaded

Biermann 2022 melanoma — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

30 samples (brain-met MBM + peripheral-met MPM, sn/sc) with private melanoma CNV: pre-Harmony same-sample fraction 0.985. Harmony(θ=10) collapses it to one continuous component (99.97% of cells; 169 detached specks removed), mixing iLISI 1.0→6.4/30, met-sites interleave (1.78/2). Every one of 28 patients traverses the shared embryonic axis end-to-end (median span 94%) — a patient-SHARED continuous gradient, not a per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

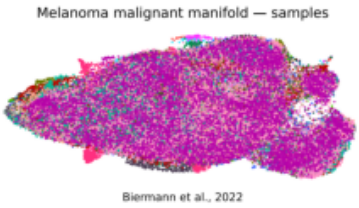
LOW-pSom, Crest-dominant melanoma (raw means Gast −0.05, pSom −0.02, Crest +0.05 — Crest the only positive module). No Gast/pSom co-expression hub (joint-high enrichment 0.91) — the two cross-repress in space (r=−0.57). Crest is an inverse TERMINAL gradient: depleted where pSom/Gast are high (joint 0.23 / 0.29), peaking at the trajectory terminus (Crest decile pt 0.60, latest; ρ=+0.32). Crest peaks at the terminus, not a Gast arm — no inversion flag.

Gate 3 — cross-repression → NA

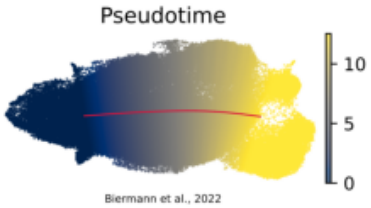
Single monotone axis — no bifurcation (curve: 2 tips, 0 forks, fold-free; a PPT tree yields only spurious forks) and no Gast/pSom co-expression Mixed hub, so there is no branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

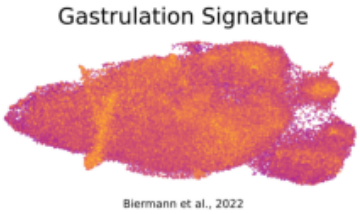
Manifold — samples



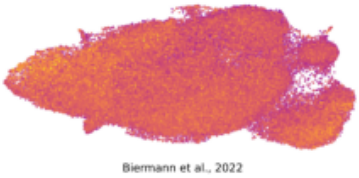
Pseudotime



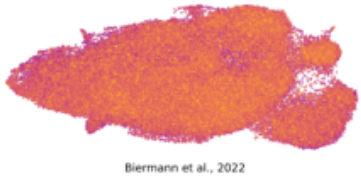
Gastrulation



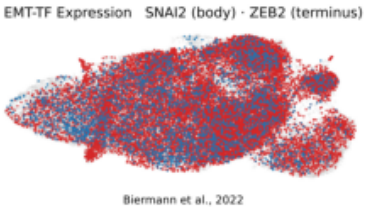
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Metastatic melanoma (Biermann et al., 2022), 10x — 30 samples (brain-met MBM + peripheral-met MPM, single-nucleus/single-cell); 136,973 cells master scope, 91,525 malignant; ≥100-cell floor keeps all 30 samples, 169 detached specks removed → 91,356 on the trajectory.

Verdict.
Non-branching pattern reproduces — G1 GO, G2 GO, G3 NA. The malignant manifold is a single monotone axis rooted at the Gast/pSom-high origin → terminal neural-crest end (pSom ρ=−0.50, Crest ρ=+0.32 along pseudotime); it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The atlas's LOW-pSom, Crest-dominant case — Crest the only positive module, the neural-crest identity of a neural-crest-derived tumour. Cell-cycle confound checked: pSom tracks proliferation (r=+0.50) but Crest, the trajectory-defining module, is decoupled (−0.33), so the axis is developmental and the whole manifold is kept. EMT terminus-loaded: ZEB2 marks the neural-crest terminus (79%), SNAI2 the body (14%).

Parameter choices

Cells / patients	91,356 / 30 (≥100 floor)	**Integration**	Harmony θ=10; PCA 20PC; spread 1.3
Island removed	169 cells (0.18%, sn micro-blobs)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**Geometry**	Linear (non-branching); 2 tips / 0 forks
EMT-TFs	SNAI2 (body) · ZEB2 (terminus); terminus-loaded		

Yost 2019 BCC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=5) integrates all 5 patients: same-sample neighbour fraction 0.94→0.41, iLISI median 1.02→3.21 of 5. One connected component (0 islands). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

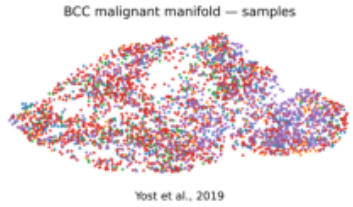
Single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.61), both oppose Crest (pSom↔Crest r=−0.84, Gast↔Crest r=−0.82). Along pseudotime Gast/pSom fall (ρ −0.43/−0.75), Crest rises to a terminal tip (decile pt=0.84); joint pSom&Crest enrichment 0.32 (<1 = inverse gradient, not a hub).

Gate 3 — cross-repression → NA

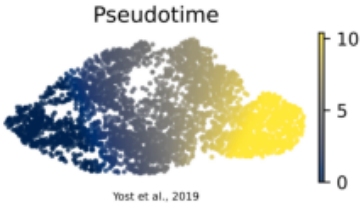
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

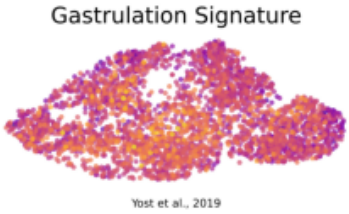
Manifold — samples



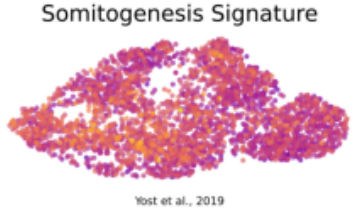
Pseudotime



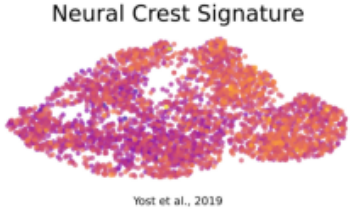
Gastrulation



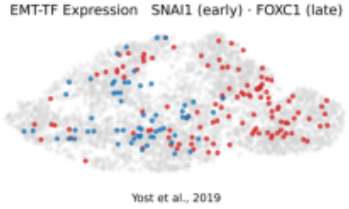
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous basal cell carcinoma (Yost et al., 2019), 10x — 3,391 malignant cells over 5 patients (5/16 kept at ≥110-cell floor); single low-cycling body (~11% cycling). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs SNAI1 (early/root) → FOXC1 (late/terminus).

Parameter choices

Sample threshold	110 cells (geo-mean of cliff 254→48, ratio 5.29) → 5/16, 3,391 cells	**Overdispersed genes**	1007 (scFates find_overdispersed)
Integration	Harmony on sample θ=5 (mixing plateau; patient-specific batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	non-branching curve N=20, λ=0.3, μ=100; 2 tips / 0 forks (2.92)	**Root**	tip 2 = Gast/pSom-high, Crest-low origin (from signature order)
Module means (raw)	Gast −0.015 · pSom +0.046 · Crest −0.011 (Linear-pSom/Crest)	**EMT-TFs**	SNAI1 (early/root) → FOXC1 (late/terminus); detection-gated; FOXC/PRRX in panel

Ji 2020 cSCC — analysis summary

RUN 1 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8 patients share one continuous, sample-mixed manifold (Harmony θ=10); detached islands removed.

Gate 2 — module geography → GO (non-branching)

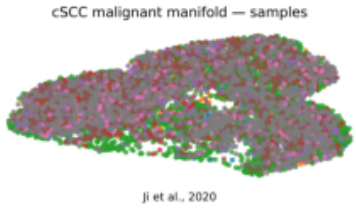
Gast+pSom co-vary (r=+0.13) and both oppose Crest (Gast~Crest=−0.70, pSom~Crest=−0.64): one developmental axis with an inverse terminal Crest gradient (non-branching).

Gate 3 — hub cross-repression → NA

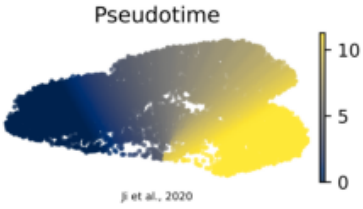
Branching-only test; manifold is non-branching (principal curve: 2 tips, 0 forks) — not applicable.

STEP FIGURES

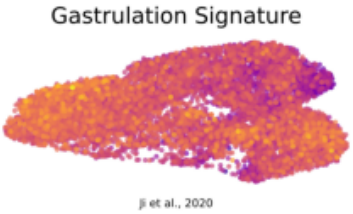
Manifold — samples



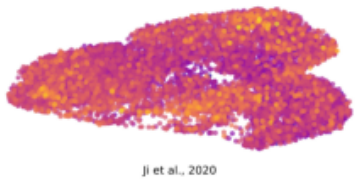
Pseudotime



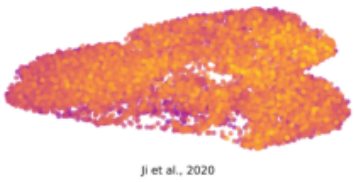
Gastrulation



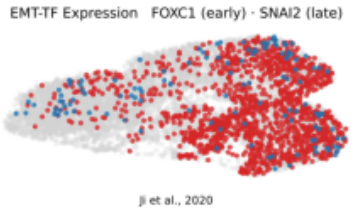
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous Squamous-Cell Carcinoma (Ji et al., 2020) — 10,531 malignant keratinocytes from 8 cSCC patients (published malignant calls; thin samples removed by a data-driven multiplicative-gap threshold). Harmony integration on patient (θ=10), principal-curve trajectory (ElPiGraph) on the whole manifold. Module means: Gast −0.032, pSom +0.042, Crest +0.023 — pSom/Crest globally stronger than Gast.

Verdict.
Non-branching — Linear pSom/Crest. A single developmental axis: a Gast/pSom-high differentiating-keratinocyte body (trajectory root) grading into a neural-crest-high basal/TSK leading edge (terminus). Along pseudotime Gast (ρ=−0.51) and pSom (ρ=−0.44) fall while Crest (ρ=+0.37) rises. The axis is verified separable from proliferation (Gast/pSom↔Crest opposition survives partialling the cell-cycle score). Gate 1 GO, Gate 2 GO, Gate 3 NA. EMT-TFs FOXC1 (early) → SNAI2 (late).

Parameter choices

Malignant cells	10,531 (8/10 patients kept)	**Sample threshold**	multiplicative-gap geomean (drops P3, P5)
Integration	Harmony on patient, θ=10 (mixing-sweep selected)	**Embedding**	PCA 20 PC · neighbours 30 · UMAP spread 1.3
Trajectory	ElPiGraph curve · Nodes=20 · λ=0.3 · μ=100	**Topology**	2 tips, 0 forks · fold-free (self-approach>1)
Root	Gast/pSom-high, Crest-low tip (data-derived)	**Modules**	Gast 76/81 · pSom 88/98 · Crest 101/104 genes
EMT-TF pair	FOXC1 (early) → SNAI2 (late)	**Scope**	whole manifold (holds low-cycling too)

Ji 2020 cSCC — analysis summary

RUN 2 OF 10

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

Patients thoroughly mixed after Harmony (θ=5, chosen on the integration sweep — mixing visually clean, 10x precedent); one connected component (99.7% of cells). Judged on the embedding, not a scalar.

Gate 2 — module geography → GO (non-branching)

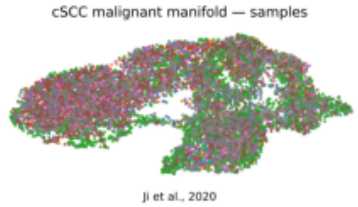
Gast & pSom overlap in the differentiating body (spatial r=+0.10) and both oppose Crest (r=−0.68 / −0.65); along pseudotime Gast declines (ρ=−0.62) and Crest rises to a terminal neural-crest / TSK edge (Crest-high decile pt=0.79). One shared developmental axis, not two competing arms.

Gate 3 — hub cross-repression → NA

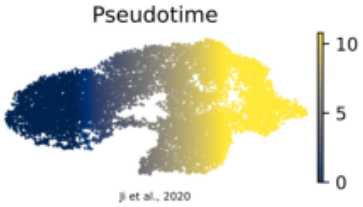
No bifurcation: the only PPT fork is the cycling compartment; the developmental axis is a single fold-free curve. Organized gene-level Gast↔pSom cross-repression (r=−0.53) IS present on the hub, but with Gast/pSom overlapping and no arm resolution this is the non-branching-with-cross-repression case (task §1), not a bifurcation.

STEP FIGURES

Manifold — samples

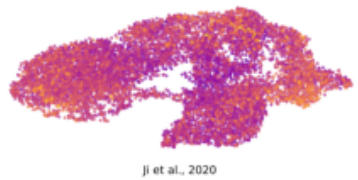


Pseudotime

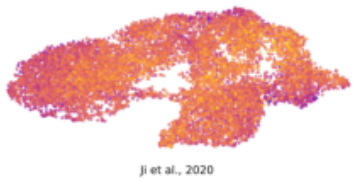


Gastrulation

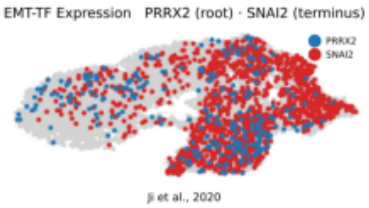
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous squamous cell carcinoma (Ji et al., 2020), 10x — 10,504 malignant cells from 8 patients (P3/P5 dropped below the count cliff).

Verdict.
The embryonic pattern reproduces as a NON-BRANCHING axis: a Gast/pSom-high differentiating origin grades to a neural-crest-high basal/TSK terminus (Crest the inverse/terminal readout). No developmental bifurcation — the sole fork is proliferation. Linear geometry; Gate 3 not applicable. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs PRRX2 (root-end, near floor) → SNAI2 (terminus).

Parameter choices

Malignant load	10,598 cells / 10 patients (publication calls)	**Sample threshold**	≥103 (geom mean of 268→40 cliff); 8/10 kept → 10,531 cells
Island removal	UMAP radius-graph (6× median NN); ~27 detached → 10,504	**Integration**	Harmony θ=5; n_pcs=20; UMAP spread=1.3
Trajectory	scFates curve (ElPiGraph) N=10 μ=100; 2 tips, 0 forks; fold-free (SA 3.0)	**Root**	Gast/pSom-high, Crest-low curve tip
Hub (context)	Mixed+Diff core; gene-level Gast↔pSom r=−0.53	**EMT-TFs**	SNAI2 (terminus); PRRX2 root-end near floor

Ji 2020 cutaneous SCC — analysis summary

RUN 3 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

θ=10 integrates 8 patients into one connected manifold; no per-sample islands.

Gate 2 — module geography → GO (branching)

Gast/pSom hub diverges into opposed Gast & pSom arms; Crest terminal on the pSom arm. A branching hub-and-spoke wishbone rooted at the Gast/pSom-high tip.

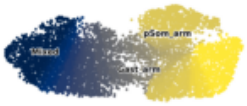
Gate 3 — hub cross-repression → GO (branching)

Mixed-hub Gast↔pSom mutual r = −0.47 (Puram −0.49); null 0.01±0.09, p<0.003.

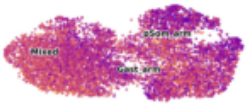
STEP FIGURES

Manifold — samples

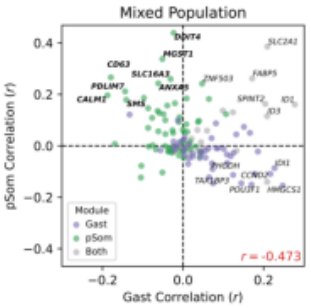
Trajectory / root



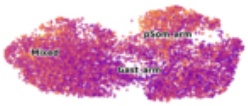
Gastrulation



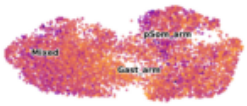
Gate 3: r = −0.47



Somitogenesis



Neural Crest



EMT-TF expression

Dataset.
Cutaneous squamous cell carcinoma (Ji et al., 2020), 10x — 8/10 patients retained; 7,085 malignant developmental cells (10,531 pre-scope).

Verdict.
Embryonic branching pattern REPRODUCES — Gast/pSom hub, two cross-repressing arms, Crest terminal on the pSom arm (r=−0.47 vs Puram −0.49). Gate 1 GO / Gate 2 GO / Gate 3 GO. A Gast/pSom hub diverges into opposed Gast and pSom arms with the neural-crest terminus on the pSom arm; EMT-TFs PRRX2 (Gast arm) / SNAI2 (pSom arm terminus).

Parameter choices

Threshold	100 cells/sample (clean 6.70× count gap 40→268); keeps 8/10 patients (10,531 cells)	**Genes**	1,443 overdispersed (scFates); PCA 50 PC
Integration	Harmony θ=10 (10x precedent); branch survives strong integration	**Cycling scope**	3,446 cells in 5 high-cycling (>50%) Leiden lobes removed → 7,085-cell fork re-embedded (Filbin)
Embedding	kNN n_pcs=20 / n_neighbors=30; UMAP spread=1.3	**Trajectory**	scFates PPT tree Nodes=60 λ=3000 σ=0.5; 4 tips / 2 forks (hub-and-spoke Y)
Hub	fork-Leiden(res 0.4) max joint Gast+pSom = 'Mixed' (1,513 cells)	**Root**	Gast/pSom-high, Crest-low tip (root_score 0.99)
EMT-TFs	PRRX2 (Gast arm) / SNAI2 (pSom arm terminus)		

Ji 2020 cSCC — analysis summary

RUN 4 OF 10

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

Harmony(θ=5) on sample over 20 PCs: patients thoroughly intermixed, one connected kNN component, one continuous body. A cycling lobe (35% of cells) is a proliferation confound → scoped to the developmental compartment for the trajectory.

Gate 2 — module geography → GO (non-branching)

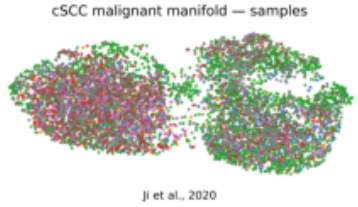
Modules present and structured, but NON-branching: Gast & pSom OVERLAP (smoothed r=+0.11) and JOINTLY oppose Crest ((Gast+pSom)–Crest r=−0.88). No Gast-arm vs pSom-arm divergence — a principal-tree Y has both terminal arms leaning the same way (Gast<pSom); Crest = inverse terminal gradient. Matches the linear Sun/Kürten references, not the branching Puram hub-and-spoke.

Gate 3 — hub cross-repression → NA

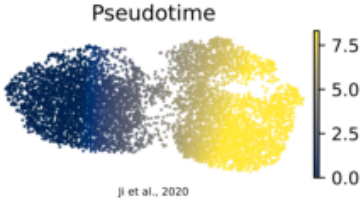
Not applicable — no bifurcation. (Context: module cross-repression on the Mixed convergence zone is r=−0.50, near the Puram anchor ≈−0.49 — the §1 'some cross-repression may persist' clause within the overlap case, not sufficient for branching without two spatially-resolved arms.)

STEP FIGURES

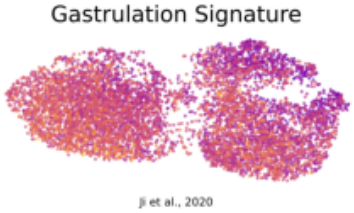
Manifold — samples



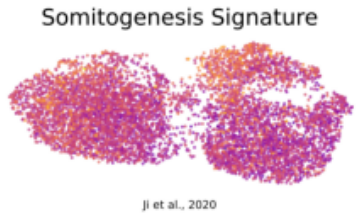
Pseudotime



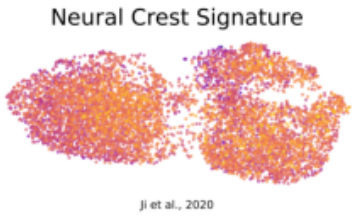
Gastrulation



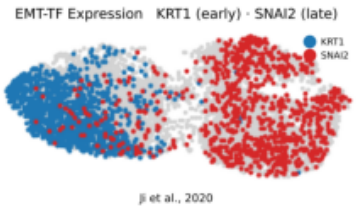
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10,598 malignant cells (8/10 samples ≥100 → 10,531; developmental compartment 6,897 after scoping the cycling lobe).

Verdict.
Embryonic programs reproduce as a NON-BRANCHING gastrulation/somitogenesis → neural-crest gradient (Gate 1 GO / Gate 2 GO non-branching / Gate 3 NA) — not the Puram-type bifurcation. Pseudotime gradient Gast ρ=−0.50, pSom ρ=−0.39, Crest ρ=+0.31. EMT/lineage TFs KRT1 (early) → SNAI2/SLUG (late).

Parameter choices

Min-cell threshold	≥100 (clean 6.7× count gap 268→40; keeps 8/10)	**Integration**	Harmony θ=5 (swept 0/2/5/10, visual mixing); PCs=20, no scaling
Cycling scope	per-Leiden cycling fraction >0.5 (3,634 cells; Gast-independent)	**Geometry**	NON-BRANCHING — principal curve (ElPiGraph), 2 tips / 0 forks, fold-free (ratio 2.97)
Root	Mixed (Gast/pSom-high, Crest-low origin — a tip, not the Crest-high terminus)	**Annotation**	data markers (KRT1/10, MMP10/PTHLH/LAMC2, TP63) + Ji 2020 subtypes (TSK, Tumor-KC)
EMT-TFs	KRT1 (early) vs SNAI2/SLUG (late)		

Ji 2020 cutaneous SCC — analysis summary

RUN 5 OF 10

GO / NO-GO GATES

Gate 1 — integration & continuity →GO

Harmony (θ=5) merges 8 samples into one connected kNN component; continuous, no sample islands.

Gate 2 — module geography →GO (branching)

Gast/pSom-high hub diverges into a Gast-arm and a pSom-arm; Crest a terminal readout toward the tips.

Gate 3 — hub cross-repression →GO (branching)

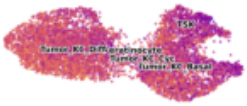
Hub Gast↔pSom module r = −0.37 (perm p = 0.007), competitive; Puram −0.49 / Filbin −0.34 anchors.

STEP FIGURES

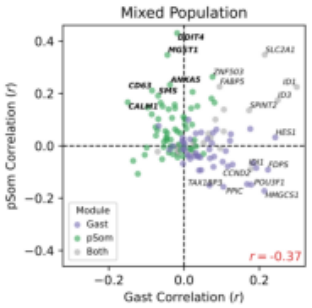
Manifold — samples

Trajectory / root

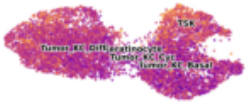
Gastrulation



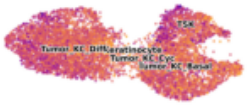
Gate 3: r = −0.37



Somitogenesis



Neural Crest



EMT-TF expression

Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10,531 malignant cells over 8 samples after thresholding (8 of 10; clean 6.7× per-sample count gap, keep ≥100). Developmental compartment 6,800 cells (3,731 cycling scoped out).

Verdict.
The embryonic pattern reproduces as a BRANCHING hub-and-spoke — a Gast/pSom-high hub diverging into competing Gast and pSom arms with hub cross-repression r = −0.37, the Puram signature on squamous biology. Gate 1 GO / Gate 2 GO / Gate 3 GO. EMT-TFs SNAI2 (50%, late/terminal) + PRRX2 (9%, root-end).

Parameter choices

Min-cell threshold	≥100 per sample (6.7× count gap 40→268); drops P3, P5	**Overdispersed genes**	1,443 (scFates find_overdispersed)
Embedding	PCA 50 → Harmony(sample, θ=5) → kNN n_pcs=20 (scree elbow PC17), UMAP spread=1.3	**Trajectory / geometry**	scFates PPT tree (branching), Nodes=50 λ=2000 σ=0.5; dev compartment, fold-free
Root (hub)	Gast/pSom-high, Crest-low convergence node (Crest late = terminal)	**Hub identity**	Hub milestone, 1,195 cells; module mutual r = −0.37 (p=0.007)
Crest arm	terminal readout toward arm tips (ρ_pt = +0.25); not on the cross-repression axis	**EMT-TFs**	SNAI2 (50%, late/terminal) + PRRX2 (9%, root-end); SNAI1/ZEB1 near 10x floor

Ji 2020 cSCC — analysis summary

RUN 6 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/10 samples (≥103 cells; data-derived gap 40→268, r=6.7) intermix on one continuous manifold — 0 detached islands (UMAP radius-graph). Cell cycle (35%) is an orthogonal compartment, not a separate lineage.

Gate 2 — module geography → GO (non-branching)

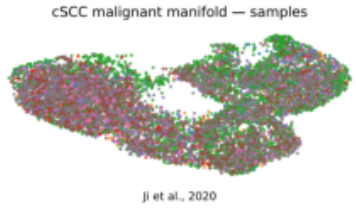
Gast/pSom co-vary (r=+0.11) and decline along pseudotime (ρ=−0.48 / −0.44); Neural Crest anti-correlated (r=−0.69 / −0.63), rising to a terminal readout (ρ=+0.35; crest-high decile mean pt=0.72). One dominant signature axis (67% of variance).

Gate 3 — hub mutual exclusivity → NA

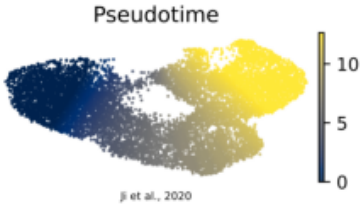
Single monotone axis — every trajectory tip is pSom-leaning (no Gast-dominant arm), so there is no bifurcation and no Mixed hub; no branch pair to test for mutual exclusivity.

STEP FIGURES

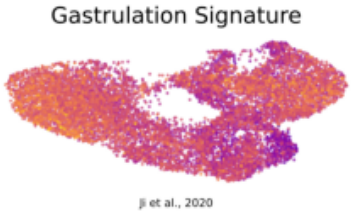
Manifold — samples



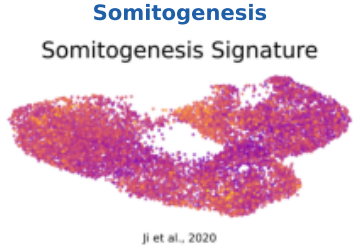
Pseudotime



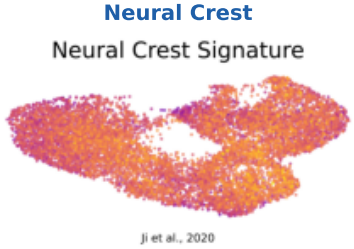
Gastrulation



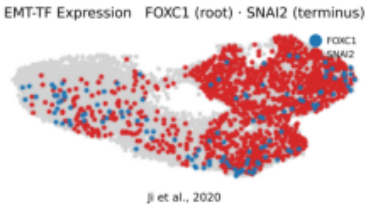
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10 patients / 48,164 cells (atlas); malignant only: 10,598 loaded, 10,531 across 8 samples (≥103) enter the trajectory. Non-branching — Linear, Gast/pSom → terminal Neural Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Ji 2020's malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (linear), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Cell cycle is orthogonal to the embryonic axis (pt ρ≈0), so the whole manifold is kept — the reference relationship holds in every compartment. EMT-TFs FOXC1 (early/root) → SNAI2 (late/terminus); canonical SNAI1/ZEB1/TWIST near the 10x detection floor.

Parameter choices

Min cells/sample	≥103 (8/10; gap 40→268, r6.7)	**Integration**	Harmony(sample) θ=5; PCA 20PC
Root node	tip 2 = Gast/pSom-high, Crest-low	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Scoping	none — whole manifold	**Geometry**	Linear (non-branching); no hub
Subtypes	Ji 2020 calls + markers	**EMT-TFs**	FOXC1 (early) → SNAI2 (late)

Ji 2020 cutaneous SCC — analysis summary

RUN 7 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Cell-cycle regressed out (de-novo S/G2M scores), Harmony on sample (θ=2), n_pcs=20. Embedding is sample-integrated (all 8 patients interleave; 30-NN cross-sample fraction 0.63 vs 0.78 max) and one continuous body (single radius-graph component). Without CC regression a spurious cycling axis dominates a lobe — the confound the prior linear read never removed.

Gate 2 — module geography → GO (branching)

Basal Gast/pSom hub diverges into a Gast-leaning differentiation arm (KRT1/10; Gast +0.00 vs hub −0.04) and a pSom-leaning TSK / leading-edge arm (pSom +0.06). Crest is a terminal readout enriched on the pSom (TSK) arm and depleted at the differentiation tip — uncoupled from the Gast/pSom axis (acceptable).

Gate 3 — hub cross-repression → GO (branching)

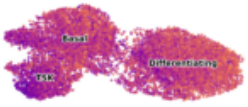
On the basal hub (n=3,756), Gast-loading and pSom-loading genes anti-correlate: mutual r = −0.38 (Puram anchor −0.49; Filbin branching −0.34). Competitive cross-repression at the branch point.

STEP FIGURES

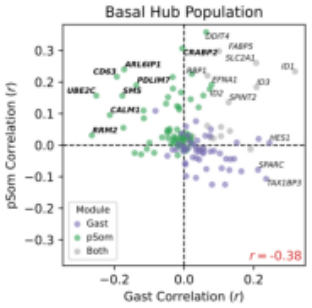
Manifold — samples

Trajectory / root

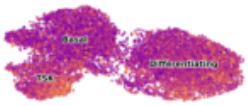
Gastrulation



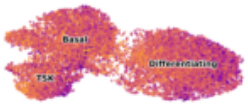
Gate 3: r = −0.38



Somitogenesis



Neural Crest



EMT-TF expression

Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10,523 malignant cells (8/10 patients). Geometry: BRANCHING (hub-and-spoke).

Verdict.
Pattern REPRODUCES (branching). A basal progenitor hub diverges into a keratinocyte-differentiation arm (Gast) and a TSK / leading-edge EMT arm (pSom) with hub cross-repression r = −0.38; Crest is a terminal readout on the pSom arm. Gate 1 GO / Gate 2 GO / Gate 3 GO — the third branching case in the atlas. EMT-TFs SNAI2 / PRRX2.

Parameter choices

Threshold	keep samples ≥103 (gap-finding 40→268, ~6.7×) → 8/10 patients, 10,598 → 10,531 → 10,523 (islands)	**Cell cycle**	de-novo S/G2M scores regressed out of the over-dispersed matrix (clean-embedding step)
Embedding	1,443 over-dispersed genes, PCA 50 → Harmony(sample, θ=2), n_pcs=20, UMAP spread=1.3	**Trajectory**	scFates PPT tree (Nodes=80, λ=600, σ=0.4) on X_umap; root = basal-max fork
Hub	Basal progenitor cluster (branch-point / root convergence zone); subtypes by markers + Ji calls	**EMT-TFs**	SNAI2 / PRRX2 (Gast/pSom arms)

Ji 2020 cutaneous SCC — analysis summary

RUN 8 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Malignant cells from 8/10 samples (≥100 cells) form one continuous, sample-mixed manifold (iLISI≈4.0; single kNN component; 13 detached cells removed). Cell-cycle regression was required — it is the largest program and otherwise dominates the embedding.

Gate 2 — module geography → GO (branching)

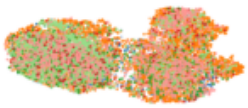
The embryonic modules resolve into a branching wishbone: a Gast/pSom co-expression hub diverges into a Gast-leaning (Basal) arm and a pSom-leaning (Differentiated) arm. Crest is a terminal readout at the tips, uncoupled from Gast/pSom.

Gate 3 — hub cross-repression → GO (branching)

Within the 4,888-cell Mixed hub, per-gene Gast and pSom loadings mutually anti-correlate: r = −0.54 (Puram anchor ≈ −0.49) — competitive cross-repression, the bifurcation signature.

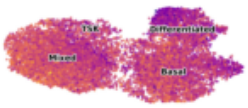
STEP FIGURES

Manifold — samples

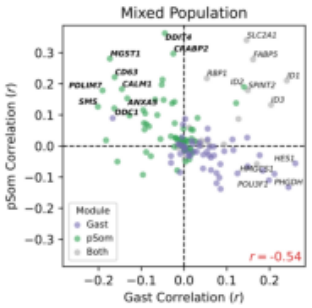


Trajectory / root

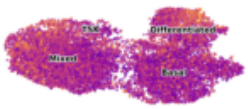
Gastrulation



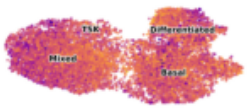
Gate 3: r = −0.54



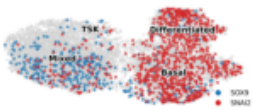
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10 patients, 48,164 cells; 10,598 malignant → 10,518 after QC. Subtypes Mixed / Basal / Differentiated / TSK.

Verdict.
Pattern reproduces — all three gates GO. Branching wishbone rooted at the Gast/pSom hub; competitive Gast↔pSom cross-repression (r=−0.54); Crest terminal, uncoupled. Gate 1 GO / Gate 2 GO / Gate 3 GO. EMT-TFs SOX9 (hub/basal), SNAI2 (Differentiated arm).

Parameter choices

Min cells/sample	≥100 (gap 40→268, 6.7×; keeps 8/10, drops P3, P5)	**Cell-cycle regression**	S + G2M scores regressed — KEY to a clean manifold
Island removal	UMAP radius-graph, largest component (−13 cells)	**Integration**	Harmony on sample, θ = 2, 20 it; n_pcs = 20
Embedding	UMAP min_dist = 0.4, spread = 1.0	**Trajectory**	scFates PPT tree, Nodes = 60, λ = 2000, σ = 1.5
Geometry / root	Branching (3-tip wishbone); root = Mixed hub (node 16, a tip)	**Hub / Gate-3 r**	Mixed (4,888 cells); r = −0.54
EMT-TFs	SOX9 (hub/basal), SNAI2 (Differentiated arm) — Snail/Zeb/Twist ~absent		

Ji 2020 cutaneous SCC — analysis summary

RUN 10 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/10 samples form one continuous, sample-mixed manifold (iLISI≈4.0; single kNN component; 13 detached cells removed). Cell-cycle regression required.

Gate 2 — module geography → GO (branching)

A branching wishbone: three lobes (Basal, Differentiated, Mixed) meet at one convergence with a smooth monotone pseudotime (r≈0.96). Rooted at the Basal tip (COL17A1^{+}/TP63^{+} progenitor).

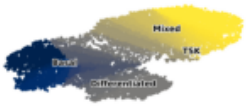
Gate 3 — cross-repression → GO

On the Basal root population (3,524 cells), Gast↔pSom gene loadings mutually anti-correlate: r = −0.41 (permutation p≈0.0005) — within range of branching anchors (Puram −0.49, Filbin −0.34).

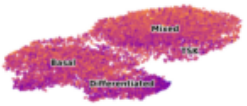
STEP FIGURES

Manifold — samples

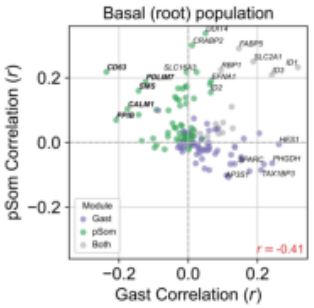
Trajectory / root



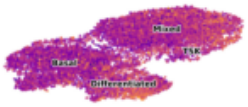
Gastrulation



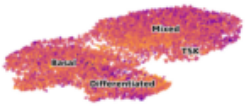
Gate 3: r = −0.41



Somitogenesis



Neural Crest



EMT-TF expression

Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10 patients, 48,164 cells; 10,598 malignant → 10,518 after QC. Subtypes Mixed / Basal / Differentiated / TSK. Branching (hub-and-spoke wishbone).

Verdict.
Branching — clean 3-tip wishbone rooted at the Basal progenitor tip, competitive Gast↔pSom cross-repression (r=−0.41, p≈0.0005) at the Basal root; Crest terminal. All three gates GO. EMT-TFs SOX9 / SNAI2.

Parameter choices

Min-cell threshold	≥100 (gap 40→268; keeps 8/10)	**Cell-cycle regression**	S + G2M regressed — key to clean manifold
Integration	Harmony on sample, θ=2; n_pcs=20	**Embedding**	UMAP min_dist=0.4, spread=1.0
Trajectory	scFates PPT N=40, λ=700, σ=0.8 (3-tip wishbone)	**Root**	Basal tip (COL17A1^{+}/TP63^{+} progenitor)
Gate-3 pop / r	Basal root (3,524 cells); r=−0.41, p≈0.0005	**EMT-TFs shown**	SOX9 (basal) / SNAI2 (Differentiated arm)